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MIL-STD-3087(AV)
16 June 2026

DEPARTMENT OF DEFENSE
STANDARD PRACTICE

ARMY AIRWORTHINESS QUALIFICATION
ENGINE CONTROL SYSTEM
AND ACCESSORY COMPONENTS

AMSC 10633 FSC ENVR

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FOREWORD

1. This standard (STD) is approved for use by the Combat Capabilities Development Command (DEVCOM), Aviation & Missile Center (AvMC), Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense (DoD). Unless otherwise stated in contractual documentation, 'Airworthiness Authority' as used in this document refers to the office of the delegated Airworthiness Authority within the invoking DoD Command.

2. This standard defines requirements for the bench qualification of engine control system components and engine accessories.

3. This standard is intended for application to manned or unmanned, fixed- or rotary-wing military aircraft.

4. The testing protocols and procedures defined herein are to be followed by engine manufacturers (and control system and accessory component suppliers) in order for components to be deemed airworthy. Tailoring of this standard is allowable and expected, in accordance with the guidance provided in 6.5.

5. Additional platform-level or system integration requirements for an engine model and/or its control and accessory components are not addressed herein.

6. Comments, suggestions, or questions on this document should be addressed to:

   U.S. Army
   Combat Capabilities Development Command
   Aviation & Missile Center
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SUMMARY OF CHANGE MODIFICATIONS

(Referenced by: 6.7)

CHANGE NOTICE: Initial – Effective 16 June 2026

Paragraph Number	Modification Description of change

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1. SCOPE

1.1 Scope.

This standard specifies requirements for the qualification of engine control system components and engine accessories. It contains both an overview of the entire qualification process as well as detailed requirements for specific qualification testing. These requirements apply when preparing airworthiness qualification plans (AQP) for the specified components.

The testing protocols and procedures defined in these AQPs are to be followed by engine manufacturers (and control system and accessory component suppliers) in order for components to receive a preliminary flight rating (PFR), and/or achieve a qualification test (QT) rating for production release.

Users are cautioned that the test requirements defined herein do not address additional platform-level or system integration requirements for an engine model and/or its control and accessory components, which may be specified in an AQP or statement of work (SoW). Flight testing, for example, is sometimes required for the qualification of a new engine control system component but is beyond the scope of this document. Examples of additional platform-level qualification test requirements include aircraft-level electromagnetic compatibility (EMC) and electromagnetic vulnerability (EMV) tests.

1.2 Classification.

Not used.

1.3 Applicability.

This standard is applicable to all engine control system and accessory components intended for application on manned or unmanned, fixed- or rotary-wing military aircraft.

1.4 Navigation.

In addition to the Table of Contents (ToC), this document is extensively hyperlinked. Within the document itself, most references to paragraphs, tables, figures, appendices, wherever mentioned, are hyperlinked so that users can go directly to those references.

Also, under each paragraph heading, table title, or figure title, are listed the locations within the standard where they are mentioned (these are also hyperlinked). Use of these hyperlinks is strongly encouraged in order to assure that all paragraphs are reviewed relevant to any particular requirement.

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2. APPLICABLE PUBLICATIONS

2.1 General.

The publications listed in this section are specified in sections 4 or 5 of this standard. This section does not include publications cited in other sections of this standard, or recommended for additional information, or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet specified requirements of publications cited in sections 4 or 5, whether or not they are listed here.

2.2 Government publications.

2.2.1 Specifications, standards, and handbooks.

The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise stated in sections 4 or 5 of this standard, or in a solicitation or contract, the latest issue of these publications is to be used.

(Copies of these publications are available online at https://quicksearch.dla.mil)

INTERNATIONAL STANDARDIZATION AGREEMENTS

AFLP-3747 (restricted)	Guide Specifications (Minimum Quality Standards) for Aviation Turbine Fuels (F-24, F-27, F-34, F-35, F-37, F-40 and F-44)

DEPARTMENT of DEFENSE SPECIFICATIONS

MIL-PRF-23699	Lubricating Oil, Aircraft Turbine Engine, Synthetic Base, NATO Code Numbers O-152, O-154, O-156, and O-167
MIL-PRF-46170	Hydraulic Fluid, Rust Inhibited, Fire Resistant, Synthetic Hydrocarbon Base, NATO Code No. H-544
MIL-PRF-5606	Hydraulic Fluid, Petroleum Base; Aircraft, Missile, and Ordnance, NATO Code H-515
MIL-DTL-5624	Turbine Fuel, Aviation, Grade JP-5 (NATO F-44)
MIL-E-7016	Electric Load and Power Source Capacity, Aircraft, Analysis of
MIL-PRF-7024	Calibrating Fluids, Aircraft Fuel System Components
MIL-DTL-83133	Turbine Fuel, Aviation, Kerosene Type, JP-8 (NATO F-34) and NATO F-35
MIL-PRF-83282	Hydraulic Fluid, Fire Resistant, Synthetic Hydrocarbon Base, Metric, NATO Code Number H-537
DOD-PRF-85734	Lubricating Oil, Helicopter Transmission System, Synthetic Base
MIL-PRF-87257	Hydraulic Fluid, Fire Resistant, Low Temperature, Synthetic Hydrocarbon Base, Aircraft and Missile

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DEPARTMENT of DEFENSE STANDARDS

MIL-STD-1553	Digital Time Division Command/Response Multiplex Data Bus
MIL-STD-202	Electronic and Electrical Component Parts: Method 112, Seal Method 301, Dielectric Withstanding Voltage
MIL-STD-2169	High-Altitude Electromagnetic Pulse (HEMP) Environment
MIL-STD-461	Requirements for the Control of Electromagnetic Interference Characteristics of Subsystems and Equipment
MIL-STD-464	Electromagnetic Environmental Effects Requirements for Systems
MIL-STD-704	Aircraft Electric Power Characteristics
MIL-STD-810	Environmental Engineering Considerations and Laboratory Tests

DEPARTMENT of DEFENSE HANDBOOKS

MIL-HDBK-704-1	Guidance for Test Procedures for Demonstration of Utilization Equipment Compliance to Aircraft Electrical Power Characteristics (Part 1 of 8 Parts)
MIL-HDBK-704-8	Guidance for Test Procedures for Demonstration of Utilization Equipment Compliance to Aircraft Electrical Power Characteristics 28 Vdc (Part 8 of 8 Parts)

2.2.2 Other Government publications.

The following other Government publications form a part of this standard to the extent specified herein. Unless otherwise stated in sections 4 or 5 of this standard, or in a solicitation or contract, the latest issue of these documents is to be used.

DEPARTMENT of TRANSPORTATION

(Copies of these publications are available from: https://www.FAA.gov)

FAA Advisory Circular AC 20-136	Aircraft Electrical and Electronic System Lightning Protection
FAA Advisory Circular AC 33.17	Engine Fire Protection §33.17
FAA Advisory Circular AC 33.28	Guidance Material for 14 CFR §33.28, Engine Control Systems
FAA Order 8110.105	Airborne Electronic Hardware Approval Guidelines

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GENERAL SERVICES ADMINISTRATION

(Copies of these publications are available from: https://quicksearch.dla.mil)

A-A-52624	Antifreeze, Multi Engine Type

UNITED STATES ARMY

(Copies of these publications are available from: U.S. Army, Combat Capabilities Development Command, Aviation & Missile Center, Attn: FCDD-AMA-P, Building 4488, Redstone Arsenal, AL 35898-5000.)

AMACC	Army Military Airworthiness Certification Criteria
AV-E-8593	General Specification for Turboshaft Aircraft Engines
TOP 01-2-603	Test Operations Procedure (TOP), 01-2-603 Rotorcraft Laboratory

2.3 Non-government publications.

The following publications form a part of this standard to the extent specified herein. Unless specified otherwise in sections 4 or 5 of this standard, or in a solicitation or contract, the latest issue of these documents is to be used.

AMERICAN SOCIETY for TESTING and MATERIALS (ASTM)

(Copies of these publications are available from: American Society for Testing and Materials, 100 Barr Harbor Drive, PO Box C700, West Conshohocken, PA 19428-2959, or https://www.astm.org)

ASTM D1655	Standard Specification for Aviation Turbine Fuels
ASTM D5001	Standard Test Method for Measurement of Lubricity of Aviation Turbine Fuels by the Ball-on-Cylinder Lubricity Evaluator (BOCLE)

INTERNATIONAL STANDARDS

(Copies of these publications are available from: International Organization for Standardization, ISO Central Secretariat, Chemin de Blandonnet 8, CP 401, 1214 Vernier (Geneva), Switzerland, or https://www.iso.org or https://webstore.ansi.org)

ISO 12103-1	Road Vehicles – Test Contaminants for Filter Evaluation Part 1: Arizona Test Dust (grade A4 coarse – grade A2 fine)

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RADIO TECHNICAL COMMISSION for AERONAUTICS, INC. (RTCA)

(Copies of these publications are available from: RTCA Inc., 1150 18th NW, Suite 910, Washington, DC 20036, or https://www.rtca.org)

DO-160	Environmental Conditions and Test Procedures for Airborne Equipment
DO-178	Software Considerations in Airborne Systems and Equipment Certification
DO-254	Design Assurance Guidance for Airborne Electronic Hardware

ARINC INDUSTRY ACTIVITIES

(Copies of these publications are available from: https://aviation-ia.sae-itc.com)

ARINC 429	Mark 33 Digital Information Transfer System (DITS)

SAE INTERNATIONAL

(Copies of these publications are available from: SAE International, 400 Commonwealth Drive, Warrendale, PA 15096-0001, or https://www.sae.org)

AIR6108	Engine Electronic Unit, Fire and Overheat Design Guide
AMS2518	Thread Compound, Anti-Seize, Graphite-Petrolatum
ARP1797	Aircraft and Aircraft Engine Fuel Pump Low Lubricity Fluid Endurance Test
ARP1827	Measuring Aircraft Gas Turbine Engine Fine Fuel Filter Element Performance
ARP4024	Aircraft/Engine Fuel Pump Net Positive Suction Pressure Performance Test and Evaluation
ARP4028	Aircraft/Engine Fuel Pump Two Phase (Slugging Flow) Inlet Performance Test and Evaluation
ARP492	Aircraft Engine Fuel Pump Cavitation Endurance Test
AS1055	Fire Testing of Flexible Hose, Tube Assemblies, Coils, Fittings, and Similar System Components
AS1241	Fire Resistant Phosphate Ester Hydraulic Fluid for Aircraft
AS4111	Validation Test Plan for the Digital Time Division Command/Response Multiplex Data Bus Remote Terminals
AS4273	Fire Testing of Fluid Handling Components for Aircraft Engines and Aircraft Engine Installations

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2.4 Order of precedence.

Unless otherwise noted herein or in a contract, in the event of a conflict between the documents referenced herein and the contents of this standard, the text of this standard takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

Note, when test and/or performance requirements specified in higher-level engine or component design documents conflict with those defined herein, the Airworthiness Authority, in conjunction with the supplier, should harmonize the requirements and revise documentation to match.

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3. DEFINITIONS

3.1 Terminology.

(Referenced by: 5.1.7)

General definition of the terminology and nomenclature found within this standard is provided below. Specific definition may vary with contextual usage.

airworthiness	The property of an air system configuration to safely attain, sustain, and complete flight in accordance with approved usage limits.
Airworthiness Authority	A U.S. Government agency having engineering cognizance over a particular aircraft system, subsystem, or component and responsibility for determining the capability of that aircraft system, subsystem, or component to function satisfactorily when used within prescribed limits. Also includes any foreign authority whose airworthiness approval has been accepted by one or more of the DoD's airworthiness authorities. As used herein, it refers to the office of the delegated Airworthiness Authority within the invoking DoD Command, the Using Service, or similar authority.
airworthiness qualification plan	Requirements for airworthiness qualification developed by the Airworthiness Authority when requested by the procuring activity.
airworthiness qualification specification	The AQS defines the contractor's obligation to conduct specific analyses, reviews, tests, surveys, and demonstrations to fulfill the requirements and objectives specified in the AQP.
airworthiness release	A technical document that provides operating instructions and limitations necessary for safe flight of an aircraft system, subsystem, or component.
application system software	The software package that performs the main functions of the control system or component. It receives processed input signals from the hardware operating system (OS), processes the information, and outputs command signals to the hardware operating system to control output drivers that interface with electromechanical or electrohydraulic effectors.
closed-loop test	Testing wherein the target hardware is utilized within a simulated real-time engine and airframe environment. Control system performance requirements can be validated with highly repeatable test cases and airframe/engine mission profiles.
complex electronic hardware	Electronic hardware that is not considered simple. A hardware item is simple if a comprehensive combination of deterministic tests and analyses can ensure correct functional performance under all foreseeable operating conditions with no anomalous behavior.
component improvement program	A design, development, and qualification program undertaken with the express purpose of improving the performance and/or reliability of an existing system, subsystem, or component. The item to be developed may be intended as a direct replacement for an existing item for the purpose of addressing obsolescence or cost issues.

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component limiting temperature	The maximum allowed surface temperature, established by the engine supplier, for any given engine subsystem or component. This limit is to be respected by the airframe integrator when designing engine nacelle cooling provisions, so that as installed in the airframe, the engine component surface temperatures do not exceed their design limits. Typically, a fully instrumented thermal survey is conducted during flight testing in order to determine if any limiting temperature is being exceeded.
control mode	Each defined operational state of the engine control system in which the crew can exercise satisfactory engine control, which may involve evaluation in the aircraft/rotorcraft.
critical design review	The transition point between detailed design and fabrication of a configuration item (CI) or aggregate of configuration items. The primary focus is on the completed detailed design documentation and draft production specifications. The detail design hardware drawings are reviewed and approved before the contractor manufactures test items.
data item description	A standardization document that defines the data content, preparation instructions, format, and intended use of data required of a contractor.
design assurance test	Any test which is part of the design and development process of an item, the purpose of which is to assess component performance, make design improvements, and reduce qualification test failure risk. Neither test plan approval nor satisfactory test results are required.
documentation	Any media that provides a record of the design or design process that can be reviewed.
engine component	Items of equipment, furnished as part of and qualified with the engine, whose size, conformation, and dynamic and static characteristics are essential to attain the engine performance specified in the engine specification. Fuel pumps, engine controls, variable guide vane actuators, anti-icing valves, and temperature/pressure/speed/torque sensing systems or devices are included in this category. Components may require separate qualification, acceptance test, or adjustment. Note that 'component' as used herein does not refer to the major rotating subsections or modules, including accessory gearbox, that comprise the core engine.
engine control system	Any system or device that controls, limits, or monitors engine operation.
engineering cognizance	The technical awareness and knowledge of the design, function, and performance sufficient to determine prescribed limits required for safe operation and continued airworthiness.
failure reporting and corrective action system	A system put in place by the supplier to track component failure data during any phase of a component's life in order to discern any systemic failure trends, report them to the user, and institute corrective actions (design and/or quality) as necessary to meet performance and reliability specifications.

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flight critical	Anomalies involving the equipment that could result in permanent partial disability, severe injury, or severe occupational illness that may result in hospitalization of at least three personnel, loss exceeding $1M (but less than $10M), or reversible severe environmental damage.
flight test	The engine is installed in an aircraft that is then tested in flight to acquire verification and validation data not obtainable through engine system-level ground test.
(formal) qualification test	A hardware test (QT) or a software test (FQT) that is required to be completed in order for the Using Service or Airworthiness Authority to determine whether a configuration item complies with allocated requirements for that item. The suite of QT and FQT tests is part of the set of activities required for the engine to receive a qualification test rating.
full-authority digital engine control	An engine control system whose primary functions are provided by digital electronics and in which the electronic control unit (ECU) has full-range authority over the engine power or thrust.
life cycle environmental profile	A design and test decision baseline document outlining real-world, platform-specific, environmental conditions that a subsystem or component will experience during service-related events (e.g., transportation, storage, operational deployment/use) from its release from manufacturing to the end of its useful life.
maximum continuous power	An engine rating condition at which the engine is capable of operating continuously.
mission	The period beginning with the start of the engine prior to flight and ending with the shutdown of the engine at the completion of the flight.
nonconformance	The failure of an item to meet a defined characteristic or process.
operating system software	The software package (a.k.a. computer software configuration item [CSCI]) that interfaces with and services the application software (AS). It interacts with, and controls, the unit hardware, and processes input signals to, and output signals from, the application system software. It is responsible for scheduling real time operating tasks and controlling hardware interfaces.
operational flight program	The operational flight program (OFP) is made up of the operating system and the application software. Each of these pieces may comprise several CSCIs.
platform	The airframe application in which the engine and its control system are to be used.
preliminary design review	The preliminary design review (PDR) represents the approval to begin detailed design. The primary focus is on the adequacy of top-level design documentation for hardware and software configuration items. The PDR is a check to verify that the allocated baseline requirements for the engine, including components, have been addressed and that they can most likely be met or exceeded by the proposed functional design.

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preliminary flight rating	This milestone establishes the acceptability of the engine and its control system to power the aircraft throughout its full envelope. The engine is not required to meet full verification requirements for durability and reliability and is not required to be the final production configuration.
prescribed limits	The full authorized range or envelope of operating, environmental, and sustaining criteria or characteristics for the safe and reliable use of the aircraft system, subsystem, allied equipment, or component as determined by analysis, tests, and operating experience.
primary mode	The mode for controlling the engine under normal operation; often referred to as the 'normal mode'.
qualification test rating	This milestone establishes the acceptability of the engine for low-rate production release. It is the sum of analysis, demonstration, and test activities accomplished on engines and components submitted for qualification to confirm the suitability of an engine model for production and service use.
standard	A document that establishes uniform engineering or technical criteria, methods, processes, and practices.
standard practice	Standard practices specify procedures on how to conduct non-manufacturing functions that, at least some of the time, are obtained via contract from private sector firms.
statement of airworthiness qualification	A document establishing qualification status and airworthiness release that is issued in conjunction with the Airworthiness Qualification Substantiation Record (AQSR) normally completing an airworthiness qualification program.
tailoring	The process by which individual requirements (sections, paragraphs, or sentences) of the selected specifications, standards, and related documents are evaluated to determine the extent to which they are most suitable for a specific system and equipment acquisition, and the modification of these requirements to ensure that each achieves an optimal balance between operational needs and cost.
test method standard	A standard that specifies procedures or criteria for measuring, identifying, or evaluating qualities, characteristics, performance, and properties of a product or process.
Using Service	The particular branch of the Government or Department of Defense (DoD) which will be operating the aircraft system and which is responsible for the airworthiness (i.e., Airworthiness Authority) of the platform and all of its subsystems.
validation	The process of evaluating system and hardware/software requirements to determine compliance with specified performance.
verification	The process of evaluating hardware/software design and implementation to determine compliance with system and hardware/software requirements.

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3.2 Acronyms and abbreviations.

AC	advisory circular
A/D	analog-to-digital
AGB	accessory gear box
AHSS	abnormal high limit steady-state
AIR	aerospace information report
AM	amplitude modulation
AMACC	Army military airworthiness certification criteria
AMSC	Acquisition Management Systems Control (number)
AQP	airworthiness qualification plan
AQS	airworthiness qualification specification
AQSR	airworthiness qualification substantiation record
ARINC	Aeronautical Radio, Inc.
ARP	aerospace recommended practice
AS	aerospace standard / application software
ASD	acceleration spectral density
ASTM	ASTM International
ATP	acceptance test procedure
AV	Combat Capabilities Development Command, Aviation and Missile Center
AvMC	Aviation and Missile Center
AWR	airworthiness release
BB	broadband
BOCLE	ball-on-cylinder lubricity evaluator
BR	best range (for a given fuel load)
BTP	build-to-print
BTS	build-to-specification
CCDC	Combat Capabilities Development Command
CDRL	contract data requirements list
CE	conducted emissions

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CEH	complex electronic hardware
CERT	combined-environment reliability test
CFR	Code of Federal Regulations
CI	configuration item
CIP	component improvement program
CLT	component limiting temperature
CN	change notice
COTS	commercial off-the-shelf
CS	conducted susceptibility
CSCI	computer software configuration item
CW	continuous wave
DAL	design assurance level
DAT	design assurance test
dB	decibel
DEVCOM	Combat Capabilities Development Command
DID	data item description
DoD	Department of Defense
DTL	detail
E^3	electromagnetic environmental effects
ECP	engineering change proposal
ECU	electronic control unit
ELSS	emergency low limit steady-state
EMC	electromagnetic compatibility
EMI	electromagnetic interference
EMP	electromagnetic pulse
EMS	engine model specification
EMV	electromagnetic vulnerability
ESS	environmental stress screening
FAA	Federal Aviation Administration

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FADEC	full-authority digital engine control
F/D	frequency-to-digital
FM	frequency modulation
FMU	fuel metering unit
FP	fire proof
FPI	fluorescent penetrant inspection
FQT	formal qualification test
FR	fire resistant
FRACAS	failure reporting and corrective action system
g (or G)	gravity
GF	gunfire
GI	ground idle
HALT	highly accelerated life testing
HASS	highly accelerated stress screening
HEMP	high-altitude electromagnetic pulse
HDBK	handbook
HMU	hydromechanical metering unit
HPM	high-power microwave
HSA	hot spot analysis
IAW	in accordance with
ICD	interface control document
INR	initial nuclear radiation
I/O	input/output
IOC	initial operational capability
ISO	International Organization for Standardization
LRU	line replaceable unit
LVDT	linear variable differential transformer
MCP	maximum continuous power
MTBF	mean time between failure

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NaCl	sodium chloride
NATO	North Atlantic Treaty Organization
NB	narrowband
NPSP	net positive suction pressure
OEM	original equipment manufacturer
OFP	operational flight program
OS	operating system
P/N	part number
PC	printed circuit
PDR	preliminary design review
PFR	preliminary flight rating
PLD	programmable logic device
PMA	permanent magnet alternator
PRF	performance or pulse repetition frequency
PSD	power spectral density
PTIT	power turbine inlet temperature
QT	qualification test
RAP	resonance assessment profile
RDG(T)	reliability development and growth (test)
RE	radiated emissions
RMS	root mean square
RS	radiated susceptibility
RT	real time
RTCA	Radio Technical Commission for Aeronautics
RVP	Reid vapor pressure
SAE	Society of Automotive Engineers
SFC	specific fuel consumption
SL	system level or service life
SLCD	sea-level, cold-day

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SLSD	sea-level, standard-day
SOT	simulated operational test
SoW	statement of work
SP	standard practice
SRD	Systems Readiness Directorate
SRS	shock response spectrum
STD	standard
TBD	to be determined
ToC	table of contents
TRR	test readiness review
TT	test time
TVP	true vapor pressure
USDA	United States Department of Agriculture
UUT	unit under test
V/L	vapor-to-liquid (ratio)
V/m	volts per meter
WSD	wear scar diameter

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4. GENERAL REQUIREMENTS

4.1 Overview.

This section provides an overview of component qualification. It describes the various program phases, the types or categories of testing to be accomplished in each phase, and the overall objectives of each group of tests or test category. An overall 'roadmap' of this process, cross-referenced to the paragraphs in this standard, is illustrated in FIGURE 1. Detailed test requirements, corresponding to the paragraphs below, are specified in section 5.

4.2 Program phases.

4.2.1 Development.

Generally, the majority of control system component development programs are undertaken as part of engine development, however in some instances ongoing component improvement programs (CIP) result in suppliers proposing significant upgrades to existing components, or replacing them outright with an entirely new item. This is particularly true for electronics due to part obsolescence. No matter the development vehicle, whether part of engine development or stand alone, all components must be qualified for service use.

Engine components must pass a series of tests designed to evaluate their ability to function acceptably and reliably when operating within prescribed physical and electromagnetic environments. This series of tests is typically delineated in a qualification specification, prepared by the equipment or component supplier, in response to a Using Service-prepared qualification plan.

4.2.1.1 Airworthiness qualification plan.

Prior to undertaking any component development program, a stand-alone airworthiness qualification plan (AQP) is generally prepared by the Using Service or Airworthiness Authority. The AQP outlines the specific test and analysis requirements that the component (or components) must successfully meet to be granted a preliminary flight rating and a qualified rating status for service use.

The test requirements to be included in the AQP are extracted from this standard, specifying only those tests and analyses pertinent to the component (or components) to be qualified. TABLE I lists the tests specified in this standard and can therefore be used as a convenient checklist to identify all the tests applicable to a specific component. In some cases, such as a complete engine development program, identification of the specific components to be developed may not yet be firmly established, in which case the test matrix tables in their generic form (e.g., see TABLE VII) can be utilized directly for engine contractor guidance.

4.2.1.2 Airworthiness qualification specification.

The airworthiness qualification specification (AQS) is the contractor's response to the qualification requirements specified by the AQP. The AQS identifies each of the components to be developed and qualified, and the specific applicable test requirements and test methods to be utilized for each of those components. The extent to which qualification by similarity and/or analysis is to be performed in lieu of actual testing is also indicated. Once approved by the Airworthiness Authority and the contractor, the AQS forms the basis for the qualification program.

The essential elements of an AQS can be satisfied by completing Part 1 of the test planning documentation specified in data item description (DID) DI-NDTI-81895 (6.3).

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4.2.2 Preliminary flight rating.

Flight testing is often an integral part of the qualification program, however, prior to allowing any component to be flight tested an airworthiness release must be approved for limited use of that component on the platform to be flown. Generally, this requires completion of a subset of the qualification test suite (often referred to as PFR) to be completed ahead of full qualification (referred to as QT).

Consequently, some of these 'pre-qualification' tests, for example PFR vibration and explosive atmosphere, may be acceptable for qualification if they are properly conducted and the component changes between completion of flight test and the beginning of qualification are documented, and then judged to be immaterial to the outcome of those tests.

In addition to component-level tests, there are system-level, and sometimes vehicle-level, tests conducted to assess overall electromagnetic compatibility of the airframe and engine components. This testing may require contractor support to assist with system operation and assessment of results.

4.2.3 Qualification.

In order for any engine control or accessory component to achieve a qualification test rating, it must be tested in accordance with (IAW) Airworthiness Authority-approved test plans and procedures. Any testing begun prior to receipt of approval of the pertinent test plan/procedure is typically at the contractor's risk.

If a particular test was successfully completed in support of PFR testing, the contractor may request a test waiver, if the component to be qualified is unchanged. If there are differences between the component to be qualified and that which successfully passed PFR testing, the contractor may prepare and submit a hardware reconciliation report for consideration, which assesses the differences as being immaterial to test outcome.

The environmental tests specified in documents such as MIL-STD-810, RTCA/DO-160, AV-E-8593, and other standards have been refined over many years of experience, and they have shown themselves to be valuable benchmarks in determining if a component is suitable for flight test and initial fielding. However, passage of qualification tests is no guarantee of reliable field service, it is simply a hurdle that must be cleared for initial fielding.

Significant weight should be given to actual in-service experience with a component, if such is available. Service use and experience is easily the equivalent of qualification testing. The exception is the case where the service environment experienced is deemed to be significantly less stressful than the intended environment of a new application.

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4.3 Test categories

4.3.1 Simulated operational test.

The simulated operational test (SOT/see 5.3) is meant to uncover design shortcomings or weaknesses that might cause early failures of the component during initial operational capability (IOC) fielding in a service environment. It can be thought of as a minimal reliability development and growth test (RDGT), but should not be considered a substitute for a long term combined-environment reliability test (CERT).

To this end, unit operation over a simulated usage cycle is conducted for several hundred hours, while subjecting it to vibration; extreme temperatures and rapid temperature transitions; contaminated fluids (fuel or air); low lubricity fuel; and fuel pumping cavitation (not all are applicable). During these tests, the unit under test (UUT) is expected to perform as per specification, with some allowance for wear if appropriate (for example, a fuel pump may only have to meet service limits by the end of the test). In all cases, component performance must be continuously monitored and recorded so that any degradation or deviation from specified performance can be determined.

Because the component is often subjected to similar temperature extremes as those used for the environmental temperature tests, contractors sometimes request waiving of the hot/cold/cycling thermal environmental tests. However, if this is the case, the SOT must be performed on the same UUT as the other performance environmental tests (to establish cumulative damage), and furthermore must be performed in the same order (that is, as the first test). Because of the large number of environmental tests required, coupled with the several hundred hours duration of the SOT itself, it is impractical to perform all of these tests on a single unit; therefore, using this approach is discouraged.

4.3.2 Environmental tests.

(Referenced by: 4.4.1, APPENDIX E)

The tests in this category subject the component to the default MIL-STD-810 or actual (measured) environments expected to be encountered during operation. The tests can be divided into three broad groups:

a. A relatively universal ‘series’ of tests meant to be conducted sequentially on a single unit, and which are applicable to most component types (see 5.4)

b. Special tests that are somewhat unique to the qualification of engine control system components and engine accessories, but are still applicable to many of the component types (see 5.5)

c. Tests that are very component-specific and therefore generally only apply to a single component type (see 5.6)

The tests in the first group are meant to be conducted sequentially on one to three units maximum so that the cumulative effects of environmental exposure can be assessed. Consequently, the order of the conducted tests, and on which units, is important (see TABLE II) since the cumulative damage amassed on the UUT may influence whether it passes or fails subsequent tests. Therefore, tests should be performed in the order specified unless the reasons for changing the order are compelling and special relief is granted by the Airworthiness Authority.

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These tests can be further divided into two major sub-groups; those that are conducted while the unit is operating, and those that are conducted with the unit in a non-operating state. For the 'operating' test sub-group, the component is expected to meet its performance specification (this can be the engine model specification, purchase specification, or acceptance test procedure [ATP], for example) while being exposed to the test environment. This means that the UUT must be fully operational during the test and pertinent parameters recorded such that its performance under duress can be measured and assessed. It is not acceptable to perform these tests on a non-operational unit, or on an operating unit but without monitoring it for 'performance' during the test. APPENDIX E discusses 'operating' and methods that can be employed to monitor for 'performance' during tests.

For the non-operating test sub-group, the component is not required to be operating during application of the test environment. However, it must still pass post-test ATPs in order to demonstrate acceptable functionality after exposure to the environment; and in some cases, specific to the individual test procedure, periodic performance checks may still be required.

The special tests and the component-specific tests are not generally required to be performed on a single unit nor in any particular sequence. These tests typically have a wider latitude for tailoring to the individual component and its usage on the designated engine/airframe platform.

4.3.3 Electromagnetic environmental effects (E^3).

The tests in this category (see 5.7) are meant to subject the *system* to an aperiodic electromagnetic environment wherein some deviation from specified performance may be allowable (for example, a 0.5% to 1.0% peak-to-peak variation in engine power) when the system is subjected to the test environment. The objective is to assemble a complete engine control system and subject it to the specified events or conditions while observing the effects on the control system component or engine accessory. The effect the component has upon the aircraft, in terms of conducted emissions (CE) and/or radiated emissions (RE), is also evaluated.

To that end, two approaches have been used and have been found acceptable. The first is to simulate an engine operating at various specified conditions and observe its operation while subjecting the actual control system hardware and operational flight program software to the environment (closed loop). The second approach is to install specialized software in the electronic control unit and utilize the software to capture any anomalous behavior of input or output signals for later analysis using a combined control system/engine/airframe model to predict performance resulting from the recorded anomalous behavior (open loop). Both methods have their pros and cons, and have been used successfully in the past.

The tests in this category can be run concurrently, since there is no requirement to perform the tests on the same group of components. Practically speaking, the requirement to assemble a complete system in order to perform the test is the limiting factor. Consequently, these tests are typically run sequentially, with electromagnetic interference (EMI) completed first, any identified design improvements incorporated, and then lightning testing conducted.

As an aside, there is often confusion between the E^3 *design* requirements and the E^3 *verification* requirements. This document deals with the test methods used to verify design performance; it does not specify those requirements. In the case of E^3, engine control systems must meet all of the relevant requirements specified in MIL-STD-464 from a design perspective. Verification that the design meets those requirements is the province of MIL-STD-461, as tailored herein.

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4.3.4 System / Software verification and validation.

For Army Aviation, system safety requirements are defined in AMACC 14.0 while software and programmable hardware airworthiness requirements are defined in AMACC 15.0. Other DoD aviation components may have their own defined requirements.

For engine controls, all software and/or programmable hardware must be developed, verified, and validated in accordance with the most stringent design assurance level (DAL) specified by RTCA/DO-178 and/or RTCA/DO-254, as described in AMACC 15.0 (Software Airworthiness) and its subsections. This includes the application of Federal Aviation Administration (FAA) Order 8110.105B in differentiating simple versus complex electronic hardware.

In some specific instances where the software or programmable hardware is designed to have minimal or no effect on engine operation, it may be acceptable to use a DAL lower than 'A'. In order to justify such departures, and in addressing commercial off-the-shelf (COTS) complex electronic hardware (CEH) or software components, a system safety assessment as described in AMACC 14.0 (System Safety) and its subsections must be performed and submitted for approval.

As part of system verification and validation, special tests are also conducted to thoroughly understand the effects of common-mode signal failures (5.8.2) and airframe power interrupts (5.8.3) on engine control system behavior.

It should be noted that system operation during power interrupts is generally not well defined from a design point of view, and that the practical way to assess whether or not the system is robust (particularly where multiple channels are concerned) is to test the system in a closed-loop environment. Additional on-engine testing may be warranted depending on the results of closed-loop testing, particularly in the starting regime where engine model fidelity may not be adequate to fully evaluate control performance.

4.3.5 Reliability development and growth.

While successful SOT and environmental qualification tests should ensure a minimally acceptable level of field reliability for a component, reaching the predicted 'mature' level of reliability requires a proactive approach to 'grow' the component to those levels. Highly accelerated life testing (HALT), vibration endurance testing, and combined-environment reliability testing are common methods utilized to achieve this objective (see 5.9).

To be most effective, HALT should be conducted during the component development phase so that there is sufficient time to incorporate design changes prior to the start of qualification. Vibration endurance testing, while not part of actual qualification, should be conducted prior to completion of the qualification program to determine if any change to the design should be made to enhance reliability. CERT, because of its real-time nature, is most often conducted coincident with initial fielding since it is a long-duration testing program not usually compatible with a development and qualification schedule.

Additionally, another HALT should be conducted post qualification so that a highly accelerated stress screening (HASS) test can be developed for deployment during the initial production phase. If no significant design changes were introduced after the HALT conducted during component development, the data gathered during that test may be sufficient for the purpose of designing a production stress screen.

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4.4 Testing alternatives.

4.4.1 Previous component approval.

Engine control and accessory components requiring testing, as specified herein, may have some tests waived at the option of the Airworthiness Authority, if the component has been previously approved by another recognized Airworthiness Authority for use on another engine or airframe. All such components for which a test waiver is requested must conform to substantially the same parts list and configuration as the approved component.

As stated previously in 4.3.2, some tests may be required simply because they make up part of a suite or battery of tests to be performed on the unit under test. The need to demonstrate robustness to cumulative damage may override the fact that the component may have completed a particular test previously, or could be verified by similarity or analysis.

In order to receive consideration for a test waiver, a similarity argument report must be prepared and submitted to the Airworthiness Authority for approval.

4.4.2 Alternate test methods or approaches.

The tests and test methods specified in the AQP constitute the requirements that must be met for engine control system and accessory components to achieve an experimental flight airworthiness release and/or a qualified rating status. Civilian certification tests and test methods that provide equivalent design and initial production fielding assurance may be substituted for specific identified military qualification requirements.

Notwithstanding any stated or assumed equivalency, all proposed changes to the tests specified in the AQP require approval from the Airworthiness Authority.

4.4.3 Component replacement programs.

The requirements described in this standard, as a whole, are meant to define the qualification regimen for newly designed and developed engine control system line replaceable unit (LRU) components and accessories. For cases where in-service components are being replaced, they usually fall into one of the three categories described in the subparagraphs which follow.

4.4.3.1 Reliability improvement.

Components and/or systems that have been previously qualified and are in current service may need to undergo subsequent qualification as part of a ‘component improvement program’. These programs are typically undertaken when the component is suffering an unacceptable level of field reliability.

Standard qualification tests, as presented herein, may provide a benchmark for initial flight suitability but do not guarantee that the component will meet its predicted or stated reliability goal. Therefore, qualifying the CIP component to the same tests as the initial design provides little to no insight as to any improvement being made. In fact, the ‘improved’ component may actually be worse than the current design since the qualification tests are absolute hurdles and do not indicate the margin by which the initial component exceeded those test limits.

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Hence for CIP, it is prudent to qualify the component via comparison tests that are specifically designed to provoke the particular failure mode or wear mechanism that is the reason for the improvement program. With that goal in mind, a series of specialty tests (at the piece-part, sub-assembly, or LRU level) should be developed so that relative improvement can be demonstrated on the current parts side-by-side with the new parts. This testing should be performed using a statistically meaningful number of samples of both the old and the proposed new design.

This is not to suggest that some overall unit-level qualification to the requirements of this standard is not necessary, particularly when a large number of changes are being made to the component, or the interaction of the several changes being made cannot be fully assessed by individual part tests.

4.4.3.2 Second-source development.

Components are sometimes chosen to have a second-source brought on line for reasons of cost and/or availability. In this case, a new supplier will be manufacturing the component to a design or procurement specification (i.e., build-to-specification [BTS]). So even though the function is meant to be identical to the original, the component must still go through unit-level qualification to at least the same level as did the original.

If qualification of the original part was several or more years earlier, the AQP should be developed using the latest versions of test procedures and any new tests that have been imposed since that time, as defined herein. There may be valid arguments to not increase the scope of the qualification effort over that previously conducted, but these must be strong technical arguments, not simply to reduce qualification costs or risks.

For components for which qualification was more recent, the opposite holds true; consideration should be given to developing an AQP using the original suite of tests, with an exception for new tests that may be pertinent to the component in question.

Second-source components occasionally are build-to-print (BTP) when the engine manufacturer holds the design rights. For these cases, some of the original qualification tests may be waived since the testing will generally focus on mechanical stress tests, such as vibration, shock, and temperature, to assure internal part manufacturing tolerances are such that the unit functions properly under these conditions. For example, it would not be expected that tests that impinge on the external housing such as sand & dust, rain, fungus, etc. would be needed, since it would be unchanged (design and materials) from the originally qualified item.

Regardless of the type of second-source component, BTS or BTP, back-to-back unit-level testing is required to confirm functionality equal to or ‘better’ than the original. Simply passing a set of original qualification tests does not guarantee equivalent functional performance, since qualification tests do not always stress the limits of the performance of the actual ‘as-designed’ component.

In a similar vein, the performance specification, post qualification of the second-source, may require updating to capture the ‘as-built’ performance of both the original and second-source components, so that any future potential source has the complete set of requirements needed to duplicate the performance of the original component.

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4.4.3.3 Electronic parts obsolescence.

Components and/or systems for which key electronic parts have become obsolete and are no longer available are increasingly more common, and represent a third class of components to be qualified.

Because the original equipment manufacturer (OEM) is unchanged and these efforts are expected to be relatively frequent (whether to replace a few simple parts that have gone obsolete, or to replace a microprocessor or other specialty integrated circuit), targeted testing based on initial qualification may be appropriate.

Changes to an ECU or other electronic component that are considered to be Class II engineering change proposals (ECP) due to their limited scope, normally do not require re-qualification, instead relying on supplier-developed design assurance testing (DAT). However, there are instances where the number of Class II changes, over time, become so numerous that re-test of the unit becomes necessary to re-establish a qualification baseline for future changes.

Class I ECPs usually involve more extensive changes to printed circuit (PC) board layouts and parts changes, but even if the change only involves a single part, such as a microprocessor or complex gate array, the unit will require a combination of supplier DATs and certain unit-level qualification tests aimed at verifying electrical and mechanical integrity of the part change.

As is the case for BTS/BTP replacements where back-to-back functionality must be verified, the same holds true here. It is not possible to fully verify component performance specification compliance solely on the basis of repeating qualification tests due to their limited ability to measure the degree to which a component may exceed specified performance, or in the case where actual performance of the component is not fully captured by requirements specifications to begin with.

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5. DETAILED REQUIREMENTS

(Referenced by: 4.1)

5.1 Global testing requirements.

5.1.1 Test readiness review.

In order to assure, to the greatest extent possible, that tests are properly conducted according to the approved test plans/procedures, the contractor shall convene a test readiness review (TRR) prior to all test execution. If testing involves many separate procedures, the TRR can be held in a piecemeal fashion; a part 1 to discuss the test plan in general and a part 2 to review in detail the test procedure(s) just prior to execution. For reference, the content of a typical TRR is provided in H.2.8**.**

5.1.2 Acceptance test or calibration.

(Referenced by: TABLE II)

Engine components shall be acceptance tested in accordance with procedures developed by the contractor and approved by the Airworthiness Authority, as specified in 5.10.1. All component acceptance test procedure results shall be recorded and provided to the Government in test reports, as necessary, to substantiate component compliance with performance specifications.

Components not requiring an ATP shall be tested or calibrated under normal operating conditions to demonstrate satisfactory functioning and compatibility with other system components.

Once component testing has begun, the component shall not be acceptance tested unless specifically authorized in the approved test plan/procedure. Recalibration or adjustment of the component at any time after the start of testing is prohibited and, if done, shall invalidate all prior testing at the sole discretion of the Airworthiness Authority.

5.1.3 General test procedures.

All components shall be cleaned of oil, grease, or corrosion prevention compounds used for preparation for storage, prior to the start of testing. Test assemblies or components shall be subjected to operating loads simulating those encountered on the engine or airframe. Sufficient instrumentation shall be provided to indicate the performance of each component and to indicate that the functional relationships of components are maintained as required by the applicable performance specification.

Functional checks shall be performed at the end of each test or group of tests and at other times, at the option of the contractor, to verify that the component still meets new-part or allowable service limits, and that the function is unimpaired. All components shall be supplied with such fluids as they normally handle or contact, except components normally in contact with fuel may sometimes be supplied with test fluids as specified for the individual tests for safety reasons. When non-specified fluids are used for testing purposes, attention shall be paid to the viscosity of the test fluids, particularly when leakage or seepage/weepage failure criteria are in effect.

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5.1.4 Re-test, disassembly, and inspection.

(Referenced by: TABLE II)

Upon completion of specific tests, as identified in the approved test plan/procedure, the component ATP shall be repeated utilizing the same configuration-controlled procedure as was used for pre-test. Components not subjected to an ATP shall be tested or calibrated under normal operating conditions to demonstrate satisfactory functioning and compatibility with other system components.

All component ATP results shall be recorded and provided to the Government in test reports, as necessary, to substantiate component compliance with performance specifications.

After the planned battery of tests has been completed for each unit in the test program, the unit shall be disassembled and inspected for indications of failure, impending failure or excessive wear. Note that all disassemblies and inspections shall be specified in the approved test plan/procedure. Unplanned disassemblies or inspections shall only be conducted upon the specific authorization of the Airworthiness Authority.

5.1.5 Test success criteria.

The component tests shall be considered to be successful when, in the judgment of the Airworthiness Authority:

a. Component performance and functions were within established limits. For components with software or firmware, there were no hard faults, soft faults, channel swaps, loss of communication or microprocessor resets.

b. There was no fluid leakage from any component other than that of a nature and rate specified in the component specification.

c. There was no hang-up or hesitation of any component or part.

d. Acceptance tests or recalibrations indicate that the component was within its new-part limits, or service limits if allowed.

   Note, in some instances success criteria may also require comparison of pre- and post-test ATP measured values. For example, an ATP may require the resistance of a solenoid coil to measure between 50 and 100 ohms. A pre-test value of 55 ohms and a post-test measured value of 95 ohms would ‘pass’ the ATP limits, but this large change in pre- and post-test resistance values clearly indicates a failure of some type. Test success criteria must consider these types of additional criteria, beyond simply passing the ATP.

e. The component teardown inspection showed no indication of failed, excessively worn or distorted parts, or impending part failure.

f. Component dimensions and tolerances measured post-test, when compared with similar measurements made prior to the test, were deemed acceptable for continued use.

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5.1.6 Power supply transient application: generic.

(Referenced by: 5.4.9, 5.6.4.4e, 5.6.4.4g)

Power supply transients, only when specified to be applied for a particular test (e.g., Explosive Atmosphere, Power Compatibility), shall consist of the application of over voltages and the application of spike voltages as specified below. These transients (derived from MIL-STD-704A, notice 1) are only applicable to components that are directly connected to airframe power. The parameters specified in subparagraph 'a' pertain to the setup of the transient generator used to apply the voltage spikes to the UUT. The peak voltage may be substantially lower when the actual UUT is connected.

a. Voltage spikes: apply 5 of each to the 28 Vdc steady-state input (see H.2.5)
   (1) +600 Volts open circuit for 10 µsecs, +300 Volts into 50 Ohms, 12 Amperes max.
   (2) -600 Volts open circuit for 10 µsecs, -300 Volts into 50 Ohms, -12 Amperes max.
   (3) +200 Volts open circuit for 50 µsecs, +100 Volts into 50 Ohms, 4 Amperes max.
   (4) -200 Volts open circuit for 50 µsecs, -100 Volts into 50 Ohms, -4 Amperes max.

b. Over voltages for 28 Vdc systems consisting of 28 Vdc → Over Voltage → 28 Vdc:
   (1) 80 Vdc for at least 0.050 seconds.
   (2) 60 Vdc for at least 0.50 seconds.
   (3) 40 Vdc for at least 2.0 seconds.
   (4) 35 Vdc for at least 3.0 seconds.

c. Over voltages for alternating current systems consisting of:
   (1) 180 Volts root mean square (RMS) for at least 0.10 seconds.
   (2) 160 Volts RMS for at least 0.50 seconds.
   (3) 140 Volts RMS for at least 1.5 seconds.
   (4) 125 Volts RMS for at least 5.0 seconds

5.1.7 Component limiting temperature: generic.

(Referenced by: 5.3.3d, 5.3.6a, 5.4.3, 5.4.3.1, 5.4.4, 5.4.5, 5.4.9, 5.5.2.1, 5.5.2.2, 5.5.2.3, 5.6.1d, 5.6.2, TABLE XV)

When environmental tests specify that the maximum test temperature is the component limiting temperature (CLT), or when structural/mechanical tests require the CLT to be applied simultaneous with the test stressor (e.g., vibration), the guidelines below apply. Note that CLT is defined as the measured external surface temperature of the component that must not be exceeded when the engine is in its installed environment (see definition of term in 3.1). CLTs are usually found in the engine model specification (EMS), the engine/airframe interface control document (ICD), and/or the component design specification.

In some cases, for instance when a component axially spans more than one engine section, the CLT may be defined at more than one location along the component's longitudinal axis, and these temperatures will likely be different (in some cases significantly so). Additionally, a component that mounts externally but penetrates the engine case will also be exposed to different temperatures (an Np sensor, for example). Consequently, the ability to execute such a test, using more than one CLT, can be problematic, and sometimes completely impractical.

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In those instances, the engine manufacturer/component supplier, in concert with the Airworthiness Authority, should agree on the surface temperature to be used for the test. Consideration should also be given to increasing the primary test stressor (vibration, load, speed, etc.) if achieving a suitable test temperature is impractical.

When choosing the test temperature, it is important to understand the methodology that was used to define the CLT, and what that CLT represents. It may be the absolute maximum allowed surface temperature that can be tolerated before exceeding some internal part limit (e.g., solenoid coil wire), or it might be a temperature that is defined for other purposes such as component lifing (i.e., not a functional limit). In such cases where there exists 'headroom' between the specified CLT and the functional temperature limit (if they are not the same), consideration should be given to testing at the functional limit instead of the EMS-specified CLT.

For components that handle fluid of some type (e.g., fuel, oil, hydraulic fluid), it may be the case that the fluid temperature dominates the external ambient temperature when it comes to establishing the maximum temperature of internal componentry, due to the much higher thermal capacity of the fluid. In such instances, running the vibration test with the fluid at its maximum operational temperature may be a preferable alternative to running the test at the CLT.

Component bench qualification to a temperature higher than the CLT means that, if during the flight test engine thermal survey, the component surface temperature is found to exceed the CLT, but not the temperature used for qualification, there will be no need to repeat the affected qualification test to prove reliable performance at the higher-than-expected surface temperature.

5.2 Tests required for PFR.

(Referenced by: 5.7.1,TABLE III, J.2.3)

In order for the platform to receive an airworthiness release for flight test, components shall have been tested in accordance with the requirements of TABLE III. Components shall conform to the same parts list and configuration as those planned for use for flight test. Any testing begun prior to receipt of official Airworthiness Authority approval of the associated test plan is at the contractor's risk.

A standard suite of EMI tests, as specified in 5.7.1, shall be completed in order to receive an airworthiness release. Other specific system/software verification and validation tests, to be accomplished for PFR, shall be proposed by the contractor and approved by the Airworthiness Authority.

At the Airworthiness Authority's discretion, some or all of the testing conducted for PFR may be accredited towards the full system qualification requirements specified in 5.4, 5.5, 5.6, 5.7 and 5.8.

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5.3 Simulated operational test.

(Referenced by: 4.3.1, 5.10.3, 5.10.4.1, TABLE IV, J.2.3)

The following series of tests apply to the fuel system, oil system, ignition system, engine anti-icing system, and engine control system, including sensing and actuation components. However, not every test listed below is applicable to every component (refer to TABLE IV). For each component to be qualified, a single serialized unit shall be used for all of the specified testing. The tests shall be conducted in the following order:

1) Low-lubricity fuel
2) High temperature
3) Room temperature and contamination
4) Low temperature
5) Temperature-vibration cycling
6) Engine fuel system cavitation endurance and low temperature operation
7) Disassembly and inspection

Tests shall be conducted on groups of related components so arranged and interconnected as to simulate their normal relationship and function on the engine or airframe. However, subassemblies or components of a system may be tested separately if such separation does not prevent simulation of the complete function of the components or subassemblies. Insofar as practicable, components shall be positioned in their normal orientation as mounted on the engine or airframe. All shaft-driven accessories shall be operated under the maximum allowable axial and angular misalignment conditions specified at the gearbox drive pad.

Note that disassembly of the component is specifically prohibited until completion of the entire battery of tests to be performed. Disassembly, calibration, or other adjustments of the component shall invalidate all prior testing. Only component acceptance test procedures are allowed to be performed between test segments, not including calibrations or other adjustments which might be part of the ATP.

Pre- and post-test ATPs or functional checks of the component shall be conducted, as applicable, and the component shall be operating and verified for specified performance throughout the test, except for periods when the component is specified to be ‘off’.

If a component ATP includes field service limits in addition to new-part limits, then the field service limits may be considered in determining pass/fail at the conclusion of the series of tests (as well as for any intermediate ATPs) if approved by the Airworthiness Authority.

5.3.1 Operational cycle.

(Referenced by: 5.9.3)

A component operational cycle shall be defined by the contractor and submitted to the Airworthiness Authority for approval. General guidance for defining the operational cycle is provided below. Note that this is not the environmental profile to be applied, which is specified in the individual test subparagraphs, 5.3.2 through 5.3.7.

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5.3.1.1 Cycle definition: components excluding ignition system.

(Referenced by: 5.3.3a, 5.3.4a, 5.3.5, 5.3.6, APPENDIX A)

The component operational test cycle shall be consistent with the following general guidelines:

a. Each component input signal shall pass through its complete defined range of operation at least once during each cycle.

b. Pilot controlled inputs, such as a power demand signal, shall be varied in single step changes over their total range and shall not be changed again until all output variables have reached their steady-state values.

c. Engine supplied input signals shall be varied in their usual relation to each other, and shall be consistent with component outputs (e.g., fuel flow, guide vane position, etc.).

d. Input variables substantially independent of other control inputs, such as pressure altitude, oil temperature, air media temperature, etc., shall be varied at a rate faster or slower than each other so that every input signal eventually will have functioned at each value of the independent variables. This requirement is particularly important for mechanical assemblies when input signal combinations can result in overdriving components or cause binding of parts when the component is exposed to environmental extremes. Additional guidance is provided in H.2.2.

e. When manual or automatic transfer from one mode of operation to another, or from one channel to another is provided, the manual or automatic means shall be used to demonstrate mode and channel transfers during the operational cycle. All components within the UUT or LRU shall be functionally cycled, even if identically redundant.

f. Functions or features of the component which would not normally be activated by the simulated operational cycle (e.g., overspeed prevention) shall be activated at least once every ten (10) operational cycles.

The ‘basic’ operational cycle is generally defined as the time it takes the slowest changing signal or parameter to complete a single cycle, with all other signals or parameters being varied at non-multiple faster rates of change. However, there may be instances when some of the independent variables may vary at a rate slower than the basic cycle for practical purposes, but the number of these signals or parameters should be minimized, and the rates should not be less than one-third the basic cycle rate as defined herein.

Since the test requires that a minimum number of operational cycles or minimum time be completed, whichever is the longer duration, the basic operational cycle should be defined to be on the order of ten minutes to keep the test execution time manageable.

The test plan shall include a complete list of inputs to be cycled, the corresponding ranges for each input, and the cyclic profile to be followed by each input. Disturbing functions such as variations in fuel pressure and media airflow shall be included in the list of inputs, if appropriate to the component type. Input and output parameters versus time shall be recorded at a rate, resolution, and accuracy suitable for detection of deviation from performance specifications (e.g., ATP), throughout the test.

For electronic control units or similar components, an alternative to continuously varying input signals as described in a., above, may be considered by the Airworthiness Authority, depending on the monitoring approach utilized for the unit under test. A discussion of alternate approaches is provided in APPENDIX A.

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5.3.1.2 Cycle definition: ignition system.

(Referenced by: 5.3.3d, 5.3.4d, 5.3.5d)

Ignition system components shall be tested in accordance with one of the applicable operational cycles detailed below. For the purpose of these tests, the minimum and maximum voltages and frequencies are those that correspond to the extreme conditions permitted on the engine for satisfactory functioning of the ignition system. The ignition system shall be tested with the same number of ignitors connected as used on the engine.

The ignitors shall be installed in a suitable chamber and the chamber then purged with air or nitrogen at a rate specified in the test plan. The chamber pressure shall be regulated to simulate the internal engine pressure from minimum to maximum at which the ignition system is expected to function, over the entire engine operating envelope.

Applied Voltage Sequence	Ignition System Design Type			
	Light-off Duty Cycle		Continuous-Duty Cycle	
	On	Off	On	Off
1) Nominal	1 minute	1 minute	60 minutes	20 minutes
2) Maximum	1 minute	1 minute	40 minutes	20 minutes
3) Minimum	1 minute	1 minute	30 minutes	20 minutes
4) Nominal	1 minute	53 minutes	40 minutes	10 minutes

5.3.2 Low-lubricity fuel.

(Referenced by: 5.3.1)

Fuel system assemblies, or components normally requiring fuel for lubrication, shall undergo a simulated mission operational test for at least 100 hours. The test shall be in accordance with ARP1797, except that the components shall be subjected to twenty segments of five hours of operating time followed by one to five hours (at contractor discretion) of non-operating time.

The fuel system components shall be subjected to an acceptance test prior to and after the low lubricity test. Performance shall conform to new part standards, unless specifically authorized by the Airworthiness Authority to utilize field service limits (if any are defined).

The fuel used for the test shall be American Society for Testing and Materials (ASTM) D1665 Jet A, NATO F-24, or JP-8, degraded to provide a lubricity wear scar diameter (WSD) between 0.85 mm and 0.96 mm as determined by ASTM D5001. Alternatively, it is acceptable to use MIL-PRF-7024 Type II calibration fluid, degraded to meet the WSD requirement.

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5.3.3 High temperature.

(Referenced by: 5.3.1)

Each test assembly or component shall undergo testing in accordance with the following subparagraphs.

a. Engine components (excluding electronic/electromechanical assemblies, components with no moving parts, or ignition systems). Operationally cycle the component in accordance with 5.3.1.1 for a period of 100 hours or 600 operational cycles, whichever is the longer duration. During this cycling, ambient and fluid temperatures shall be varied as follows:

(1) Maintain the ambient temperature at 85°C (185°F) for 60 minutes. Then increase the temperature, within five minutes, to the specified component limiting temperature, and maintain this temperature for 120 minutes. Return the ambient temperature to 85°C (185°F) within five minutes. This ambient temperature profile shall be repeated continuously while the component is being operationally cycled. The temperature profile shall be adjusted so that it is not an integral multiple of the operational cycle.

(2) For components normally in contact with fuel, the fuel shall be NATO F-24, JP-8 or Jet A, but with an aromatic content of at least 25 percent. Toluene may be added to the fuel to meet this requirement. Fuel or oil at the component's inlet shall be held at its maximum allowable temperature, as specified in the engine model or component specification.

(3) Other fluids used for cooling or control purposes shall be maintained at their maximum allowable temperatures.

(4) Components utilizing bleed air or requiring pneumatic input signals shall be subjected to air at pressure, temperature, and flow values corresponding to those occurring throughout the range of engine operation for sea-level, hot-day conditions. The profiles for these parameters shall not be integral multiples of one another so that the component is exposed to various combinations of these parameters during operational cycling.

b. Electronic/electromechanical assemblies. The high temperature test shall not be performed, since it is included as part of the combined temperature-vibration test (5.3.6).

c. Components with no moving parts. The high temperature test shall not be performed, since it is included as part of the combined temperature-vibration test (5.3.6).

d. Ignition system (includes exciter). Prior to the start of the operational test, the exciter shall be 'aged' IAW 5.3.6e. Preconditioning. The ignition system shall then be operated either for 100 light-off duty cycles or 25 continuous-duty cycles in accordance with 5.3.1.2 at the specified maximum component limiting temperature (5.1.7).

A 30-minute shutdown following each light-off duty cycle, or a two-hour shutdown following each continuous-duty cycle, shall be performed. During the entire shutdown period, the air temperature surrounding the unit shall correspond to the highest stagnant air temperature expected after a ground engine shutdown at 55°C (131°F) ambient with no special cooling, such as forced ventilation or rotation of rotors, prior to the shutdown. At the conclusion of testing, checks shall be made of insulation resistance, over voltage capability, ignitor output energy, and spark rate.

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5.3.4 Room temperature and contamination.

(Referenced by: 5.3.1, 5.3.7)

Each test assembly or component shall undergo testing in accordance with the following subparagraphs.

a. Engine components (excluding electronic/electromechanical assemblies, components with no moving parts, or ignition systems). Each test assembly or component shall be operationally cycled in accordance with 5.3.1.1 for a period of 300 hours or 1800 operational cycles, whichever is the longer duration. Operational cycling shall be accomplished in six 50-hour (or 300 operational-cycle) segments.

(1) Test assemblies normally in contact with fuel shall be supplied with fuel conforming to NATO F-24, JP-8 or Jet A. Control of ambient or fuel temperature is not required.

During the first forty-five (45) hours of each test segment, contaminate the fuel downstream of the fuel tank with at least the concentration of contaminant specified in TABLE V. During the remaining five hours of each segment, contaminate the fuel with at least the concentration of contaminant specified in TABLE VI. The solid contaminant shall not be recirculated.

The test assembly shall remain idle for at least 18 hours following the second and fifth 50-hour test segment, and for at least 6 hours following the first, third, and fourth 50-hour segment. During these non-operating times, the unit shall not be drained of any of the contaminated test fuel that would normally remain in the unit.

During testing, upstream fuel filters may only be used if furnished with the engine fuel system. These may be serviced as recommended by the engine manufacturer. Unless otherwise authorized by the Airworthiness Authority, change out intervals shall be no more often than the time period needed to accumulate the total contaminant obtained during the 12-hour fuel filter capacity test of 5.6.7.

(2) Components utilizing bleed air or requiring pneumatic input signals shall be subjected to air at pressure, temperature, and flow values corresponding to those occurring throughout the range of engine operation for sea-level, standard-day (SLSD) conditions. The profiles for these parameters shall not be integral multiples of one another so that the component is exposed to various combinations of these parameters during operational cycling.

If the component utilizes engine bleed air, then during the first hour and for one hour after each succeeding ten (10) hours of testing, this air shall be contaminated as follows:

(a) Three parts per million engine lubricating oil by mass.

(b) A salt concentration of 0.2 parts salt (NaCl) per million parts of air by mass (salt shall be introduced using a 4.0 percent water solution).

(c) Distilled water to saturate the air at 52°C (126°F) at an ambient absolute pressure of 14.7 pounds per square inch (psi).

(d) International Organization for Standardization (ISO) 12103-1, grade A2, Fine Test Dust, 1.46 x 10^{-4} pounds (lbs.) of dust per pound of air.

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(3) Test assemblies normally in contact with oil shall be supplied with oil conforming to MIL-PRF-23699. Control of ambient or oil temperature is not required. Prior to the start of the test, the oil in the engine oil tank (simulated or actual) shall be filled with the specified total amount of oil IAW the EMS. If the supply tank is simulated, the amount of usable oil shall mimic that available in the actual engine oil reservoir.

The oil shall then be contaminated with a 50/50 mix of ISO fine (A2) and coarse (A4) test dust at a concentration of 1.0 milligram per 1 US quart of oil. Self-contained or separate oil filters that are part of the engine assembly under test shall be utilized.

Midway through each 50-hour segment, the test shall be halted, one quart extracted from the tank and replaced with one quart of new oil contaminated as above. During any test segment, if an engine oil filter impending bypass indication is triggered, the supplier shall determine if the test segment can be completed without going into actual bypass. If the test is stopped to change the filter, the supply tank will be topped off with fresh contaminated oil to the fill line, and the remainder of the test segment completed.

The test assembly shall remain idle after each 50-hour test segment for a minimum of 4 hours, except for the last segment. During these non-operating times, the assembly shall not be drained of any of the contaminated test oil that would normally remain in the unit.

b. Electronic/electromechanical assemblies. The room temperature test shall not be performed, since it is included as part of the combined temperature-vibration test defined in 5.3.6.

c. Components with no moving parts. The room temperature test shall not be performed, since it is included as part of the combined temperature-vibration test defined in 5.3.6.

d. Ignition system (includes exciter). The ignition system shall be operated at an ambient temperature between 16°C (61°F) and 38°C (100°F). Throughout the test, broadband random background vibration shall be applied. Unless otherwise specified, the vibration spectrum to be applied is 0.02 G^2/Hz over a bandwidth of 10 to 2000 Hz for airframe-mounted components, and 0.03 G^2/Hz over a bandwidth of 10 to 2000 Hz for engine-mounted components.

The system shall be operated in accordance with either the light-off duty cycle or the continuous-duty cycle specified in 5.3.1.2, depending on the ignition system type.

For the light-off duty type, the system shall be operated for 300 cycles.

For the continuous-duty type, the system shall be operated for 75 cycles.

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5.3.5 Low temperature.

(Referenced by: 5.3.1)

Each test assembly or component shall be soaked at an ambient temperature of lower than -54°C (-65°F) for a minimum of ten (10) hours. Upon completion of the 10-hour soak, the ambient temperature of -54°C (-65°F) shall be maintained while the test assembly or component is operationally cycled. The component shall be cycled in accordance with 5.3.1.1 for a total of 20 hours or 120 operational cycles, whichever is the longer duration (applies to subparagraph a. only). The test shall consist of ten 2-hour (or 12-cycle) test segments, with a 1-hour non-operating period between test segments.

a. Engine components (excluding electronic/electromechanical assemblies, components with no moving parts, or ignition system).

   (1) For test assemblies or components normally in contact with fuel, fluid conforming to MIL-DTL-5624W, grade JP-4, (or an approved test fluid with equivalent characteristics) shall be present in each test assembly or component. For test assemblies or components normally in contact with oil, oil conforming to MIL-PRF-23699 shall be present in each test assembly or component.

   (2) Prior to each test segment, the test fluid temperature shall be reduced to below -54°C (-65°F) for a minimum of one hour, after which operational cycling may begin. While cycling, fluid temperatures may rise as anticipated in service operation under similar ambient conditions. However, if -34°C (-29°F) is reached before completion of a test segment, the test shall be stopped and restarted when the fluid temperature has been reduced to below -54°C (-65°F) for a minimum 15-minute period.

   (3) Components utilizing bleed air or requiring pneumatic input signals shall be subjected to air at pressure, temperature, and flow values corresponding to those occurring throughout the range of engine operation defined for sea-level, cold-day in the EMS. The profiles for these parameters shall not be integral multiples of one another so that the component is exposed to various combinations of these parameters during operational cycling.

b. Electronic/electromechanical assemblies. The low temperature test shall not be performed, since it is included as part of the combined temperature-vibration test (5.3.6).

c. Components with no moving parts. The low temperature test shall not be performed, since it is included as part of the combined temperature-vibration test (5.3.6).

d. Ignition system (includes exciter). The ignition system shall be tested at an ambient temperature of -54°C (-65°F). The system shall be operated in accordance with either the light-off duty cycle or the continuous-duty cycle specified in 5.3.1.2, depending on the ignition system type.

   For the light-off duty type, the system shall be operated for 12 cycles, followed by a 10-hour minimum inoperative soaking period, and then a final 12 cycles.

   For the continuous-duty type, the system shall be operated for 3 cycles, followed by a 10-hour minimum inoperative soaking period, and then a final 3 cycles.

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5.3.6 Combined temperature-vibration.

(Referenced by: 5.3.1, 5.3.3b/c/d, 5.3.4b/c, 5.3.5b/c, FIGURE 2)

Each electronic/electromechanical assembly (not including ignition exciters), or other components without moving parts, shall undergo operational cycling in accordance with 5.3.1.1 while being exposed to combined temperature-vibration. Test duration shall be a minimum of 300 hours or 100-600 operational cycles, as discussed in 'Temperature-vibration environmental cycles' below. Prior to the start of the operational test, all electronic assemblies shall be 'aged' in accordance with subparagraph e. 'Preconditioning'.

The assembly or component shall be monitored at all times to assure compliance with performance specifications. However, specification compliance is only required when the unit is operating between the minimum and maximum allowable operating temperatures defined in the engine model or component specification. Power up or operation within the non-operating temperature ranges shall not result in damage to the equipment, nor result in the equipment indicating any failure condition to other subsystems. Once the ambient temperature is within the specified operating limits, the equipment shall meet its performance requirements without requiring power cycling or other means of reset.

The contractor shall propose an ambient and cooling media (if applicable) temperature profile that generally replicates component usage in the intended engine/airframe application. The profile shall be approved by the Airworthiness Authority. The following general guidelines shall be followed in defining the ambient profile, an example of which is provided in FIGURE 2 for a typical engine-mounted ECU or similar equipment.

It should be noted that some components that fall into the category of 'no moving parts' may or may not require the imposition of an operational cycle during the SOT, for the reason of design simplicity. An example would be a magnetic speed sensor where the target wheel is not part of the sensor itself. Attempting to drive the target wheel independently of the sensor would be difficult to simulate with any fidelity because of the vibratory and temperature response differences between the mounting locations of these two independent components.

In the above case, subjecting the sensor to the environmental temperature-vibration profile, while continuously monitoring for shorts or opens, would be sufficient. The same would hold true for similar components without moving parts such as temperature sensors, electrical harnesses, heat exchangers, etc. However, other simple components may be more easily subjected to 'operation' such as a pressure sensor and the like, since superimposing a pressure cycle on the environmental cycle is far simpler to effect.

Temperature-vibration environmental cycles require a significantly longer period than that of an operational cycle, therefore allowance must be made regarding the number of operational cycles required for those components relying solely on the environmental cycle to verify 'performance'. Due to environmental profile variability, the number of operational cycles shall be proposed by the supplier and agreed by the Airworthiness Authority. This should not be construed as alleviating the supplier from making a good-faith effort to define a realistic operational cycle for all components, regardless of type.

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a. Air temperature. Vary the ambient temperature between the specified minimum non-operating temperature and the component limiting temperature (5.1.7) or the non-operating temperature, whichever is the higher. If the maximum non-operating temperature is not specified, it shall correspond to the highest stagnant air temperature reached, in the vicinity of the UUT, after a ground engine shutdown at 55°C (131°F) ambient, with no cooling prior to engine shut down.

(1) The ambient temperature cycle shall begin and end at room temperature.

(2) Steady-state temperature dwells shall occur in the following order.

1) Room	2) Maximum operating
3) Maximum non-operating	4) Maximum operating
5) Room	6) Minimum operating
7) Minimum non-operating	8) Minimum operating

(3) All steady-state dwells shall be 15 minutes, minimum, once the component has reached stabilization. Stabilization shall be defined as the component temperature, measured at 1 inch from the coldest surface, being within 3°C (5.4°F) of the target temperature, or the component temperature changing by no more than 3°C (5.4°F) over a 5-minute period, whichever is achieved first.

(4) Transitions between operating temperatures shall be 5°C (9°F) per minute, minimum. If the equipment is engine mounted, transitions in the increasing direction shall occur at 10°C (18°F) per minute, minimum.

(5) Transitions between operating and non-operating temperatures shall be 2°C (3.6°F) per minute, maximum.

b. Electrical power. If the component is electrically powered, power shall be applied and removed in accordance with typical component usage.

(1) During the ‘hot’ half of the temperature profile, applied voltage shall be at the maximum steady-state value of 31.5 Vdc for MIL-STD-704 low voltage DC systems. If the component under test is powered from another source, the applied power shall be no less than the maximum allowable specified in the component specification. If multiple sources power the component, the applied power shall be that which maximizes internal heating.

(2) During the ‘cold’ half of the temperature profile, applied voltage shall be at the minimum specified value (typically 16 Vdc for non-electrically started engines or 12 Vdc for electrically-started engines). If the component under test is powered from another source, the applied power shall be no greater than the minimum allowable specified in the component specification. If multiple sources power the component, the applied power shall be that which minimizes internal heating.

(3) When the maximum non-operating temperature is reached (simulating a soak back condition), electrical power shall be removed.

(4) At the beginning of the transition from the minimum operating temperature to the minimum non-operating temperature, electrical power shall be removed to assist in cooling.

(5) At the beginning of the transition from the non-operating temperature to the operating temperature, electrical power shall be applied in accordance with b(1) or b(2).

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c. Cooling media. If liquid or air is used for cooling the component, the following shall apply:

(1) Maintain the cooling media at its maximum specified temperature and minimum specified flow during the 'hot' half of the temperature profile.

(2) Maintain the cooling media at its minimum specified temperature and maximum specified flow during the 'cold' half of the temperature profile.

d. Vibration. During all portions of the test, except for those times defined in subparagraphs (1) and (2) below, broadband random background vibration shall be applied. The vibration spectrum to be applied is 0.0133 g^2/Hz over a bandwidth of 10 to 2000 Hz for airframe-mounted components, and 0.020 g^2/Hz over a bandwidth of 10 to 2000 Hz for engine-mounted components. Vibration shall be applied to the component in each of its three mutually orthogonal axes, split equally over the test time period.

(1) During the steady-state dwell at the specified maximum non-operating temperature of the component, or during transitions to/from that condition.

(2) During the steady-state dwell at the specified minimum non-operating temperature of the component, or during transitions to/from that condition.

e. Preconditioning. Electronic assemblies, and electromechanical components containing circuit boards or similar, shall be preconditioned for a period of 3 days (72 hours) without interruption (performance monitoring is not required) IAW with step (1). If the electronic assembly is hermetically sealed, this will conclude the preconditioning phase. For non-hermetically sealed assemblies, additional preconditioning step (2) shall be executed.

After completion of preconditioning, a room temperature ATP (full or partial) is permitted to confirm unit performance IAW the component specification. Notwithstanding the time taken to conduct the ATP, operational testing must begin within 24 hours of completion of the step (2) humidity preconditioning, if applicable.

(1) The unit under test shall have maximum specified electrical power applied for 72 hours at a surrounding air temperature of 55°C (131°F), after which power shall be removed.

(2) The electronic assembly shall be placed, unpowered, in a humidity chamber. Temperature shall then be increased to 55°C (131°F) and humidity to 95% RH over an 8-hour period, and shall remain at those conditions, uninterrupted, for 80 hours. Temperature and humidity will then be returned to ambient conditions over a final 8-hour period, for a total time of 96 hours.

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5.3.7 Engine fuel system cavitation endurance and low temperature operation.

(Referenced by: 5.3.1)

The purpose of this test is to evaluate the ability of the engine fuel system pumping elements to resist deterioration due to cavitation over prolonged periods of time, and to evaluate fuel system operation under maximum viscosity conditions. The portion of the aircraft fuel system from the fuel tank to the engine fuel inlet, and the engine fuel system from the engine fuel inlet to the engine fuel nozzles, shall be included in the test assembly. Note that any miscellaneous components used in the test assembly up to the engine fuel inlet are not under test; they are only being utilized to accurately simulate the conditions at the fuel pump inlet. For that reason, it may be acceptable to simulate these airframe components with approval from the Airworthiness Authority.

Except as stated below, the test procedure shall be in accordance with ARP492. The test parameters listed below are default values. The contractor is allowed to tailor these parameters (e.g., inlet fuel pressure, inlet fuel temperature, etc.) to match the specified depressed fuel inlet capability of the engine fuel system. Changes shall require the approval of the Airworthiness Authority prior to conducting the test.

If the component under test has not gone through the room temperature and contamination test defined in 5.3.4, then prior to the start of this test, the system shall be ‘aged’ by having fuel passed through it at the maximum continuous engine fuel flow rate for five hours while contaminated with at least the amount specified in TABLE VI.

Clean fuel shall be used to conduct the endurance portion of the test. Each cycle consists of 45 minutes at the maximum continuous speed and flow required by the engine at the altitude corresponding to the fuel tank pressure, followed by 15 minutes at the minimum idle speed and flow required by the engine at those same conditions. Backpressure to the engine main fuel pump shall be in accordance with the operating conditions.

The test comprises two parts (hot and cold fuel), as described on the following page. The hot fuel testing is intended to evaluate the engine fuel system endurance under cavitation conditions. The cold fuel testing, on the other hand, is intended to demonstrate that the engine fuel system is able to operate satisfactorily under maximum allowed fuel viscosity conditions.

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a. Part one (hot fuel). A restriction shall be inserted upstream of the engine fuel inlet to control fuel pressure at the inlet to whichever of the following conditions represents the highest absolute inlet pressure: (a) 1.0 psi above the true vapor pressure (TVP) of the fuel, (b) that fuel pressure corresponding to a vapor-to-liquid ratio of 1.0, (c) 35% of fuel tank pressure, or (d) 2.5 psia. This restriction shall be set prior to the 45-minute maximum continuous speed and flow portion of each cycle, and shall not be readjusted for the 15-minute idle speed and flow portion of the test cycle.

Twenty (20) cycles shall be run at each condition in the table below, with the fuel tank pressure and engine fuel inlet temperature set in accordance with the table, using the specified fuel type.

Test Condition	Fuel Type	Fuel Tank Pressure (psia)	Fuel Inlet Temperature (°C/°F)
a	NATO F-24 (iaw AFLP-3747) or Jet A (iaw ASTM D1655) w/military additives or JP-8 (iaw MIL-DTL-83133)	14.6	71/160
b		13.7	
c		9.3	
d		13.6	57/135
e		11.4	
f		7.0	
g		11.9	43/109
h		9.7	
i		5.4	
j	JP-4 (iaw MIL-DTL-5624W)	11.0	29/84
k		8.9	
l		5.4	

b. Part two (cold fuel). A restriction shall be inserted upstream of the engine fuel inlet to control fuel pressure at the inlet to the higher of 35 percent of the specified fuel tank pressure or 2.5 psia. This restriction shall be set prior to the 45-minute maximum speed and flow portion of each cycle, and shall not be readjusted for the 15-minute idle speed and flow portion of the test cycle.

Twenty (20) cycles shall be run at each condition in the table below. Fuel temperature at the engine fuel inlet shall be controlled in order to maintain fuel viscosity in accordance with the table.

Test Condition	Fuel Type	Fuel Tank Pressure (psia)	Fuel Inlet Temperature
a	JP-8 (iaw MIL-DTL-83133)	14.5	To maintain viscosity ≥ 12 centistokes (Cs)
b		8.7	
c		4.4	

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5.4 Environmental series tests.

(Referenced by: 4.3.2, 5.2, 5.10.3, 5.10.4.1, TABLE VII, J.2.3)

For the environmental series battery of tests specified in TABLE VII, two or three serialized units should be used to complete the test sequence. In all cases, the tests conducted on each unit, and the sequence of those tests, shall be as specified in TABLE II. Any proposed deviation from the specified sequence requires approval by the Airworthiness Authority. If any of the environmental tests are approved to be met by analysis or similarity, the remaining tests shall still be performed in the order shown. In general, requirements are to be met via testing rather than by similarity/analysis, so that damage is accumulated on the UUT from one test to the other as they are executed in the series.

If the component under test is hermetically sealed, it shall not be disassembled for inspection until the last test of the series has been completed. At that time, the component shall be inspected for defects or damage which may have occurred during the testing. Prior to disassembly, verification of hermetic seal integrity shall be performed IAW MIL-STD-202-112, test condition A. Failure of the hermetic seal shall disqualify that component.

Hermetically sealed components are exempt from certain tests as stated in the individual test requirement descriptions (also noted in TABLE II and TABLE VII). If a hermetically sealed component must undergo explosive atmosphere testing (see 5.4.9 for requirements), then prior to the test the hermetic cavity shall be penetrated to simulate a latent failure, and then the breach resealed if the component is planned for further testing.

Note that unit disassembly between any of the tests sequenced 1 through 6 for UUT1 of TABLE II is prohibited. Doing so without prior authorization from the Airworthiness Authority may invalidate all testing on that unit. Only external visual or fluorescent penetrant inspections (FPI) are allowable.

5.4.1 Default environmental parameters.

(Referenced by: 5.4.2, 5.4.7, 5.4.9)

When specific environmental test parameters for ambient temperature and altitude are not identified herein, nor in applicable aircraft, engine model, or component specifications, then the following test limits shall be utilized as defaults.

Default Environment Parameter		Operational	Non-Operational
Low Temperature		-54°C (-65°F)	-54°C (-65°F)
High Temperature	**Airframe Mounted**	+85°C (+185°F)	+85°C (+185°F)
	Engine Mounted	+149°C (+300°F)	+174°C (+345°F)
Low Pressure	**Altitude and Temperature**	6,096 m (20,000 ft) and -25°C (-13°F)	15,240 m (50,000 ft) and -54°C (-65°F)

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5.4.2 Temperature: high / low / shock (transient).

(Referenced by: 5.4.3.1, 5.4.3.2, TABLE II NOTE 3/)

Components shall be subjected to temperature testing in accordance with subparagraphs a, b, and c, below. The three tests shall be combined into a single temperature profile. Components shall be subjected to pre- and post-test ATPs or functional checks, as applicable, and be operating and verified for specified performance throughout the test, except as noted.

a. High temperature. In accordance with MIL-STD-810H(C1), Method 501.7, Procedures I (storage/non-operating) and II (operating); constant temperature method.

b. Low temperature. In accordance with MIL-STD-810H(C1), Method 502.7, Procedures I (storage/non-operating) and II (operating).

c. Temperature shock (transient). In accordance with MIL-STD-810H(C1), Method 503.7, Procedure I-C, with the number of cycles increased to five (minimum).

The test shall comprise two (2) long-duration steady-state temperature cycles (a. and b. above, combined) followed by five (5) shorter duration transient cycles (c. above). If the total time period for the five (5) transient cycles is insufficient to completely determine UUT compliance with performance requirements, the number of cycles shall be increased as necessary to allow this determination to be made.

For the steady-state temperature profile, each dwell shall be maintained for two (2) hours minimum after the component temperature, measured at 1 inch from the coldest surface (high temperature) or hottest surface (low temperature), is within 3°C (5.4°F) of the ‘dwell’ target temperature. For the subsequent transient temperature cycles, the dwell temperature shall be maintained for ten (10) minutes minimum after the component is within 3°C (5.4°F) of the ‘dwell’ target temperature, as measured above.

The minimum temperature ramp rate for the steady-state profile is 5°C (9°F) per minute for airframe-mounted components, and 10°C (18°F) per minute for engine-mounted components. For the transient cycles, the respective ramp rates are 15°C (27°F) and 20°C (36°F) per minute. During the high-rate transient cycles, unit accuracy may not meet specified requirements, in which case widening the pass/fail limits may be necessary with Airworthiness Authority concurrence. However, when the measured UUT temperature rate of change slows to 5°C per minute (airframe-mounted components) or 10°C per minute (engine-mounted components), signal conditioning accuracy shall meet specified requirements.

For electronic assemblies powered by the airframe electrical bus, power shall be left ON during the simulated hot soak back temperature rise portion of the steady-state profile since power may not be removed coincident with engine shutdown. For cold soak conditions below the lower operating limit, electrical power shall be removed to simulate soaks at the lower non-operating limit. After re-applying electrical power at the minimum non-operating temperature (simulating aircraft power up), performance testing shall be resumed when the ambient temperature reaches the minimum operating temperature specified for the component.

FIGURE 3 is an example of a combined temperature profile for an airframe-mounted electronic unit which has a specification-defined lower operating limit of -40°C (-40°F) and an upper operating limit of 71°C (160°F), with undefined non-operating limits. In this example, the default table in 5.4.1 would be utilized to determine the unspecified non-operating limits for test purposes.

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If the component also conveys fluid (fuel, oil or air), then the maximum temperature to which it is exposed may not occur at maximum (max) ambient nor at max fluid temperature, but rather at some combination of the two. The contractor shall provide an analysis that establishes the ambient/fluid temperature combination that results in the worst-case UUT internal temperature(s), which shall then be used as the high temperature operating (and/or non-operating) condition as appropriate to the component type.

For the steady-state low temperature portion of the test, fluid temperature shall be lowered to the minimum specified operating/non-operating value, whichever is lower, while the UUT ambient temperature is transitioning from hot to cold. When the unit is operating, the fluid temperature at the inlet to the component may be allowed to rise (depending on the component type), but by no more than 20°C (36°F) from its initial starting temperature. If exceeded, then the test shall be stopped until the fluid temperature is returned to its minimum value for at least 15 minutes, at which time the test can be re-started.

For the transient portion of the test, fluid temperatures should generally rise and fall with the ambient air temperature. It is recognized that the rate at which the fluid temperatures can be increased or decreased is a function of the fluid reservoir quantity and the heating/cooling capacity of the heat exchange equipment. The contractor shall make all reasonable efforts to control fluid temperature to the minimums and maximums specified in the component specification.

FIGURE 4 is an example of a temperature test profile for a fuel metering unit that meets the intent of this requirement. Note that while no specific fuel or oil is mandated for this test, the fuel/oil must be compatible with an ambient temperature cycle that covers an extreme range of operating and non-operating temperatures, while at the same time not inhibiting the ability of the unit to meet its performance requirements (due to viscosity effects, etc.). For this purpose, acceptable fuels include JP-4 and MIL-PRF-7024 Type II calibration fluid. Use of a synthetic oil conforming to MIL-PRF-23699 is also recommended for its extended operating temperature range.

In almost all cases, the operating and non-operating temperature for an engine-mounted component is the same, unless it is being actively cooled while the engine is running (e.g., a fuel pump). Loss of cooling when the engine shuts down may result in the component temperature increasing as a consequence of engine ‘soak back’, and then after some minutes dropping back down below operating max temperature.

If an actively cooled component is also electrically powered by the aircraft (e.g., an ECU), even when the engine is shut down the component will still be operating. Therefore, its temperature will likely continue to increase until it reaches some equilibrium point, which may very well be above electrical part limits, possibly resulting in failure.

All of these non-operating temperatures are well above any ‘storage’ temperature that the component will be subjected to during its lifetime. Therefore, in complying with the 810 temperature test methods cited in this section, it is not appropriate to use storage temperatures or the diurnal cycle in determining the ambient test temperature to use when the engine is assumed to be not running.

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5.4.3 Vibration.

Components may be tested as assemblies or as individual units; however, they shall simulate installation orientation or other environmental or physical details, such as electrical harness 'pigtails' to the first engine tiedown point, pneumatic or hydraulic connections or fittings, engine-supplied brackets, pneumatic ducting, etc.

During the test, the component shall be subjected to its maximum limiting temperature (discussed in 5.1.7) and shall be operating to evaluate any functional effects of the vibration on performance.

For vibration testing that is performed at a temperature less than the CLT, the vibration profile shall be increased to compensate for the difference in material properties between the test temperature and the specified CLT, for that component. The determination of the increase in vibration level should use industry standard methodologies, and shall be concurred by the Airworthiness Authority.

Tightening of component mounting hardware at any time during the test, without Airworthiness Authority approval, may result in invalidation of the entire test. The contractor will make every effort to design the component mounting fixture so that it can be rotated for each test axis, rather than the component itself, in order to avoid violating the prohibition against tightening mounting hardware. Match marking should be used to allow visual determination of loosening of mounting hardware or subcomponents of the unit under test.

5.4.3.1 Vibration and high temperature: preliminary flight rating.

(Referenced by: 5.4.3.2, TABLE III NOTE 6/)

Components shall be subjected to a vibration test in order to:

1) Verify functional performance of the component when exposed to traditional resonance dwells and sinusoidal cycling.
2) Determine that there are no unacceptable component structural or operational resonance modes.

For PFR only, if the component cannot be functionally tested at its maximum limiting temperature (5.1.7) while being vibrated, a separate functional test shall be performed to verify acceptable performance at the limiting temperature. The test shall precede the vibration test, and will generally conform to the requirements of 5.4.2, except the steady-state and transient test cycles need not go below room temperature, and only two transient cycles are required instead of five.

Components equipped with integral vibration isolators, or attached to vibration-isolated brackets or mounting frames, may utilize new isolators at the start of each axis of vibration testing, with the approval of the Airworthiness Authority. Replacement of the isolators at any other time is not allowed. If the isolators are 'hidden' or uninspectable when the component is mounted on the engine, then isolator changeout between test axes is strictly prohibited.

Test Category b.1 (specified in 5.4.3.1.1 and 5.4.3.1.2) requires a resonance search and a maximum of four (4) 30-minute dwells prior to sinusoidal cycling. However, there may be instances when more than four significant resonances are found, in which case consideration shall be given to increasing the number of dwells. Dwell frequencies shall be proposed by the contractor and concurred by the Airworthiness Authority. Regardless of the number of dwells performed, a minimum of one (1) hour of sinusoidal cycling must be conducted on a per axis basis.

For test Category c (specified in 5.4.3.1.3) there is no requirement for a resonance search and subsequent dwells; only sinusoidal cycling is required (at a sweep rate of ½ octave per minute vs 1 octave per minute for Category b.1).

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5.4.3.1.1 Engine-mounted components.

(Referenced by: 5.4.3.1, 5.4.3.1.4)

Engine-mounted components shall be tested IAW MIL-STD-810C, Method 514.2, Category b.1, Procedure I, Part 1, Curve L (turbine) or Curve F (reciprocating). If normally equipped with vibration isolators, Part 2, Curve AR (turbine) or Curve B (reciprocating) is also applicable. If the engine is intended for use in a helicopter, then the Curve L test amplitudes will be increased over the 5 Hz to 33 Hz frequency range to match Curve M of Figure 514.2-3. Test duration shall be IAW Table 514.2.II, with test profiles IAW Figure 514.2-2.

5.4.3.1.2 Fixed-wing airframe-mounted components.

(Referenced by: 5.4.3.1, 5.4.3.1.4)

Components located in the fuselage or wing area of a fixed-wing aircraft shall be tested IAW MIL-STD-810C, Method 514.2, Category b.1, Procedure I, Part 1, Curves G, H or J (turbine) or Curves C, D or E (reciprocating). If normally equipped with vibration isolators, Part 2, Curve AR (turbine) or Curve B (reciprocating) is also applicable. Test duration shall be IAW Table 514.2-II, with test profiles IAW Figure 514.2-2.

5.4.3.1.3 Helicopter airframe-mounted components.

(Referenced by: 5.4.3.1, 5.4.3.1.4)

Components mounted in the fuselage of a rotary-wing aircraft shall be tested IAW MIL-STD-810C, Method 514.2, Category c, Procedure I, Part 1, Curve M, and if normally equipped with vibration isolators, Part 2, Curve B is also applicable. Test duration shall be IAW Table 514.2-III, with test profiles IAW Figure 514.2-3.

5.4.3.1.4 Known application environment.

(Referenced by: 5.4.3.2)

In the context of PFR testing, a ‘known’ vibration environment is very unlikely unless the component is being mounted on the engine of an existing fielded airframe, as might be the case for a component improvement program. Installing a new engine in an existing airframe, or an existing engine in a new airframe, does not qualify as a ‘known’ environment because of the influence each has on the other’s vibration characteristics, as stated in MIL-STD-810H(C1), Method 514.8, Annex D, 2.11e.

Nevertheless, if the airframe- and engine-induced vibration environment is known, then the contractor may request that the test amplitudes specified by 5.4.3.1.1 and (5.4.3.1.2 or 5.4.3.1.3) be tailored to replicate the measured vibratory environment. Sufficient in-flight measurement data must be provided so that the Airworthiness Authority can concur with the proposed test profile.

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5.4.3.2 Vibration: qualification.

(Referenced by: 5.4.3.2.2, TABLE II NOTE 3/)

Components shall be subjected to a vibration test in order to verify performance-to-specification when exposed to the expected 'worst-case' (aka 'functional') vibration environment encountered during service usage. Either a sine-on-random or a narrowband-on-random, or both, test profiles shall be applied, depending on the mounting location of the equipment under test.

The duration of the functional vibration test shall be four (4) hours per axis in order to demonstrate that the component can function satisfactorily at 'worst' case vibration levels for a minimum 4-hour mission. Functional performance of the component (i.e., fully operational with all inputs and outputs cycled) shall be monitored, recorded, and verified throughout the entire test period. At the conclusion of testing in all three axes, a calibration or ATP shall be conducted to fully verify component functionality.

Note that if there is no PFR program phase, then the specified PFR vibration test is also required for qualification. In this case, it shall be conducted before the random vibration test(s) in order to meet the objectives specified in 5.4.3.1.

Components equipped with integral vibration isolators, or attached to vibration-isolated brackets or mounting frames, shall use the same isolators that were mounted on the unit during the temperature test of 5.4.2.

Replacement of the isolators at any time is specifically prohibited during functional performance testing. However, if the test cannot be completed due to excessive isolator deterioration, as determined by the contractor, authorization to replace them must be received from the Airworthiness Authority. If replacement of isolators occurs during the test, it must be fully documented in the test report, including time at change out, detailed photographs of condition, and identification of the position on the component of each isolator.

The Procedure I shock test of 5.4.4 is recommended to be conducted in conjunction with the functional performance vibration test since the same equipment is typically used for both tests. After each axis of vibration is completed, the shock test for that axis should be conducted before moving on to the next axis of vibration.

Two possibilities for qualification are described in the subsections below: "known application environment" and "unknown application environment"; the distinction being explained in 5.4.3.1.4. The preferred approach is the known application environment procedure, which is consistent with MIL-STD-810H(C1), Task 402 ("Life Cycle Environmental Profile-LCEP").

The contractor is encouraged to conduct vibration qualification in accordance with the requirements of 5.4.3.2.2, if possible. Consequently, during initial flight testing the actual installed vibration environment for each component should be ascertained so that when qualification testing is undertaken, it can be performed accordingly [refer to MIL-STD-810H(C1), Method 514.8, Annex D, Section 2.3.e(1)].

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5.4.3.2.1 Unknown application environment.

(Referenced by: 5.4.3.2.2, 5.4.4, 5.4.5, 5.9.2, K.3, TABLE K-II NOTE 3/)

Vibration shall be IAW MIL-STD-810H(C1), Method 514.8, Procedure I, Appendix D, Categories 13, 14 and/or 22. The requirements of Category 13 apply to components mounted in propeller-driven aircraft, while Category 14 applies to those mounted in helicopters. If the component is turbine engine mounted, the requirements of Category 22 are additional to those of Category 13 or Category 14.

The default requirements for Categories 13, 14, and 22 are separately tabulated in TABLE VIII, TABLE IX, and TABLE X, respectively. Absent actual measured engine vibration data (both airframe and engine induced), these tables represent the default vibration test spectrum to be used when qualifying any engine-mounted component. Note that these test profiles are modified versions of those found in MIL-STD-810H(C1); reasons for which are discussed in G.3.1.

For Categories 13 and 14, the low frequency sine tones shall be synchronously swept over their expected operational ranges throughout the test. Because of the narrow range of operation, the tones should be swept linearly from minimum to maximum over a period of no less than 20 seconds, rather than logarithmically as is done when sweeping over several octaves. If equipment dictates that logarithmic sweeps must be used, they shall be no faster than 0.5 octaves/minute. The default sweep ranges shall be +/- 5% of the center frequency for Category 13, and +/- 10% of the main rotor center frequency and harmonics for Category 14.

For Category 22 testing, the narrowband notional test frequencies listed in the table below are to be adjusted based on the expected or actual operating ranges of the gas generator and power turbine for the specific engine in question. Engines with more than one gas generator rotor may require additional narrowband excitations to cover the operating ranges of the different spools. Unless otherwise specified, for components mounted on reciprocating engines the broadband and narrowband functional levels specified for Category 22 shall be increased by 50% and the spike frequency bands adjusted, as required, to the engine operating speed ranges.

Category 22 Narrowband ‘Spike’ Frequencies and Amplitudes		
NB Spike	**Frequency Range**	**Default ASD (g^2/Hz)**
f_0	Gas Generator: Ground Idle (GI%) to (GI+10)%	1.0
f_1	Gas Generator: 90% to 105%	1.0
f_2	Power Turbine (PT): 95% to 105%	1.0
f_3	PT: Single Engine GI to Dual Engine GI or Accessory Gear Box: Dominant Frequency -10/+5%	1.0

When two or more vibration categories are applicable to the component, they can be combined into a single test profile to be applied to the component for the specified test duration, or they can be applied sequentially to the same serial number unit, each for the specified test duration (see G.3.3).

Note that default Category 14, as tabulated in TABLE IX, is not applicable to the H-47, H-60, and H-64 airframes, since the actual vibration is considered ‘known’ as defined by TOP 01-2-603A. This is discussed in the following subsection.

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5.4.3.2.2 Known application environment.

(Referenced by: 5.4.3.2, 5.4.4, 5.4.5, 5.9.2, K.3, TABLE K-II NOTE 3/)

If the vibration environment is known, the contractor should define both the expected 'worst-case' vibration environment to be encountered during service (e.g., out-of-balance engine) and the 'typical' (e.g., mission-weighted) vibration environment expected over the life cycle of the equipment. The contractor shall provide substantiating data for the defined vibration levels and obtain concurrence from the Airworthiness Authority. Testing shall be IAW the procedures and durations specified in 5.4.3.2 and 5.4.3.2.1.

For the H-47, H-60, and H-64 helicopters, the vibration environment is deemed to be 'known' as a result of the publication of TOP 01-2-603A (Test Operations Procedure – Rotorcraft Laboratory Vibration Test Schedules) in 2023, resulting in the Category 14 default no longer being applicable to these three platforms, as stated in MIL-STD-810H(C1).

The vibration test profiles for these three aircraft are specified in TABLE XI, TABLE XII, TABLE XIII, and TABLE XIV. Note that those test curves have been adjusted (as explained in APPENDIX G) to account for the difference between an accelerated vibration profile used for life testing, as stated in TOP 01-2-603A, vs a lower amplitude worst-case vibration profile needed for functional testing.

5.4.4 Shock.

(Referenced by: 5.4.3.2, K.1)

Components shall be subjected to an operational shock test in accordance with MIL-STD-810H(C1), Method 516.8, Procedure I. The shock test spectrum shall be in accordance with Figure 516.8-2 (Test SRS) and Table 516.8-III, or Figure 516.8-3 (Terminal peak sawtooth) and Table 516.8-IV (Flight Vehicle Materiel). Tests may be conducted in conjunction with the functional vibration test defined in 5.4.3.2.1 or 5.4.3.2.2, by running vibration then shock before changing test axes. Components shall be subjected to pre- and post-test ATPs or functional checks as applicable, and shall be operating and verified for specified performance during the test. Because the duration of the applied shock is under 100 milliseconds, it may not be possible to determine compliance with all performance specifications, in which case selected monitoring may be necessary.

Components located in a crew compartment, or near a system or other component critical to occupant safety (e.g., a fuel tank), shall be subjected to an impact test in accordance with MIL-STD-810H(C1), Method 516.8, Procedure V, if it is determined that it could become a hazard if it breaks free of its mounts during a crash. The shock test spectrum shall be IAW Figure 516.8-2 (Test SRS) and Table 516.8-III, or Figure 516.8-3 (Terminal peak sawtooth) and Table 516.8-XIII. The component need not be operating during the test nor pass a post-test ATP or functional check. With the approval of the Airworthiness Authority, a dummy unit may be used if it duplicates the mounting arrangement, overall mass, and moments of inertia of the actual component.

During the above test(s), the component shall be subjected to its maximum limiting temperature, as specified in the engine model or component specification (see 5.1.7).

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5.4.5 Gunfire shock.

(Referenced by: K.1)

Components shall be subjected to gunfire shock testing in accordance with MIL-STD-810H(C1), Method 519.8, Procedure I or Procedure II, when specified by the SoW or other contractual vehicle. The specific procedure to be utilized shall be proposed by the contractor and agreed by the Airworthiness Authority. Components shall be subjected to pre- and post-test ATPs or functional checks, as applicable, and shall be operating at their component limiting temperature (as specified in the engine model or component specification, see 5.1.7) and verified for specified performance during the test.

Gunfire shock testing should be performed based on actual acquired data whenever possible. However, if the gunfire shock test profile is unknown at the time of component qualification, the UUT shall be tested to an estimated gunfire vibration profile determined IAW Procedure III, as defined in Annex C of Method 519.8.

If no specific platform/gun combination is specified in the SoW, the contractor shall define a default test profile by using a similar approach to that used in preparing example profiles for the H-60 platform (see K.3). Note that gunfire energy is highly sensitive to the distance between the gun and the mounting location of the equipment under test, so the example profiles are only pertinent to the indicated locations on the profile plots (generally the main engine/transmission area).

Gunfire is an independent source of vibration to be applied to the component in addition to other specified sources, either simultaneously or in separate tests. Engine-mounted equipment is subject to three simultaneous sources of vibration (e.g., Categories 13/14, 22, and gunfire shock) while off-engine components are subject to two sources (e.g., Category 13/14 and gunfire shock). This applies to the functional tests described in 5.4.3.2.1 and 5.4.3.2.2, as well as vibration endurance described in 5.9.2.

5.4.6 Acceleration.

Components shall be subjected to an acceleration test in accordance with MIL-STD-810H(C1), Method 513.8, Procedure II. The test levels shall be in accordance with Table 513-II; however, it is required that the highest level listed in the table be applied in each direction in each axis so that the component is qualified independent of mounting orientation. If the test is not conducted at the component's maximum limiting temperature, then the test *g*-level shall be increased by 50%.

Components shall be subjected to pre- and post-test ATPs or functional checks as applicable, and shall be operating and verified for specified performance during the test.

For components which contain engine-driven rotating elements (e.g., fuel metering unit [FMU], fuel pump, permanent magnet alternator [PMA], etc.), it is generally not necessary to drive these elements for this test as long as other elements that may be affected by application of a sustained acceleration load can still be operated in some manner such that their effect on unit performance can be verified. Notwithstanding the above, any testing wherein the component is proposed to be partially non-operational shall be specifically concurred by the Airworthiness Authority.

Recorded test data shall include the applied acceleration *g*-level (or equivalent) and the performance signals versus time. The data shall show acceleration at 0 *g*, reaching the specified value, remaining there for at least the specified test time, and then returning to 0 *g*.

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If a component has no moving parts, or cannot be practically tested using a centrifuge, the acceleration requirement shall be satisfied via Procedure IV (sine burst testing), immediately following the shock test, utilizing the same test setup. For efficacy, it is recommended that the sequence of pre-test resonance scan, vibration test, post-test resonance scan, shock, and sine burst be conducted on each component axis before moving to the next axis of test.

The sine burst waveform shall conform to the example illustrated in 810H(C1), Figure 513.8-1, except it shall comprise 30 cycles, with 10 cycles at peak amplitude and an equal number of cycles for the ramp up and ramp down to the peak. The peak amplitude shall be determined as described in the first paragraph of 5.4.6. The sine burst frequency shall be no greater than 1/3 of the first resonant frequency of the component, as determined by a resonance scan (pre- or post-vibration, or pre-sine burst, as applicable). The sine burst waveform shall be applied to each axis of the component under test a minimum of five (5) times.

5.4.7 Low pressure (altitude).

Electronic assemblies and electromechanical components shall be subjected to a low-pressure test in accordance with MIL-STD-810H(C1), Method 500.6, Procedure II (operation) and, if environmentally sealed, Procedure I (storage). Components shall be subjected to pre- and post-test ATPs or functional checks, as applicable, and shall be operating and verified for specified performance if being tested in accordance with Procedure II. Hermetically sealed components are exempt from this test.

Unless otherwise specified, the non-operational and operational altitudes and temperatures utilized for Procedures I and II, respectively, are defined in 5.4.1. If the component is mounted in an area where the external temperature environment is higher than the default value for Procedure II, the test temperature shall be the worst-case maximum. Note, if Procedure II is required, altitude shall be cycled three (3) times, uninterrupted, instead of the single cycle specified in the procedure.

5.4.8 Rain.

Components shall be subjected to a rain test in accordance with MIL-STD-810H(C1), Method 506.6, Procedure I (blowing rain). Electronic assemblies shall be tested to Procedure I, if mounted on the engine, or Procedure III (drip), if located in a portion of the airframe that may be exposed to condensation or rain ingress. Components shall be subjected to pre- and post-test ATPs or functional checks, as applicable, but need not be operating, with the exception of electronic components which shall be verified for specified performance during the test.

Components shall be inspected for water ingress at test completion, including the removal of air cavity covers that do not require further disassembly nor re-calibration or adjustment of any kind. Hermetically sealed components are exempt from this test.

5.4.9 Explosive atmosphere.

(Referenced by: 5.4, TABLE II NOTE 8/, TABLE VII NOTE 9/)

Engine-mounted components are generally exempt from an explosive atmosphere requirement, except for those which have fuel passing through and have internal air cavities which can accumulate an explosive fuel-air vapor mixture. Exemption is granted since cooling airflow through the engine compartment normally prevents buildup of an explosive atmosphere, so there is no external source for such an atmosphere to penetrate the component (the source must be from inside the component itself). Additionally, the surface temperature of the engine hot section routinely exceeds the auto-ignition temperature of JP-4, so component external surface temperatures are moot.

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Off-engine mounted components, which cannot plausibly be surrounded by an explosive fuel-air vapor mixture even under local failure conditions, are also exempt from explosive atmosphere. See H.2.4 'decision tree' for guidance in determining requirement applicability.

All non-exempt components capable of creating a spark (including those which are hermetically sealed), shall be subjected to explosive atmosphere testing IAW MIL-STD-810H(C1), Method 511.7, Procedure I. Components shall be subjected to pre-test ATPs or functional checks as applicable, but need not pass any post-test checks, unless further testing is planned for the unit.

The test shall be conducted at sea level and at the default operational altitude specified in 5.4.1, or the equipment specification, whichever is greater (but not to exceed 12,192 meters [m] 40,000 feet [ft]). During the test, the component surface temperature shall be maintained at its specified component limiting temperature (5.1.7). The component shall be operating continuously, with the applied input voltage and output loads at their maximums. During each altitude test sweep, all make and break contacts, if any, shall be operated at least ten times.

If a component utilizes airframe power, power supply transients as described in 5.1.6 shall be applied to the component during the test. Ten (10) repetitions of group a. and group b. power supply transients shall be applied during each test altitude sweep, at least four (4) of which shall take place during operation of make and break contacts, if any.

Component air cavity covers shall be loosened and hermetic seals penetrated to permit ingress of the explosive atmosphere. Ignition components or systems shall be operated continuously. Ignitors shall be mounted so as to prevent the spark-producing electrode from coming into contact with the explosive atmosphere.

Additionally, all non-exempt components (even those which are hermetically sealed) shall be shown by hot spot analysis (HSA) that no external or internal constituent part, which can plausibly be surrounded by an explosive mixture, can exceed the minimum auto-ignition temperature of JP-4 (+230°C/+446°F) under worst-case conditions (including failures). If the minimum margin is less than +20°C (+36°F), the contractor shall submit the analysis to the Airworthiness Authority.

5.4.10 Fungus.

Components shall be subjected to a fungus test in accordance with MIL-STD-810H(C1), Method 508.8, or evidence provided in an analysis report which indicates that all exposed materials of the component are fungus inert or have been previously tested.

Analyses, when provided in accordance with 5.10.4.3, shall list all parts and chemical-based substances (including glues, sealants, etc.) employed in the design of the component, and shall include supplier certifications as to the non-nutrient character of the constituent materials. Components which have drain holes or other types of ports to the ambient environment are not considered sealed, and therefore the parts internal to the component shall also be analyzed for fungus resistance, as required above. For electronic units, any parts that are conformally coated need not be analyzed, however the conformal coating itself must be fungus resistant.

Components shall be subjected to pre- and post-test ATPs or functional checks as applicable, but need not be operating during the test.

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5.4.11 Humidity.

Components shall be subjected to a humidity test in accordance with MIL-STD-810H(C1), Method 507.6, Procedure II (Aggravated Cycle). The test cycle shall be in accordance with Figure 507.6-7 with temperature and humidity levels as defined in Table 507.6-IX. Components shall be subjected to pre- and post-test ATPs or functional checks as applicable, but need not be operating during the test, other than as specifically required by the procedure. Hermetically sealed components are exempt from this test.

5.4.12 Salt fog.

Components shall be subjected to a salt fog test in accordance with MIL-STD-810H(C1), Method 509.8, Procedure I – Corrosion Screening (two cycles). Components shall be subjected to pre- and post-test ATPs or functional checks as applicable, but need not be operating during the test.

When whole-engine salt fog testing is conducted for qualification, the individual component test may be waived if petitioned by the engine manufacturer and agreed by the Airworthiness Authority.

5.4.13 Sand and dust.

Components shall be subjected to a sand and dust test in accordance with MIL-STD-810H(C1), Method 510.7, Procedures I (blowing dust) and II (blowing sand), in any order. Test media composition shall conform to MIL-STD-810H(C1), Method 510.7, 4.2.1.3 (sand), and 4.1.1.4d(1) or (2) (dust).

Procedure I testing shall be conducted at the component's specified component limiting temperature for a total duration of six (6) hours with the component operating and verified for specified performance throughout the test period. The unit shall be rotated so that each vulnerable surface (maximum of 6) faces the blowing dust stream for an equal amount of time. The 6-hour ambient temperature portion of the test is not required.

Procedure II testing shall be conducted at an ambient air temperature of +49°C (+120°F) for 90 minutes per vulnerable surface. The component need not be operating during this test, but shall be heated to its CLT prior to the beginning of the test. If more than one vulnerable surface is to be tested, component pre-heating shall be repeated prior to each 90-minute test period.

All components shall be subjected to pre- and post-test ATPs or functional checks, as applicable.

5.4.14 Contamination by fluids.

Engine-mounted components, or those located in the engine compartment, shall be subjected to an external contamination test in accordance with MIL-STD-810H(C1), Method 504.3, 4.5.5, Step 3, b. Intermittent Contamination, or evidence provided in an analysis report which indicates that all exposed materials of the component are unaffected by the specified fluids, or have been previously tested. Components shall be subjected to pre- and post-test ATPs or functional checks, as applicable, but need not be operating during the test.

Unless otherwise stated by the Airworthiness Authority, the fluids identified in TABLE XV represent the minimum set of test fluids that shall be applied individually at the temperature indicated, with the UUT at the specified external steady-state temperature.

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5.5 Special tests.

(Referenced by: 4.3.2, 5.2, TABLE XVI, J.2.3)

For the special component tests specified in TABLE XVI, testing can be performed on any suitable unit or units, provided that the features, functions or characteristics that are being verified conform to those of the component to be qualified. If the unit proposed for test is not identical to the parts list and configuration of the component to be qualified, the contractor shall provide justification that the differences are not germane to the outcome of the specific test. Any such justification shall require approval by the Airworthiness Authority.

5.5.1 Overspeed and containment.

For any engine component containing rotating elements (fuel/oil pumps, hydromechanical metering units [HMU], starters, inlet particle separator blowers, etc.), testing shall be conducted to verify that the rotating elements will not fail at the specified engine speeds, or that the parts will be contained if they do fail. Components shall be tested in their fully operational state and oriented in the same manner as when mounted on the engine. If not tested at their CLT, test speeds shall be increased to account for degraded material properties using industry-standard methodology. For alternators, see 5.6.2 for specific test requirements.

The component shall be operated at a speed which corresponds to:

a. 122 percent of the maximum allowable steady-state engine speed, for 5 minutes, if the component drives or is driven by the gas generator accessory gearbox.

b. 105 percent of the highest speed attained during a worst-case engine overspeed event, for 1 minute, if the component is driven by the power turbine accessory gearbox.

If the component cannot achieve the speeds stated in a. or b. without having a failure, then containment of the failure must be demonstrated during test a. or b., as applicable.

At the completion of the test, the component shall be disassembled and examined for evidence of mechanical damage or impending failure. If impending failure is indicated, the contractor shall repeat the test with the rotating elements ‘seeded’ to produce an actual failure, at which time containment shall be demonstrated.

5.5.2 Pressure tests.

(Referenced by: 5.10.3, 5.10.4.1, J.2.3)

5.5.2.1 Pressure cycling.

(Referenced by: 5.5.2.2)

For any engine component pressurized by fuel, oil or air, testing shall be conducted to demonstrate that the component can withstand 15,000 cycles between the minimum and the maximum design operating pressures. Dwell time at each pressure shall be no less than one (1) second.

During pressure cycling, the component shall be held at its maximum limiting temperature (5.1.7). The test may be conducted at a lower temperature if the applied pressure is increased to compensate. Any such request by the contractor shall be supported by an analysis justifying the pressure/temperature combination, which shall be submitted to the Airworthiness Authority for approval.

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Components shall be subjected to pre- and post-test ATPs or functional checks as applicable, but need not be operating during the test unless required in order to perform the test. During, and at the completion of the test, the component shall not exhibit any leakage from gaskets or other sealing surfaces.

Any component being tested in accordance with this paragraph shall also be tested in accordance with 5.5.2.2 (proof pressure) and 5.5.2.3 (burst pressure). These three tests shall be performed on the same serial number unit, in the following order: 1) pressure cycling, 2) proof pressure, 3) burst pressure. Disassembly of the component or removal of any pressurized cavity cover after the pressure cycling test completion is specifically prohibited; only external visual inspection or FPI shall be allowed.

5.5.2.2 Proof pressure.

(Referenced by: 5.5.2.1, 5.5.2.3)

For any engine component pressurized by fuel, oil or air, testing shall be conducted to demonstrate that the component can continue to operate in accordance with performance specifications after exposure to 1.5 times (fuel/oil) or 2.0 times (air) the design maximum operating pressure, for a period of 10 minutes. At the end of the test period, pressure is returned to the design maximum operating pressure and held for an additional 5 minutes. This test shall be performed on the same unit that has first undergone the pressure cycling test of 5.5.2.1.

The component shall be at its maximum limiting temperature (5.1.7) during application of the test pressure. The test may be conducted at a lower temperature if the applied pressure is increased to compensate. Any such request by the contractor shall be supported by an analysis justifying the pressure/temperature combination.

Components shall be subjected to pre- and post-test ATPs or functional checks, as applicable, but need not be operating unless required to perform the test. During the test, and for the 5-minute period immediately following, the component shall not exhibit any seepage or leakage from gaskets or other sealing surfaces, and there shall not be any visually observable permanent deformation.

Disassembly of the component or removal of any pressurized cavity cover after test completion is not allowed. If disassembly is required in order to reconfigure the unit to allow it to be tested further (ATP or burst test), specific Airworthiness Authority approval to do so shall be requested.

5.5.2.3 Burst pressure.

(Referenced by: 5.5.2.1)

For any engine component pressurized by fuel, oil or air, testing shall be conducted to demonstrate that the component can safely withstand 2.0 times (fuel/oil) or 2.5 times (air) the design maximum operating pressure for a 1-minute period. At the end of the test period, pressure shall be returned to the design maximum operating pressure and shall be held for an additional period of 5 minutes. This test shall be performed on the same unit that has first undergone the proof pressure test of 5.5.2.2.

The component shall be at its maximum limiting temperature (5.1.7) during application of the test pressure. The test may be conducted at a lower temperature if the applied pressure is increased to compensate. Any such request by the contractor shall be supported by an analysis justifying the pressure/temperature combination, and shall be submitted to the Airworthiness Authority for approval.

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During the test, and for a 5-minute period immediately following, the component shall not exhibit any leakage from, or fracture of, the pressurized cavities. Seepage or leakage from gaskets or other sealing interfaces is allowable.

Unless otherwise indicated, no post-test ATP or functional check is required since the component shall be considered to be non-serviceable after the test,

The component shall be disassembled after completion of this test and inspected for indications of impending failure using non-destructive techniques. The component shall not be reassembled and shall be held in bonded stores for possible government inspection.

5.5.3 Fire test (resistance / proof).

Lines, fittings, and components which convey flammable fluids shall be tested as specified in AS1055 and AS4273 while conveying fluids at the worst-case combination of flow rate, pressure, and temperature possible over the complete operating envelope of the engine. Worst-case, in this context, refers to the operating condition at which the component being tested is least able to resist burn through or fluid leakage.

The component shall be rated ‘fire resistant’ if during a 5-minute flame application period, followed by a 5-minute quiescent period, there are no measurable leaks. The component shall be rated ‘fire proof’ if during a 15-minute flame application period, followed by a 5-minute quiescent period, there are no measurable leaks.

All fire tests shall be conducted using NATO F-24 for fuel components, or a suitable substitute such as JP-8 or Jet A, as approved by the Airworthiness Authority. Fuel shutoff (if the component incorporates such a feature) shall be initiated at the end of the five-minute flame impingement period, and shall remain in the shutoff state throughout the quiescent period. It is also acceptable for fuel to terminate as a direct result of component failure at any time during the test. Any interconnections from the engine firewall to the component shall be included in the test, and shall be considered ‘under test’ no matter who has responsibility for the interconnections.

If any engine control electronic assembly is located in a designated fire zone, it shall be tested to verify conformance to a fire-resistant rating [refer to FAA AC 33.28-3, 6-2 Guidance: Engine Control System Failures, i. Local events, (8)]. During the 5-minute flame application period, the engine control system shall continue to operate the engine in accordance with the requirements of the engine model or component specification, or cause the engine to fail safe (engine shutdown, or fail fixed if so designed). Additional general guidance for test conduct can be found in section 8, Verification of Compliance, Society of Automotive Engineers (SAE) Aerospace Information Report (AIR) AIR6108.

Oil/hydraulic tanks/reservoirs shall be rated fire proof and shall be compliant with testing requirements as specified in AC 33.17-1B [9. OTHER CONSIDERATIONS, 9.1 Flammable Fluid Tanks]. The initial condition for the test shall be the maximum allowed fluid temperature, as specified in the EMS, and flow/pressure commensurate with that temperature.

The general component operational procedure to be followed, once the fire test begins, is detailed below. It is intended to simulate an engine fire being detected while at maximum rated power, followed by a crew-initiated power reduction to flight idle. This is then followed by five minutes at idle, which allows the crew sufficient time to determine a course of action, including activation of fire suppression systems and engine shut down.

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Systems to be rated fireproof must continue to be exposed to a flame with the engine now shut down, with either zero rotational speed of the accessory gearbox and its pad-driven pumps (fuel, oil or hydraulic), or the lowest expected windmilling speed, if applicable. Note that accessory gearbox (AGB)-driven pumps that continue to rotate due to windmilling, without the means to shut off their source of flammable fluid, shall require a fire proof rating, as well any downstream components that might be pressurized by said pump. Note that system testing of the AGB and attached oil or fuel related components must follow the same general procedure as delineated below.

Changes to the following notional procedure shall be concurred by the Airworthiness Authority.

a. Flame impingement at time = 0 seconds.

b. Fifteen (15) seconds after flame impingement begins (fire detection and action taken), flow shall be linearly reduced over a 5-second period to the lowest flow/pump rpm (fuel or oil) specified in the EMS for a SLSD Flight Idle condition.

c. At the end of five (5) minutes elapsed time, fuel shutoff capability shall be demonstrated, and the fuel and/or oil pump rpm shall then be linearly reduced to zero over a 10-second period, or to the lowest expected ‘windmilling’ speed, if applicable to an ‘as-installed’ engine.

d. For a fire-resistant rating, the flame shall then be removed and the component under test observed for an additional five (5) minutes to determine if during the 10-minute period there was imminent component detachment, or any leakage or dripping material sufficient to cause re-ignition, as determined by the Airworthiness Authority.

e. For a fire-proof rating, flame impingement shall continue for an additional ten (10) minutes, after which the flame shall be removed and the component under test observed for an additional five (5) minutes to determine if during the 20-minute period there was imminent component detachment, or any leakage or dripping material sufficient to cause fire intensification or re-ignition, as determined by the Airworthiness Authority.

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5.5.4 Pressure wash.

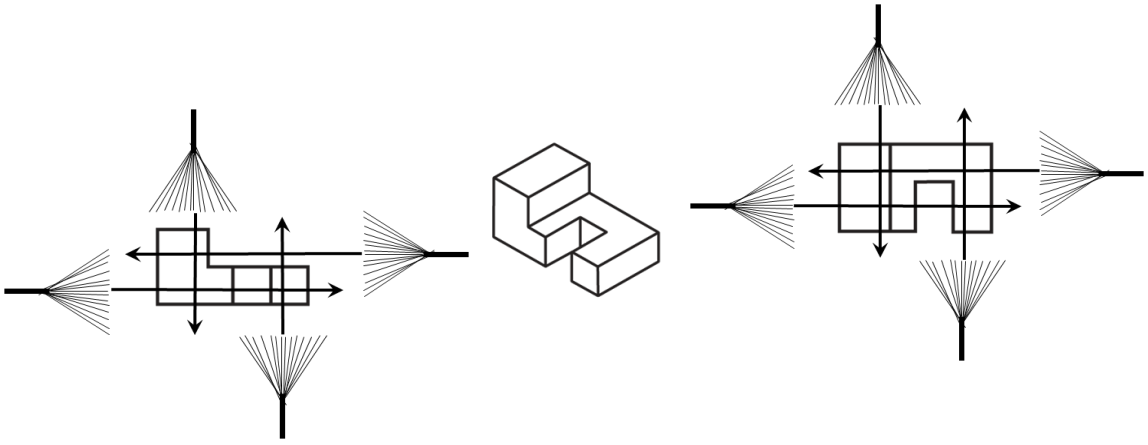
All engine components mounted in a location where they may be subject to water impingement as a result of engine washing shall be verified to maintain air cavity seal integrity. Components located in areas of the engine which are protected from water wash impingement, or those without air cavity seals, are exempt from this requirement.

The engine (or an engine carcass), with all engine-mounted components installed, shall be subjected to a pressure wash test in accordance with the general requirements specified below. At the contractor's option, this test may be conducted on individual components, or groups of components, provided that the test set up creates a reasonable facsimile of the actual installation of the component on the engine. Components need not be operating but shall be subjected to pre- and post-test ATPs or functional checks, as applicable. Note that the post-test ATP should be conducted as soon as practicable after steps g and/or h so that functional effects of water ingress can be determined.

a. The engine inlet and exhaust may be covered.

b. Each accessible surface of the component, as installed on the engine, shall be sprayed from a distance of 18 to 24 inches to the spray nozzle.

c. Each surface face shall be swept across a total of 8 times; twice each coming from the left, right, top and bottom directions, as illustrated below.



d. Pressure shall be maintained between 2500 psi and 3000 psi throughout the test with a 45° fan spray nozzle.

e. Spray wand sweep rate across the component shall be no faster than one (1) inch per second.

f. At test completion, without removing the components from their mounts, they shall be dried using lint free towels so that there is no dripping or standing water visible.

g. Covers or other component features which lead to air cavities shall be removed (if accessible while still mounted) and the cavity examined for evidence of water ingress.

h. Components shall then be removed and disassembled to the extent necessary to determine water ingress.

The contractor shall document the quantity of water in each cavity, the possible effects of the water on component operation, assess the long-term impact on operational reliability, and propose possible design changes to prevent water ingress. The Airworthiness Authority shall determine if modifications to the component or its maintenance procedures are required.

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5.6 Component-specific tests.

(Referenced by: 4.3.2, 5.2, TABLE XVII, J.2.3)

In addition to those already specified, certain additional tests shall be conducted for specific component types as identified in TABLE XVII. Tests in the 'SL' component category column are meant to be conducted at the system-level, either on an actual engine or using an engine/airframe integration model of a fidelity suitable to the test being performed.

Testing can be performed on any suitable unit, provided that the features, functions or characteristics that are being verified conform to those of the component to be qualified. If the test unit is not identical to the parts list and configuration of the component to be qualified, the contractor shall provide justification that the differences are not germane to the outcome of the specific test. Any such justification shall require approval by the Airworthiness Authority.

5.6.1 Oil Reservoir: cyclic and proof pressure.

The filler cap and other fittings shall be installed and the tank mounted in a manner similar to its on-engine configuration. The entire test assembly shall then be subjected to the following tests:

a. Cyclic fatigue test. The oil tank shall be cycled between its design minimum and maximum differential pressure limits at no more than four (4) times per minute for a minimum of 15,000 cycles. During the first 7,500 cycles, the oil reservoir shall be at the nominal oil operating temperature, and during the last 7,500 cycles at the maximum oil operating temperature. Throughout this cycling, there shall be no leakage, nor detrimental deformation of the oil reservoir, filler cap, or fittings.

b. Proof pressure test. Upon successful completion of the cyclic fatigue test, the same oil tank shall be subjected to a proof pressure test of 2.0 times the design maximum differential pressure limit (and the minimum differential pressure limit, if opposite in sign). Proof pressure shall be held for a minimum of 10 minutes, followed by 5 minutes at the design maximum differential pressure, with the wall of the oil reservoir at the maximum oil operating temperature. There shall be no leakage at any time, nor any permanent deformation of the oil reservoir, filler cap, or fittings.

c. Valve test. If the oil reservoir assembly incorporates a pressurizing valve or pressure relief valve, these also shall be tested to demonstrate proper functioning. The contractor shall specify in the test plan the procedure that will be used to demonstrate this functionality.

d. Proof pressure test for PFR. The tank shall be subjected to both a positive and a negative differential pressure, of the same magnitude, for a period of 30 minutes each, at the component limiting temperature (5.1.7). The differential pressure magnitude shall be two times the absolute value of either (1) or (2) below, whichever is greater:

   (1) Maximum design differential pressure.

   (2) Difference between sea level atmospheric pressure and the atmospheric pressure at the engine service ceiling.

   The requirement to apply both a positive and a negative differential pressure is applicable to all tank designs, including vented in the case of failed or blocked pressure equalization valves or circuits.

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5.6.2 Alternator: load and overspeed / containment.

(Referenced by: 5.5.1)

The alternator shall be subjected to pre- and post-test ATPs or functional checks, as applicable, for tests a. and b. only. All tests shall be conducted with the alternator at its component limiting temperature (5.1.7), and at full-rated electrical power.

a. Load test. The alternator shall be operated at a speed corresponding to 115 percent of the maximum allowable steady-state mechanical speed of the driving engine rotor, for a one-hour period. At completion of the test, there shall be no evidence of mechanical or electrical damage, or impending failure.

b. Overspeed. The alternator shall be operated at a speed corresponding to 122 percent of the maximum allowable steady-state mechanical speed of the driving engine rotor, or the maximum allowable transient speed of the driving engine rotor, whichever is greater, for a minimum of five (5) minutes to demonstrate structural integrity.

If the alternator cannot achieve the speeds stated in a. or b. without having a failure, then containment of the failure must be demonstrated during test a. or b., as applicable.

At the completion of the two tests, the component shall be disassembled and examined for evidence of mechanical damage or impending failure. If impending failure is indicated, the contractor shall repeat the overspeed test with the rotating elements ‘seeded’ to produce an actual failure, at which time containment shall be demonstrated.

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5.6.3 Heat exchanger: cycling and proof pressure.

Heat exchangers for cooling or heating of engine fluids shall be subjected to the following tests. If the heat exchanger assembly incorporates a bypass valve, regulator, or indicating feature, additional tests shall be conducted to demonstrate proper functioning of these features.

a. Flow, pressure, and temperature cycling test. The heat exchanger shall be subjected to a flow, pressure, and temperature cycling test for 15,000 cycles. The test cycle shall comply with the following requirements:

(1) Oil shall be induced into the unit at a flow rate, pressure, and temperature, determined from design data, based on engine intermediate rated power at SLSD conditions.

(2) Air (or fuel) shall be induced at the same time into the oil cooler at a rate, pressure, and temperature commensurate with the condition defined in a(1).

(3) The unit shall then be allowed to stabilize and the outlet conditions of the oil circuit recorded. Stabilization is defined as the outlet temperature changing no more than ±1°C (±1.8°F) over 5-minute period.

(4) The above measured outlet condition a(3) shall be used to determine the start and end of each test cycle, as outlined below.

(a) Oil and air (or fuel), shall be induced into the unit, simultaneously, at the given inlet conditions as stated in a(1) and a(2) above.

(b) These conditions shall be maintained until the oil outlet temperature is within ±5°C (±9°F) of the temperature recorded in a(3).

(c) When this point is established, the oil circuit shall be returned to a zero flow and pressure state. The air (or fuel) circuit shall remain ‘on’ to speed the cooling process of the oil cooler.

(d) When the air (or fuel) outlet temperature is within ±5°C (±9°F) of the air (or fuel) inlet temperature, the oil circuit shall be turned ‘on’ IAW a(4)(a) to begin a new cycle.

At test completion, there shall not be any evidence of internal or external leakage, nor permanent deformation of components.

b. Proof pressure. Upon successful completion of the cycling test, the same heat exchanger shall be subjected to proof pressure testing. Each side of the heat exchanger shall be individually pressurized to two (2.0) times its maximum working pressure, twice, and held two (2) minutes for each pressure application. During the application of pressure to one side, the other side shall be empty and at atmospheric pressure. There shall not be any evidence of external leakage, nor internal leakage into the ‘dry’ side. The heat exchanger shall be at its maximum limiting temperature during the test.

c. Closed-loop coolant system. If fluid cooling utilizes a closed-loop system, test conditions a(1) and a(2) are not applicable. Oil shall be induced at an elevated temperature and coolant at a reduced temperature such that stabilization condition a(3) is at least 10°C (18°F) lower than the inlet oil temperature. The oil ‘on’ time should be NLT one (1) minute in duration, as should the oil ‘off’ time. Conditions shall be coordinated with the Airworthiness Authority in order to achieve an acceptable test cycle.

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5.6.4 Electronic unit.

Throughout this section specifically, and in this standard generally, when airframe power system tests are described in detail, the descriptions and/or requirements are based on the assumption that 28 Vdc is the airframe power source. If the airframe power source is other than 28 Vdc, testing equivalent to that described in 5.6.4.2, 5.6.4.4, and 5.6.4.5 shall be performed. This shall include all test methods (relevant to the installation) which are specified in the subpart of MIL-HDBK-704 applicable to that power source. The applicable subpart is defined in MIL-HDBK-704-1. All proposed testing shall be approved by the Airworthiness Authority.

5.6.4.1 PMA-induced damage.

A PMA neutral short-to-ground test shall be performed to demonstrate that a directly connected electronic unit is neither damaged nor causes unacceptable engine control system behavior if the neutral of the alternator (not normally brought out to a connector) shorts internally to the alternator case, and thereby to engine ground. Note that this is a test of components powered by the PMA, not a test of the PMA itself; the PMA is test enabling hardware only.

If concurred by the Airworthiness Authority, an analysis by the supplier may be accepted in lieu of test, in which case the supplier shall submit an analysis conforming to the requirements of 5.10.4.3.

5.6.4.2 Electrical loads analysis: engine.

(Referenced by: 5.6.4)

An analysis, supported by test data, shall be prepared to establish the worst-case power consumption for the component, if it utilizes airframe power as either a primary or backup power source. The analysis shall verify that the component maximum current draw does not exceed the circuit breaker capacity under the worst-case combination of loads, ambient temperature, and failure of one power source, if redundant. Supporting data shall include a ‘load measurements’ test performed in accordance with MIL-HDBK-704, LDC101.

The analysis shall fully explain all assumptions that were made in defining the ‘worst-case’ operating conditions that will produce: maximum in-rush current, maximum steady-state current, maximum current during engine start, and maximum transient current. Notwithstanding specific data gathered IAW LDC101, the contractor shall make any measurements necessary to support the analysis assumptions.

The results of this analysis (performed at the engine level) are intended to assist in the electrical integration of the engine with the airframe by:

a. Selecting/recommending wire sizes and circuit protection devices for supplying airframe electrical power to the engine, and

b. Facilitating the electrical load analysis of the airframe, as governed by MIL-E-7016.

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5.6.4.3 Overheat.

Overheat testing shall be conducted on all engine control electronic assemblies, unless the component is mounted in a location where the surrounding ambient air temperature, during external-environment failure conditions, cannot exceed the maximum ambient air temperature to which the component is qualified/certified [refer to FAA AC 33.28-3, 6-2 Guidance: Engine Control System Failures, i. Local events, (7)]. Waiving of this test under this exclusion shall be supported by an analysis approved by the Airworthiness Authority.

However, if the engine control electronic assembly incorporates hardware or software that initiates hardware shutdown/shutoff, or other defined accommodation in response to excessive internal temperatures, then this test shall be performed even if the external ambient temperature cannot exceed the defined qualification/certification temperature under environmental failure conditions.

Engine control electronic assemblies shall be tested to verify conformance to the fail-safe requirements of the engine model specification. The assemblies shall be subjected to pre-test ATPs or functional checks, as applicable, and shall be operating in a 'closed-loop' mode during the test using an engine/airframe model in order to assess the effects of the overtemperature condition on the engine. The tested components need not pass a post-test ATP or functional check.

Exact procedural conduct of this test is not defined herein, however general guidance can be found in section 8.3, Verification of Overheat Compliance, of SAE AIR6108, and in H.2.1. Specific test plans/procedures shall be agreed between the contractor and the Airworthiness Authority.

5.6.4.4 Power compatibility.

(Referenced by: 5.6.4)

A test shall be performed if the component utilizes airframe power as either a primary or backup power source to ensure that it is compatible with standard platform electrical characteristics as defined by MIL-STD-704. By utilizing the test procedures specified below, the component will be tested to the worst-case requirements of MIL-STD-704A through 704F/CN1. In addition to the standard platform electrical characteristics, the EMS or SoW may define additional requirements in the case where an intended platform fails to meet 704 standards. Unless otherwise stated, tests shall be performed using only airframe 28 Vdc power, whether it is designated as the primary or backup source. Note that the 'LDC' procedures referenced below are from MIL-HDBK-704-1, Table I, Power groups, Low Voltage DC (28 Vdc), with the specific LDC procedures listed in Table IX. Procedure execution is detailed in MIL-HDBK-704-8.

Should the airframe power type belong to one of the other Table I power groups, the supplier shall propose equivalent testing. All proposed testing shall be approved by the Airworthiness Authority.

a. Steady-state voltage. Test to MIL-STD-704C emergency low limit steady-state (ELSS) at 16.0 volts-direct current (Vdc) and 704F/CN1 abnormal high limit steady-state (AHSS) at 31.5 Vdc in accordance with LDC401.

   NOTE: The LDC102 and LDC301 "Steady State Limits for Voltage" tests are both enveloped by the LDC401 test when the specified ELSS and AHSS limits defined above are utilized for test execution. If narrower voltage limits are used for the LDC401 test, then testing IAW LDC102 and LDC301 may also be required.

b. Voltage distortion. Test to 704A requirements in accordance with LDC103, except use 704F/CN1 requirements for test conditions E, F and K.

c. Ripple voltage. Test to 704A requirements in accordance with LDC104.

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d. Voltage transients: normal. Test to 704A requirements in accordance with LDC105.

e. Voltage transients: abnormal. Test to 704A requirements in accordance with LDC302. In addition, include any untested over voltages specified in 5.1.6b. or c. as applicable for a DC or an AC electrical power system.

f. Voltage transients: starting. Test to 704F/CN1 requirements in accordance with LDC501. This is typically down to 12 Vdc for engines that use an electrically powered starter, or down to 16 Vdc otherwise.

g. Voltage spikes. Apply voltage spikes in accordance with 5.1.6a. utilizing the method defined by RTCA/DO-160, Section 17.

h. Power interrupts: basic. Test to 704F/CN1 requirements in accordance with LDC201.

NOTE: LDC201, "2 Validation Criteria" states, "The utilization equipment must maintain the specified performance during power interrupts." This should not be construed to mean that the equipment must tolerate interrupts without affecting performance, unless it is so indicated in the equipment specification.

i. Power interrupts: supplementary. Apply the following set of electrical interrupts to the UUT: 2, 4, 6, 8, 9, 10, 11, 12, 14, 16, 20, 30, 50, 100, 200, and 500 milliseconds, and 1, 2, 5, 7, and 10 seconds.

j. Power failure. Test to 704A requirements in accordance with LDC601.

k. Power phase reversal. If the component does not utilize keyed connectors for those harnesses that carry airframe 28 Vdc power, test in accordance with LDC602.

l. Power up. The component shall be successively powered up in 1.0 Vdc increments between the steady-state voltage limits defined in subparagraph a. The test shall be repeated twice; once with all output loads at the worst-case minimum specified resistance and once at the worst-case maximum specified resistance.

m. Voltage sweep (normal range). After dwelling five (5) minutes at the MIL-STD-704C ELSS voltage, sweep at 0.5 Vdc per minute to the 704F/CN1 AHSS voltage, and then dwell for five (5) minutes. Sweep back to the ELSS voltage at 0.5 Vdc per minute. Perform sweep test with:

(1) no engine-supplied power.

(2) engine-supplied power equivalent to that provided at the minimum engine ground idle speed.

(3) engine-supplied power equivalent to that provided at the maximum allowed gas generator speed.

n. Voltage sweep (under range). Starting at 0 Vdc, ramp or step voltage at +0.1 Vdc per second up to the ELSS voltage. After five (5) minutes at ELSS, ramp or step voltage at -0.1 Vdc per second to 0 Vdc; the test may be stopped when the UUT fully powers down. Perform sweep test with:

(1) no engine-supplied power.

(2) engine-supplied power equivalent to that provided at the minimum engine ground idle speed.

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5.6.4.5 Short circuit protection.

(Referenced by: 5.6.4, 5.8.5a)

For any short circuit condition specified in subparagraphs a. through d., the condition shall only affect the shorted interface. Subsequent to the removal of the short circuit condition, the component shall be undamaged and fully functional.

a. Any two adjacent pins within the component connector(s).

b. Any component connector pin and ground.

c. Any two pins (including non-adjacent) of any input sensor, output effector, or other input/output device or subsystem which interfaces with the component.

d. Any two wires in a twisted-pair, twisted-triplet, twisted-quadruplet, etc. (shielded or unshielded), which is not covered by conditions a., b., or c.

Additionally, a short of 28 Vdc or PMA power to any pin or to the connector shell (in the same connector) shall not result in the propagation of a permanent failure in the component beyond the interface or circuit affected by the short circuit condition.

5.6.4.6 Data bus specification compliance.

Electronic units which incorporate data bus communication interfaces shall be verified for compliance with protocol specifications.

a. MIL-STD-1553. Data bus remote terminal functionality in accordance with SAE AS4111.

b. ARINC 429. Data bus hardware characteristics in accordance with Aeronautical Radio, Inc. (ARINC) specification 429, Part 1, Appendix A.

c. Other data bus type (e.g., EADIN - engine area distributed interconnect network). The contractor shall propose a verification plan for Airworthiness Authority review and approval.

5.6.4.7 Crystal oscillator temperature performance.

Electronic units which incorporate crystal oscillators shall undergo a slow-sweep temperature test to verify their ability to meet performance requirements at every ‘steady-state’ temperature point, from minimum to maximum, as specified in the engine model or component specification. The UUT shall be tested twice IAW the following cycle:

a. Soak at the minimum specified non-operating temperature for at least three (3) hours with power OFF.

b. Be powered ON and the temperature linearly ramped to the maximum operating temperature at a rate not to exceed 0.25°C (0.45°F) per minute.

c. Be powered OFF and then soaked at the maximum specified non-operating temperature for at least three (3) hours.

d. Ramped down to the maximum operating temperature and held at that temperature for thirty (30) minutes.

e. Be powered ON and the temperature linearly ramped to the minimum operating temperature at a rate not to exceed 0.25°C (0.45°F) per minute.

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5.6.5 Fuel pump.

5.6.5.1 Net positive suction pressure.

Each engine fuel pump shall be tested in accordance with SAE ARP4024 to determine if it meets or exceeds engine model or component specification requirements for operation at fuel inlet pressures in close proximity to fuel true vapor pressure. Both total and static fuel pressure above true vapor pressure (TVP) shall be measured. Before, during, and immediately after the test, fuel samples shall be taken to calculate the fuel true vapor pressure IAW ARP492. Key fuel properties include the fuel Reid vapor pressure (RVP) and the fuel distillation slope at the 10% distilled point.

5.6.5.2 Self priming.

A test shall be conducted to demonstrate that the engine fuel system is capable of specified self-priming requirements. Full prime is reached when fuel pump output flow meets or exceeds specified engine light-off fuel flow. The test shall be conducted with fuel-wetted internal pump surfaces. 'Wetted' is defined as a pump which has been completely filled with fuel and then the unit rotated once in the direction of each fuel inlet and discharge interface, such that each uncapped interface is pointed down, allowing all residual fuel to drain from that interface.

The portion of the aircraft fuel system from the fuel tank to the engine fuel inlet shall be included or simulated for this test. The portion of the engine fuel system from engine fuel inlet to the engine positive displacement fuel pump inlet (if they are not the same), shall be included in the test assembly. Note that any components used in the test assembly up to the engine fuel inlet are not under test; they are only being utilized to accurately simulate the conditions at the fuel pump inlet.

The performance criteria described below are default values. The contractor is allowed to tailor these parameters to match the priming capability specified for the engine fuel system under test. These changes shall require approval from the Airworthiness Authority prior to test conduct.

Test conditions/performance criteria:

Fuel line ID **Initial test state**	0.625 inches Air in line between fuel tank and engine fuel inlet
Vertical lift	6 feet
Fuel temperature	135°F (57°C)
Tank altitude	5,000 feet
Fuel type	NATO F-24 (iaw AFLP-3747), or equivalent
Time to reach full prime	30 seconds (max) at minimum cranking speed

5.6.5.3 Bubble ingestion.

If the fuel pumping system is required to continuously ingest 'slugging' flow at the engine fuel inlet without discharge flow interruption, the capability shall be demonstrated using the default parameters as delineated below. Test specifics may be altered if the EMS, SoW, or component specification dictate otherwise.

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The flow structure shall be generally composed of alternating gas/liquid 'slugs', each of which is twelve (12) inches in length when viewed in a transparent vertical pipe of the same internal diameter as that of the engine inlet fuel line. The gas phase shall be derived from the pumped fuel. The test shall be conducted in accordance with SAE ARP4028.

The continuous-ingestion performance test shall be run for fifteen (15) minutes at each condition, a through h, defined in the table below.

Test Condition	Fuel Type	Fuel Inlet Temperature (°C/°F)	Fuel Tank Pressure (psia)	Pump Speed & Fuel Flow (at specified power/alt.)
a	NATO F-24 (iaw AFLP-3747) or Jet A (iaw ASTM D1655) w/military additives or JP-8 (iaw MIL-DTL-83133)	71/160	14.7	Maximum Continuous
b				Ground Idle
c			10.1	Maximum Continuous
d				Ground Idle
e	JP-4 (iaw MIL-DTL-5624W)	29/84	14.7	Maximum Continuous
f				Ground Idle
g			10.1	Maximum Continuous
h				Ground Idle

5.6.5.4 Vapor/liquid (V/L) capability.

A test shall be conducted to demonstrate that the engine fuel pumping system is capable of continuously ingesting fuel with a vapor-to-liquid ratio as specified in the engine model or component specification. The test procedure shall be IAW SAE ARP492.

If V/L requirements are undefined in the component performance specification or user-specified requirements document(s), testing shall demonstrate the maximum capability of the pumping system, starting at a minimum V/L of 0.45. Once the maximum stable V/L is reached for a given flow condition, the V/L limit shall be maintained for one (1) hour before moving to the next flow condition. Note that testing beyond a V/L of 1.0 is not required.

V/L capability shall be demonstrated at three (3) flow conditions utilizing JP-4, or alternatively Jet B, at 29°C (84°F) as measured at the engine fuel inlet. The flow points shall be equivalent to Idle, 75% max continuous, and max continuous for engine performance as specified at 4,000 feet/95.0°F.

5.6.6 Ignition system: fouling tests.

The following tests are applicable to all ignition systems in order to demonstrate that the ignition system is capable of consistently starting the engine under fouling conditions. The applied exciter voltage for the test shall be set to the minimum defined in the engine model specification.

5.6.6.1 Carbon deposits.

The spark ignitor of the ignition system test assembly shall demonstrate the specified sparking rate and energy delivery with spark gaps covered, filled, or bridged with a generous application of an amorphous carbon/oil mixture IAW SAE AMS2518, or equivalent. The minimum EMS-specified input voltage shall be applied to the exciter for no less than ten (10) minutes.

Sparking rate and energy delivery shall be monitored and recorded at the beginning, during, and at the end of the 10-minute test period.

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5.6.6.2 Water ingestion.

With the spark ignitor positioned in a manner simulating the mounted position in the engine, the specified sparking rate and energy delivery shall be demonstrated for ten (10) minutes continuous while the electrode is immersed in water to a depth of one (1) inch. Prior to starting the test, the ignitor shall be soaked for a minimum period of one (1) hour in distilled water, up to the specified depth. Immediately thereafter, the minimum EMS-specified input voltage shall be applied to the exciter to begin the 10-minute test period.

Sparking rate and energy delivery shall be monitored and recorded at the beginning, during, and at the end of the 10-minute test period.

5.6.7 Fuel filter element capacity / efficiency.

(Referenced by: 5.3.4a(1))

Filter capacity, if not defined in the equipment specification, shall meet the default requirements specified in the table. Verification shall be in accordance with the dirt capacity test as defined in SAE ARP1827 as modified for the fuel type, mass flow, fluid temperature, and contaminant injection default parameters defined below, or in the component specification. If the EMS, SoW, or other procurement document also requires a filter element efficiency test, it shall conform to ARP1827 test methods.

Filter rating	As specified in the EMS or 25 microns, absolute
Contaminant	TABLE VI (heavy dirt)
Fuel type	NATO F-24 (iaw AFLP-3747) or Jet A (iaw ASTM D1655) w/military additives or JP-8 (iaw MIL-DTL-83133)
Filter capacity before bypass	12 hours at platform BR (best range) cold day conditions
Operating parameters	Test plan to specify the following BR cold day test parameters: • Altitude • Fuel temperature at altitude • Engine power and rated specific fuel consumption (SFC)
Impending to actual bypass	4 hours at maximum continuous power (MCP) at sea-level, cold-day (SLCD) conditions

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5.7 Electromagnetic environmental effects.

(Referenced by: 4.3.3, 5.2, 5.7.5, 5.10.3, 5.10.4.1, H.2.3)

Engine controls and accessories shall meet their performance requirements when subjected to the electromagnetic environmental effects test requirements defined by MIL-STD-461, as tailored herein. These tests are summarized in TABLE XVIII.

System testing shall be performed on the entire engine control system, or portions thereof, in order for those components to be qualified. Units or components used for testing shall be aircraft representative. This includes cable harnesses, cable breaks (for aircraft bulkheads), connectors, drain or ground wires (at their maximum length), and mechanical installation. Electrical bonding of components to the engine (or airframe structure) shall be representative of levels historically achieved in the field, over the service life of the component/engine, as discussed in 5.7.5.

Components that include software shall have either the operational flight program installed or utilize specialized input/output (I/O) interface software to allow monitoring and recording of any signal upsets or anomalies.

5.7.1 Electromagnetic interference.

(Referenced by: 5.2, TABLE XVIII)

All engine electrical and electronic controls and accessories shall meet their performance requirements when subjected to the susceptibility and emission requirements of MIL-STD-461, as tailored herein, for a flight critical system. Unless otherwise specified, RS103 shall be tailored to the modulations, field strength levels, and frequency ranges defined in TABLE XIXa and TABLE XIXb.

For TABLE XIXa, testing shall be conducted up to six (6) dB above the specification level, limited only by the capability of the laboratory. As described in TABLE XIXa, the field strength for “AM” (amplitude modulation) signals pertains to the average field strength of the modulated signal, as opposed to the peak as defined in MIL-STD-461.

For TABLE XIXb, testing shall focus on active electronic devices such as the engine control unit and components containing computing, amplifying, rectifying, and circuitry implemented with solid-state components. These devices shall be radiated at different angles (not just head-on). Testing shall also focus on the leading portion of the wiring that interfaces to these devices, along with the interface connector, and any location along the wiring harness where assembly junctions could compromise shielding effectiveness. The leading portion of wiring and assembly junctions shall be radiated in a direction perpendicular to their features, as much as practical.

The full field levels of TABLE XIXb are applicable to flight critical/essential and safety critical subsystems and components. These levels should also be considered for application to mission critical/essential systems, otherwise the requirement for the peak field strength is a minimum of 200 V/m RMS (and should be combined into TABLE XIXa as an additional modulation during test). This also applies to flight testing in a restricted environment for any system, when approved by the government.

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For all EMI tests, the setup shall locate the engine firewall connectors (the transition between the engine interface harnesses and airframe wiring) at the front of the test bench, and route the first two (2) meters of airframe wiring after the firewall connectors along the front of the test bench. The airframe side of the firewall connectors shall be a test point for CS114, CS115, CS116, RS103, and indirect-effects lightning testing. Alternatively, routing non-overbraided, aircraft-representative wiring directly to the engine interface connectors, without simulating the firewall connection, is permissible when approved by the Airworthiness Authority.

These requirements shall be met for engine operation throughout the environmental envelope, and for all control system operating modes. TABLE XVIII identifies the tests from MIL-STD-461, Table V, applicable to propulsion systems. Those identified with an 'X' are the standard suite of tests that shall be completed before first flight, as discussed in 5.2.

5.7.2 Electromagnetic pulse.

All engine electrical and electronic controls and accessories shall meet their performance requirements when subjected to the electromagnetic pulse (EMP) requirements of MIL-STD-461, RS105, Radiated Susceptibility, Transient Electromagnetic Field. Note that the rise time is the salient portion of the waveform.

High-Altitude Electromagnetic Pulse (HEMP, 5.7.6), if applicable, should be applied to the entire EMI setup (as much as practical), using both voltage and current (through the ground plane). Other methods of injecting the transient can be considered. If EMP is combined with HEMP, then the electronic devices under focus are also tested in different orthogonal orientations above the test table (with their chassis tied to the ground plane using a ground strap).

The equipment shall also meet the requirements of CS116 up to a 150% over test (Ip calibrated up to 10 Amperes IAW Figure CS116-2 in MIL-STD-461, but allowed to rise up to 15 Amperes during the test). See H.2.6 for test discussion.

5.7.3 Lightning.

All engine electrical and electronic controls and accessories shall meet their performance requirements when subjected to the indirect effects of simulated 200k ampere lightning discharges when tested in accordance with RTCA/DO-160, Section 22, or alternatively, MIL-STD-461, Method CS117, when approved by the Airworthiness Authority.

Test Method CS117 of MIL-STD-461 may also be used for ancillary equipment that is not flight critical/essential, mission critical/essential, safety critical, or any system that is installed in a "well protected" to "moderate protected" lightning environment (as described in AC 20-136B and RTCA/DO-160, Section 22, Levels 1 through 3), when approved by the Airworthiness Authority.

The contractor shall specify applicable waveforms and voltage/current levels based on the location of the component in relation to the engine structure. Interfaces to unprotected wiring and systems on the aircraft shall be included, with waveforms and levels appropriate for the interface to be specified by the contractor. The contractor-specified waveforms and voltage/current levels shall be coordinated with the airframe manufacturer and submitted to the Airworthiness Authority for approval. Typical test waveforms and levels for H-60 and H-64 platforms are provided in APPENDIX D, as an example.

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All electrical connector contacts which interface directly with aircraft wiring should be pin injection tested. Although not a formal Army airworthiness requirement as stated in NOTE 1/ at the bottom of APPENDIX D.3, it is still required for FAA certification. If the component does not contain any active electronic parts, a dielectric withstand voltage (hi-pot) method may be used to demonstrate pin injection test requirements, as described in RTCA/DO-160, Section 22, Paragraph 22.5.1 (see also MIL-STD-202-301 for reference). All components that contain active electronic parts shall be subjected to bulk cable injection tests.

5.7.4 Personnel-borne electrostatic discharge.

All engine electrical and electronic controls and accessories shall meet their performance requirements after being subjected to simulated static electricity discharges using Test Method CS118 of MIL-STD-461, tailored accordingly.

5.7.5 Electrical bonding analysis.

(Referenced by: 5.7)

The contractor shall generate an analysis that describes their approach to electrical bonding between components on the engine, and electrical bonding at the interface to the aircraft. This analysis shall be approved by the SRD prior to conducting any of the testing in 5.7.

The engine shall meet all the electromagnetic test requirements levied in 5.7 and its subparagraphs with the electrical bonding levels and methodologies which are practically achieved over the service life of the engine.

A detailed discussion of electrical bonding is provided in H.2.3.

5.7.6 High-altitude electromagnetic pulse.

(Referenced by: 5.7.2)

When applicable by contract, engine electronic controls and assemblies, when subjected to a high-altitude electromagnetic pulse (HEMP) as defined in MIL-STD-2169, shall maintain normal engine operation. If normal operation cannot be maintained during the event, the control system shall temporarily enter a fail-safe mode and then resume normal operation at event termination, or if so designed, permanently enter a fail-fixed mode if control of the engine is no longer possible.

5.7.7 Initial nuclear radiation.

When applicable by contract, engine electronic controls and assemblies, when subjected to initial nuclear radiation (INR), shall maintain normal engine operation. If normal operation cannot be maintained during the event, the control system shall temporarily enter a fail-safe mode and then resume normal operation at event termination, or if so designed, permanently enter a fail-fixed mode if control of the engine is no longer possible. The definition of the nuclear environment will be provided separately via applicable contract.

5.7.8 High-power microwave.

When applicable by contract, engine electronic controls and assemblies, when subjected to a high-power microwave (HPM) environment, shall maintain normal engine operation. If normal operation cannot be maintained during the event, the control system shall temporarily enter a fail-safe mode and then resume normal operation at event termination, or if so designed, permanently enter a fail-fixed mode if control of the engine is no longer possible. The definition of the HPM environment will be provided separately via applicable contract.

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5.8 System verification and validation tests.

(Referenced by: 5.2, 5.10.3, 5.10.4.1)

All components or systems incorporating software shall be tested in accordance with the requirements specified in the subsections below.

5.8.1 Engine control system.

Tests shall be conducted to verify and validate both engine control performance and engine monitoring software (if supplied). This includes testing of modules to ensure that each module functions as designed, and hardware/software integration tests to verify proper interaction between software components.

Tests shall be conducted in a closed-loop bench test environment with engine model simulation, or on an actual engine, or a combination of both, to demonstrate satisfactory performance and behavior of the entire engine control system. In particular, power interrupts occurring during the start region shall be evaluated during PFR to determine the effects on channel swapping, start times and start aborts. These tests can be a subset of the cases prepared under 5.8.3. Test cases shall be approved by the Airworthiness Authority to assure adequate coverage.

The requirements of the engine model specification shall be verified by test and analysis in accordance with RTCA/DO-178, RTCA/DO-254, and software standard practices as described in AMACC 15.0 (Software Airworthiness) and its subsections, or other DoD-component standards.

5.8.2 Common-mode multiple signal failure.

(Referenced by: 4.3.4)

Testing shall be conducted to demonstrate that the engine control system can safely accommodate simultaneously occurring multiple signal failures due to a common cause (a connector backing off or ballistic damage to a harness branch).

It shall be demonstrated by test that acceptable engine control system performance and/or behavior is maintained:

a. for any single connector being disconnected.

b. for any electrical harness branch being severed.

Each test case defined in a. and b. shall be executed at max continuous power. Test cases defined in a. shall also be conducted at an engine speed above light off and below ground idle, or below the speed at which the control system is electrically self-sufficient.

The test shall be conducted in a closed-loop bench test environment or on an actual engine, or a combination of both. If the signal group failure is expected to have a more pronounced effect at an engine operating condition other than those specified above, these should be tested in place of the specified conditions with the concurrence of the Airworthiness Authority.

The actual physical connector can be disconnected, or the group of signals associated with that single connector or branch opened simultaneously (defined as all pins open within 50 msec maximum). If a connector or branch contains only one signal or parameter, then it need not be tested as part of this requirement; nevertheless, all input/output signals shall be verified for proper failure accommodation. This test requirement applies to all electrical connectors and/or harness branches within the engine control system, not just those on the electronic control unit.

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The test plan shall include, as a minimum:

(1) A detailed electrical schematic of the engine control system showing all harness branches and connectors that contain any signal going to and from the control unit.

(2) For each ‘disconnect’ or ‘sever’ test case, the signal group to be opened.

(3) Expected results (if known *a priori*) for each failure test case.

Sample test matrices are provided in APPENDIX B.

5.8.3 Complex power interrupts.

(Referenced by: 4.3.4, 5.8.1, APPENDIX C)

A comprehensive test of engine control system response to airframe power bus interrupts shall be conducted. The Airworthiness Authority will provide guidance/assistance in preparing the necessary test cases based on the engine control system architecture and the electrical power sources utilized.

The engine control system shall tolerate and safely recover from single, multiple, and overlapping interrupts of varying durations when:

a. applied to each channel and both channels, with channels in both healthy and failed conditions (assuming a failed alternator).

b. applied during the engine start region, prior to alternator power being available.

A test case matrix of engine operating conditions, control system operating states, and interrupt timing and duration shall be prepared by the contractor, assisted by the Airworthiness Authority. For each test case, the expected control system response and recovery shall be described in detail. Sample matrices are included in APPENDIX C for a dual-channel symmetric control system and a dual-channel asymmetric control system to illustrate the level of complexity that may be encountered in defining the test cases.

5.8.4 Helicopter drive system torsional stability.

Linear and non-linear torsional stability analyses of the combined engine and aircraft rotor system shall be performed prior to first flight, or whenever any engine or aircraft system component which may affect torsional stability is modified.

The linear analysis determines the open loop frequency response of a linearized model of the engine/rotor system. It shall be performed at all operating conditions. Both single-engine and dual-engine stability shall be assessed. All engine control loops (e.g., power turbine governor, torque limiter, temperature limiter, etc.) shall be evaluated. The linearized analysis shall encompass the loci of operating conditions with the least stability margin (open loop gain and open loop phase).

The non-linear analysis determines torsional stability in the time domain. It is used to evaluate stability for larger perturbations to the engine/rotor system. It is permissible, often preferable, to substitute actual on-wing torsional test data for non-linear simulated torsional analysis data.

Typical data presented for both types of analyses (linear and non-linear) is described in H.2.9.

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5.8.5 Fault injection.

(Referenced by: APPENDIX F)

Testing shall be conducted to demonstrate that the engine control system can safely accommodate singly occurring and sequential failures of I/O signals, as well as tolerate high resistance and low capacitance between signal pins, and between pins and airframe grounds. Generalized pass/fail criteria are listed in F.3.3.

The contractor shall prepare closed-loop test cases and procedures to validate that the system/software responds as intended when failures are injected into the power, communication, input, and output signals of the engine control system. The test plan shall be submitted to the Airworthiness Authority for approval, and shall evaluate the following general failure categories/types:

a. Single point failures. Each individual input or output signal shall be failed (open and short). Opens shall be single wire only, while shorts shall be in compliance with the requirements of 5.6.4.5. Additional short combinations may be required, as defined by the system I/O and wiring configuration. APPENDIX F provides examples of short combinations that may require testing.

b. Sequential failures.

   (1) For three (3) or fewer signals that are backups of one another, fail the signals sequentially, in each order permutation, to verify expected signal selection.

   (2) For four (4) or more signals that are backups of one another, one test case shall fail signals in the order of their selection priority (from highest priority to lowest priority, all the way to default). For the other test cases, the three (3) signals most likely to fail shall be tested in each order permutation (6), with the remaining backup signals sequentially failed, in varying order, as part of the six (6) tested permutations.

c. Intermittent failures. Fail open each input or output signal for a duration that is less than the fault detection/confirmation time for the failed signal.

d. Data bus failures. Fail each system data bus for several different durations beginning with a minimum of 0.5 seconds and a maximum of 10 seconds.

e. Slow failures. Fail each signal both upwards and downwards to simulate a ‘slow’ ramp to the out-of-range limit for the signal.

f. Resistance, continuous (steady-state) electrical impedance.

   (1) Analog interface signals to the aircraft (including discretes) shall be immune to shorts with resistances greater than 10 kOhms between the connector pins supporting a common function, and 40 kOhms between each pin and airframe ground.

   (2) Thermocouple signals shall be immune to shorts with resistances greater than 1.5 megohms between connector pins supporting the same function, and 1.5 megohms between each pin and airframe structural ground.

g. Capacitance, continuous (steady-state) electrical impedance.

   (1) Analog interface signals to the aircraft (including discretes) shall be immune to capacitances less than 0.2 µF between each connector pin and the shell.

   (2) Thermocouple signals shall be immune to capacitances less than 0.001 µF between each pin and connector shell.

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5.9 Reliability development and growth tests.

(Referenced by: 4.3.5, TABLE XX)

Component types identified in TABLE XX shall undergo reliability growth testing in accordance with the requirements specified below, unless exempted by the Airworthiness Authority.

5.9.1 Highly accelerated life testing.

(Referenced by: TABLE XX NOTE 1/)

Components shall undergo HALT prior to the start of qualification in order to identify and correct any inherent design weaknesses. The component shall be exposed to increasing stresses, applied in step increments, of temperature, vibration, and voltage, individually and combined, until the operating and destruct limits of the component (i.e., the fundamental limit of the technology) are determined. In order to differentiate between reaching the operating limit and the destruct limit, the test should involve "backing down" stress levels to ascertain whether the component is able to resume operation or has become permanently disabled. Efficacy of any design changes shall be evaluated in a 'verification' HALT.

It is recommended that applied vibration take the form of broadband random with a frequency bandwidth of 10 to 2,000 Hz. The initial applied vibration step should have an acceleration spectral density (ASD) of no less than 0.01 g^2/Hz (4.5 g_{RMS} equivalent), up to a maximum of 0.03 g^2/Hz (7.7 g_{RMS} equivalent). From this initial step, it is also recommended that subsequent steps do not exceed 5-7 g_{RMS}, gradually reducing to 2-3 g_{RMS} as the number of steps increases. A balance should be struck between reaching the destruct limit quickly vs having sufficient resolution in the step size to assign a value to the destruct limit with some level of confidence.

The contractor may also choose to superimpose other vibration spectra onto the broadband random, such as MIL-STD-810H(C1) Category 14 sine tones and/or Category 22 narrowband random 'spikes' to mimic qualification requirements. If so, the tone and narrowband spike frequencies and amplitudes should be in line with the guidance provided in APPENDIX G. Step change size in terms of vibration energy should still follow the guidance in the previous paragraph in order to balance test time with test resolution.

Although the addition of other test spectra to the broadband random is acceptable, the contractor should keep in mind that HALT data is used to inform a production HASS, and that the typical HASS consists of a broadband random spectrum without sine tones or narrowband random spikes. Consequently, inclusion of these elements during HALT may complicate determination of a suitable HASS vibration profile/level, thus requiring additional test time

A detailed test plan shall be prepared by the contractor and submitted for approval.

5.9.2 Vibration: endurance.

(Referenced by: 5.4.5, K.3)

Prior to completion of qualification, components designated in TABLE XX shall undergo a vibration endurance test, equivalent in duration to the EMS- or SoW-specified life for the component, or 5,000 hours if unspecified. The vibration test spectrum shall represent the average or 'typical' engine over its useful life. To reduce test duration to a manageable number of hours, test time can be 'accelerated' by increasing the amplitude of the typical vibration level so that equivalent damage is accrued, as discussed in APPENDIX G.

The test shall be IAW 5.4.3.2.1 or 5.4.3.2.2 (whichever paragraph under which the 4-hour qualification test was conducted) with the test time adjusted to demonstrate the specified life.

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5.9.3 Combined-environment reliability test.

Components shall undergo a CERT for a minimum of 5,000 hours, to be completed before the start of full-rate production. A minimum of 1,500 hours shall be accumulated on each of two units, with the remaining hours accumulated on a maximum of three additional units.

An operational cycle, similar to the one used for SOT (5.3.1), shall be defined, along with the environmental profile to be applied. At a minimum, vibration, temperature, voltage (if applicable), and altitude environments shall be combined, with humidity applied during non-operating periods. A process to develop a suitable test profile is discussed in MIL-STD-810H(C1), Method 520.5, Procedure III. However, the limitation on the number of test cycles as described in the procedure is not applicable to this CERT; test duration shall be as specified above.

A test plan shall be prepared by the contractor and submitted to the Airworthiness Authority for concurrence. The test plan shall describe the proposed test failure reporting and corrective action system (FRACAS) to be utilized during the CERT.

5.9.4 HALT for highly accelerated stress screening.

Post-qualification HALT shall be conducted for all electronic units to provide data to develop a preliminary HASS test for initial operational capability or pilot lot production.

A detailed test plan shall be prepared by the contractor and submitted to the Airworthiness Authority for concurrence. However, if no significant design changes were introduced after the HALT conducted during component development, the data gathered during that test may be sufficient for the purpose of designing the production stress screen.

During the early phases of a development program, the HASS will not yet have been developed, yet there will still be a need to perform some level of environmental stress screening (ESS) on electronic units or assemblies prior to acceptance test. The ESS should follow typical industry practices (such as NAVMAT P-9492) based on the complexity (and criticality) of the component to be qualified. It is expected that the supplier will submit the proposed ESS as an integral part of the ATP, or as a separate process document to be executed prior to ATP.

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5.10 Documentation.

All data shall be submitted or recorded in accordance with the following subparagraphs, with the exception of software or programmable hardware, which shall conform to RTCA/DO-178 and/or RTCA/DO-254 guidelines and be in general compliance with software standard practices as described in AMACC 15.0 (Software Airworthiness) and its subsections, or other DoD component standards. Additional documentation guidance is provided in APPENDIX J.

5.10.1 Acceptance test procedures.

(Referenced by: 5.1.2)

All controls and other external components utilized for flight test or qualification purposes shall be tested in accordance with their governing ATP(s). General guidance describing the contents of an ATP can be found in DI-RELI-80322 (6.4), but it has been canceled and is therefore not mandatory. However, this does not relieve the contractor of the requirement to prepare an ATP with equivalent content needed for approval by the Airworthiness Authority.

All component performance specifications (e.g., ATPs) shall contain upward pointing traceability matrices to the documents from which the performance requirements are derived. There shall be no acceptance or other performance criteria for which an authorizing source is not specified, which in some cases may be manufacturer best practice if not traceable to a higher level requirement specification.

5.10.2 Pass / fail criteria definition and traceability.

For each component to be qualified, pass/fail criteria for every input or output signal or parameter, or other performance characteristic, shall be defined and submitted to the Airworthiness Authority as part of the component test plan. These criteria shall be traceable to the component ATP and/or other specification which defines required component performance.

5.10.3 Test plans and procedures.

For each component to be qualified, overall test plans shall be prepared in accordance with Part 1 of DI-NDTI-81895 (6.3). Part 1 should be submitted to the Airworthiness Authority for approval prior to preparation and/or submittal of any Part 2 individual test procedures.

Once the Part 1 test plan summary is approved by the Airworthiness Authority, individual test procedures can be prepared in accordance with Part 2 and submitted for approval. Upon approval, the individual test procedures can be included within the previously approved Part 1 summary as uniquely identifiable test procedure subsections.

The above requirement applies to the group of tests listed under 5.3, 5.4, and 5.5.2. Other component test plans can be submitted singularly or grouped as agreed by the contractor and the Airworthiness Authority. Note that the E^3 plan (5.7) shall be prepared IAW DI-EMCS-80201 (6.4), and the software test plans (5.8) IAW AMACC 15.0 for DEVCOM AvMC; other DoD components may require different test plan submittal requirements.

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5.10.4 Reports.

5.10.4.1 Test reports.

For each component to be qualified, test reports shall be prepared in accordance with the requirements of DI-NDTI-81897 (6.3). The report may be submitted after the completion of all testing, or after any single test or group of tests, at the contractor's discretion. If the contractor chooses to submit the test report in piecemeal fashion, the first submittal should include items 2 through 20 of the DID, completed to the maximum extent possible, with the test report(s) included in item 21. Additional reports can be submitted as available. Any test requirements satisfied via analysis or similarity shall be referenced by document number in the appropriate section of the test report, but shall not be included (the report shall only include the outcome of actual tests).

The above requirement applies to the group of tests listed under 5.3, 5.4, and 5.5.2. Other component test reports can be submitted singularly or grouped as agreed by the contractor and the Airworthiness Authority. Note that the E^3 test report (5.7) shall be prepared IAW DI-EMCS-80200 (6.4), and the software test reports (5.8) IAW AMACC 15.0 for DEVCOM AvMC; other DoD components may require different test report submittal requirements.

5.10.4.2 Similarity argument reports.

(Referenced by: J.2.4.2)

Any component to be granted qualification by 'similarity' shall be similar to a qualified and/or certified component. The qualified and/or certified component to which similarity is being claimed shall have passed tests that are the same as, or similar to, the test requirements specified in the AQP, or shall be in service use on a similar engine and/or platform combination.

Each report shall contain, but is not limited to, the items listed in DI-NDTI-81896 (6.3). A similarity argument summary for each component should be provided early on, containing items 2 through 15 of the DID, as applicable. For reference, FIGURE J-1 illustrates a 'flowchart' based on the content requirement of the DID. Airworthiness Authority concurrence with the summary information will minimize the contractor's risk of non-approval of the similarity report when finally submitted.

5.10.4.3 Analysis reports.

(Referenced by: 5.4.10, 5.6.4.1, J.2.4.2, FIGURE J-1)

Verification of component qualification requirements by theoretical analysis is allowable. Each report shall contain, but is not limited to, the items listed in DI-NDTI-81896 (6.3), as applicable. Certifications and use of standard parts, materials, and processes may be employed as part of verification by analysis. The theoretical basis and the application of the theory to a particular component should be mutually agreed upon by the Airworthiness Authority and the contractor well in advance of actual AWR or qualification testing.

For reference, FIGURE J-1 illustrates a 'flowchart' based on the content requirement of the DID. Airworthiness Authority concurrence with the summary information will minimize the contractor's risk of non-approval of the analysis report when finally submitted.

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6. NOTES

This section contains information of a general or explanatory nature.

6.1 Intended use.

Determination of applicable test requirements for engine control and accessory components intended for military unique air vehicle propulsion or auxiliary power applications.

6.2 Acquisition requirements.

Not applicable.

6.3 Associated data item descriptions (DID).

(Referenced by: 4.2.1.2, 5.10.3, 5.10.4.1, 5.10.4.2, 5.10.4.3)

This standard has been assigned an Acquisition Management Systems Control (AMSC) number authorizing it as the source document for the following DIDs. When it is necessary to obtain the data specified in the 'Referenced by' column in the table below, the applicable DIDs must be listed on the contract data requirements list (CDRL), DD Form 1423.

Number	Title	Referenced by
DI-NDTI-81895	Engine Control System Component Test Plan Requirements	*4.2.1.2* *5.10.3*
DI-NDTI-81896	Engine Control System Component Similarity/Analysis Report	*5.10.4.2* *5.10.4.3*
DI-NDTI-81897	Engine Control System Component Test Report Requirements	*5.10.4.1*

The above DIDs were current as of the date of this standard. ASSIST should be researched at https://quicksearch.dla.mil to ensure that only current and approved DIDs are cited on DD Form 1423.

6.4 DIDs for contract consideration.

(Referenced by: 5.10.1, 5.10.3, 5.10.4.1)

Although this standard is not the source document for the following DIDs, when it is necessary to obtain the associated data specified in the 'Referenced by' column of the table below, the applicable DIDs must be listed on the contract data requirements list (DD Form 1423).

Number	Title	Referenced by
DI-EMCS-80200	Electromagnetic Interference Test Report (EMITR)	*5.10.4.1*
DI-EMCS-80201	Electromagnetic Interference Test Procedures (EMITP)	*5.10.3*
DI-RELI-80322	Quality Conformance Inspection and Test Procedures (canceled)	*5.10.1*

The above DIDs were current as of the date of this standard. ASSIST should be researched at https://quicksearch.dla.mil to ensure that only current and approved DIDs are cited on DD Form 1423.

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6.5 Tailoring guidance.

(Referenced by: FOREWORD)

The test requirements specified herein should be tailored in the AQS to the specific component(s) to be qualified, and may be further modified in the AQP in recognition of procedures and/or processes unique to the engine or control system contractor, or component supplier. All such tailoring is to be concurred by the Airworthiness Authority.

6.6 Subject term (key word) listing.

Airworthiness, Army
Engine, aircraft
Flight rating, preliminary
Plan, qualification
Qualification, component
Software, qualification
Control system, engine
FADEC, full-authority
Gas turbine engine, aircraft
Propulsion
Reciprocating engine, aircraft

6.7 Change notations.

The margins of this standard will be marked with vertical lines to indicate substantive modifications when generated by a change notice. This is for convenience only and users are therefore cautioned that no guarantee is made as to the accuracy of these change identifiers.

The **SUMMARY OF CHANGE MODIFICATIONS** will describe all of the changes made since the last release of this standard, with the exception of those made to correct non-substantive typographical errors, grammar, or minor clarification of requirements.

6.8 Standard creation / technical assistance.

	DEVCOM AvMC, SRD, Propulsion Division
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TABLE I. Component test applicability check list.

(Referenced by: 4.2.1.1)

COMPONENT:			'X' APPLICABLE TEST REQUIREMENT		
5.3	**Simulated operational test**		5.6.3	Heat exchanger: cycling and proof pressure	
5.3.1	Operational cycle definition		5.6.4	Electronic control unit	
5.3.1.1	All components excluding ignition sys.		5.6.4.1	PMA-induced damage	
5.3.1.2	Ignition system		5.6.4.2	Electrical loads analysis^{PFR}	
5.3.2	Low-lubricity fuel		5.6.4.3	Overheat^{PFR}	
5.3.3	High temperature		5.6.4.4	Power compatibility^{PFR}	
5.3.4	Room temperature and contamination		5.6.4.5	Short circuit protection	
5.3.5	Low temperature		5.6.4.6	Data bus specification compliance	
5.3.6	Combined temperature-vibration		5.6.4.7	Crystal oscillator temp. performance	
5.3.7	Engine fuel system cavitation endurance		5.6.5	Fuel pump	
5.4	**Environmental series tests**		5.6.5.1	Net positive suction pressure	
5.4.1	Default environmental parameters		5.6.5.2	Self priming	
5.4.2	Temperature: high / low / shock (transient)		5.6.5.3	Bubble ingestion	
5.4.3	Vibration		5.6.5.4	Vapor/liquid (V/L) capability	
5.4.3.1	Vibration and high temperature^{PFR}		5.6.6	Ignition system: fouling tests	
5.4.3.2	Vibration: qualification		5.6.6.1	Carbon deposits	
5.4.4	Shock (crash safety)^{PFR}		5.6.6.2	Water ingestion	
5.4.5	Gunfire shock		5.6.7	Fuel filter element capacity/efficiency	
5.4.6	Acceleration		**5.7**	**Electromagnetic environmental effects**	
5.4.7	Low pressure (altitude)		5.7.1	Electromagnetic interference^{PFR}	
5.4.8	Rain		5.7.2	Electromagnetic pulse	
5.4.9	Explosive atmosphere^{PFR}		5.7.3	Lightning	
5.4.10	Fungus		5.7.4	Personnel -borne electrostatic discharge	
5.4.11	Humidity		5.7.5	Electrical bonding analysis	
5.4.12	Salt fog		5.7.6	High-altitude electromagnetic pulse	
5.4.13	Sand and dust		5.7.7	Initial nuclear radiation	
5.4.14	Contamination by fluids		5.7.8	High-power microwave	
5.5	**Special tests**		**5.8**	**System verification and validation tests**	
5.5.1	Overspeed and containment		5.8.1	Engine control system^{PFR}	
5.5.2	Pressure tests		5.8.2	Common-mode multiple signal failure	
5.5.2.1	Pressure cycling		5.8.3	Complex power interrupts	
5.5.2.2	Proof pressure		5.8.4	Helicopter drive sys torsional stability^{PFR}	
5.5.2.3	Burst pressure		5.8.5	Fault injection	
5.5.3	Fire test (resistance / proof)^{PFR}		**5.9**	**Reliability development and growth tests**	
5.5.4	Pressure wash		5.9.1	HALT (prior to QT start)	
5.6	**Component-specific tests**		5.9.2	Vibration: endurance (prior to QT completion)	
5.6.1	Oil reservoir: cycling and proof pressure^{PFR}		5.9.3	CERT (prior to full-rate production)	
5.6.2	Alternator: load and overspeed / containment		5.9.4	HALT for HASS (prior to pilot lot completion	

NOTES: PFR superscript – Test required for flight AWR if applicable to the component type. PFR tests are a subset of the QT tests and may require repeating during QT, if deemed necessary by the Airworthiness Authority.

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TABLE II. Environmental-series test sequence.

(Referenced by: 4.3.2, 5.4, TABLE VII NOTE 10/)

		Number of Units Under Test					
		UUT1	UUT1	UUT2	UUT1	UUT2	UUT3
Test Requirement	**Paragraph**	**Test Sequence Number[1]**					
Pre-Test	5.1.2	IAW Airworthiness Authority-approved supplier ATP					
Pre-Test	Hermetic	IAW Airworthiness Authority-approved supplier procedure					
Vibration: Sine Sweep	5.4.3.1	If not completed for PFR, test is precursor for qualification					
High Temperature	5.4.2a	1	1		1		
Low Temperature	5.4.2b	1	1		1		
Temp (Transient)	5.4.2c	1	1		1		
Vibration: Qualification[3]	5.4.3.2	2	2		2		
Impact Shock	5.4.4	2	2		2		
Gunfire Shock	5.4.5	2	2		2		
Acceleration	5.4.6	3	3		3		
Low Pressure (Altitude)[7]	5.4.7	4	4		4		
Rain[7]	5.4.8	5a	5a		5a		
Inspection (only)	5.4.8	5b	5b		5b		
Hermiticity Verification[6]	MIL-STD-202-112A	H1	H1		H1		
Explosive Atmosphere[8]	5.4.9	6	6		6		
Disassemble & Inspect	5.1.4	NH[4]	7		7		
Fungus	5.4.10	7		1			1
Humidity[7]	5.4.11	8		2			2
Salt Fog	5.4.12	9		3		1	
Sand and Dust	5.4.13	10		4		2	
Contam. by Fluids[2]	5.4.14	11		5		3	3
Hermiticity Verification[6]	MIL-STD-202-112A	H2		H1		H1	H1
Disassemble & Inspect	5.1.4	12		6		4	4

NOTES: 1/ In any column, tests with the same sequence number can be executed in any order prior to the next higher sequence number in that column.

2/ For Contamination by Fluids, the test can be executed on either unit, in the given sequence number order.

3/ If component qualification requires functional vibration testing IAW 5.4.3.2, it must be preceded by high/low/shock (transient) temperature testing IAW 5.4.2.

4/ The legends “H” and “NH” indicate the class of components under test; hermetic and non-hermetic. Disassembly & Inspection is to be performed in the order shown, depending on component class.

5/ Not used.

6/ Hermeticity verification (H1 and H2) shall be performed during the qualification test sequence IAW the specified test procedure. Other proposed test methods must be capable of detecting a leak rate of 10^{-5} atm cm^3/s, and shall require approval by the Airworthiness Authority.

7/ Test is waived for hermetically sealed components.

8/ If required to be tested, hermetically sealed components shall have the seal penetrated prior to the test, and resealed afterwards if the unit is to undergo further testing. Testing is generally waived for engine-mounted components, unless they contain fuel and have air cavities in which fuel-air vapor can accumulate. See 5.4.9 and H.2.4 for EA applicability.

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TABLE III. PFR test matrix.

(Referenced by: 5.2, REMARKS)

Component Test Requirement and Method of Compliance – PFR Test Matrix *(Paragraph 5.2)*			Component Category/Classification (see H.2.7)																			
				Electronic Unit			Fuel Handling				Oil Handling			Sensors		Actuators						
Test Requirement	Para.	Method or Source	System Level (SL)	Exciter & Ignitor	Analog or Digital	ECU with Software	Main Pump	Boost Pump	FMU/HMU	Filter	Pump/Filter	Tank	Heat Exchanger	Simple	Complex	Electro-Hydraulic	Electro-Mechanical	Electro-Pneumatic	Electrical Harness	Alternator or Starter	Fuel/Oil Pipes, Fittings, Hoses
Vibration & High Temp: PFR[6]	5.4.3.1	MIL-STD-810C, Met 514, P1, Cat b.1 and c		X[1]	X[1]	X[1]	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X[10]
Shock	5.4.4	MIL-STD-810H(C1) Method 516, P5			X[2]	X[2]															
Explosive Atmosphere	5.4.9	MIL-STD-810H(C1) Method 511, P1		X[9]	X	X			X	X						X				X	
Fire Test	5.5.3	SAE AS1055 (FP) SAE AS4273 (FR)				FR[3]	FR	FR	FR	FR	FP[8]	FP	FP[8]	FRP[5]	FRP[5]	FR					FP[4]
Oil Reservoir	5.6.1d	This Standard										X									
Electrical Loads Analysis[7]	5.6.4.2	This Standard + MIL-HDBK-704, LDC101		X	X	X			X						X	X	X	X			
Overheat	5.6.4.3	This Standard + FAA AC33.28	X			X															
Power Compatibility	5.6.4.4	This Standard	X			X															
EMI	5.7.1	MIL-STD-461 + This Standard	X																		
System/Software Verif. & Valid.	5.8.1	RTCA/DO-178/254 + AMACC 15.0	X		X	X															
Torsional Stability Analysis	5.8.4	This Standard	X			X															

NOTES: 1/ If airframe mounted, test IAW Category c only, 2/ If in crew compartment or near a system critical to crew safety, 3/ If in fire zone, 4/ FR rating for fuel lines, hoses and fittings, 5/ If sensors penetrate engine areas containing oil or fuel (such as a tank), and therefore act as 'fittings' or plugs, they are required to be fire proof or resistant, according to the component to which they are attached. 6/ If the test is not conducted at CLT, then a partial high-temperature test IAW 5.4.3.1 is required prior to vibration. 7/ Applicable to components directly connected to airframe electrical power subsystem. 8/ Fire proof, unless pump can be shut off or cannot pressurize downstream components at windmilling speed, and filter or heat exchanger cannot discharge a hazardous quantity of oil. 9/ Not applicable to ignitors. 10/ Vibration requirement can be satisfied via on-engine testing (RAP test, etc.), instead of a separate component bench test.

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TABLE IV. Qualification test matrix: simulated operational test.

(Referenced by: 5.3)

Component Test Requirement and Method of Compliance – SOT Matrix *(Paragraph 5.3)*			Component Category/Classification (see H.2.7)																				
			System Level (SL)	Electronic Unit			Fuel Handling				Oil Handling			Sensors		Actuators			Electrical Harness	Alternator or Starter	Fuel/Oil Pipes, Fittings, Hoses
				Exciter & Ignitor	Analog or Digital	ECU with Software	Main Pump	Boost Pump	FMU/HMU	Filter	Pump	Tank/Heat Exchanger	Filter	Simple	Complex	Electro-Hydraulic	Electro-Mechanical	Electro-Pneumatic			
Test Requirement	**Para.**	**Method or Source**																					
Operational Cycle Definition	5.3.1	This Standard	**As Required**																				
Low Lubricity Fuel	5.3.2						X	X	X							X							
High Temperature	5.3.3						X	X	X	X	X		X			X		X					
Room Temperature & Contamination	5.3.4						X	X	X	X	X		X			X		X					
Low Temperature	5.3.5						X	X	X	X	X		X			X		X					
Combined Temp-Vibration	5.3.6			X	X	X						X		X	X		X		X	X			
Engine Fuel System Cavitation Endurance	5.3.7						X	X															

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TABLE V. Fuel contamination mixture: light.

(Referenced by: 5.3.4a(1))

Contaminant	Particle Size (microns)	Quantity (grams/1,000 gallons)
Ferroso-Ferric Iron Oxide (Fe_3O_4, black color, Magnetite)	0-5	1.0
Ferric Iron Oxide (Fe_2O_3, Hematite)	0-5 5-10	5.0 1.0
Prepared dirt IAW ISO 12103-1 (Arizona test dust - A4 coarse)	---	2.0
Cotton Linters	Staple below 7 (USDA grading standards SRA-AMS 180 and 251)	0.02

TABLE VI. Fuel contamination mixture: heavy.

(Referenced by: 5.3.4a(1), 5.3.7, 5.6.7)

Contaminant	Particle Size (microns)	Quantity (grams/1,000 gallons)
Ferroso-Ferric Iron Oxide (Fe_3O_4, black color, Magnetite)	0-5	1.5
Ferric Iron Oxide (Fe_2O_3, Hematite)	0-5 5-10	27.0 1.5
Crushed Quartz	300-420 150-300	1.0 1.0
Prepared dirt IAW ISO 12103-1 (Arizona test dust - A4 coarse)	---	8.0
Cotton Linters	Staple below 7 (USDA grading standards SRA-AMS 180 and 251)	0.1
Crude Naphthenic Acid (not required if service fuel is used)	---	0.03% by volume
Salt water prepared by dissolving salt in distilled water or other water containing not more than 200 ppm of total solids	4 parts by weight of NaCl 96 parts by weight of H_2O	0.01% by volume (entrained)

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TABLE VII. Qualification test matrix: environmental series.

(Referenced by: 4.2.1.1, 5.4)

Component Test Requirement and Method of Compliance - Environmental Series[10] *(Paragraph 5.4)*			Component Category/Classification (see H.2.7)																			
				Electronic Unit			Fuel Handling				Oil Handling			Sensors		Actuators						
Test Requirement	Para.	Method or Source	System Level (SL)	Exciter & Ignitor	Analog or Digital	ECU with Software	Main Pump	Boost Pump	FMU/HMU	Filter	Pump	Tank/Heat Exchanger	Filter	Simple	Complex	Electro-Hydraulic	Electro-Mechanical	Electro-Pneumatic	Electrical Harness	Alternator or Starter	Fuel/Oil Pipes, Fittings, Hoses
Default Environmental Parameters	5.4.1	This Standard	**As Required**																			
High Temperature	5.4.2a	MIL-STD-810H(C1) Method 501.7, P1&P2		X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
Low Temperature	5.4.2b	MIL-STD-810H(C1) Method 502.7, P1&P2		X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
Temperature Shock (Transient)	5.4.2c	MIL-STD-810H(C1) Method 503.7, P1-C		X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
PFR or QT	5.4.3.1	MIL-STD-810C, Method 514.2, P1, Cat b.1 or c		X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X[11]
Vibration: Qualification Test	5.4.3.2	MIL-STD-810H(C1) Method 514.8, P1, Cat 13 or 14 & Cat 22		X[1]	X[1]	X[1]	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X[11]
Shock[3]	5.4.4	MIL-STD-810H(C1) Method 516.8, P1		X	X	X	X	X	X	X	X	X	X	X	X	X	X	X		X	
Gunfire Shock	5.4.5	MIL-STD-810H(C1) Method 519.8, P1, P2, or P3		X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
Acceleration	5.4.6	MIL-STD-810H(C1) Method 513.8, P2		X[7]	X	X	X	X	X	X	X		X		X	X	X	X		X	
Low Pressure[8] (Altitude)	5.4.7	MIL-STD-810H(C1) Method 500.6, P2		X[4]	X[4]	X[4]											X[4]				
Rain[8]	5.4.8	MIL-STD-810H(C1) Method 506.6, P1		X[5]	X[5]	X[5]			X	X			X	X	X	X	X	X	X	X	

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TABLE VII. Qualification test matrix: environmental series – continued.

(Referenced by: 5.4)

Component Test Requirement and Method of Compliance - Environmental Series[10] (Paragraph 5.4)			Component Category/Classification (see H.2.7)																			
				Electronic Unit			Fuel Handling				Oil Handling			Sensors		Actuators						
Test Requirement	Para.	Method or Source	System Level (SL)	Exciter & Ignitor	Analog or Digital	ECU with Software	Main Pump	Boost Pump	FMU/HMU	Filter	Pump	Tank/Heat Exchanger	Filter	Simple	Complex	Electro-Hydraulic	Electro-Mechanical	Electro-Pneumatic	Electrical Harness	Alternator or Starter	Fuel/Oil Pipes, Fittings, Hoses
Explosive Atmosphere[9]	5.4.9	MIL-STD-810H(C1) Method 511,.7 P1 +Hot Spot Analysis		X[7]	X	X			X	X						X				X	
Fungus	5.4.10	MIL-STD-810H(C1) Method 508.8		X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
Humidity[8]	5.4.11	MIL-STD-810H(C1) Method 507.6, P2		X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
Salt Fog	5.4.12	MIL-STD-810H(C1) Method 509.8, P1		X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
Sand and Dust	5.4.13	MIL-STD-810H(C1) Method 510.7, P1&P2		X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
Contamination by Fluids	5.4.14	MIL-STD-810H(C1) Method 504.3, P1, 4.5.5b Intermittent		X	X[6]	X[6]	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X

NOTES:
1/ Category 22 is required only if turbine-engine mounted
2/ Deleted
3/ If in crew compartment or near critical system, Procedure V also applies
4/ If sealed design, Procedure I also applies; not applicable to ignitors
5/ If airframe mounted, utilize Procedure III instead of Procedure I
6/ If in engine compartment
7/ Not applicable to ignitors
8/ Not applicable to hermetically sealed components
9/ Engine-mounted components are generally exempt from testing, unless they are sources of an EA mixture and have air cavities in which it can accumulate (see 5.4.9)
10/ Tests in this table are to be run in sequence as defined in TABLE II
11/ Vibration requirement can be satisfied via on-engine testing, instead of a separate component bench test

Definitions:
Cat (category)
P (procedure)

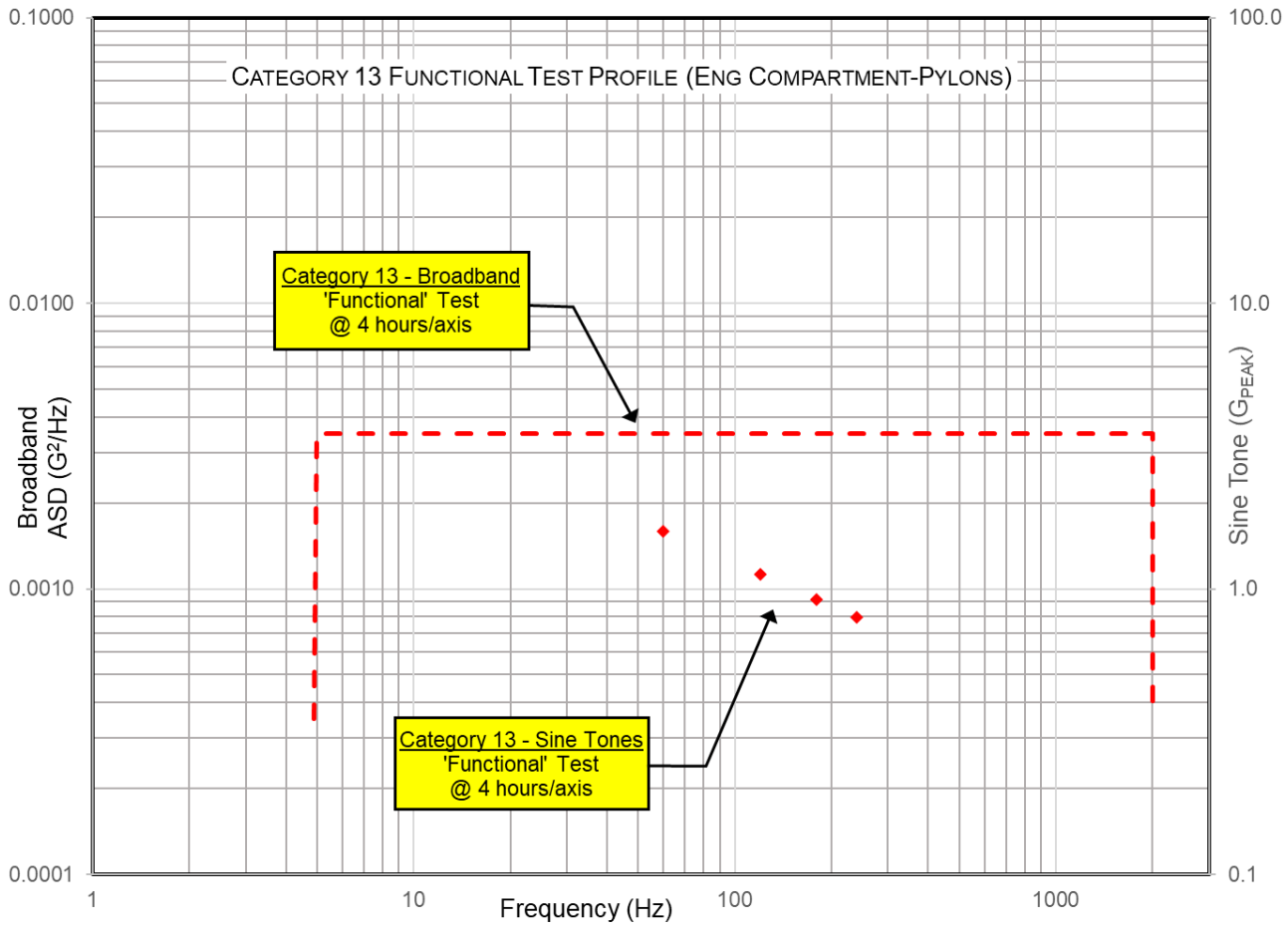
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TABLE VIII. Typical and functional vibration profiles – Category 13 (at engine).

(Referenced by: 5.4.3.2.1, G.3.1)

			Category 13 @ Engine – Default Test Profiles (Propeller Aircraft)				
			Broadband ASD (g^2/Hz)			Sine Amplitude (g_{PEAK})	
			Typical	Functional		Typical	Functional
		Test Duration	1 test hr = 1 real time hr	4 hours per axis		1 test hr = 1 real time hr	4 hours per axis
	Freq (Hz)				Freq (Hz)[1]		
1	5		0.0022	0.00350	60	1.25	1.59
2	15		0.0022	0.00350	120	0.89	1.12
3	2000		0.0022	0.00350	180	0.73	0.92
4					240	0.63	0.79

[1]*Sine tone frequencies are roughly based on C-130 propeller speed/blade count for illustrative purposes. Adjust frequencies and amplitudes as needed for specific application.*



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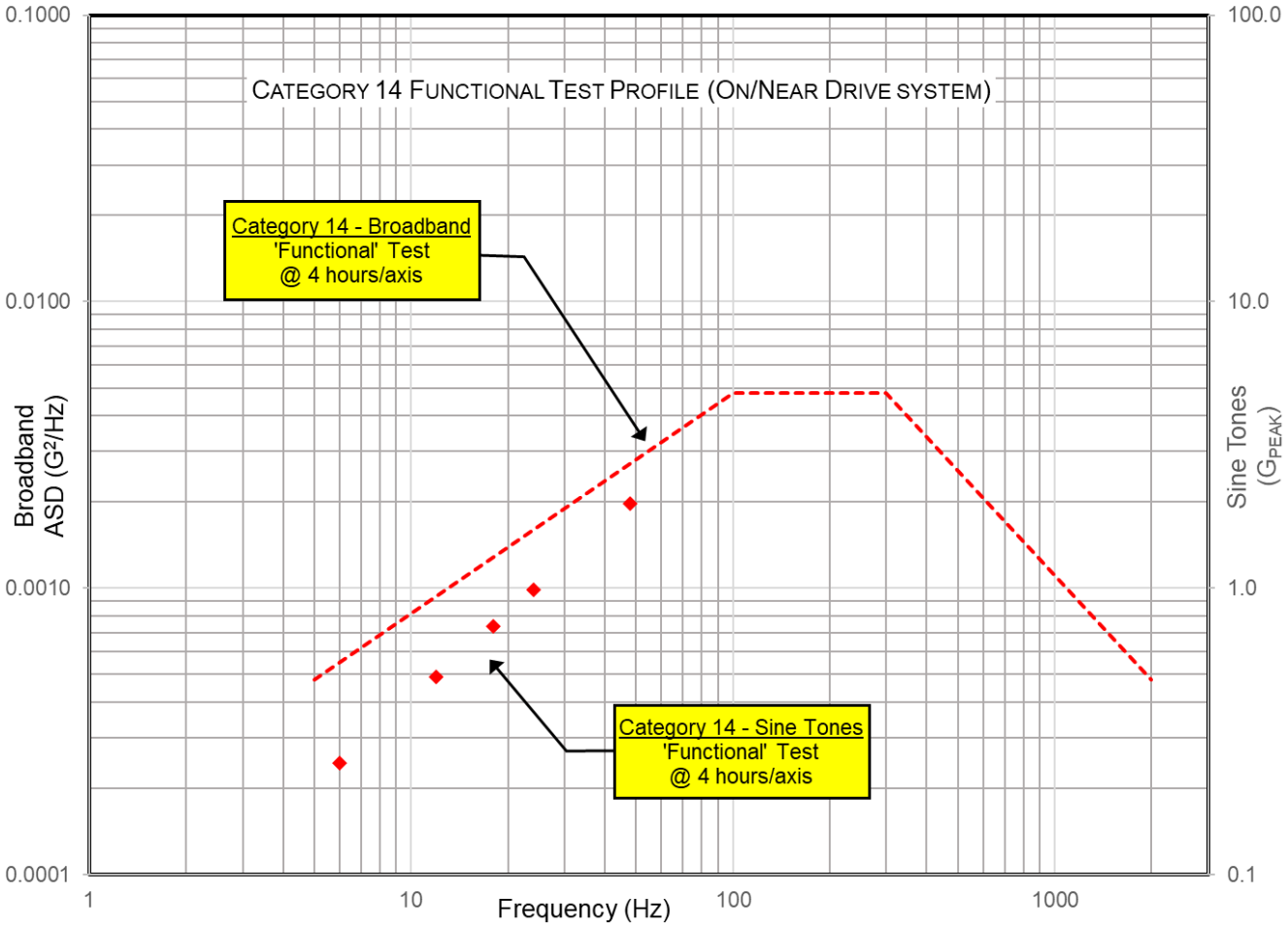
TABLE IX. Typical and functional vibration profiles – Category 14 (at engine).

(Referenced by: 5.4.3.2.1, G.3.1)

			Category 14 @ Engine – Default Test Profiles ('All' Helicopters[1])				
			Broadband ASD (g^2/Hz)			Sine Amplitude (g_{PEAK})	
			Typical	Functional		Typical	Functional
		Test Duration	1 test hr = 1 real time hr	4 hours per axis		1 test hr = 1 real time hr	4 hours per axis
	Freq (Hz)				Freq (Hz)[2]		
1	5		0.0004	0.00048	6	0.20	0.25
2	100		0.0036	0.00480	12	0.41	0.49
3	300		0.0036	0.00480	18	0.61	0.74
4	2000		0.0004	0.00048	24	0.82	0.98
					48	1.63	1.97

[1]*For the H-47, H-60, and H-64 helicopter vibration test profiles, see TABLE XI through TABLE XIV.*

[2]*Sine tone frequencies are notional for illustrative purposes only. Adjust frequencies and amplitudes as needed for specific application.*

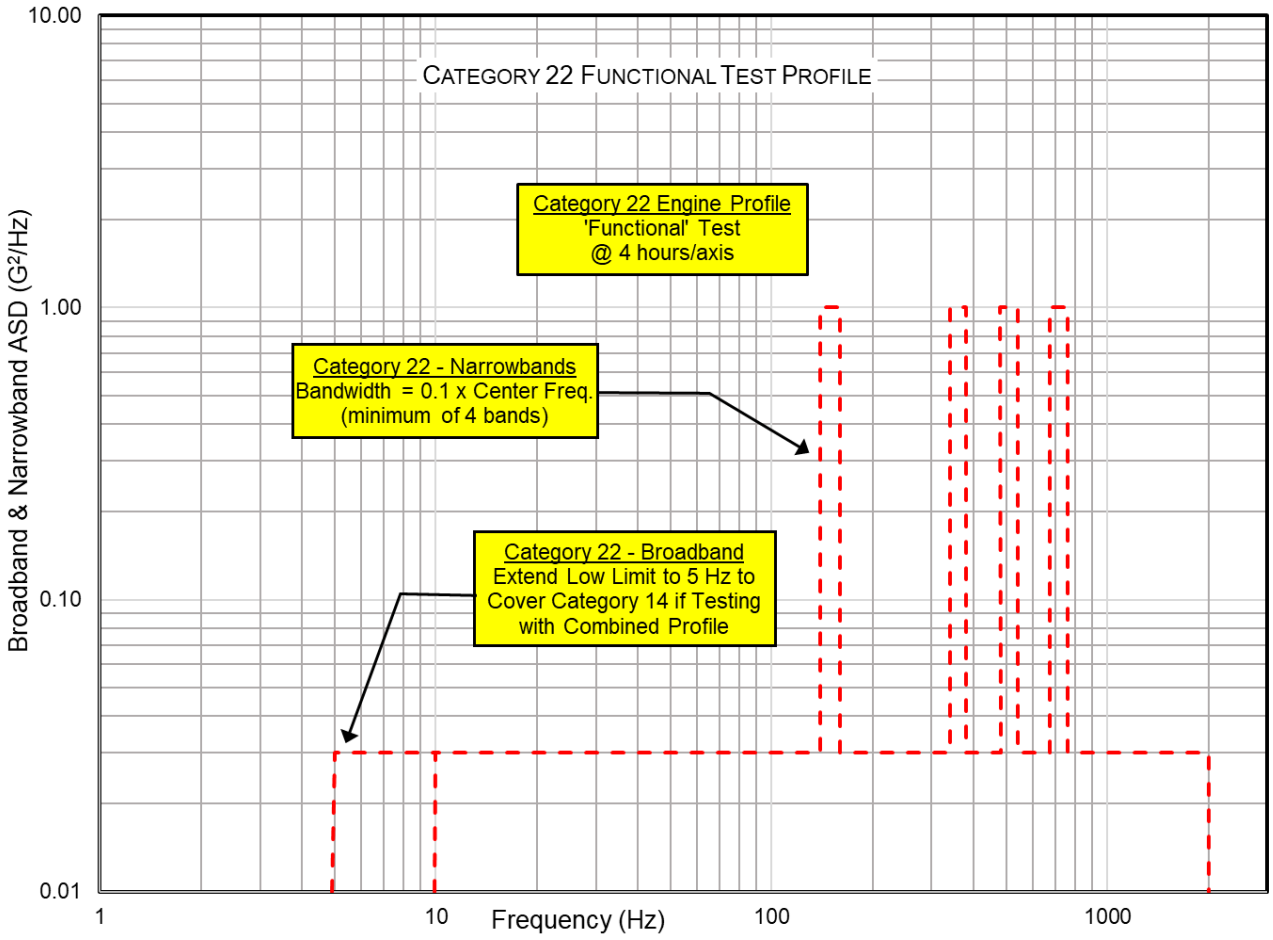


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TABLE X. Typical and functional vibration profiles – Category 22 (on engine).

(Referenced by: 5.4.3.2.1, G.3.1)

			Category 22 – Default Test Profiles (All Turbine Engines[1])				
			Broadband ASD (g^2/Hz)			Narrowband ASD (g^2/Hz)	
			Typical	Functional		Typical	Functional
		Test Duration	1 test hr = 1 real time hr	4 hours per axis		1 test hr = 1 real time hr	4 hours per axis
	Freq (Hz)				Approximate Speed Range (%)		
1	10		0.020	0.030	PT ~ Single- to dual-engine GI	0.667	1.0
2	2000		0.020	0.030	PT ~ 95 to 105%		
					GG ~ GI% to (GI+10)%		
					GG ~ 90 to 105%		
[1]*If on a reciprocating engine, increase levels by 50%*			*0.030*	*0.045*		*1.0*	*1.5*
[2]*Narrowband frequencies are notional for illustrative purposes only. Adjust frequency bands based on actual engine speed/frequency relationships.*							

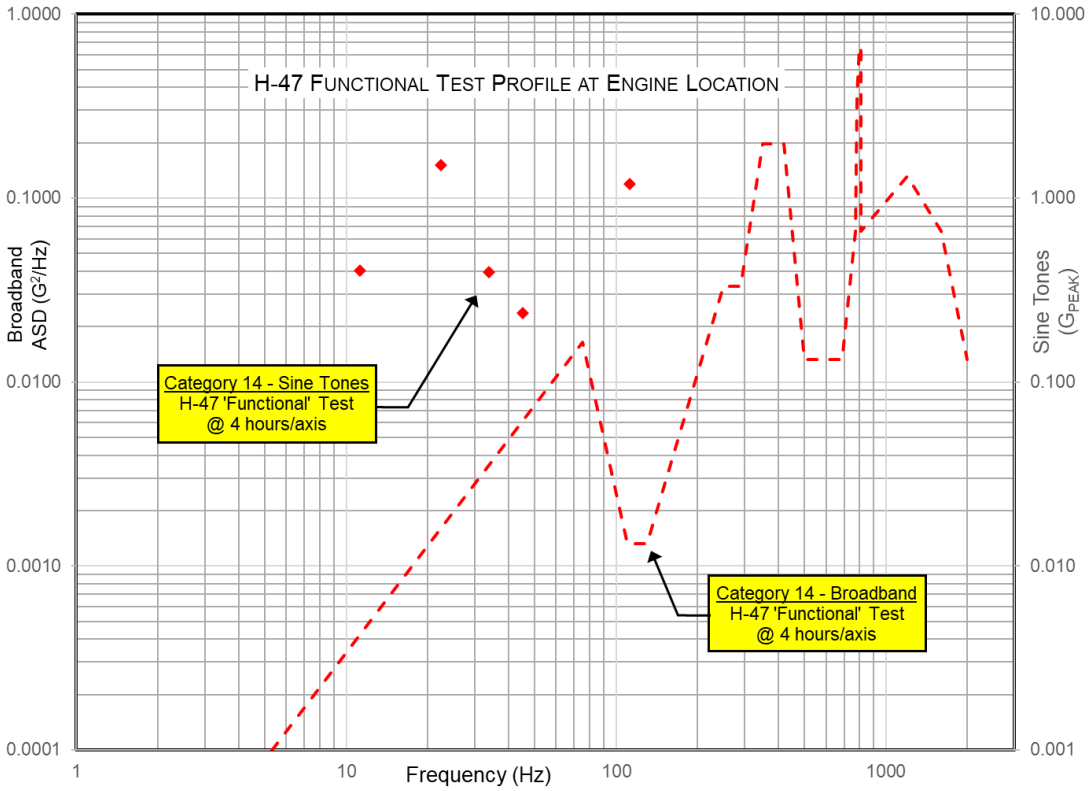


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TABLE XI. Typical and functional vibration test profiles – H-47 (at engine).

(Referenced by: 5.4.3.2.2, TABLE IX, G.3.1, G.3.2)

			Category 14 @ Engine – Default Test Profile (H-47)				
			Broadband ASD (g^2/Hz)			Sine Amplitude Ax (g_{PEAK})	
			Typical	Functional		Typical	Functional
		Test Duration	1 test hr = 1 real time hr	4 hours per axis		1 test hr = 1 real time hr	4 hours per axis
	Freq (Hz)				Freq (Hz)		
1	3		0.0000	0.00003	11.25	0.27	0.40
2	75		0.0100	0.01650	22.50	1.01	1.51
3	110		0.0008	0.00132	33.75	0.26	0.40
4	130		0.0008	0.00132	45.00	0.16	0.24
5	250		0.0200	0.03300	112.50	0.80	1.19
6	290		0.0200	0.03300			
7	350		0.1200	0.19800			
8	420		0.1200	0.19800			
9	500		0.0080	0.01320			
10	690		0.0080	0.01320			
11	770		0.0400	0.06600			
12	780		0.2200	0.36300			
13	800		0.4000	0.66000			
14	810		0.4000	0.66000			
15	811		0.0400	0.06600			
16	1200		0.0800	0.13200			
17	1600		0.0400	0.06600			
18	2000		0.0080	0.01320			

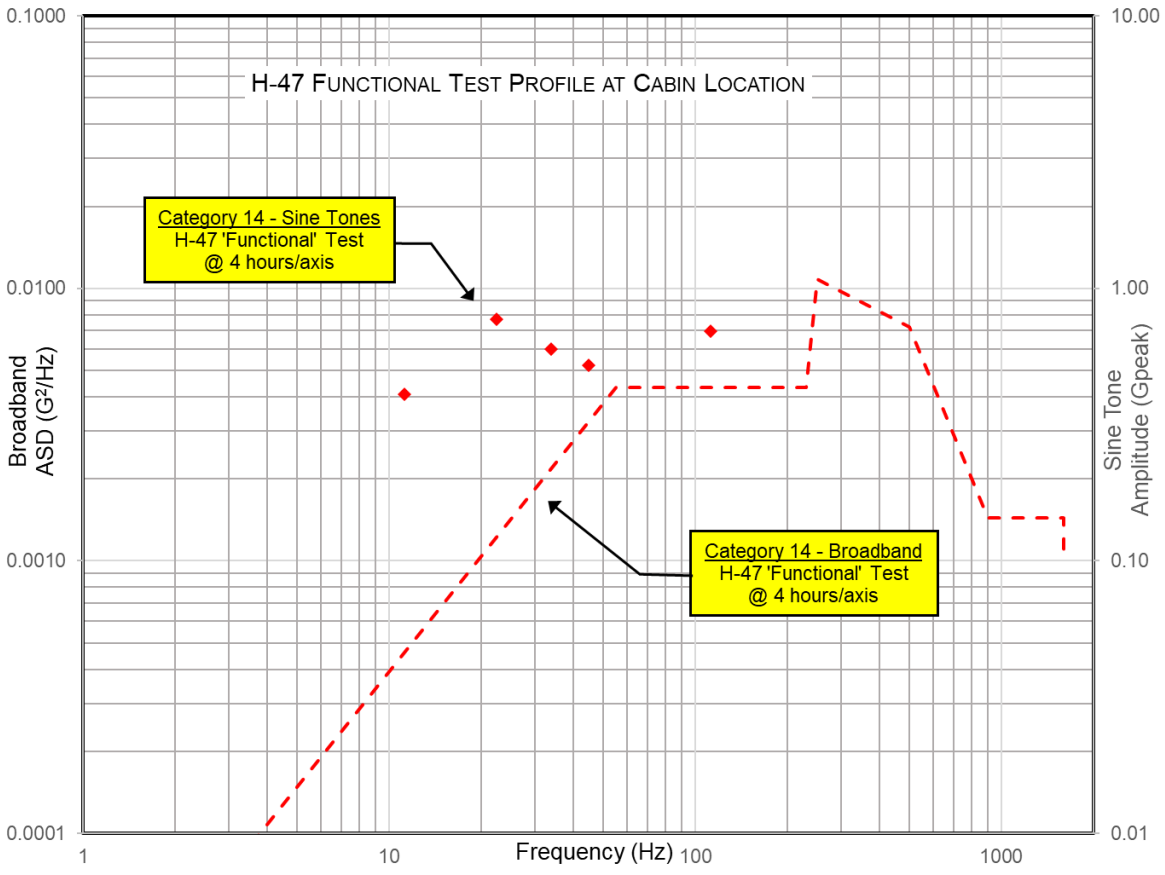


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TABLE XII. Typical and functional vibration test profiles – H-47 (in cabin).

(Referenced by: 5.4.3.2.2, TABLE IX, G.3.1, G.3.2)

			Category 14 @ Cabin – Default Test Profile (H-47)				
			Broadband ASD (g^2/Hz)			Sine Amplitude Ax (g_{PEAK})	
			Typical	Functional		Typical	Functional
		Test Duration	1 test hr = 1 real time hr	4 hours per axis		1 test hr = 1 real time hr	4 hours per axis
	Freq (Hz)				Freq (Hz)		
1	3		0.0000	0.00007	11.25	0.30	0.408
2	55		0.0024	0.00432	22.50	0.57	0.768
3	230		0.0024	0.00432	33.75	0.45	0.600
4	250		0.0060	0.01080	45.00	0.39	0.522
5	500		0.0040	0.00720	112.50	0.52	0.696
6	900		0.0008	0.00144			
7	1600		0.0008	0.00144			

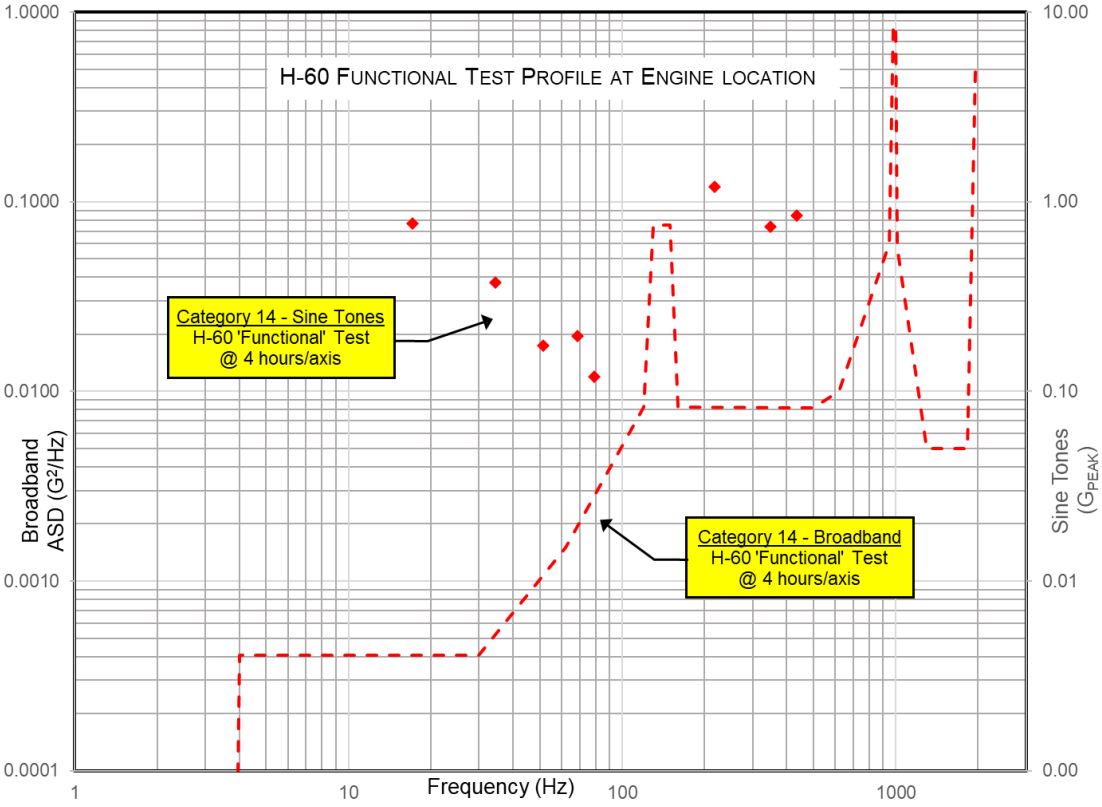


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TABLE XIII. Typical and functional vibration test profiles – H-60 (at engine).

(Referenced by: 5.4.3.2.2, TABLE IX, G.3.1, G.3.2, G.3.2.5)

			Category 14 @ Engine – Default Test Profiles (H-60)				
			Broadband ASD (g^2/Hz)			Sine Amplitude Ax (g_{PEAK})	
			Typical	Functional		Typical	Functional
		Test Duration	1 test hr = 1 real time hr	4 hours per axis		1 test hr = 1 real time hr	4 hours per axis
	Freq (Hz)				Freq (Hz)		
1	4		0.0003	0.0004	17.2	0.68	0.77
2	30		0.0003	0.0004	34.4	0.34	0.38
3	62		0.0012	0.0015	51.6	0.16	0.18
4	120		0.0066	0.0083	68.8	0.17	0.20
5	130		0.0600	0.0750	79.2	0.11	0.12
6	150		0.0600	0.0750	218.0	1.07	1.20
7	160		0.0066	0.0083	348.0	0.66	0.74
8	500		0.0065	0.0082	436.0	0.76	0.85
9	620		0.0080	0.0100			
10	950		0.0480	0.0600			
11	980		0.6800	0.8500			
12	1000		0.6800	0.8500			
13	1010		0.0460	0.0575			
14	1300		0.0040	0.0050			
15	1830		0.0040	0.0050			
16	1950		0.4000	0.5000			
17	2000		0.4000	0.5000			

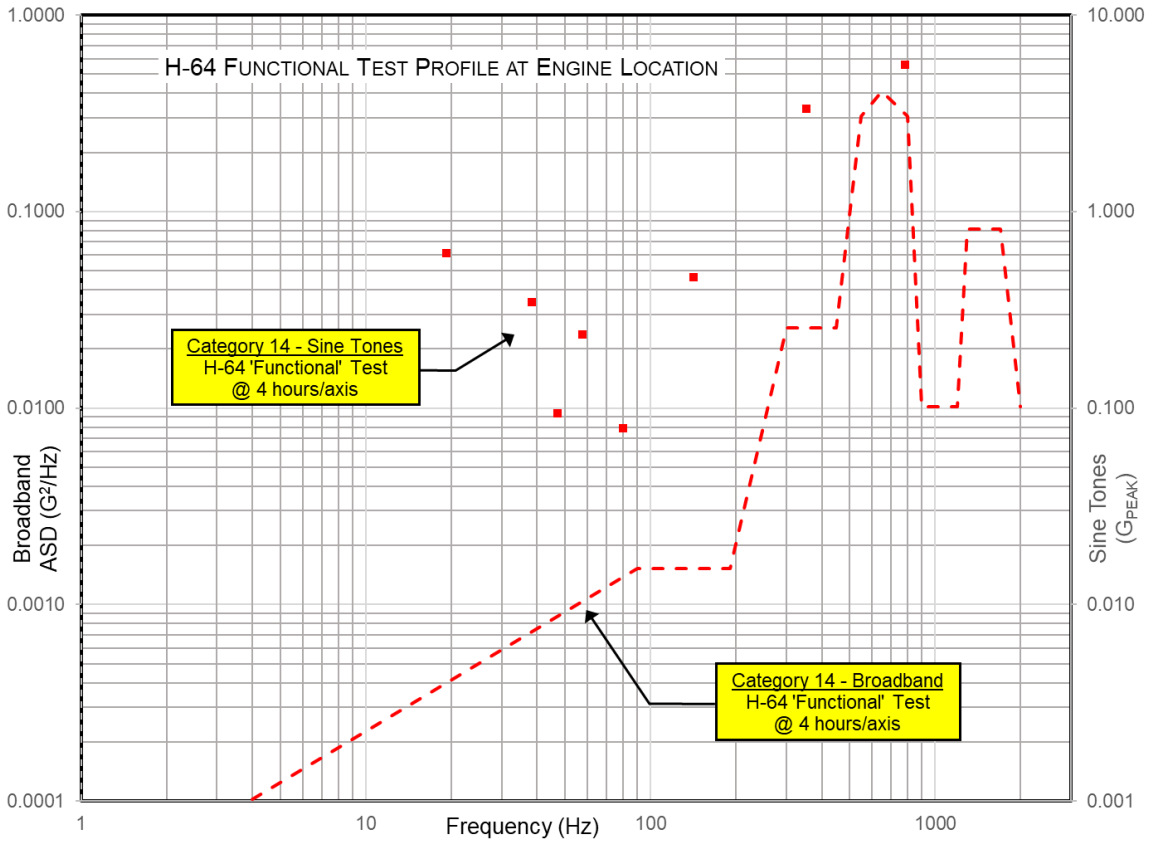


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TABLE XIV. Typical and functional vibration test profiles – H-64 (at engine).

(Referenced by: 5.4.3.2.2, TABLE IX, G.3.1, G.3.2)

			Category 14 @ Engine – Default Test Profiles (H-64)				
			Broadband ASD (g^2/Hz)			Sine Amplitude Ax (g_{PEAK})	
			Typical	Functional		Typical	Functional
		Test Duration	1 test hr = 1 real time hr	4 hours per axis		1 test hr = 1 real time hr	4 hours per axis
	Freq (Hz)				Freq (Hz)		
1	4		0.0000	0.0001	19.20	0.38	0.61
2	90		0.0006	0.0015	38.40	0.21	0.35
3	190		0.0006	0.0015	47.20	0.06	0.09
4	300		0.0100	0.0255	57.60	0.15	0.24
5	450		0.0100	0.0255	80.30	0.05	0.08
6	550		0.1200	0.3060	141.60	0.29	0.46
7	650		0.1600	0.4080	353.00	2.07	3.33
8	800		0.1200	0.3060	787.00	3.48	5.60
9	900		0.0040	0.0102			
10	1200		0.0040	0.0102			
11	1300		0.0320	0.0816			
12	1700		0.0320	0.0816			
13	2000		0.0040	0.0102			



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TABLE XV. Contamination by fluids: test media.

(Referenced by: 5.4.14)

Group	Type	Test Fluid	UUT Temp (±2°C) (±4°F)	Fluid Temp (±2°C) (±4°F)
Aviation turbine fuels	Kerosene	NATO F-24 (iaw AFLP-3747) or Jet A (iaw ASTM D1655) w/military additives or JP-8 (iaw MIL-DTL-83133)	Component Limiting Temperature (see 5.1.7)	+70°C (+158°F)
Hydraulic oils	Hydrocarbon base (synthetic)	NATO H-537 (MIL-PRF-83282) or NATO H-544 (MIL-PRF-46170)		+70°C (+158°F)
	Petroleum base	NATO H-515 (MIL-PRF-5606)		+70°C (+158°F)
	Phosphate ester base (synthetic)	SAE AS1241 contaminated with 1% water by weight (Skydrol® LD-4)		+70°C (+158°F)
Lubricating oils	Polyol ester base (synthetic)	NATO O-156 (MIL-PRF-23699)		+150°C (+302°F)
Transmission oils	Synthetic base	DOD-PRF-85734		+150°C (+302°F)
Solvents & cleaning fluids		trans-1,2-dichloroethylene (replaces 1.1.1-Trichloroethane)	+23°C (+73°F)	+23°C (+73°F)
Deicing & antifreeze fluids		Ethylene or propylene glycol mixtures (NATO S-750 [A-A-52624]) Concentration B – 60% glycol by vol.	-10°C (+14°F)	+23°C (+73°F)
Runway deicers		Potassium-acetate based solution (e.g., Cryotech E-36)		+23°C (+73°F)
Hydraulic oils	Hydrocarbon base (synthetic)	MIL-PRF-87257 (low temperature)	-54°C (-65°F)	-54°C (-65°F)

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TABLE XVI. Qualification test matrix: special tests.

(Referenced by: 5.5, TABLE XVII NOTE 4/)

Component Test Requirement and Method of Compliance – Special Tests Matrix *(Paragraph 5.5)*			Component Category/Classification (see H.2.7)																			
				Electronic Unit			Fuel Handling				Oil Handling			Sensors		Actuators						
Test Requirement	Para.	Method or Source	System Level (SL)	Exciter & Ignitor	Analog or Digital	ECU with Software	Main Pump	Boost Pump	FMU/HMU	Filter	Pump/Filter	Tank	Heat Exchanger	Simple	Complex	Electro-Hydraulic	Electro-Mechanical	Electro-Pneumatic	Electrical Harness	Alternator or Starter	Fuel/Oil Pipes, Fittings, Hoses
Overspeed and Containment	5.5.1	This Standard					X	X	X[1]		X									X[10]	
Pressure Cycling	5.5.2.1				X[2]	X[2]	X	X	X	X	X			X[2]	X[2]	X		X			X
Proof Pressure [4]	5.5.2.2				X[2]	X[2]	X	X	X	X	X			X[2]	X[2]	X		X			X
Burst Pressure[5]	5.5.2.3				X[2]	X[2]	X	X	X	X	X			X[2]	X[2]	X		X			X
Fire Test	5.5.3	SAE AS1055 (FP) SAE AS4273 (FR)				FR[6]	FR	FR	FR	FR	FP[11]	FP	FP[11]	FRP[8]	FRP[8]	FR					FP[7]
Pressure Wash	5.5.4	This Standard		X[9]	X[9]	X[9]	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X

NOTES:
1/ If mechanically driven
2/ If pressurized by fuel or air, or sealed to the atmosphere
3/ Not used
4/ Proof pressure shall follow the pressure cycling test on the same unit
5/ Burst pressure testing shall follow the proof pressure test on the same unit
6/ If in fire zone
7/ FR rating for fuel lines, hoses and fittings [see FAA AC 33.17-1B, 6.3.1].
8/ If sensors penetrate engine areas containing oil or fuel (such as a gearbox), and therefore act as 'fittings' or plugs, they are required to be fire resistant or proof, as applicable to the component to which they are attached.
9/ If engine mounted
10/ Not applicable to alternators (see TABLE XVII)
11/ Fire proof, unless pump can be shut off or cannot pressurize oil circuit at windmilling speeds, and filter or heat exchanger cannot discharge a hazardous quantity of oil.

Abbreviations:
FP (fire proof)
FR (fire resistant)

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TABLE XVII. Qualification test matrix: component-specific tests.

(Referenced by: 5.6, TABLE XVI NOTE 10/)

Component Test Requirement and Method of Compliance – Component-Specific Tests *(Paragraph 5.6)*			Component Category/Classification (see H.2.7)																			
				Electronic Unit			Fuel Handling				Oil Handling			Sensors		Actuators						
Test Requirement	Para.	Method or Source	System Level (SL)	Exciter & Ignitor	Analog or Digital	ECU with Software	Main Pump	Boost Pump	FMU/HMU	Filter	Pump/Filter	Tank	Heat Exchanger	Simple	Complex	Electro-Hydraulic	Electro-Mechanical	Electro-Pneumatic	Electrical Harness	Alternator or Starter	Fuel/Oil Pipes, Fittings, Hoses
Oil Reservoir	5.6.1a/b/c	This Standard										X										
Alternator	5.6.2																			X[4]		
Heat Exchanger	5.6.3												X									
PMA-Induced Damage	5.6.4.1				X[1]	X[1]																
Electrical Loads Analysis[5]	5.6.4.2	This Standard + MIL-HDBK-704, LDC101		X	X	X			X						X	X	X	X				
Overheat	5.6.4.3	This Standard + FAA AC33.28	X			X																
Power Compatibility	5.6.4.4	This Standard + MIL-HDBK-704	X	X[2]	X	X																
Short Circuit Protection	5.6.4.5	This Standard			X	X																
Data Bus Spec Compliance	5.6.4.6	MIL-STD-1553 + ARINC 429			X	X																
Crystal Oscillator Temp. Performance	5.6.4.7	This Standard			X[3]	X[3]																
Fuel Pump: NPSP	5.6.5.1	SAE ARP4024 SAE ARP492					X	X														

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TABLE XVII. Qualification test matrix: component-specific tests – continued.

(Referenced by: 5.6, TABLE XVI NOTE 10/)

Component Test Requirement and Method of Compliance – Component-Specific Tests *(Paragraph 5.6)*			Component Category/Classification (see H.2.7)																			
				Electronic Unit			Fuel Handling				Oil Handling			Sensors		Actuators						
Test Requirement	Para.	Method or Source	System Level (SL)	Exciter & Ignitor	Analog or Digital	ECU with Software	Main Pump	Boost Pump	FMU/HMU	Filter	Pump/Filter	Tank	Heat Exchanger	Simple	Complex	Electro-Hydraulic	Electro-Mechanical	Electro-Pneumatic	Electrical Harness	Alternator or Starter	Fuel/Oil Pipes, Fittings, Hoses
Fuel Pump: Self Priming	5.6.5.2	This Standard					X	X													
Fuel Pump: Bubble Ingestion	5.6.5.3	SAE ARP4028					X	X													
Fuel Pump: Vapor/Liquid (V/L)	5.6.5.4	This Standard					X	X													
Ignition System: Carbon Deposits	5.6.6.1	This Standard		X																	
Ignition System: Water Ingestion	5.6.6.2			X																	
Fuel Filter Capacity/Efficiency	5.6.7	This Standard ARP1827								X											
System/Software Verif. & Valid.	5.8.1	RTCA/DO-178/254 + AMACC 15.0	X		X	X															
Common-Mode Multi Signal Fail	5.8.2	This Standard				X															
Complex Power Interrupts	5.8.3					X															
Torsional Stability Analysis	5.8.4	This Standard	X			X															
Fault Injection	5.8.5	This Standard			X	X															

NOTES: 1/ If powered by alternator
2/ Not applicable to ignitors
3/ If containing a crystal oscillator
4/ Not applicable to starters (see TABLE XVI)
5/ Applicable only to components with a direct connection to the airframe electrical power bus

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TABLE XVIII. Electromagnetic environmental effects test requirements.

(Referenced by: 5.7, 5.7.1)

Para.	Test	Description[1]	Standard Suite[2]
5.7.1	**Electromagnetic Interference**		
	CE101	Conducted Emissions, Power Leads, 30 Hz to 10 kHz (Figure CE101-4, Curve #2)	**X**
	CE102	Conducted Emissions, Power Leads, 10 kHz to 10 MHz (Figure CE102-1, Basic Curve)	**X**
	CS101	Conducted Susceptibility, Power Leads, 30 Hz to 150 kHz (Figure CS101-1, Curve #2)	**X**
	CS114	Conducted Susceptibility, Bulk Cable Injection, 10 kHz to 200 MHz (Figure CS114-1, Curve #5).	**X**
	CS115	Conducted Susceptibility, Bulk Cable Injection, Impulse Excitation	**X**
	CS116	Conducted Susceptibility, Damped Sinusoidal Transients, Cables and Power Leads, 10 kHz to 100 MHz (calibration limit per Figure CS116-2, test current limit increase to 150% of Figure CS116-2, see H.2.6)	
	RE101	Radiated Emissions, Magnetic Field, 30 Hz to 100 kHz (Figure RE101-2, Navy, for emissions limit)	
	RE102	Radiated Emissions, Electric Field, 10 kHz to 18 GHz (Figure 102-3, for helicopters, 10KHz ~ 18 GHz).	**X**
	RS101	Radiated Susceptibility, Magnetic Field, 30 Hz to 100 kHz (Figure RS101-2, Army)	
	RS103	Radiated Susceptibility, Electric Field, 14 kHz to 40 GHz, as modified by 5.7.1 of this document	**X**
5.7.2	**Electromagnetic Pulse**		
5.7.3	**Lightning**		
5.7.4	**Personnel-Borne Electrostatic Discharge**		
5.7.5	**Electrical Bonding Analysis**		
5.7.6	**High-Altitude Electromagnetic Pulse**		
5.7.7	**Initial Nuclear Radiation**		
5.7.8	**High-Power Microwave**		

NOTES: 1/ All tests are to use the defined Army or Army Aircraft limits, unless stated otherwise.

2/ All tests in this table are required for component qualification. Those designated with an 'X' are the standard suite of tests to be performed for PFR.

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TABLE XIXa. Standard world-wide electromagnetic RF environment.

(Referenced by: 5.7.1, TABLE XIXb NOTE 1/)

Modulation Parameters – External to Aircraft			
Frequency Range (MHz)	**Modulation Type**	**Field Strength (V/m RMS)**	**Sample Frequencies**
0.014 – 1.99	CW, AM	200	Continuous sweep
2 – 19.9	CW, AM	↓	↓
20 – 149.9	CW, AM, FM		
150 – 249.9	AM, FM		
250 – 499.9	AM, FM		
500 – 999.9	AM, FM		
1000 – 1999.9	AM, FM		
2000 – 3999.9	AM, FM		
4000 – 7999.9	AM, FM		
8000 – 9999.9	AM, FM		
10,000 – 40,000	CW, FM	200	Continuous sweep

CW: Continuous Wave
AM: Amplitude Modulation. Modulated by 1000 Hz tone; 50% modulation
FM: Frequency Modulation
Below 1 GHz, use a 20 kHz deviation, modulated by 1 kHz tone
Above 1 GHz, use a 1 MHz deviation, modulated by a 10 kHz square wave

NOTES: 1/ Continuous sweep can be achieved using step scanning defined in Table III of MIL-STD-461.

2/ The field strength for AM modulation is defined as the average field of the modulated signal (instead of the peak field as defined in MIL-STD-461). The averaging usually performed by a field probe is acceptable for this purpose. Do not apply a special correction intended for limiting the peak of AM signals.

3/ When pulse amplitudes in TABLE XIXb are limited to 200 V/m peak RMS, the pulse modulation should be tested as an additional modulation type here, using the maximum pulse width and average PRF (pulse repetition frequency) prescribed for each frequency band.

4/ Test up to 6 dB above the spec limit, up to the capabilities of the lab. It is not the intention to damage laboratory equipment. This helps to characterize margin and uncover the most likely weaknesses which could be encountered in the field. The pass/fail criteria for meeting contract and airworthiness requirements remains the field strength listed above.

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TABLE XIXb. Pulse modulation parameters.

(Referenced by: 5.7.1, TABLE XIXa NOTE 3/)

Pulse Modulation Parameters – External to Aircraft				
Frequency Range (MHz)	**PW (µsec)**	**PRF (Hz)**	**Peak Field (V/m RMS)**	**Sample Frequencies (minimum)**
2 – 24.9	833.3	300	204	24
150 – 249.9	20.0 – 25.0	200 – 310	3120	4
250 – 499.9	25.0 – 33.0	300	2830	6
500 – 999.9	33.0	100 – 300	3480	3
1000 – 1999.9	1.0 – 2.0	670 – 1000	8420	1
2000 – 3999.9	1.0	250 – 600	21270	3
4000 – 7999.9	1.0 – 2.0	250	21270	1
8000 – 9999.9	1.0	150 – 250	21270	2
10,000 – 40,000	1.0	1000	6892	6
PRF: Pulse Repetition Frequency PW: Pulse Width				

NOTES:
1/ Bench tests of frequency range 2 -- 24.9 MHz should be performed during TABLE XIXa testing using continuous (or step-scan) sweep.

2/ Unit Under Test (UUT) to be radiated at multiple angles (or using reverberation chamber).

3/ Sample frequencies which are almost the same are considered to be one sample. Example: 2.00 GHz and 2.01 GHz are considered to be only one sample.

4/ Adding additional samples to pin-point a vulnerability does not justify the deletion of other frequencies planned for testing.

5/ Bench tests should utilize more sample frequencies than minimal, when available/practical.

6/ At the time of publication:

a/ Peak field strengths above 150 MHz can only be achieved with tuned amplifiers.

. b/ A linear amplifier may create the appearance of achieving the field strength when measuring output power and average field strength, but most likely has saturated and spread the transmitted power over multiple frequencies. Consequently, it never achieves the peak field strength required at the test frequency.

c/ Projections of ever-increasing peak field strength from a linear amplifier should be independently verified by a spectrum analyzer connected to a short wire antenna (possibly connected through attenuators).

7/ For each frequency range, the aggregated average field level of the sample frequencies (using transmitters having unique PW and PRF values) is as follows:

Frequency Range (MHz)	Average (V/m RMS)	Frequency Range (MHz)	Average (V/m RMS)	Frequency Range (MHz)	Average (V/m RMS)
2 – 24.9	102	500 – 999.9	244	4000 – 7999.9	336
150 – 249.9	200	1000 – 1999.9	200	8000 – 9999.9	336
250 – 499.9	200	2000 – 3999.9	336	10,000 – 40,000	200

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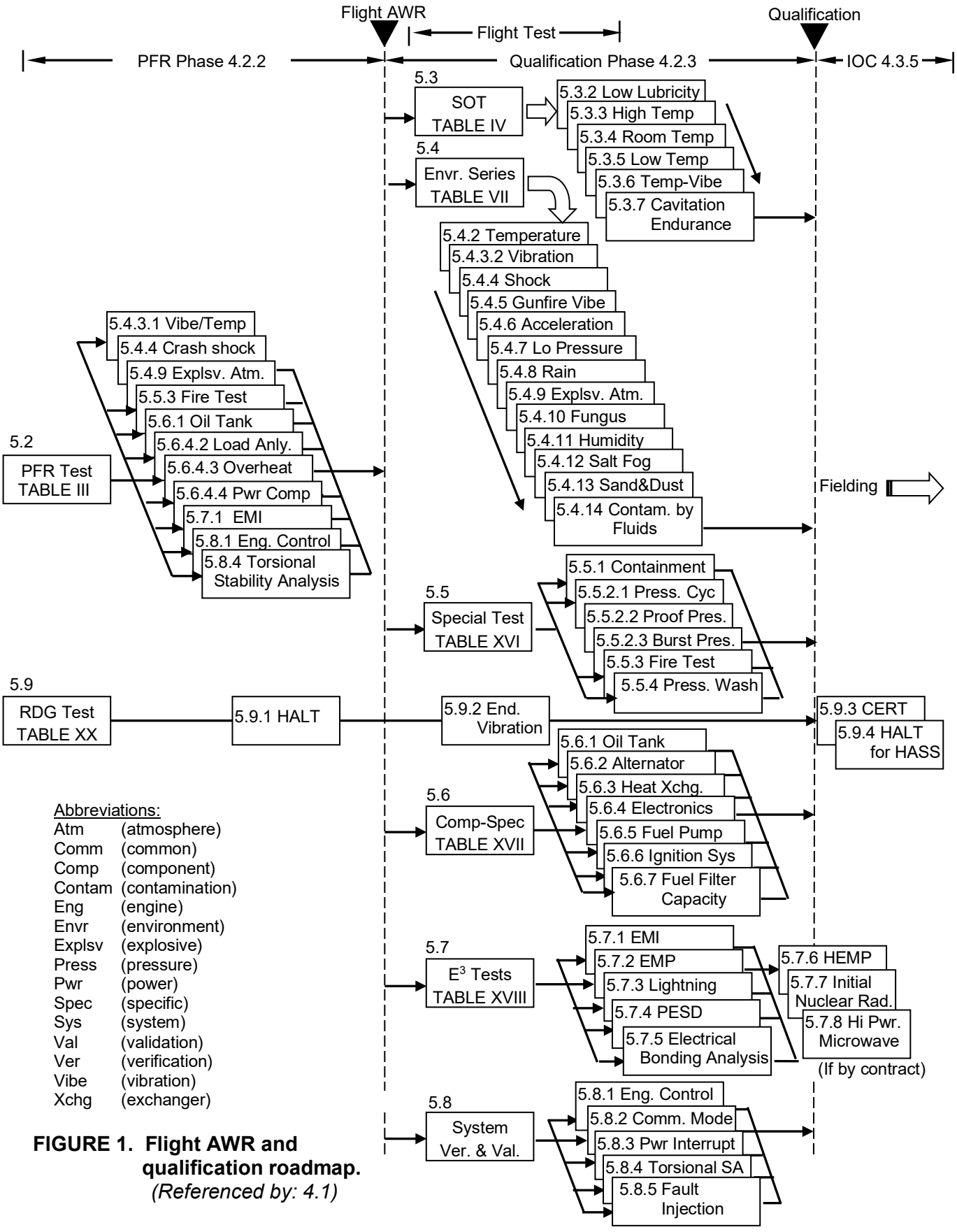
TABLE XX. Reliability and development growth test matrix.

(Referenced by: 5.9, 5.9.2)

Component Test Requirement and Method of Compliance – RDGT Matrix *(Paragraph 5.9)*			Component Category/Classification (see H.2.7)																			
				Electronic Unit			Fuel Handling				Oil Handling			Sensors		Actuators						
Test Requirement	Para.	Method or Source	System Level (SL)	Exciter	Analog or Digital	ECU with Software	Main Pump	Boost Pump	FMU/HMU	Filter	Pump	Tank	Filter	Simple	Complex	Electro-Hydraulic	Electro-Mechanical	Electro-Pneumatic	Electrical Harness	Alternator or Starter	Fuel/Oil Pipes, Fittings, Hoses
Highly Accelerated Life Testing	5.9.1	Industry Methods		X	X	X			X						X	X	X	X				
Vibration: Endurance	5.9.2	MIL-STD-810H(C1), Method 514, P1, Cat 13 or 14 & Cat 22 + Gunfire Shock, Method 519, P1, 2 or 3		X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	
Combined-Environment Reliability Test	5.9.3	This Standard		X	X	X	X	X	X		X				X	X	X	X		X		
HALT for HASS[1]	5.9.4	Industry Methods		X	X	X			X						X	X	X	X				

NOTES: 1/ Only applicable if 5.9.1 has been completed on the component

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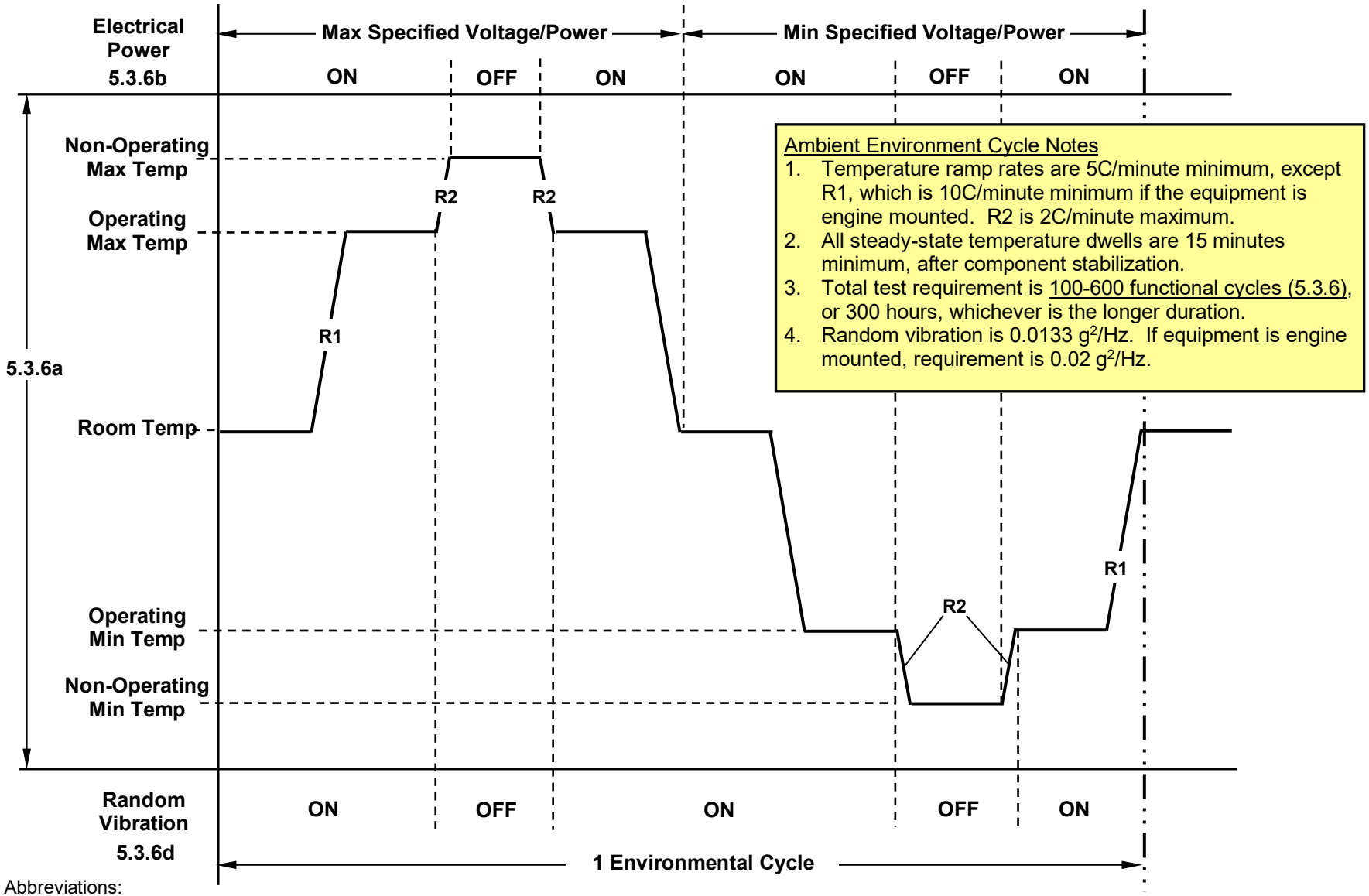


Abbreviations:

Atm	(atmosphere)
Comm	(common)
Comp	(component)
Contam	(contamination)
Eng	(engine)
Envr	(environment)
Explsv	(explosive)
Press	(pressure)
Pwr	(power)
Spec	(specific)
Sys	(system)
Val	(validation)
Ver	(verification)
Vibe	(vibration)
Xchg	(exchanger)

FIGURE 1. Flight AWR and qualification roadmap.
(Referenced by: 4.1)

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Abbreviations:
Temp (temperature)
Min (minimum)
Max (maximum)

FIGURE 2. SOT combined temperature-vibration: engine-mounted ECU example.
(Referenced by: 5.3.6)

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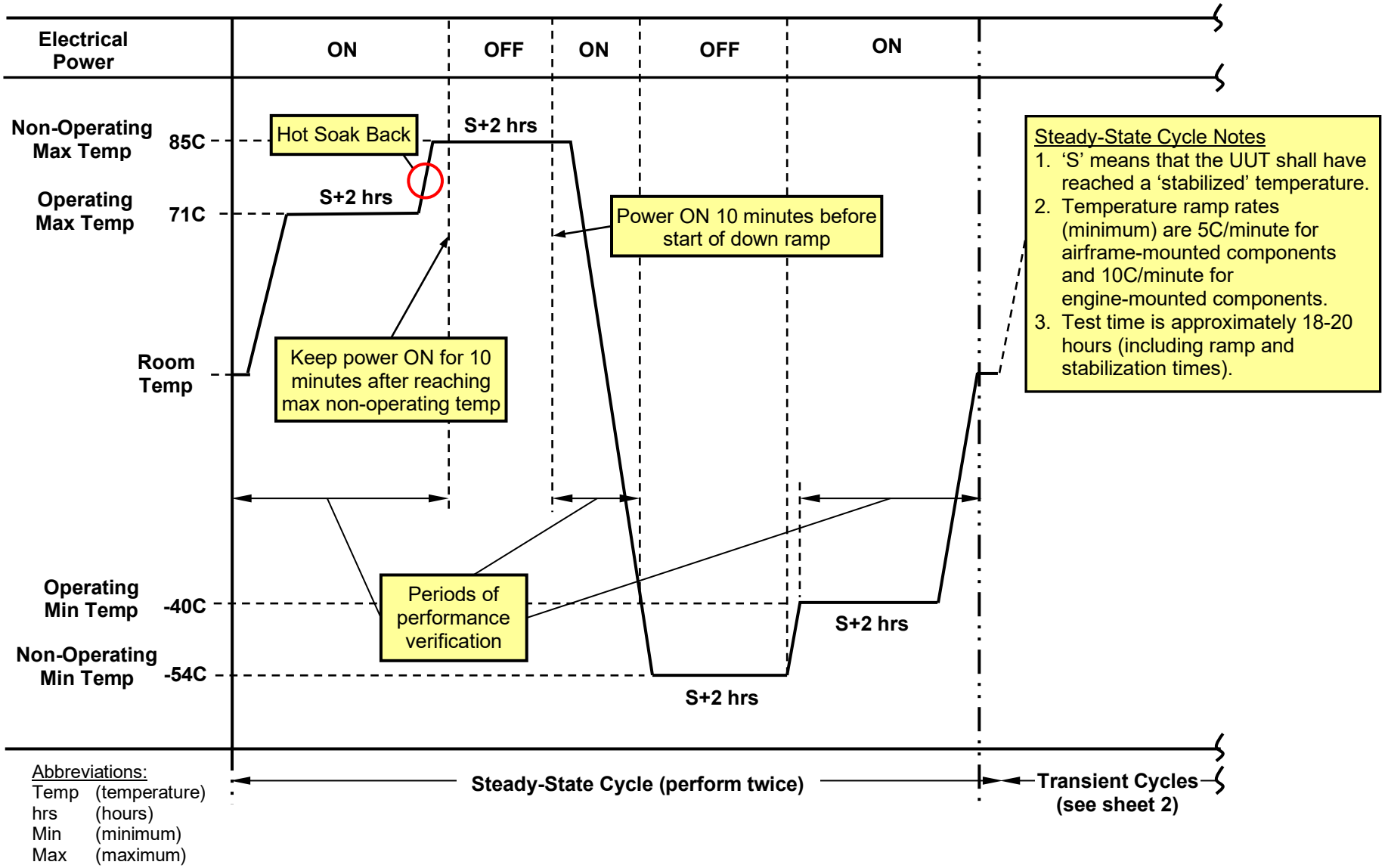
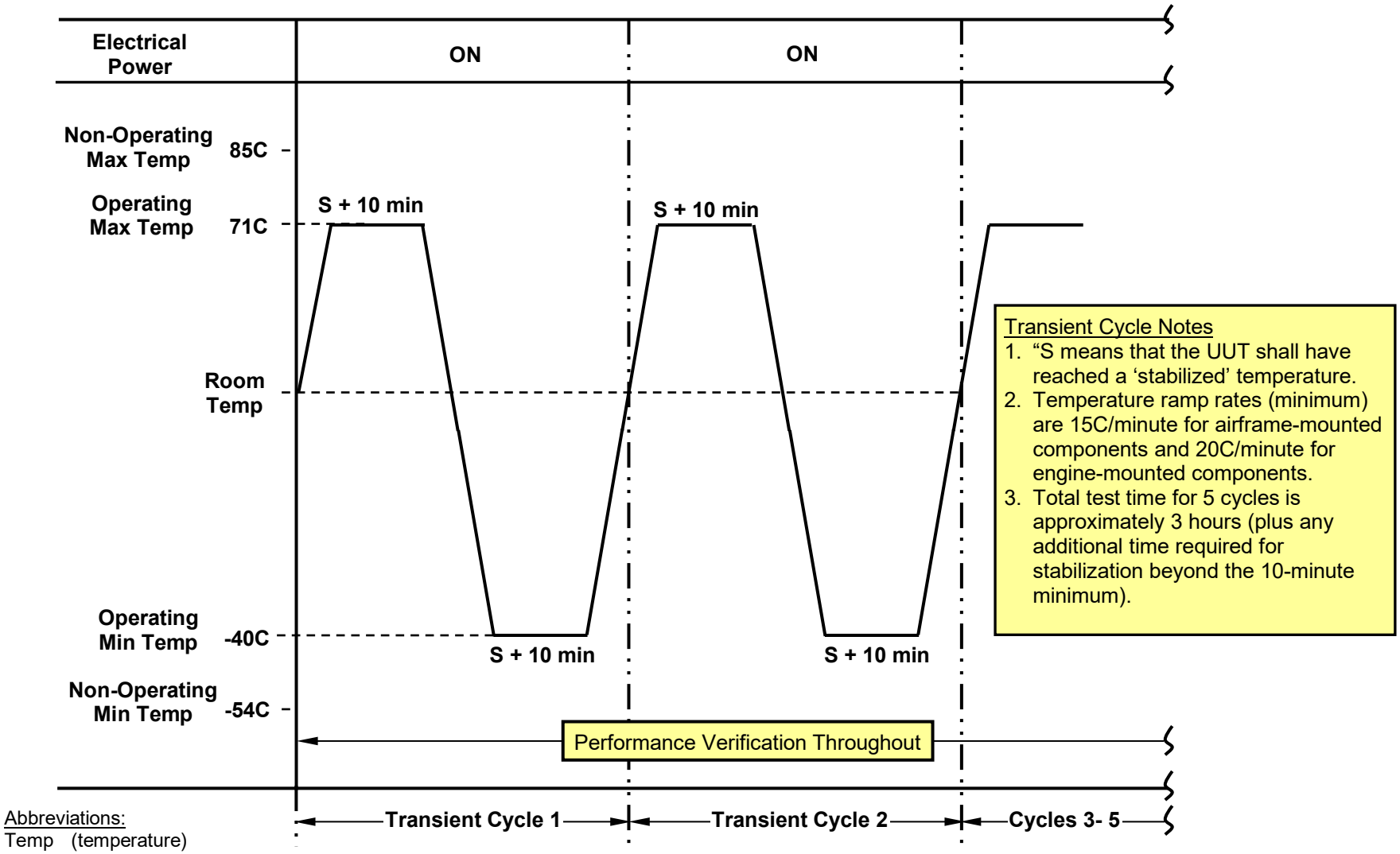


FIGURE 3. Combined high / low / shock temperature: airframe-mounted ECU example.

(Referenced by: 5.4.2 – Sheet 1 of 2)

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Abbreviations:
Temp (temperature)
Min (minimum)
min (minute)
Max (maximum)

FIGURE 3. Combined high / low / shock temperature: airframe-mounted ECU example.

(Referenced by: 5.4.2 – Sheet 2 of 2)

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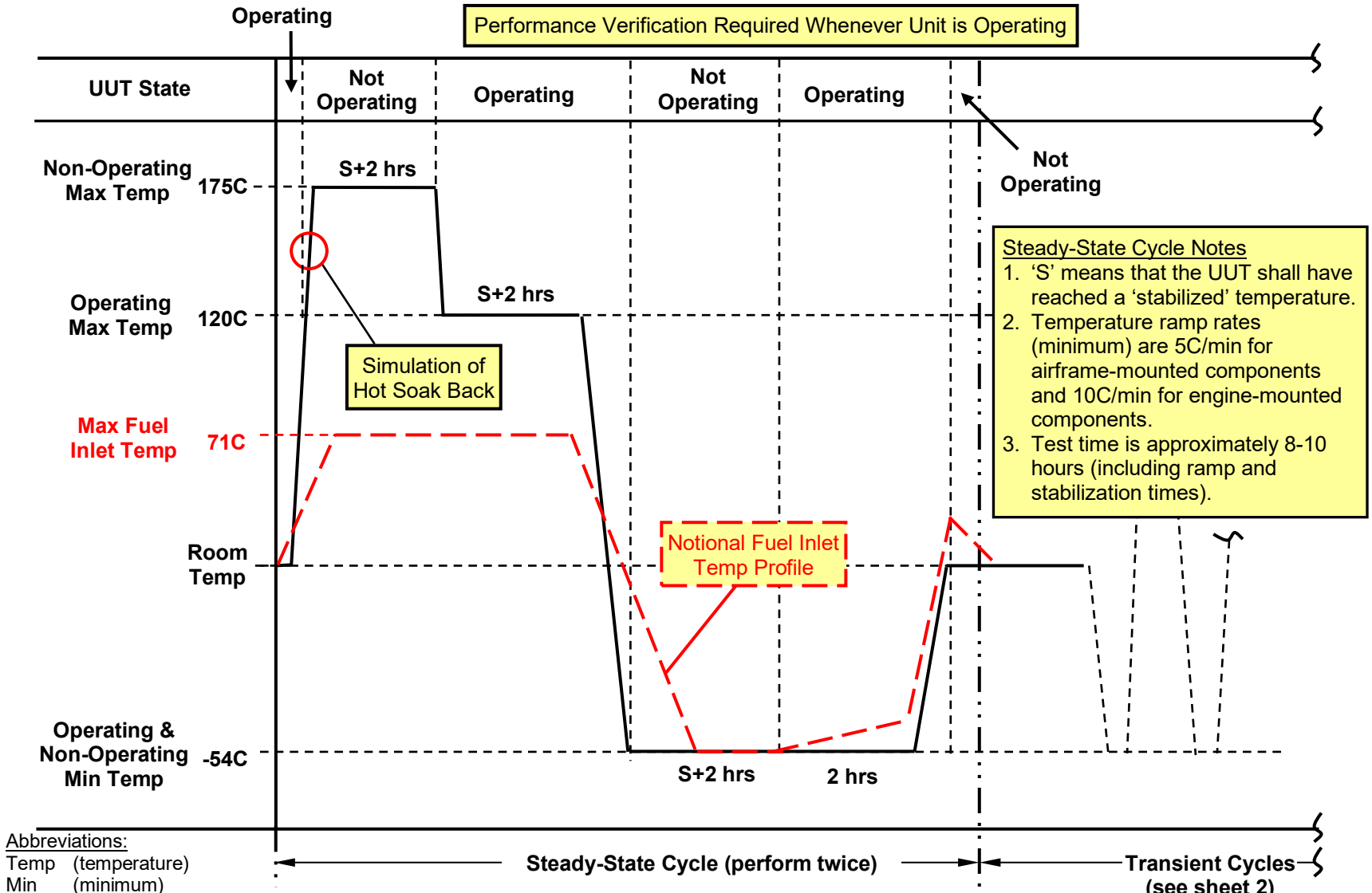


FIGURE 4. Combined high / low / shock temperature: FMU example.

(Referenced by: 5.4.2 – Sheet 1 of 2)

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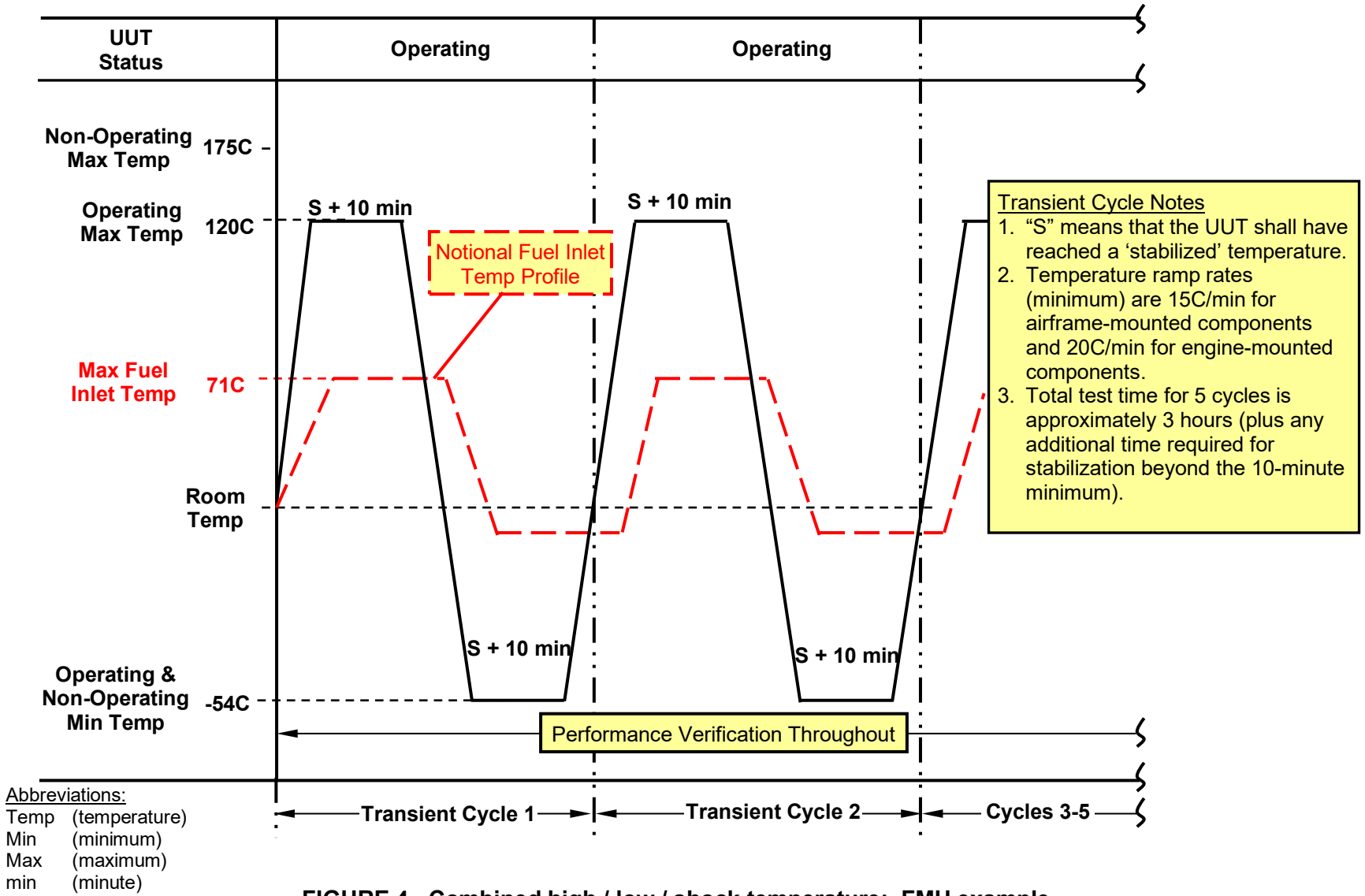


FIGURE 4. Combined high / low / shock temperature: FMU example.

(Referenced by: 5.4.2 – Sheet 2 of 2)

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SOT OPERATIONAL CYCLE for ELECTRONICS

(Referenced by: 5.3.1.1)

A.1 SCOPE

This appendix details an alternate method to define an SOT operational cycle for electronic controls using a set of 'stepped' steady-state values for each input, rather than varying them continuously as described in 5.3.1.1. The information contained herein is intended to assist the contractor in complying with the requirements defined in the referenced paragraph.

A.2 DEFINITIONS

General definitions of the terminology and nomenclature found within this appendix are provided below. Specific definition may vary with contextual usage.

headroom	The difference between a pass/fail error limit for a signal or parameter and the actual measured or predicted maximum error value.
Ng	gas generator rotor speed (also known as N1)
percent of point	percent at any given value of the parameter or signal in question
RTD	resistance temperature device (temperature measurement sensor)
set point	The value of a signal or parameter that is specified to be 'set' prior to the execution of a given test or condition.
steady-state	signal or parameter that does not vary (within a very small defined range)
T1	engine station 1 (inlet) temperature

A.3 PROCEDURE

A.3.1 General description.

The SOT cycle requires that all signals go through their complete range asynchronously during the application of the test environment so that it can be verified that there are no unexpected signal interactions (particularly in the case of mechanical components where potential interference may result in damage or jamming of parts), and that the inputs and outputs meet their required accuracies over their complete ranges of operation.

Since there are no mechanical 'interferences' possible inside an electronic unit, it may be acceptable to utilize one (or more) static set point value(s) to simplify monitoring of the signals for specified accuracy. This requires that the value chosen for the set point is the most sensitive for the circuit/signal in question. In this way, it can be reasonably assured that had the signal been cycled over its full range, it would have met its specified error limits over that range.

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A.3.2 Methodology.

In order to pick the most vulnerable signal values or set points, an analysis is required that plots both the 'as-designed' circuit accuracy from the supplier and the specified component accuracy limit (from the purchaser) versus the entire range of the signal or parameter in question. Then, the signal value chosen for the test is that value where there is the least 'headroom' between the predicted circuit accuracy and the specified limit.

A.3.2.1 Example 1.

An Ng set point is specified as 10,950 Hz with an allowed error of 0.1 % of point or 1 Hz, whichever is greater. This translates into an allowed error at the test condition of 10.95 Hz. If during the test, the reported output variation is 7.5 Hz, the Ng speed signal circuit passes. However, at 2500 Hz the allowed error is only 2.5 Hz, so a 7.5 Hz error at that signal value would be a 'fail' if the circuit had a more or less constant error. If the SOT were run with a continually cycling signal for Ng, any such errors would be uncovered.

A.3.2.2 Example 2.

Another example would be a T1 RTD, where the set point is 100 ohms and the allowed error is 1.0°C (equivalent to 0.39 ohms at 100 ohms). Without any supporting documentation, it is impossible to say if 100 ohms is the point at which the circuit exhibits the least amount of headroom between the performance of the circuit with respect to accuracy and the component limit of 1.0°C. It is entirely possible (and likely) that some other T1 value has the least headroom, but this cannot be determined without a detailed analysis.

A.3.3 Other considerations.

If the set point choices are not fully supported by analysis, then the alternative is to cycle the signals over their complete range as specified in 5.3.1.1, record the data, and then post-process it to determine pass/fail for each signal.

If an electronic control unit is itself determining pass/fail, in real time, of each of its input signals by incorporating the pass/fail limits in special ECU test software, continuously varying the inputs may be impractical to implement since pass/fail limits are static and cannot take into account a pass/fail limit which varies with a changing input signal (e.g., percent of point). In this case, the only feasible approach is to use one or more steady-state set points for each signal rather than a continuously varying signal.

In this approach, several steady-state test set conditions, each with their own pass/fail limits, are defined. Each test set condition would then be run for 30 hours with the appropriate pass/fail limits uploaded into the test software in the ECU. After 30 hours, another steady-state test condition would be set and the commensurate pass/fail limits loaded into the ECU's software. In this way, each signal or parameter would have 10 values tested, covering the complete range of operation, and they would be monitored in real time without resorting to a justification analysis or post-processing to determine pass/fail.

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SAMPLE COMMON-MODE TEST MATRICES

(Referenced by: 5.8.2, APPENDIX F)

B.1 SCOPE

This appendix details a methodology to create a set of engine control system test cases to evaluate engine/control system response to common-mode failures of input and output signals. This methodology can be applied to any level of control system complexity. Two levels of system complexity are analyzed in the figures and tables which follow, in order to illustrate the general application of the process. The information contained herein is intended to assist the contractor in complying with the requirements defined in the referenced paragraph.

B.2 DEFINITIONS

General definitions of the terminology and nomenclature found within this appendix are provided below. Specific definition may vary with contextual usage.

A/B	designator for channel ‘A’ and channel ‘B’
AI	anti-ice
Alt	permanent magnet alternator
CDP	compressor discharge pressure (also known as P3)
CiC	channel in control
Exc	excitation
Fdbk	feedback
FFIB	fuel filter impending bypass
Impend	impending
J	electrical connector ‘receptacle’ designator
LFP	low fuel pressure
LOP	low oil pressure
MGT	measured gas temperature at the power turbine (also known as PTIT)
N1	gas generator rotor speed (also known as Ng)
N2	power turbine rotor speed (also known as Np)
O/S	overspeed (of the power turbine)
OFIB	oil filter impending bypass
P	electrical connector ‘plug’ designator
P1	engine station 1 (inlet) pressure
P_{OIL}	oil pressure
Pwr	power
Q	engine torque

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Rly	relay
T1	engine station 1 (inlet) temperature
T_{OIL}	oil temperature
VG	variable geometry
WF	fuel mass flow

B.3 PROCEDURE

B.3.1 Types of common-mode failures.

Common-mode failures, as defined herein, are one of two types:

1) An electrical connector backing off due to vibration
2) Ballistic sever of a harness branch

In either of these two failure types, groups of signals are lost simultaneously. Simultaneous failures are not normally evaluated during fault detection and accommodation testing because of the virtually limitless number of possible combinations; signal failures and their effects are therefore evaluated one at a time. The two failure types defined above, however, are entirely plausible and need to be evaluated via test since fault accommodation logic is not easily analyzed when signals do not fail sequentially.

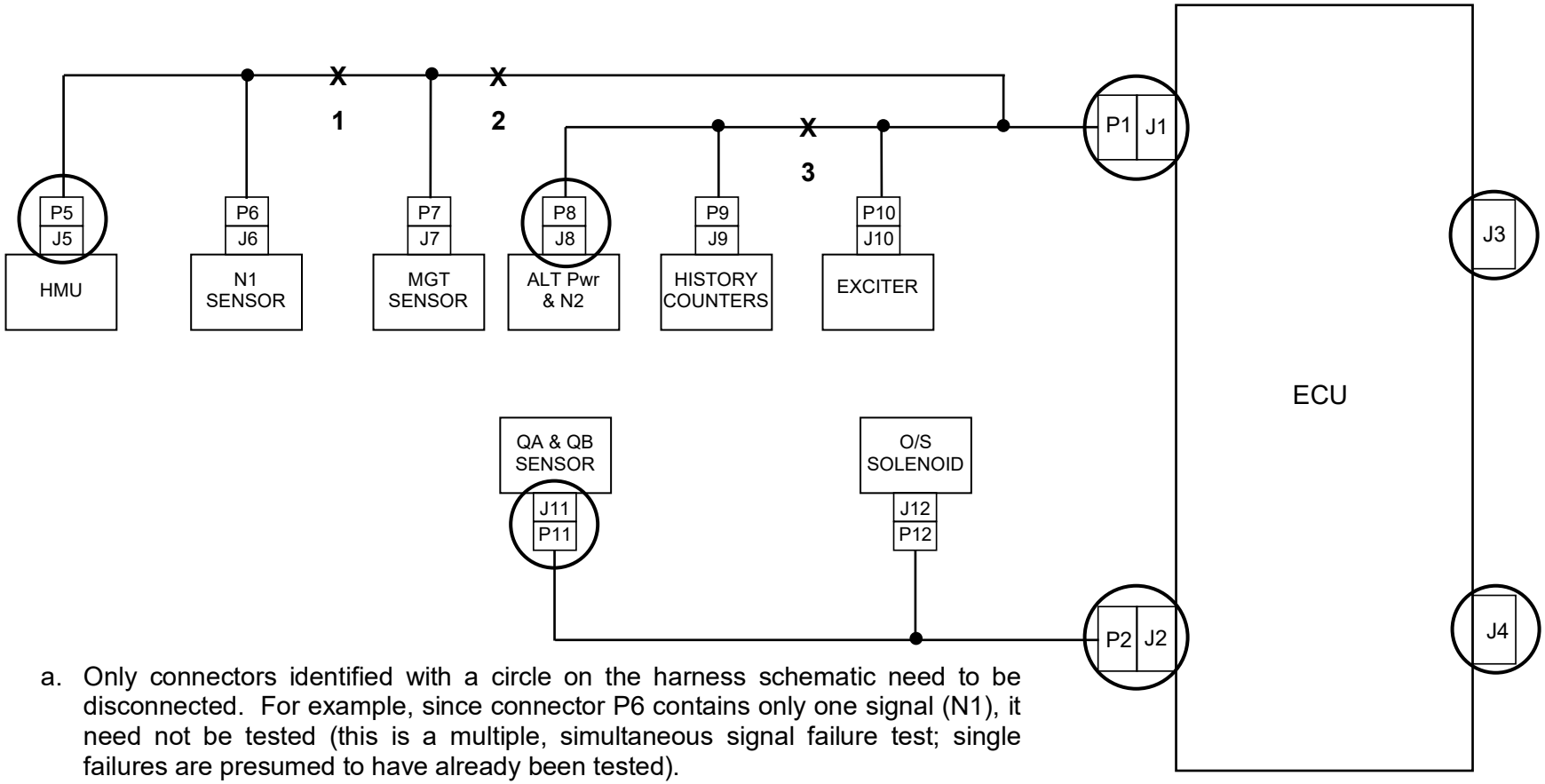
B.3.2 Methodology.

The methodology to be followed is to first develop an accurate schematic representation of the control system, showing all physical branches and connections to and from each major component of the engine control system (see FIGURE B-1 and FIGURE B-2).

Once this is completed, unique combinations of two or more signals are determined by inspection of the schematic and marked as shown in the figures. Note that if the same combination of signals exists in more than one location (because of a bulkhead pass through, for example), only one location needs to be tested since the same system response is expected no matter where the disconnect or sever occurs.

Once the locations for harness severs and connector back offs has been identified, a test case matrix (see TABLE B-I and TABLE B-II) can be developed that lists all of the test cases along with the specific signals being ‘failed’ during the execution of each of the test cases. This information is utilized as an aid in designing the test apparatus needed to effect the simultaneous ‘disconnect’ of the noted signal combinations. Note that ‘simultaneous’ here means all of the signals should be disconnected within a 50-millisecond (preferably less) window from the first signal to the last signal in the group.

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a. Only connectors identified with a circle on the harness schematic need to be disconnected. For example, since connector P6 contains only one signal (N1), it need not be tested (this is a multiple, simultaneous signal failure test; single failures are presumed to have already been tested).

b. “X” labels with numbers on the harness schematic indicate locations where a harness sever is to be simulated. These numbers match those in the table following in the “Disconnect or Sever” column.

FIGURE B-1. Low complexity single-channel system: electrical harness configuration and branching.

(Referenced by: B.3.2)

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TABLE B-I. Low complexity single-channel system: example test case matrix.

(Referenced by: B.3.2)

Test Case	Disconnect or Sever[1]	N1	History Counters	QA & QB	MGT	N2	Alt Pwr	WF Fdbk	O/S Solenoid	WF Effector
a	P5[2]							X		X
b	1	X						X		X
c	2	X			X			X		X
d	P8					X	X			
e	3		X			X	X			
f	P11			X						
g	P1	X	X		X	X	X	X		X
h	P2			X					X	
i	J3	All signals associated with J3								
j	J4	All signals associated with J4								

NOTES:

1/ “Disconnect or Sever” refers to the specific connector to disconnect, or the signal combination to be severed.

2/ Highlighted rows indicate that the connector can be disconnected, or the equivalent signals associated with that connector severed.

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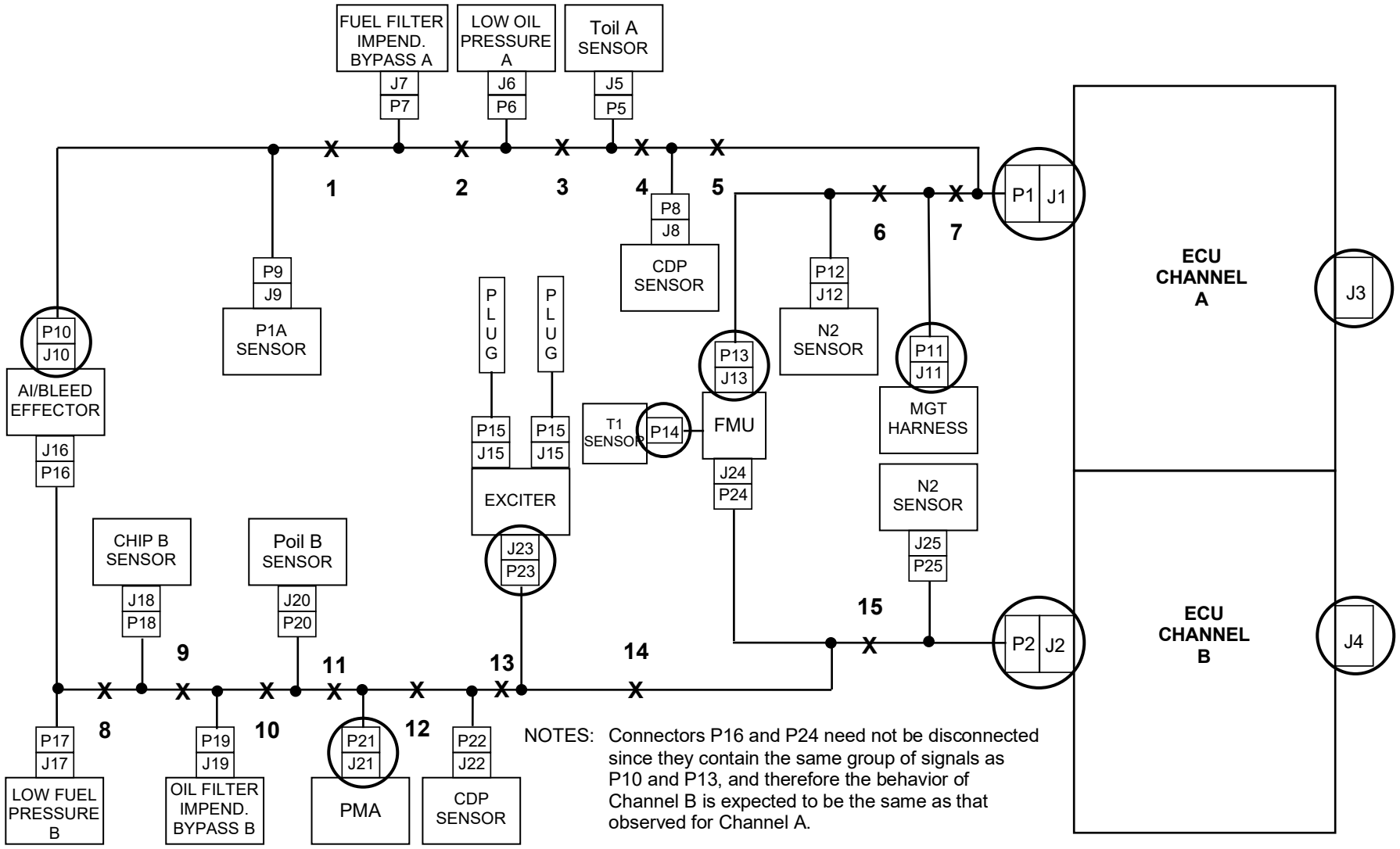


FIGURE B-2. Medium complexity dual-channel system: electrical harness configuration and branching.
(Referenced by: B.3.2)

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TABLE B-II. Medium complexity dual-channel system: example test case matrix.

(Referenced by: B.3.2)

Test Case	CIC	Disc. or Sever	N2	P1 A	T1	CDP	MGT	N1	PMA	WF Fdbk	VG Fdbk	AI Fdbk	Chips B	Poil B	Toil A	OFIB B	LOP A	LFP B	FFIB A	O/S Sol	Exc Rly	WF	VG	AI
a	A	P10										X												X
b		1		X								X												X
c		2		X								X							X					X
d		3		X								X					X		X					X
e		4		X								X			X		X		X					X
f		5		X		X						X			X		X		X					X
g		P14			A/B																			
h		P13			X			X		X	X									X		X	X	
i		6	X		X			X		X	X									X		X	X	
j		P11					A/B																	
k		7	X		X		A/B	X		X	X									X		X	X	
l		P1	X	X	A	X	A/B	X		X	X	X			X		X		X	X		X	X	X
m		J3	All signals associated with J3																					
n		P2	X		X	X		X	A/B	X	X	X	X	X		X		X		X	A/B	X	X	X
o		J4	All signals associated with J4																					
a	B	8										X						X						X
b		9										X	X					X						X
c		10										X	X			X		X						X
d		11										X	X	X		X		X						X
e		P21							A/B															
f		12							A/B			X	X	X		X		X						X
g		13				X			A/B			X	X	X		X		X						X
h		P23																			A/B			
i		14				X			A/B			X	X	X		X		X			A/B			X
j		15			X	X		X	A/B	X	X	X	X	X		X		X		X	A/B	X	X	X
k		P2	X		X	X		X	A/B	X	X	X	X	X		X		X		X	A/B	X	X	X
l		J4	All signals associated with J4																					
m		P1	X	X	A	X	A/B	X		X	X	X			X		X		X	X		X	X	X
n		J3	All signals associated with J3																					

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SAMPLE COMPLEX POWER INTERRUPT TEST MATRICES

(Referenced by: 5.8.3)

C.1 SCOPE

This appendix provides examples of engine control system test case sets designed to evaluate engine/control system response to airframe power bus interrupts of varying durations and sequences. Two types of control system architectures are shown to illustrate the level of detail necessary to meet the intent of 5.8.3. The information contained herein is intended to assist the contractor in complying with the requirements defined in the referenced paragraph.

C.2 DEFINITIONS

General definitions of the terminology and nomenclature found within this appendix are provided below. Specific definition may vary with contextual usage.

A	designator for channel 'A'
AF	airframe (referring to power source)
all power sources	Means that the test is conducted with the ECU channels powered by all of their sources, including the airframe and the dedicated alternator (if there is one).
asymmetric	Indicates that the 'control' portion of each ECU channel is different in some material way (logic, hardware, etc.).
B	designator for channel 'B'
CiC	channel-in-control
ECU operating mode	Refers to which channel is in control at the beginning of a test case.
expected outcome/effect	Defines the specific detailed behavior of the channel when recovering from an interrupt; to be determined by the control supplier.
interrupt source	Refers to which of the airframe power sources is to be interrupted; for example, AFA is the power to Channel A and AFB is the power to Channel B.
Ng	gas generator rotor speed (also known as N1)
primary and secondary channels	Generally, refers to Channels A and B, particularly in the case of an asymmetric system. However, in some fully symmetric designs, channels may alternate roles upon power up. In this case, the term Primary and Secondary refers to which channel is in control at the beginning of a test case, and which channel is in standby mode.
purpose	Indicates the overall grouping of the various test cases.
symmetric	Indicates that the 'control' portion of each ECU channel is identical/redundant, even if some of the non-control features are different/simplex (monitoring functions, for example).

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C.3 PROCEDURE

C.3.1 Objective.

Demonstration, via test, that the engine control system can tolerate and safely recover from single, multiple, and overlapping airframe power interrupts of varying durations when:

1) applied to each channel, and both channels simultaneously, with channels in both healthy and failed conditions (assuming a failed dedicated alternator).
2) applied during the engine start regime, prior to alternator power being available.

C.3.2 Methodology.

A test case matrix of engine operating conditions, control system operating states, and interrupt timing and duration is prepared by the contractor (assisted by the Airworthiness Authority if needed). For each test case, the expected control system response and recovery is described in detail, if known.

Sample test case matrices for a symmetric dual-channel FADEC system (FIGURE C-1) and an asymmetric dual-channel FADEC system (FIGURE C-2) are illustrated in TABLE C-I and TABLE C-II, respectively.

Note that when a given channel is designated ‘Primary’ for the first test case in a group of cases, then that designation refers to that same physical channel (A or B) for all subsequent test cases in that group. For example, in the symmetric system TABLE C-I, if Channel B is in control at the start of the first test case, it is referred to as the ‘Primary’ channel. Therefore, for the next test case, the ‘Secondary’ (Channel A) is in control of the engine, possibly requiring an engine shut down and restart to force the control system to alternate channels.

The 20% and 40% Ng speed points in the matrices, as mentioned in the notes below the tables, are suggested speed thresholds at which to conduct testing. The engine/control system supplier should establish these values based on speed-dependent start sequences and events that are defined in the control logic, while also taking into account the Ng speed at which the alternator is designed to be fully capable of powering the FADEC system without requiring any supplementary airframe power.

Typically, there is no need to test symmetric control systems both ways. In other words, not all of the tests in Table C-I with Channel A starting out as the channel-in-control (CiC) are repeated with Channel B as the initial CIC. This is because the hardware and software are typically identical between the two channels. However, if there is reason to believe that the response won't be identical for each channel, testing with each channel in control may be necessary to assure a safe, predictable response.

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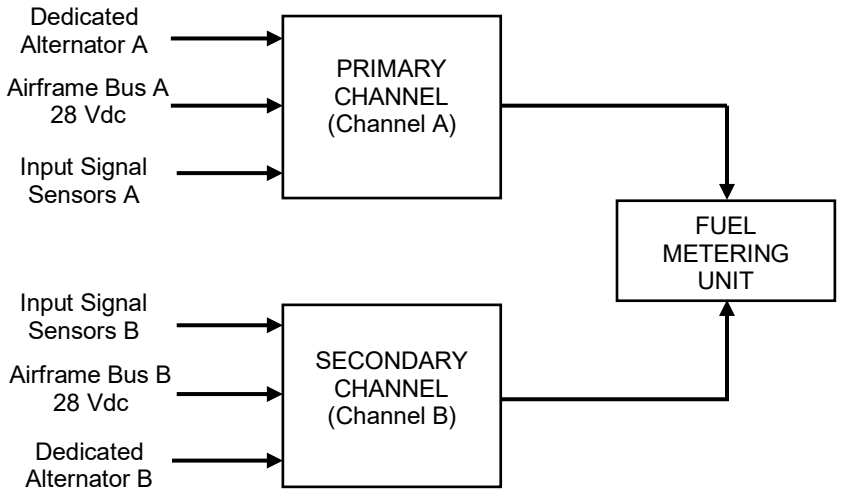


FIGURE C-1. Symmetric dual-channel FADEC system: block diagram.

(Referenced by: C.3.2)

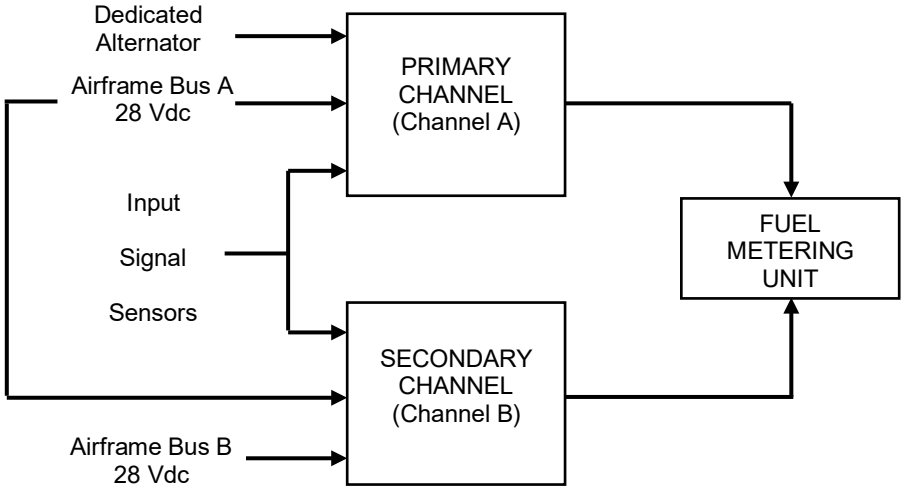


FIGURE C-2. Asymmetric dual-channel FADEC system: block diagram.

(Referenced by: C.3.2)

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TABLE C-I. Symmetric dual-channel FADEC system: test case matrix.

(Referenced by: C.3.2)

ECU Operating Mode (Engine at Cruise)	Interrupt Source	Purpose (Normal System Operation)	Expected Outcome/Effect	Milliseconds													Seconds		
				2	4	6	8	10	12	15	20	30	50	100	200	500	1	2	7
All Power Sources																			
Primary Channel w/operational Secondary Channel	AFA & AFB	Verifies that Primary/Secondary channels are immune to AF interrupts	Primary Channel: TBD Secondary Channel: TBD	X				X			X		X		X		X		X
Secondary Channel w/operational Primary Channel	AFA & AFB	Verifies that Primary/Secondary channels are immune to AF interrupts	Primary Channel: TBD Secondary Channel: TBD			X				X		X		X		X		X	

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TABLE C-I. Symmetric dual-channel FADEC system: test case matrix – Continued.

ECU Operating Mode (Engine at Cruise)	Interrupt Source	Purpose (Failed Alternator)	Expected Outcome/Effect	Milliseconds													Seconds		
				2	4	6	8	10	12	15	20	30	50	100	200	500	1	2	7
No Alt Power																			
Primary Channel w/operational Secondary Channel	AFA	Verifies that system transfers to Secondary and stays there	Primary Channel: TBD Secondary Channel: TBD	X				X			X			X		X		X	
	AFA & AFB	Verifies that system fails fixed then recovers into Primary	Primary Channel: TBD Secondary Channel: TBD		X				X				X		X		X		X
Primary Channel w/hard faulted Secondary Channel	AFA	Verifies that system fails fixed and then recovers into Primary	Primary Channel: TBD Secondary Channel: TBD	X			X					X		X		X		X	
	AFA & AFB	Verifies that system fails fixed and then recovers into Primary	Primary Channel: TBD Secondary Channel: TBD			X				X			X		X		X		X
Secondary Channel w/unhealthy Primary Channel	AFB	Verifies system transfers to Primary then recovers into Secondary	Primary Channel: TBD Secondary Channel: TBD		X				X			X		X		X		X	
	AFA & AFB	Verifies system fails fixed and then recovers into Secondary	Primary Channel: TBD Secondary Channel: TBD	X			X			X			X		X		X		X

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TABLE C-I. Symmetric dual-channel FADEC system: test case matrix – Continued.

ECU Operating Mode (Engine Starts)	Interrupt Source	Purpose (No Engine Start or Restart Anomalies)	Expected Outcome/Effect	Milliseconds													Seconds		
				2	4	6	8	10	12	15	20	30	50	100	200	500	1	2	7
ALT Available																			
Engine Start in Primary Channel w/operational Secondary Channel	AFA @ 20% Ng	Verifies that system transfers to Secondary and engine idle is reached	Primary Channel: TBD Secondary Channel: TBD	X				X			X					X		X	X
	AFA @ 40% Ng	Verifies that system transfers to Secondary and engine idle is reached	Primary Channel: TBD Secondary Channel: TBD			X				X			X		X		X	X	X
Engine Start in Primary Channel w/hard faulted Secondary Channel	AFA @ 20% Ng	Verifies that system aborts the start attempt	Primary Channel: TBD Secondary Channel: TBD				X				X			X	X		X	X	X
	AFA @ 40% Ng	Verifies that system aborts the start attempt	Primary Channel: TBD Secondary Channel: TBD		X				X			X				X		X	X
Engine Start in Secondary Channel w/unhealthy Primary Channel	AFB @ 20% Ng	Verifies that system transfers to Primary, then recovers into Secondary then reaches idle	Primary Channel: TBD Secondary Channel: TBD	X				X			X					X		X	X
	AFB @ 40% Ng	Verifies that system transfers to Primary, then recovers into Secondary then reaches idle	Primary Channel: TBD Secondary Channel: TBD			X				X			X		X		X	X	X

NOTES: 20% and 40% Ng speeds are notional only; actual values to be determined (TBD) based on speed-dependent start sequences and events, and upon the Ng speed at which the alternator is sufficient to power the FADEC system without requiring any airframe DC power.

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TABLE C-I. Symmetric dual-channel FADEC system: test case matrix – Continued.

ECU Operating Mode (Engine Off)	Interrupt Source	Purpose (Overlapping Interrupts)	Expected Outcome/Effect	Milliseconds													Seconds		
				2	4	6	8	10	12	15	20	30	50	100	200	500	1	2	7
No ALT Power																			
Primary Channel w/operational Secondary Channel	AFA then AFB	Verifies system transfers to Secondary, fails fixed, then recovers into Primary	Primary Channel: TBD			X					X			X		X	X	X	X
			Secondary Channel: TBD																
	AFB then AFA	Verifies system fails fixed then recovers into Secondary mode	Primary Channel: TBD	X				X				X			X		X	X	X
			Secondary Channel: TBD																
Secondary Channel w/operational Primary Channel	AFA then AFB	Verifies system fails fixed then recovers into Primary mode	Primary Channel: TBD		X					X			X		X		X	X	X
			Secondary Channel: TBD																
	AFB then AFA	Verifies system transfers to Primary, fails fixed, then recovers into Secondary	Primary Channel: TBD				X		X					X		X	X	X	X
			Secondary Channel: TBD																

NOTES: Overlapping interrupts are sequential, offset by ½ of the interrupt time. For example, for 100 msec, AFA then AFB means AFA interrupt occurs at time 0, then AFB interrupt occurs at time = 50 msec.

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TABLE C-I. Symmetric dual-channel FADEC system: test case matrix – Continued.

ECU Operating Mode (Engine Off)	Interrupt Source	Purpose (Multiple Interrupts)	Expected Outcome/Effect	Milliseconds													Seconds		
				2	4	6	8	10	12	15	20	30	50	100	200	500	1	2	7
No ALT Power																			
Primary Channel w/operational Secondary Channel	AFA	Verifies that system transfers to Secondary and stays there	Primary Channel: TBD										X						
			Secondary Channel: TBD																
	AFA & AFB	Verifies that system fails fixed then recovers into Primary	Primary Channel: TBD					X											
			Secondary Channel: TBD																
Secondary Channel w/operational Primary Channel	AFB	Verifies that system transfers to Primary and stays there	Primary Channel: TBD					X											
			Secondary Channel: TBD																
	AFA & AFB	Verifies that system fails fixed then recovers into Secondary	Primary Channel: TBD										X						
			Secondary Channel: TBD																

NOTES: Multiple interrupt sequence as follows: For 10 msec: three 10 msec interrupts spaced 500 msec apart. For 50 msec: three 50 msec interrupts spaced 500 msec apart.

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TABLE C-II. Asymmetric dual-channel FADEC system: test case matrix.

(Referenced by: C.3.2)

ECU Operating Mode (Engine at Cruise)	Interrupt Source	Purpose (Normal System Operation)	Expected Outcome/Effect	Milliseconds													Seconds		
				2	4	6	8	10	12	15	20	30	50	100	200	500	1	2	7
All Power Sources																			
Primary Mode w/operational Secondary Channel	AFA	Verifies that Primary/Secondary channels are immune to single AFA interrupts	Primary Channel: TBD Secondary Channel: TBD										X						
	AFA & AFB	Verifies that Primary mode is not affected when Secondary channel recovers from an interrupt	Primary Channel: TBD Secondary Channel: TBD	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
Secondary Mode w/operational Primary Channel	AFB	Verifies that Secondary channel is immune to single AFB interrupts	Primary Channel: TBD Secondary Channel: TBD										X						
	AFA & AFB	Verifies that the system fails fixed, then properly recovers into Secondary mode	Primary Channel: TBD Secondary Channel: TBD	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X

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TABLE C-II. Asymmetric dual-channel FADEC system: test case matrix – Continued.

ECU Operating Mode (Engine at Cruise)	Interrupt Source	Purpose (Failed Alternator)	Expected Outcome/Effect	Milliseconds													Seconds		
				2	4	6	8	10	12	15	20	30	50	100	200	500	1	2	7
No Alt Power																			
Primary Mode w/operational Secondary Channel	AFA	Verifies that system transfers to Secondary mode and then recovers into Primary mode	Primary Channel: TBD Secondary Channel: TBD	X				X			X			X		X		X	X
	AFA & AFB	Verifies that system fails fixed, transfers to Secondary channel, then recovers into Primary	Primary Channel: TBD Secondary Channel: TBD		X				X				X		X		X		X
Primary Mode w/hard faulted Secondary Channel	AFA	Verifies that system transfers to fail fixed Secondary channel and then recovers into Primary	Primary Channel: TBD Secondary Channel: TBD	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
	AFA & AFB	Verifies that system fails fixed, transfers to Secondary channel, then recovers into Primary	Primary Channel: TBD Secondary Channel: TBD	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
Secondary Mode w/operational Primary Channel	AFA	Verifies Secondary channel is immune to AFA interrupts and Primary channel recovers properly	Primary Channel: TBD Secondary Channel: TBD	X				X			X			X		X		X	X
Secondary Mode w/hard faulted Primary Channel	AFA	Verifies system stays in Secondary mode while Primary channel recovers from a hard fault	Primary Channel: TBD Secondary Channel: TBD		X				X			X			X		X		X
	AFA & AFB	Verifies system fails fixed and then recovers into Secondary mode while Primary channel recovers from a hard fault	Primary Channel: TBD Secondary Channel: TBD	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X

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TABLE C-II. Asymmetric dual-channel FADEC system: test case matrix – Continued.

ECU Operating Mode (Engine at Cruise)	Interrupt Source	Purpose (Failed DC Bus)	Expected Outcome/Effect	Milliseconds													Seconds		
				2	4	6	8	10	12	15	20	30	50	100	200	500	1	2	7
No AFA Power																			
Secondary Mode w/hard faulted Primary Channel	AFB	Verifies that system fails fixed and then properly recovers into Secondary mode	Primary Channel: TBD Secondary Channel: TBD	X				X			X		X	X		X	X	X	X

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TABLE C-II. Asymmetric dual-channel FADEC system: test case matrix – Continued.

ECU Operating Mode (Engine Starts)	Interrupt Source	Purpose (No Engine Start or Restart Anomalies)	Expected Outcome/Effect	Milliseconds													Seconds		
				2	4	6	8	10	12	15	20	30	50	100	200	500	1	2	7
No AFB/ALT Power																			
Engine Start in Primary Mode w/operational Secondary Channel	AFA @ 20% Ng	Verifies that system transfers to Secondary, fails fixed, recovers into Primary and reaches idle	Primary Channel: TBD Secondary Channel: TBD	X				X			X					X		X	X
	AFA @ 40% Ng	Verifies that system transfers to Secondary, fails fixed, recovers into Primary and reaches idle	Primary Channel: TBD Secondary Channel: TBD			X				X			X		X		X	X	X
Engine Start in Primary Mode w/hard faulted Secondary Channel	AFA @ 20% Ng	Verifies that system transfers to Secondary, fails fixed, recovers into Primary and reaches idle	Primary Channel: TBD Secondary Channel: TBD				X				X			X	X		X	X	X
	AFA @ 40% Ng	Verifies that system transfers to Secondary, fails fixed, recovers into Primary and reaches idle	Primary Channel: TBD Secondary Channel: TBD		X				X			X				X		X	X
Engine Start in Secondary Mode w/operational Primary Channel	AFA @ 20% Ng	Verifies that system fails fixed, recovers into Secondary and reaches idle	Primary Channel: TBD Secondary Channel: TBD	X				X			X					X		X	X
	AFA @ 40% Ng	Verifies that system fails fixed, recovers into Secondary and reaches idle	Primary Channel: TBD Secondary Channel: TBD			X				X			X		X		X	X	X
Engine Start in Secondary Mode w/hard faulted Primary Channel	AFA @ 20% Ng	Verifies that system fails fixed, recovers into Secondary and reaches idle	Primary Channel: TBD Secondary Channel: TBD				X				X			X	X		X	X	X
	AFA @ 40% Ng	Verifies that system fails fixed, recovers into Secondary and reaches idle	Primary Channel: TBD Secondary Channel: TBD		X				X			X				X		X	X

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TABLE C-II. Asymmetric dual-channel FADEC system: test case matrix – Continued.

ECU Operating Mode (Engine Starts)	Interrupt Source	Purpose (No Engine Start or Restart Anomalies)	Expected Outcome/Effect	Milliseconds													Seconds		
				2	4	6	8	10	12	15	20	30	50	100	200	500	1	2	7
No AFA/ALT Power																			
Engine Start in Secondary Mode	AFB @ 20% Ng	Verifies that system fails fixed, recovers into Secondary and reaches idle	Primary Channel: TBD Secondary Channel: TBD	X				X			X		X	X		X	X	X	X
	AFB @ 40% Ng	Verifies that system fails fixed, recovers into Secondary and reaches idle	Primary Channel: TBD Secondary Channel: TBD		X					X		X			X	X	X	X	X

NOTES: 20% and 40% Ng speeds are notional only; actual values to be determined (TBD) based on speed-dependent start sequences and events, and upon the Ng speed at which the alternator is sufficient to power the FADEC system without requiring any airframe DC power

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TABLE C-II. Asymmetric dual-channel FADEC system: test case matrix – Continued.

ECU Operating Mode (Engine Off)	Interrupt Source	Purpose (Overlapping Interrupts)	Expected Outcome/Effect	Milliseconds													Seconds		
				2	4	6	8	10	12	15	20	30	50	100	200	500	1	2	7
No ALT Power																			
Primary Mode w/operational Secondary Channel	AFA then AFB	Verifies system transfers to Secondary mode, fails fixed, then recovers into Primary mode	Primary Channel: TBD Secondary Channel: TBD			X					X			X		X	X	X	X
	AFB then AFA	Verifies system transfers to Secondary mode, fails fixed, then recovers into Primary mode	Primary Channel: TBD Secondary Channel: TBD	X				X				X			X		X	X	X
Secondary Mode w/operational Primary Channel	AFA then AFB	Verifies system stays on Secondary channel and then recovers into Secondary mode	Primary Channel: TBD Secondary Channel: TBD		X					X			X		X		X	X	X
	AFB then AFA	Verifies system stays on Secondary channel and then recovers into Secondary mode	Primary Channel: TBD Secondary Channel: TBD				X		X					X		X	X	X	X

NOTES: Overlapping interrupts are sequential, offset by ½ of the interrupt time. For example, for 100 msec, AFA then AFB means AFA interrupt occurs at time 0, then AFB interrupt occurs at time = 50 msec.

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TABLE C-II. Asymmetric dual-channel FADEC system: test case matrix – Continued.

ECU Operating Mode (Engine Off)	Interrupt Source	Purpose (Multiple Interrupts)	Expected Outcome/Effect	Milliseconds													Seconds		
				2	4	6	8	10	12	15	20	30	50	100	200	500	1	2	7
No ALT Power																			
Primary Mode w/operational Secondary Channel	AFA	Verifies that system transfers to Secondary mode and then recovers into Primary mode	Primary Channel: TBD					X					X						
			Secondary Channel: TBD																
	AFA & AFB	Verifies that system fails fixed, transfers to Secondary channel, then recovers into Primary	Primary Channel: TBD					X					X						
			Secondary Channel: TBD																
Secondary Mode w/operational Primary Channel	AFA	Verifies Secondary channel is immune to AFA interrupts and Primary channel recovers properly	Primary Channel: TBD					X					X						
			Secondary Channel: TBD																
	AFA & AFB	Verifies that the system fails fixed, then properly recovers into Secondary mode	Primary Channel: TBD					X					X						
			Secondary Channel: TBD																

NOTES: Multiple interrupt sequence as follows: For 10 msec: three 10 msec interrupts spaced 500 msec apart. For 50 msec: three 50 msec interrupts spaced 500 msec apart.

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TYPICAL LIGHTNING TEST REQUIREMENTS for FADEC SYSTEMS

(Referenced by: 5.7.3)

D.1 SCOPE

This appendix defines test requirements that have been used in the past to qualify FADEC systems for lighting environments as applicable to H-60 and H-64 platforms. The information contained herein is intended to assist the contractor in complying with the referenced paragraph; it is not meant to supersede any specific requirements defined in a SoW, EMS, component specification, or other controlling document.

D.2 DEFINITIONS

General definitions of the terminology and nomenclature found within this appendix are provided below. Specific definition may vary with contextual usage.

FADEC	full-authority digital engine control – As used herein, it pertains to the entire engine control system, not just the electronic control unit as it is sometimes misinterpreted.
IL	current limit (Amperes)
IT	test current (Amperes)
MB	multiple burst – terminology as used in RTCA/DO-160, Section 22.
MS	multiple stroke – terminology as used in RTCA/DO-160, Section 22.
level	The voltage or current level to be used for a specific test (defined in RTCA/DO-160, Section 22).
platform	Refers to the intended aircraft in which the engine and its control system will be installed. The type of aircraft, its intended operational usage, and its construction technique will dictate the lightning qualification environment.
SS	single stroke – terminology as used in RTCA/DO-160, Section 22.
VL	voltage limit (Volts)
VT	test voltage (Volts)
WF	wave form – terminology as used in RTCA/DO-160, Section 22.

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D.3 TEST LEVELS: TYPICAL

(Referenced by: 5.7.3)

Test Type	Application	RTCA/DO-160G, Section 22
Pin Injection (NOTE 1/)	28 Vdc Power Interface	WF 3/3, Level 4 WF 4/1, Level 4 WF 5A/5A, Level 4
	Other Engine Interfaces	WF 3/3, Level 3 WF 4/1, Level 3
	Other Airframe Interfaces (the levels do not limit amplitudes for bulk cable injection)	WF 3/3, Level 3 WF 4/1, Level 3 WF 5A/5A, Level 3
Single/Multiple Stroke Upset	28V Power Wire (high-side and high-return pair)	SS/MS, WF 1, Level 4 /note 1 SS/MS, WF 3, (1, 5, 10 MHz) Level 4 (NOTE 2/) SS/MS, WF 4, Level 4 /note 1 SS/MS, WF 5A, Level 4 /note 1
	Engine Wiring Harness (includes all connectors on the FADEC and Fuel Metering Unit)	SS/MS, WF 1, Level 5 SS/MS, WF 3: 1 MHz, Level 5 5, 10 MHz, Level 4 SS/MS, WF 5A, Level 4
	Aircraft Wiring Harness (inside metallic airframe)	SS/MS, WF 1, Level 3 SS/MS, WF 3 (1, 5, 10 MHz), Level 3 SS/MS, WF 5A, Level 3
	Engine Firewall Connector (on aircraft harness on aircraft-side of firewall connector)	UH-60: Same as Engine Wiring Harness AH-64: Same as inside metallic airframe
	For MS, include quantity of at least two 24-stroke waveforms as part of the 10 tests of each waveform and polarity	
Multiple Burst Upset	28V Power Wire (high-side and high-return pair)	WF 3H (1, 5, 10 MHz), Level 4 (NOTE 3/)
	Engine Wiring Harness (includes all connectors on the FADEC and Fuel Metering Unit)	WF 3H (1, 5, 10 MHz): 900-VT / 60-IL WF 6H: 320-IT (VL as high as possible)
	Aircraft Wiring Harness (inside metallic airframe)	WF 3H, Level 3 WF 6H, Level 3
	Engine Firewall Connector (on aircraft harness on aircraft-side of firewall connector)	UH-60: Same as Engine Wiring Harness AH-64: Same as inside metallic airframe

NOTES:

1/ Pin injection is not an Army airworthiness requirement. Nevertheless, it is a recommended design and test requirement for engine control system component specifications, since it helps identify marginal interface circuitry. Additionally, it is an FAA certification test requirement.

2/ Level limited to pin injection open-circuit voltage and short-circuit current magnitudes for individual connector pins or group of pins electrically tied together inside the unit under test, per pin-injection requirements. Each high-side and power pair is tested separately. The levels do not limit the amplitudes for bulk cable injection.

3/ High-side applies to individual connector pins or group of pins electrically tied together inside the unit under test. Each high-side and power pair is tested separately.

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DISCUSSION of "OPERATING" as USED in TESTING

(Referenced by: 4.3.2)

E.1 SCOPE

This appendix elaborates on the terms "operating" and "monitored" when used in establishing requirements for certain tests specified in this standard. The information contained herein is intended to assist the contractor in complying with the requirements defined in 4.3.2.

E.2 DEFINITIONS

General definitions of the terminology and nomenclature found within this appendix are provided below. Specific definition may vary with contextual usage.

COTS	commercial off-the-shelf
CP	collective pitch - The pilot-commanded total thrust requested from the engines.
ECL	engine condition lever - The pilot-commanded state for the engine (typically off, ground idle, flight idle).
ECU	electronic control unit
functionally cycled	The UUT's inputs are varied over their complete ranges so that the UUT's outputs vary over their expected values.
headroom	The difference between a pass/fail error limit for a signal or parameter and the actual measured or predicted maximum error value.
monitored	The UUT's inputs and outputs are recorded at a rate suitable to determine expected performance.
Ng	gas generator rotor speed (also known as N1)
operational	The UUT is being operated while being subjected to the test environment.
P1	ambient pressure
P3	combustor discharge pressure (CDP)
percent of point	Percent at any given value of the parameter or signal in question.
performance	The required performance for the UUT under all environmental conditions, as defined in any number of specifications, including the ATP and/or the EMS.
PMA	permanent magnet alternator
Q	engine output torque
RTD	resistance temperature device (temperature measurement sensor)
set point	The value of a signal or parameter that is specified to be 'set' prior to the execution of a given test or condition.
steady-state	A signal or parameter that does not vary (within a very small defined range).
T1	engine station 1 (inlet) temperature
T45	turbine inlet temperature

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E.3 PROCEDURE

E.3.1 General description.

For virtually all of the environmental tests to be executed, the tests are 'operational', meaning that the UUT (unit under test) is 'functionally cycled' during the applied environment while meeting its defined performance specification. This is the standard requirement for tests in accordance with MIL-STD-810 and RTCA/DO-160. The unit should demonstrate its ability to meet specified performance requirements while subjected to environmental extremes.

E.3.1.1 Operating.

'Operational' means that the UUT is powered (if applicable) and operating as it normally would if installed. For this to occur, the software (if applicable) should be the operational program that is expected to be fielded, or at least a version that is sufficient to exercise all of the hardware in the unit. It could even be special test software that is capable of monitoring inputs and outputs. In any case, the software that is running inside the unit needs to be capable of performing I/O functions such as signal conditioning, analog-to-digital (A/D) conversion, frequency-to-digital (F/D) conversion, discrete processing, output driver control, etc. and sending the processed signals out via a serial monitoring bus of relatively high bandwidth.

E.3.1.2 Performance.

''Performance' is the allowable error for input signals and output signals. This is normally defined in two places; the hardware purchase specification created by the buyer (i.e., a purchase performance specification) and the hardware ATP (acceptance test procedure) defined by the hardware supplier. In many instances, the ATP will have tighter limits than the performance specification in order for the supplier to guarantee that all production units meet the performance specification over time.

E.3.1.3 Input signals.

All ECU inputs are stimulated and all outputs loaded/driven. The inputs and the outputs are then monitored and recorded for exceedance of their allowable limits IAW the specified (most likely ATP) performance specification. The test condition might be a single or several predefined sets of fixed input signal values driving the appropriate loads. The ECU transmits its computed values for the inputs and these are compared against the test input set points. In the example of a thermocouple, one typically would use a COTS T/C simulator, pick a temperature test point, and input this mV value to the ECU. The ECU would provide a serial output in engineering units, which would then be continuously compared against the T/C simulator set point IAW the allowable error band. The same holds true for other ECU inputs; actual sensors are not required if they can be effectively simulated.

E.3.1.4 Output signals.

On the output side, the parameter would be measured and compared with the desired value. Discretes obviously have no allowable error, they need to stay in the proper commanded state. Stepper motors might be driven in a triangular waveform and this can be monitored and recorded for anomalies. It is important to note that this is an open loop test; no engine model is required. An attempt is being made to verify that under the applied environment, all ECU input signals and all ECU output signals meet their respective tolerances as specified in the purchase specification or the ATP.

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E.3.2 Methodology.

Using the above information as a starting point, each input and output signal should be listed in a table with the applicable pass/fail limits, showing where the limits are defined (paragraph of the ATP, for example) or how they are derived (a purchase specification EU limit converted to a signal limit, such as a position error in inches converted to an equivalent voltage error). These limits are generally applicable for all testing, but in some specific cases the tolerances may have to be increased where the dynamics of the applied environment greatly exceed the true environment. This typically is the case when temperature ramp rates are very high and circuits don't have sufficient time to stabilize, resulting in their inability to meet accuracy requirements (especially true for low-level signals like thermocouples).

Once a master table is created for the UUT, it can be used to define which (if not all) parameters are to be monitored and recorded for any specific environmental test. A sample signal pass/fail table is illustrated in the section below.

E.3.2.1 I/O pass/fail table.

Item	Signal	SDS Description	Nominal Setting	Tolerance	Source
1	N1A	Primary core speed	100%	±0.25%	1/
2	N1B	Secondary core speed	100%	±0.25%	1/
3	N1B(O)	Secondary core speed other engine	100%	±0.25%	1/
4	N2A	Power turbine speed A	100%	±0.25%	1/
5	N2B	Power turbine speed B	100%	±0.25%	1/
6	Nr	Rotor speed sensor	100%	±0.25%	1/
7	T1	Compressor inlet temperature	59 F	±2 F	1/
8	T45	Power turbine inlet temperature	1600 F	±15 F	1/
9	N2Sel	N2 select potentiometer	100% N2	±0.05%	1/
10	Wf	Wf metering valve position	1200 pph	±15 pph	1/
11	Q	Engine torque	70 %	±1.0%	1/
12	CP	Collective pitch	30 %	±1.0%	1/
13	ECL	Engine condition lever	60 degrees	±0.5 deg	1/
14	P1	Ambient pressure	ambient psia	±0.5 psia	1/

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15	P3	Compressor discharge pressure	75 psia	±5.0 psia	1/
16	Test Cell	Enable test cell switch	Switch closed (enable test cell)	No change	N/A
17	Load Share Select	Load share select switch	Switch closed (Q load sharing)	No change	N/A
18	Beep Up	Increase beeper	Switch closed (increase fuel flow)	No change	N/A
19	Beep Down	Decrease beeper	Switch closed (decrease fuel flow)	No change	N/A
20	O/S Test	Overspeed test switch	Switch closed (O/S test selected)	No change	N/A
21	Chan Sel	Channel A/Channel B select switch	Switch closed (Channel B)	No change	N/A
22	PAS	Power assurance test switch	Switch closed (PAS test selected)	No change	N/A
23	Spare Disc	Spare discrete	Switch closed	No change	N/A
24	Start Fuel	Start Fuel solenoid	Active (Start Fuel enabled)	No change	N/A
25	O/S SOV	Overspeed solenoid	Active (O/S Sol enabled)	No change	N/A
26	Chan Sol	Channel A/Channel B changeover solenoid	Active (Channel A)	No change	N/A
27	Bleed Sol	Compressor bleed band solenoid	Active (bleed closed)	No change	N/A
28	Ign Sol	Ignition solenoid	Active (relay energized)	No change	N/A
29	A/B Lamp	Channel A/B in control lamp	Active (Channel A in control)	No change	N/A
30	Fail Lamp	Channel fail lamp	Active (Channel failure)	No change	N/A
31	EO Lamp	Engine out lamp	Active (engine out failure)	No change	N/A
32	A/B SM A Out	Channel A/B Phase A metering valve stepper	Cycled continuously	2/	2/

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33	A/B SM B Out	Channel A/B Phase B metering valve stepper	Cycled continuously	2/	2/
34	A/B SM C Out	Channel A/B Phase C metering valve stepper	Cycled continuously	2/	2/
35	A/B SM D Out	Channel A/B Phase D metering valve stepper	Cycled continuously	2/	2/
36	A/B SM Exc	Channel A/B Excitation metering valve stepper	Active	2/	2/
37	Overspeed Status	Overspeed status to other engine	Active (O/S energized)	No change	N/A
38	Spare Disc	Spare discrete output	Active	No change	N/A
39	T45 Out	T45 temperature to cockpit	1600 F	±15.0 F	3/
40	28V A/F	Airframe 28VDC power	28V	±1.0V	4/
41	28V EM	Emergency 28VDC power	28V	±1.0V	4/
42	ALT (A)	Phase A alternator power	15.0 Vrms	+/-1 VDC	5/
43	ALT (B)	Phase B alternator power	15.0 Vrms	+/-1 VDC	5/
44	ALT (C)	Phase C alternator power	15.0 Vrms	+/-1 VDC	5/
45	GSE	Ground Support Equipment	Monitored	6/	N/A
46	1553	1553 Bus	Monitored	7/	N/A
47	CEDL	Cross-Engine Data Link	Monitored	8/	N/A

1/ Tolerances are from Engine Control System Design Specification (CSDS).

2/ Channel A (A SM) and Channel B (B SM) stepper motor phases cycled continuously during the tests and monitored with their corresponding BIT signals. Excitations are also verified with their BIT signals.

3/ There is no CSDS requirement for this output, use CSDS requirement from similar input.

4/ Airframe and Emergency 28V input tolerance is from ATP.

5/ Alternator tolerance is from ATP for 28VDC output of 3-phase rectifier.

6/ The GSE communications link will be checked while power is applied to the ECU to verify proper communications.

7/ The 1553 bus communications will be monitored by test software which allows a PC to be both a Bus Controller and a Bus Monitor. This software will be monitoring the bus activity when the ECU is powered and when it is not powered.

8/ The CEDL will be tested while power is applied. The CEDL output is wrapped back to the input where the ECU compares its current state with the state of the other engine.

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E.3.2.2 Input signal selection.

The next step is to define the test stimulus to be applied. For the battery of environmental tests, usually one or two sets of input signal values are defined (those that may be particularly sensitive to the environment or points where the signal conditioning/measurement errors are known to be close to the allowable limits). These input conditions are set using very accurate (1/5 or better of the error band) test equipment (typically, the ATP rig is used) with the UUT transmitting the conditioned/measured value via a monitoring bus of some type (high speed UART, 1553, etc.). The transmitted value is then recorded and monitored to assure that it does not deviate from the ATP set point by more than the allowable error defined in the pass/fail criteria table illustrated in E.3.2.1.

E.3.2.3 Output signal selection.

Output conditions are also similarly defined. For example, a torque motor drive may be set to output a fixed value (say 20 mA) and monitored using an external signal conditioner (again in the ATP rig) to verify that it stays within + 1 mA throughout the test (if this was the limit). Other analog driver signals are monitored similarly. Discretes like solenoids are typically set to both states (sequentially) and monitored for inadvertent state change. Stepper motor drivers are more troublesome, but in some cases a drive pattern is established that results in a fixed-rate up and down sweep that can be recorded and monitored for anomalous behavior.

E.3.2.4 Simple sensors/devices.

Testing of simple sensors such as N1, T1, T45, P1, P3, etc. is typically stand alone; that is, no ECU is involved. This doesn't preclude testing several different sensors at the same time, if the applied test environments are the same.

For testing simple items like speed sensors, temperature sensors or pressure sensors, the approach depends entirely on how the device functions and how it is constructed. The purpose of the test is key to deciding how best to establish that the UUT meets its ‘performance’ specification and how much ‘functional’ cycling is required. Below are examples of testing approaches utilized for some different sensor types.

a. Speed sensor: For a speed sensor comprising redundant coils, operation does not require the use of a spinning target wheel, since the target is not necessary to demonstrate that the unit can withstand the environment. Typically, one would continuously monitor each coil by passing a known current through the coil. This current would then drive a monitor/fault warning circuit that would assure that at no time during the applied environment was there an intermittent open/short condition.

   At the conclusion of the testing, the pre-test and post-test ATPs would be compared to assure that the measured coil resistance value after the test was nearly identical to the pre-test value to verify that no permanent damage had occurred. For example, while an ATP limit for coil resistance might be 300 ohms $\pm$ 75 ohms, the pre/post comparison limit for coil resistance might be the initial ATP value $\pm$ 10 ohms.

   In other words, a sensor that passes the pre- and post-test ATP with coil values of 250 ohms and 350 ohms, respectively, would be a failure under the comparison requirement, since there is no reason for the coil resistance to have changed due to the applied environment (other than some kind of failure).

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b. RTD: The requirements would be for the coil resistance to track the surrounding temperature (either the ambient in the case of a test where it is uncontrolled or the test chamber temperature when applicable). This means the external environment temperature needs to be monitored and recorded so that resistance vs temperature can be plotted over the test duration. Here, a pre- and post-test coil resistance comparison would not be necessary since the ATP by itself would have a very tight limit on coil resistance to begin with.

c. Strain gauge type pressure sensor: Pneumatic pressure would have to be varied during an SOT so that the output could be compared with the input pressure. Here, the fact that a strain gauge element is utilized, the only way to acceptably assure performance over the environment is by having a controlled variable pressure, whether continuously varied or as a series of discrete set points. For regular environmental tests, two set points would be sufficient (ambient and a higher set point, for example).

E.3.2.5 SOT functional cycle.

For the SOT, the only real difference from the environmental suite of tests is that the inputs and outputs are “cycled” throughout their ranges during the test, as opposed to using one or two test stimulus sets. Inputs don’t have to follow a dynamic profile (which would actually make determination of performance very difficult), instead they can be stepped through defined steady-state condition ‘sets’ which can be more easily monitored against the allowable tolerance for each input/output signal value.

Following is an example of an SOT functional cycle for an ECU. It is an asymmetric cycle that results in virtually all combinations of input and output values being set over a 60-minute period.

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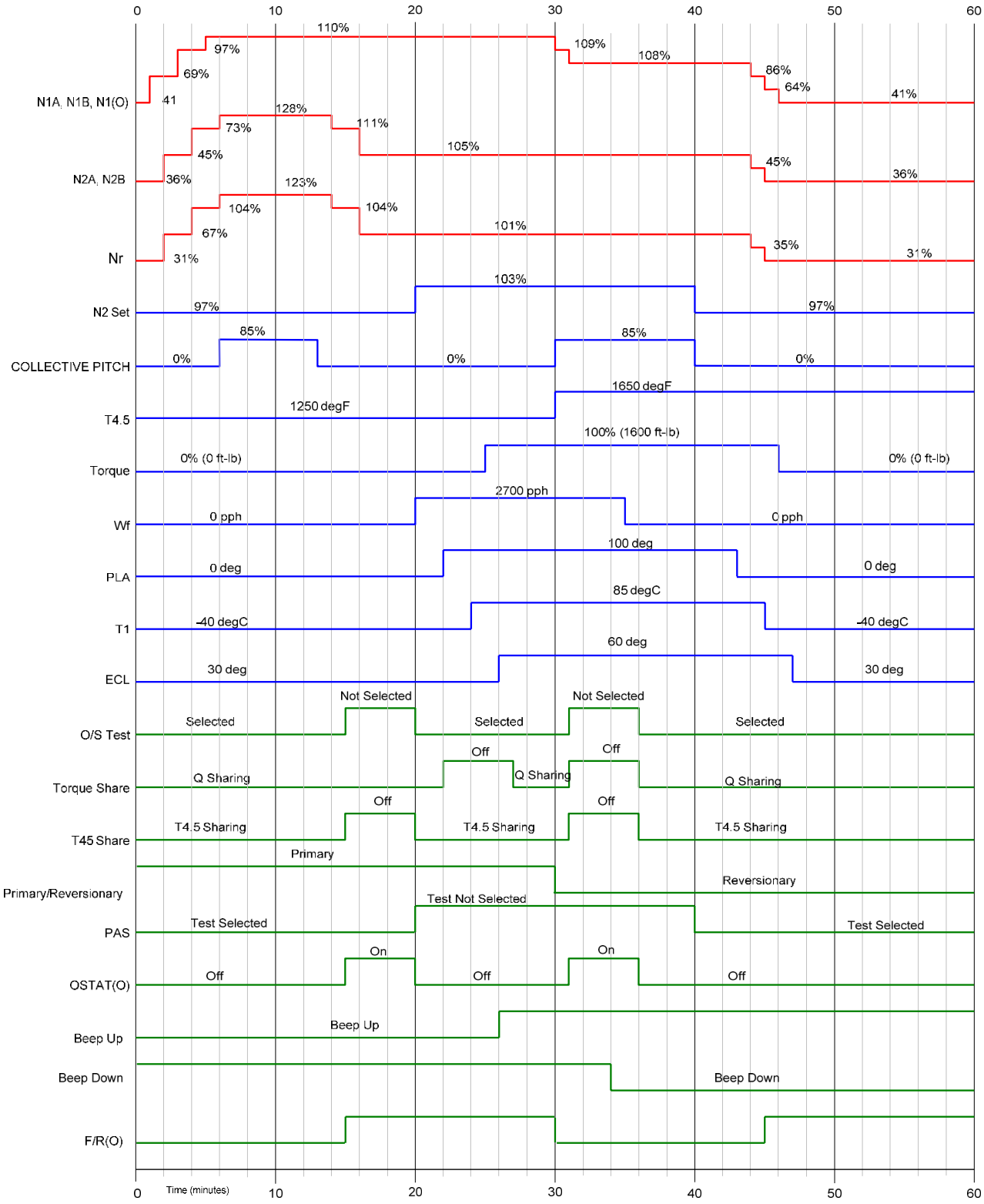


Figure E-1. Example Input / Output Signal Functional Cycle Definition for an ECU.

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FAULT INJECTION TESTING

(Referenced by: 5.8.5a)

F.1 SCOPE

This appendix provides examples of test case types that have been used in the past to evaluate engine/control system response to failures of input and output signals in order to meet requirements specified in this standard. The information contained herein is intended only to assist the contractor in complying with the requirements defined in the referenced paragraph.

F.2 DEFINITIONS

General definitions of the terminology and nomenclature found within this appendix are provided below. Specific definition may vary with contextual usage.

electronic control unit (ECU)	The main engine control, which may comprise more than one channel for redundancy. Sometimes referred to as the 'FADEC', which is usually reserved for the entire engine control system, not just the ECU.
formal qualification test (FQT)	Pertaining to software qualification testing of the ECU and its embedded software. Complete stem-to-stern testing of the software after completion of dry runs to eliminate any coding or test procedure errors.

F.3 PROCEDURE

F.3.1 Types of signal failures.

Input/output signal failures, as defined herein, are grouped into one of five general types:

1) single point, 2) sequential, 3) intermittent, 4) data bus, and 5) slow drifts.

In all of these failure types, signals are failed one at a time to determine if the engine control system response is as specified by the relevant top-level system/software documentation. In all cases, the expected response should be easily determinable based on the specified requirements. It should be noted that in the case of common-mode failures (discussed in APPENDIX B), expected response is not easily analyzed when signals do not fail sequentially.

Signal failures focus on wiring faults and malfunctions of sensors and electronic circuitry. Wiring faults focus on conductor breaks and shorts between adjacent wires, shields, and the airframe structure. These faults can arise from wear, incorrect construction and installation, and ballistic damage. Sensor malfunctions consider hard breaks, calibration drifts and sensor detachment (such as a position sensor becoming loose, causing it to slip or detach).

F.3.2 Methodology.

Each of the signal failure types listed above can be tested using typical methods of introducing opens and shorts, intermittents, and 'slow' drifts. While such testing can be accomplished with an engine running in a test cell, for safety and test repeatability reasons it is preferable to utilize a closed-loop simulation environment as is typically used for software FQT. The following subsections provide examples of testing which meets the intent of 5.8.5.

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F.3.2.1 Single point failures.

a. Shorts between all wire combinations within unshielded twisted pairs, triplets, quadruplets, quintuplets, or sextuplets or more.

(1) Pairs: 1 combination.

(2) Triplets: 4 combinations.

(3) Quadruplets: 11 combinations.

(4) Quintuplets: 26 combinations.

(5) Sextuplets or more: 26 combinations, maximum, based on lack of outcome surety.

b. Shorts of individual unshielded wires under a common bundle shield.

(1) Between any combination of two, three, four, or five wires within the bundle.

(2) Between any six or more wires; maximum number of combinations same as five wires.

c. Short of any one wire to ground.

d. Shorts between any two wires terminated at adjacent pins or contacts in an electrical connector.

e. Shorts between any two wires in the connector of a sensor, output effector, or other external input or output device, or subsystem, which interfaces with the ECU. This is limited to simple single-function devices; it does not apply to connectors supporting multiple functions or interfacing to other electronic assemblies.

f. Open circuit of any one wire.

NOTES:

1/ Because of overlapping situations, execution of all the tests listed above would likely result in repetition of some short or open test conditions, in which case they would not need to be performed repeatedly.

2/ For the purpose of this requirement, twisted wiring is defined as having one twist per 6 inches or tighter. The short circuit tests between individual wires within twisted wiring do not apply to wires containing gradual twists of one twist per several feet (as commonly applied to an overall wiring harness).

3/ Most twisted wiring configurations in avionics applications are either twisted pairs or twisted triplets. Twisted wiring containing 5 or more wires is uncommon, however for twisted wiring containing more than 5 wires, tests should be limited to no more combinations than the number based on 5 wires (or less).

4/ If a large bundle of twisted wiring is encountered, they can be sorted by common inputs and outputs containing identical circuitry. Completely test one sample of wiring associated with each type of circuit (including shorts with wires associated with neighboring circuits) and apply similarity to all the other inputs or outputs containing the same circuitry. However, the consequence of faulting the different functions, associated with the identical circuitry, will still need to be analyzed.

5/ Large bundles of wiring can be divided into more manageable groups by the use of shielding, which provides both electromagnetic interference (EMI) protection and a grounded wall. A wire with an exposed gap in the insulation will contact the grounded shield before contacting another wire on the other side of the shield.

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F.3.2.2 Sequential failures.

a. For groups of signals (two or more) that are backups for one another, fail each sensor in the group one after the other in the sequence in which they back each other up. Repeat the test if there is more than one backup sequence for the group. Repeat the test with up to three different random sequences of failures. The end result of these sequences should be setting a default value, using a last valid reading, or failing to a defined safe state.

b. Choose groups of three unrelated input signals and then fail the group members in each possible sequence (as above) to determine system behavior.

c. Repeat a and b until all signal groups are tested.

F.3.2.3 Intermittent failures.

Open or short a signal wire for a duration less than the specified time to confirm the failure.

a. Assert failure, then remove the fault.

b. Assert failure, remove the fault for 0.5 seconds, assert failure, then remove fault.

c. Same as b., but with three sequential failures.

d. Short a wire to ground for 10 seconds minimum, then remove short.

F.3.2.4 Data bus failures (each unique bus, one at a time).

a. Interrupt each non-redundant aircraft interface data bus for the following intervals, then re-establish communication.

0.5, 3.0, 10.0, and 60.0 seconds

b. Interrupt each redundant aircraft interface data bus, all channels at once, for the following intervals, then re-establish communication.

0.5, 3.0, 10.0, and 60.0 seconds

F.3.2.5 Slow drift failures (redundant signals).

Introduce a drift to one channel of a redundant sensor signal such that the value seen by the unit under test is “sensor initial value + drift”. Begin with ‘equal’ redundant signals, then

a. cause a slow positive measurement drift in one sensor channel until the channel delta exceeds the specified failure detection threshold, then continue drifting until out of range.

b. cause a slow negative measurement drift in one sensor channel until the channel delta exceeds the specified failure detection threshold, then continue drifting until out of range.

F.3.2.6 Slow drift failures (closed-loop position).

Substitute a sensor signal with one that drifts, up or down, independent of any closed-loop feedback mechanism, thereby simulating a sensor that has become disconnected from whatever servomechanism or actuator that it is supposed to be measuring. Begin with the sensor reading the correct position, then

a. cause a slow upward drift in the sensor channel until the failure logic detects an ‘open-loop’ condition.

b. cause a slow downward drift in the sensor channel until the failure logic detects an ‘open-loop’ condition.

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F.3.3 Generalized pass/fail criteria.

(Referenced by: 5.8.5)

a. All test cases shall result in safe engine operation.

b. A fault on one control channel shall not propagate to any other control channel.

c. A fault to the wiring of one function shall not propagate to any other function having independent wiring.

d. Fault detection and accommodation features shall perform as intended.

e. The control system shall not be damaged by the following wiring faults:

(1) Open circuit of any wire.

(2) Short circuits to chassis ground or engine structure.

(3) Short circuits between any two (or more) wires that are not used for 28 Vdc aircraft electrical power or PMA wiring.

f. Short circuits between any wire and 28 Vdc aircraft electrical power or PMA wiring shall result in continued safe engine operation.

F.3.4 Other considerations.

While the above subsections provide examples of test cases meeting the intent of 5.8.5, contractors are cautioned that these exact tests are not necessarily required since test case design is system architecture and software dependent. The Airworthiness Authority will work with contractors to prepare test plans and approve test cases which satisfy the test requirements specified in the standard.

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VIBRATION TEST PROFILES for H-47, H-60, and H-64

(Referenced by: 5.4.3.2.2, 5.9.1, 5.9.2)

G.1 SCOPE

This appendix discusses the derivation of vibration test profiles for the H-47, H-60, and H-64 platforms based on flight test data gathered by the Aviation & Missile Center, Redstone Arsenal, Huntsville, Alabama. A detailed example of how one of these vibration test profiles (H-60) was developed is provided; the same methodology being used for the H-47 and H-64.

The information contained herein is intended to assist the contractor in complying with the requirements defined in the referencing paragraphs below the appendix title, hence the reason for the extensive details which follow. The aim is to provide sufficient information to allow the contractor to tailor the vibration test profiles for shorter or longer service life requirements and test times, and to handle qualification for multiple platforms and vibration sources.

G.2 DEFINITIONS

General definitions of the terminology and nomenclature found within this appendix are provided below. Specific definition may vary with contextual usage.

accelerated endurance	A test that has been 'accelerated' to reduce the time needed to prove the endurance of equipment to 'typical' vibration over its service life (i.e., one hour of test time equates to many hours of in-service use at typical vibration levels).
ASD	Acceleration spectral density for random vibration, expressed in units of g^2/Hz, where g is the acceleration due to gravity (9.81 m/sec^2).
A_X	Peak amplitude of sine tones expressed in units of *g*.
broadband	Random vibration spectrum that is applicable over the complete frequency range of interest for a component.
endurance	The average vibration level that a component is expected to 'endure' over its specified service lifetime. Also referred to as a 'typical', 'mission-weighted', or 'reliability' vibration level. One test time hour equates to one in-service hour.
exaggeration	The multiple that test levels are above typical (endurance) levels is referred to as the 'exaggeration' factor. The exaggeration factor is computed using the time compression algorithms (Miner's rule) specified in MIL-STD-810H(C1), section 2.2, page 514.8A-4 (Eq. 1) and page 514.8A-5 (Eq. 4). Over exaggerating may lead to unrepresentative failures, consequently the test levels (under normal circumstances) should be limited to not greater than: (1) Sinusoids: 2.0 times the functional level. The test time per axis can be calculated as follows: $\text{Test Time}_{\text{per axis}} = \text{Service Life}_{\text{hours}} \times (g_{\text{PEAK Typical Level}} / g_{\text{PEAK Test Level}})^m$ (2) Random profiles: 4.0 times the functional level. The test time per axis can be calculated as follows: $\text{Test Time}_{\text{per axis}} = \text{Service Life}_{\text{hours}} \times (g^2/\text{Hz}_{\text{Typical Level}} / g^2/\text{Hz}_{\text{Test Level}})^{m/2}$

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functional	Also known as 'worst-case' levels. These are the highest levels that a component is exposed to in operational service (max allowable engine vibration, exceedance of which requires engine replacement). The component needs to function properly when exposed to these vibration levels. These worst-case levels factor into the constraints used to upper bound the exaggeration factors.
m-Factor	Material exponent (m) used in Miner's Rule equations. Typical values are provided in Table 514.8F-III of MIL-STD-810H(C1). This standard uses a default of 7.125 for sine tones and 8.0 for random vibration. These default values represent a component with a 50/50 mix of steel and aluminum alloy piece parts (see m-Factor cross plot in G.3.1.4). The contractor may propose values based on the material composition of the component under test, with supporting rationale.
narrowband	Random vibration spectrum, or 'spike', that is applicable over a very narrow portion of the complete frequency range of interest for a component.
random	Vibration profile that is non-deterministic, comprising all sine tones of varying frequencies and amplitudes over a specified bandwidth (expressed in g^2/Hz).
service life	The expected operational usage of the equipment, expressed in engine hours, before the equipment becomes uneconomical to repair due to a failure rate inflection point.
sinusoid	A pure sine tone vibration profile, expressed in peak *g*, used to represent the dominant frequencies resulting from major rotating airframe or engine components. The sine tones may 'wander' about a small band to represent engine or airframe rotor/propeller speed ranges.
typical	The 'typical' level of vibration expected to be encountered over the service life of the equipment. Also known as 'mission-weighted', 'service' or 'endurance' level. This level is the baseline from which exaggerated levels are computed to allow for an accelerated endurance test.
worst-case	The vibration level which the equipment needs to withstand over a short period of time, while still meeting all operational performance requirements. This is not an 'exaggerated' level; one test-time hour per axis at worst-case equates to one in-service unit-hour at worst-case.

G.3 GUIDANCE

The topics listed below are discussed with the intent of clarifying vibration testing as specified in this standard. Some of this information is the result of answering contractor questions during component development programs, while other information is elucidation for the benefit of those not familiar with the topic in question.

G.3.1 Vibration profile adjustment factors

G.3.2 Derivation of Category 14 profiles for H-60

G.3.3 Combining and splitting vibration profiles for test purposes

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G.3.1 Vibration profile adjustment factors.

(Referenced by: 5.4.3.2.1, G.3.2.2, G.3.2.5)

The subparagraphs that follow discuss this table and its various entries, as well as the derivation or source of those entries. Note that not all of the table entries are discussed since they are explained in greater detail in the identified sources.

MIL-STD-810H(C1)	**1**	**Category**	**13**	**14**		**22**
	2	**Application**	Propeller Aircraft	Rotary-Wing Aircraft		Turbine Engines
	3	**Method 514.8 Procedure I**	Sine on Broadband Random			Narrowband on Broadband
	4	**Test Spectrum Source**	810H(C1), Fig 514.8D-2 Table 514.8D-II	810H(C1), Fig 514.8D4 Tables 514.8D-IIIa/IIIb	TOP 01-2-603A	810H(C1), Fig 514.8D-12
	5	**Spectrum Type**	Accel Endurance 1 TT hr = 1000 RT hrs (514.8D-5, 2.2e)	Accel Endurance 4 TT hr = 2500 RT hrs (514.8D-9, 2.3g)	Accel Endurance 4 TT hrs = 2500 RT hrs (C/D/E.5.2b)	Functional (514.8D-25, 2.11f)
	6	**Applicability**	All Aircraft	All except for: H-47/H-60/H-64	H-47/H-60/H-64	All Turbine Engines
	7	**Equipment Location**	Fig 514D-3 Engine Nacelle or Pylons	Fig 514.8D-5 On/Near Drive System Elements		On Engine
Adjustment Factors: to Derive Typical Amplitude	**8**	**ASD Broadband Random (g^2/Hz)**	0.22 x End.(R4) (514.8D-5, 2.2d)	0.18 x End.(R4) [514.8D-8, 2.3e(4)]	0.20 x End.(R4) (C/D/E.6.3b)	2/3 x Func.(R11)
	9	**L_o Narrowband Random (g^2/Hz)**	0.22 x 0.6 (R4) (514.8D-5, 2.2d)	Not Applicable		2/3 x Func.(R12)
	10	**Sine A_x (g peak)**	$S_0 = 0.162\sqrt{f_0}$ (Table 514.8D-II, note 3)	0.34 x End.(R4) [514.8D-8, 2.3e(4)]	0.447 x End.(R4) (C/D/E.6.3b)	Not Applicable
Adjustment Factors: to Derive Functional Amplitude	**11**	**ASD Broadband Random (g^2/Hz)**	0.35 x End.(R4) (514.8D-5, 2.2d)	0.24 x End.(R4) [514.8D-8, 2.3e(5)]	@ Cabin Location H-47: 0.36 x End.(R4) @ Engine Location H-47: 0.33 x End.(R4) H-60: 0.25 x End.(R4) H-64: 0.51 x End.(R4) (Tables C-7/D-7/E-7)	Functional (R4)
	12	**L_o Narrowband Random (g^2/Hz)**	0.35 x 0.6 (R4) (514.8D-5, 2.2d)	Not Applicable		Functional (R4)
	13	**Sine A_x (g peak)**	$S_0 = 0.205\sqrt{f_0}$ (Table 514.8D-II, note 3)	0.41 x End.(R4) [514.8D-8, 2.3e(5)]	@ Cabin Location H-47: 0.60 x End.(R4) @ Engine Location H-47: 0.58 x End.(R4) H-60: 0.50 x End.(R4) H-64: 0.72 x End.(R4) (Tables C-7/D-7/E-7)	Not Applicable
Resulting Default Test Profiles	**14**	**Typical & Functional Test Profiles**	TABLE VIII	TABLE IX	TABLE XI, TABLE XII TABLE XIII, TABLE XIV	TABLE X
	15	**Typical Test Duration**	1 test time hour/axis = 1 real time hour/axis. Unit to be fully operational IAW performance specifications.			
	17	**Functional Test Duration**	4 hours/axis = 4 real time hours/axis at worst-case vibration levels. Unit to be fully operational IAW performance specifications.			
	18	**Accelerated Endurance**	Calculate test level and duration using 'typical' vibration profiles (rows 8, 9, 10) adjusted in amplitude IAW Equations 1 and 2 listed in G.2 'exaggeration'.			
	19	**m-Factor**	This standard uses default 'm' values of 7.125 for sinusoidal vibration and 8.0 for random vibration. These values represent a component comprising a 50/50 mix of steel and aluminum alloy piece parts. Suppliers may propose using other 'm' factors based on the composition of the component to be tested.			

NOTES: 1/ TT = test time 2/ RT = real time 3/ (R4) = Row 4 of this table; similarly for other row references

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G.3.1.1 MIL-STD-810H(C1): rows 1-7.

These 7 rows define the MIL-STD-810H(C1) vibration test categories applicable to the components that are the subject of this standard, as well as identifying the location in 810H(C1) or TOP 01-2-603A where the baseline test profiles can be found (row 4).

Of particular importance is row 5 that identifies the vibration profile type and the page/paragraph in 810H(C1) where this is identified. Notice that for Categories 13 and 14, the profiles are 'accelerated' endurance, that is, they are not 'real-time'. The damage that is accrued is accelerated as listed in the applicable cell. For example, for Category 14, 4 test-time (TT) hours per axis using the test profile identified in row 4 is the equivalent of 2500 real-time (RT) hours per axis at the typical level of vibration for that category.

For Category 22, the test spectrum is not accelerated like the others; it is defined as a 'worst-case' envelopment of turbine engine measured data, and hence is appropriate for functional testing of a component where the component needs to withstand the worst-case engine vibration that could occur for any given engine, for at least the duration of a single mission; i.e., real time worst-case or functional (interchangeable terminology, herein).

In both cases (worst-case and accelerated), the profiles need to be reduced to a typical, baseline level, from which various accelerated endurance profiles can then be calculated using the Miner's rule equations listed in G.2 'exaggeration'. The two following sections describe the factors used to reduce the endurance amplitude to typical (G.3.1.2) and endurance amplitude to functional (G.3.1.3), as well as the source of those factors. Use of factors other than those listed (which are considered default) is allowable, based on using reliable/defensible vibration data for any given engine or airframe application.

G.3.1.2 Adjustment factors (to derive typical amplitude): rows 8-10.

These rows define the factors which are used to reduce the vibration levels to those which would be used for a 1-to-1 hourly service life test. The level is considered to be 'typical', 'average', 'mission-weighted', etc.; essentially what the component will experience over a lifetime of usage, on a real-time basis.

For example, for Category 13 the endurance spectrum defined in row 4 is simply multiplied by 0.22, as specified in row 8, to determine the broadband ASD values. For reference, the location in 810H(C1) where this factor can be found is listed in the cell in parentheses. Similarly, the narrowband L_O value is determined by multiplying the 810H(C1) value of 0.6 by the same 0.22 scale factor (row 9), which is then converted into an equivalent sine tone amplitude (S_0) in row 10 using the noted formula.

Category 14 calculations for the two groups of helicopters are similar. However, the factors are sourced differently; from 810H(C1) for 'all' helicopters and from TOP 01-2-603A for the H-47, H-60, and H-64.

Lastly, for Category 22 there are no adjustment factors listed in 810H(C1) to reduce the functional profile to typical, or increase it to accelerated endurance. In this case, a multiplier of 0.67 is being used to convert functional to typical for both broadband and narrowband values.

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G.3.1.3 Adjustment factors (to derive functional amplitude): rows 11-13.

Conversion of the accelerated endurance profiles to functional test profiles is accomplished in the same manner, as explained in the previous section to convert to typical test profiles. Obviously, the factors used are greater since worst-case vibration levels are greater than the average or typical levels.

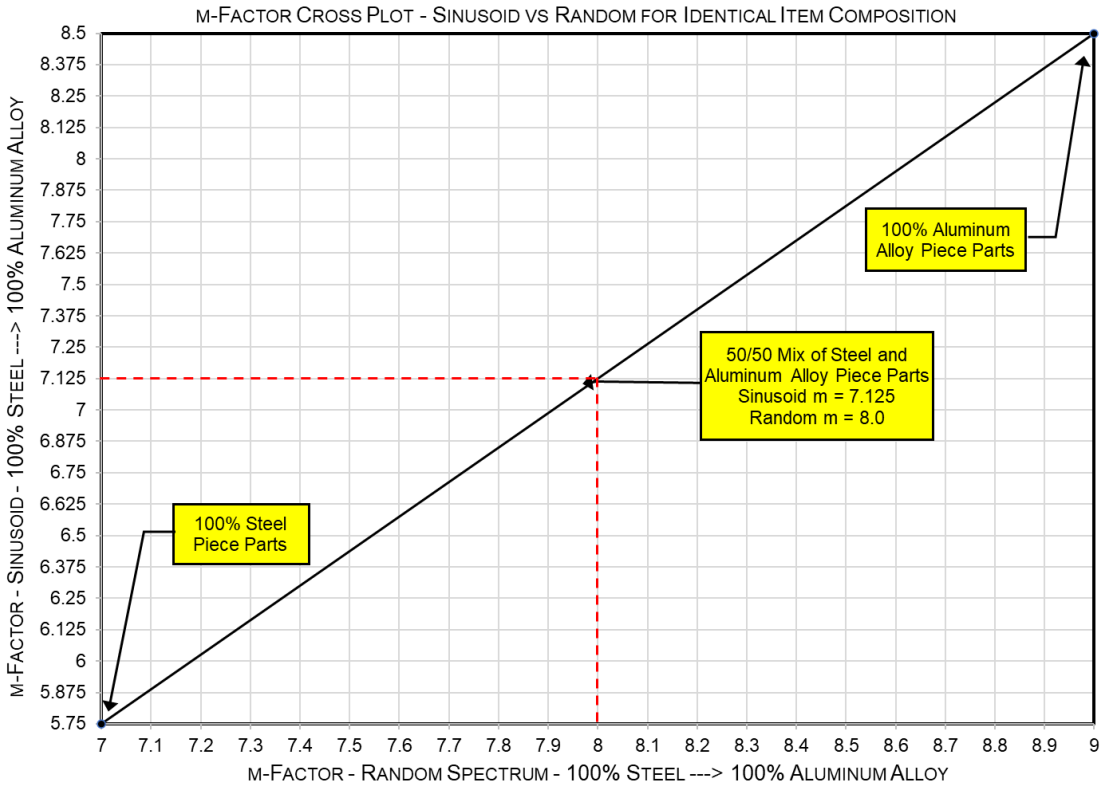
One difference to note is that Category 14 for the H-47, H-60, and H-64 platforms uses unique factors for each in order to reduce the endurance amplitude to a functional level. The first factor listed in rows 11 and 13 is the factor to use for the H-47 when the equipment is mounted in the cabin area. The other listed factors immediately below represent reduction factors for the engine area of all three helicopters. The vibratory environment for other areas of these platforms is included in the TOP.

G.3.1.4 Typical / functional profiles and accelerated endurance calculation: rows 14-19.

(Referenced by: G.2 'm-Factor', G.3.3, K.2 'm-Factor')

These last rows point to specific tables and sections in this standard where the default vibration profiles for both typical and functional testing can be found. The derivation of these defaults is based on both MIL-STD-801H(C1) and TOP 01-2-603A, as discussed in this appendix subsection, and summarized in the table.

It is recommended that absent any specific knowledge of the composition of the component to be subjected to the endurance vibration test, that the default 'average' m-Factors (as specified in row 19 of the table) be used when calculating the endurance profile. Use of different factors can be proposed, based on the specific materials of the component under test, otherwise the default values are acceptable as is. Note that the m-Factors can be picked off of the plot below for any assumed ratio of steel to aluminum parts.



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G.3.2 Derivation of Category 14 default profiles for H-60.

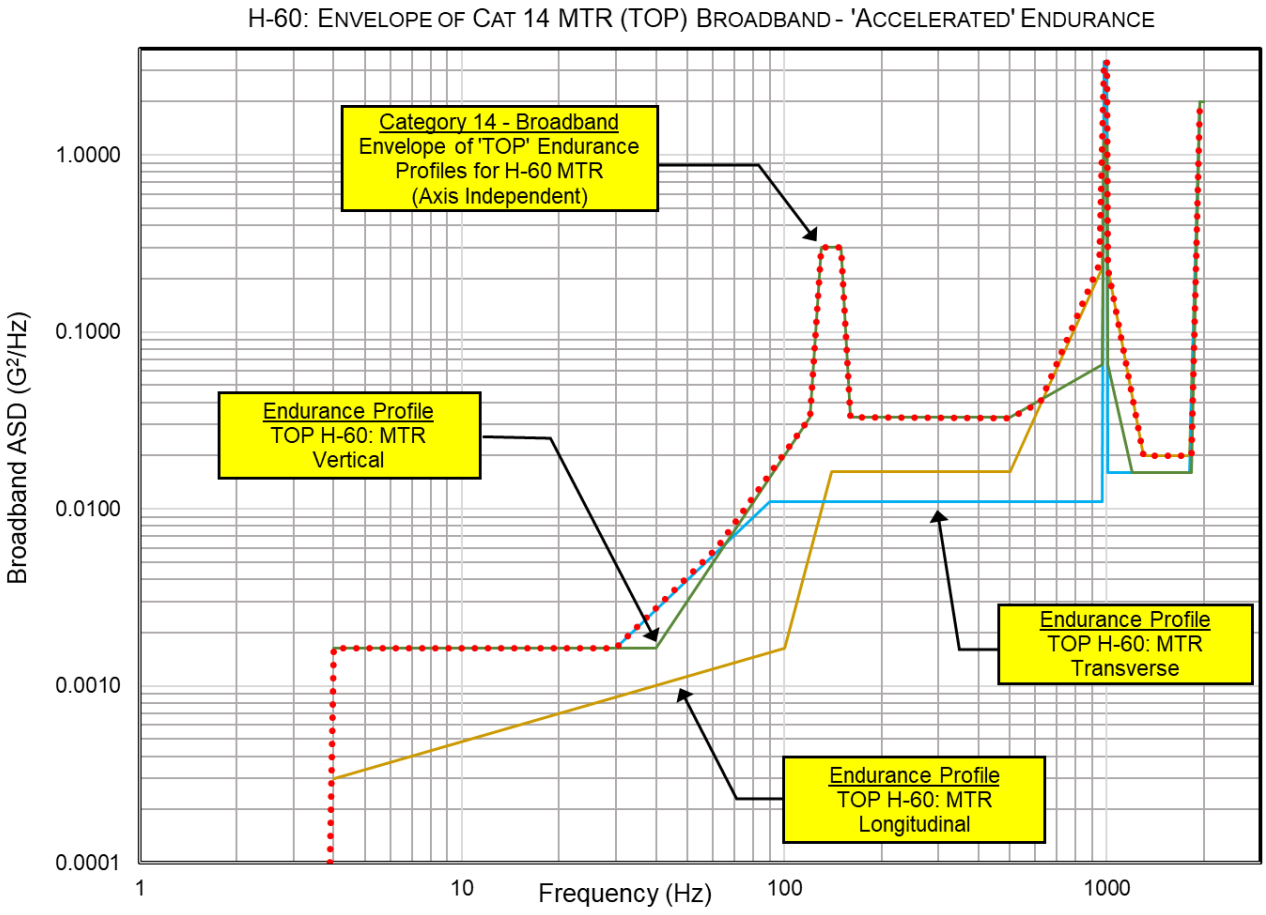
The subsections which follow illustrate the process that was used to derive the H-60 platform vibration profile for Category 14 (TABLE XIII), using the data provided in TOP 01-2-603A. Derivation of the Category 14 vibration profiles for the H-47 (TABLE XI / TABLE XII) and H-64 (TABLE XIV) platforms followed the same process.

G.3.2.1 Step 1.

(Referenced by: G.3.2.4)

The vibration data provided in the TOP document (stated as already accelerated) was reviewed to find the profiles at or nearest to the engine/drivetrain location. In the case of the H-60, the applicable data is Figure D-8, Main Transmission Region (MTR).

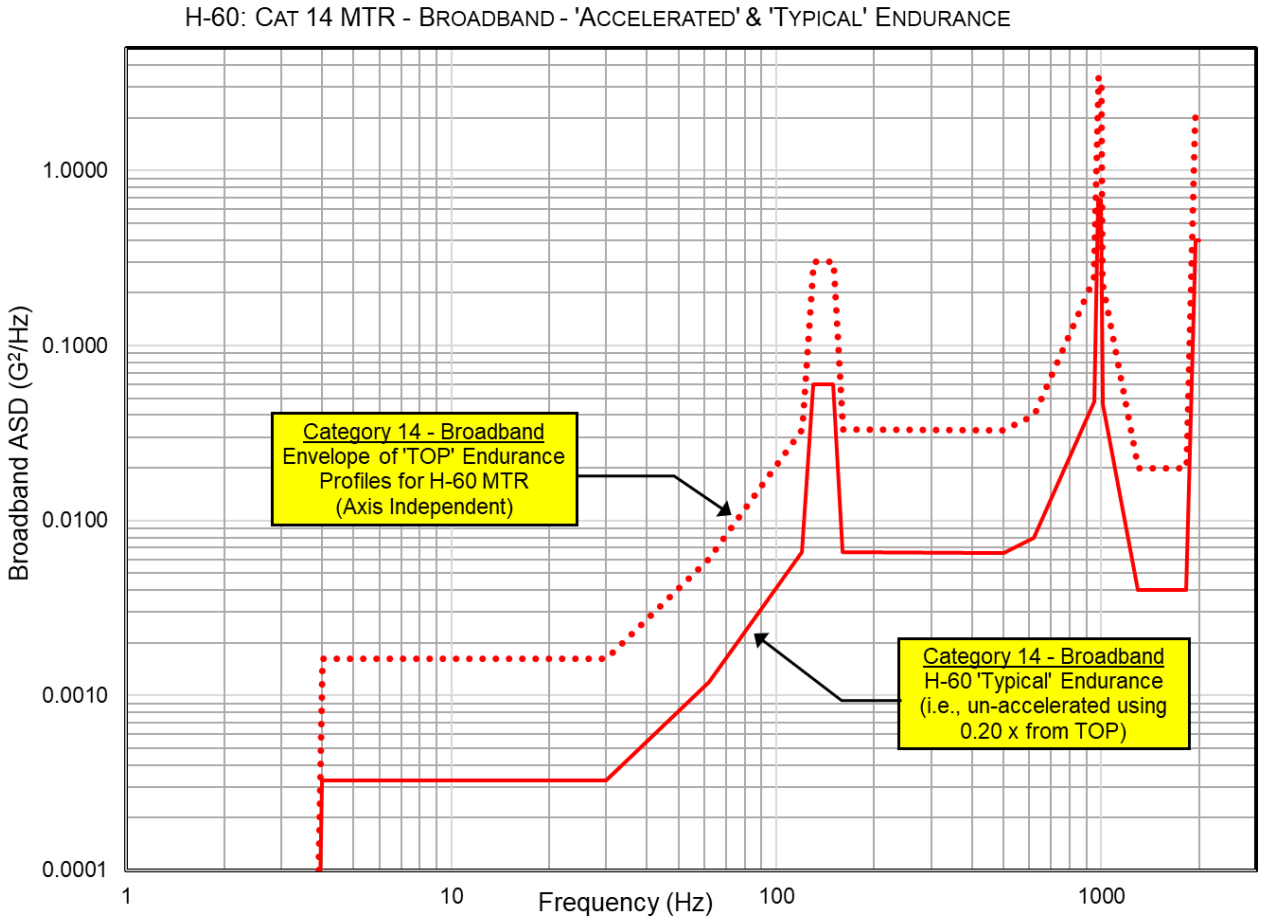
The individual axis profiles (longitudinal/transverse/vertical) were then plotted and an 'envelope' (shown as the dotted red line) overlaid on the maximums over the entire frequency range of 4 Hz to 2000 Hz. This resultant profile represents the TOP axis-independent accelerated endurance vibration profile that, as stated in TOP D.5.2b, demonstrates 2500 hours of service life when run for 4 hours/axis.



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G.3.2.2 Step 2.

In accordance with Row 8 of the table in G.3.1, scaling the accelerated endurance profile by 0.20 results in the un-accelerated endurance, or 'typical' profile shown by the solid line in the plot below. It is this typical profile which forms the basis for developing an accelerated endurance profile that may be using different values for the m-Factor because of the material composition of the component to be tested, as well as different required service life hours.

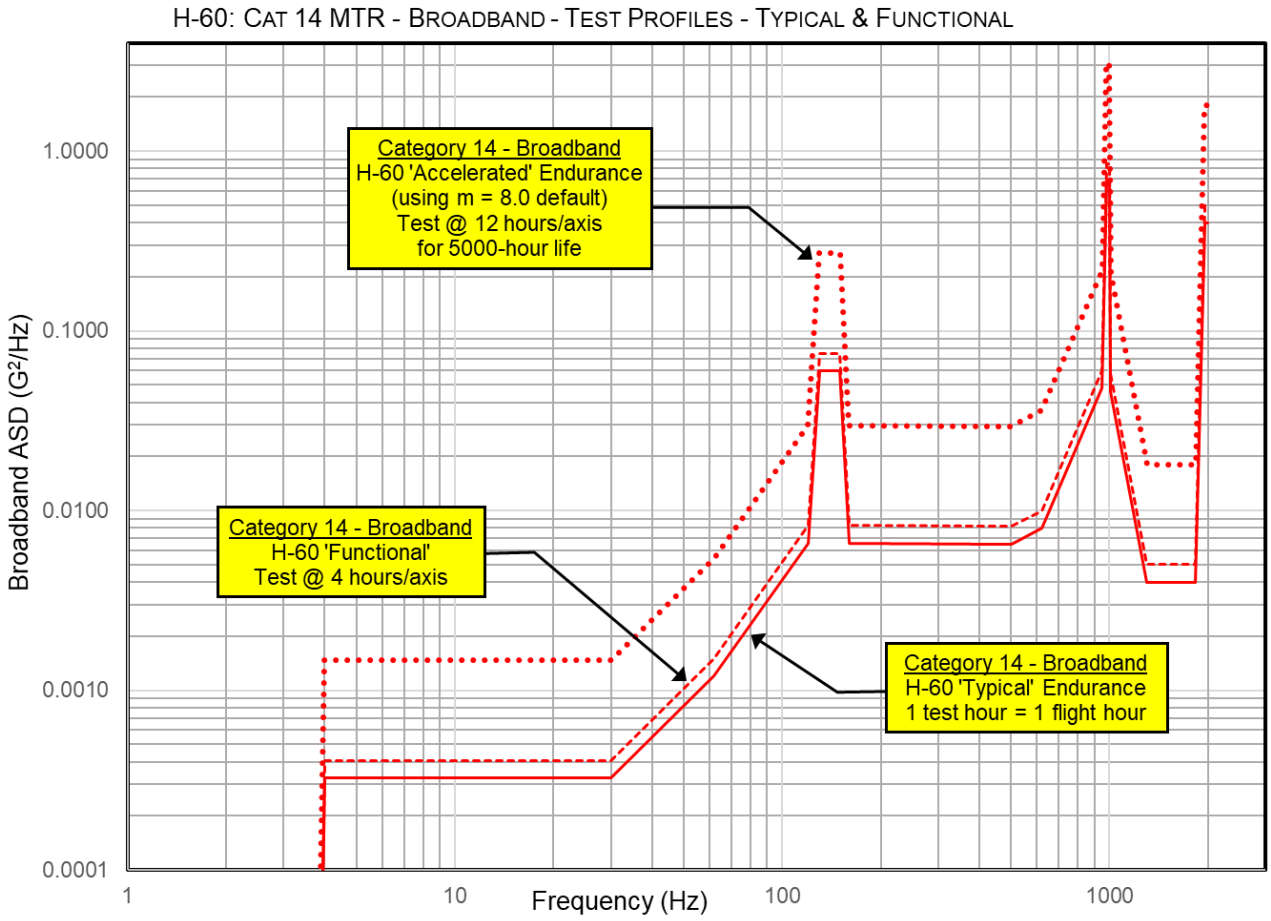


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G.3.2.3 Step 3.

Similar to step 2, the accelerated endurance profile can be brought back to a ‘worst-case’ or ‘functional’ level by scaling it by 0.25 (Table G.3.1, Row 11). This level, represented by the dashed line in the figure below, is the vibration that the component needs to tolerate for at least a single mission, which is defined herein as four (4) hours; hence a test time of 4 hours/axis.

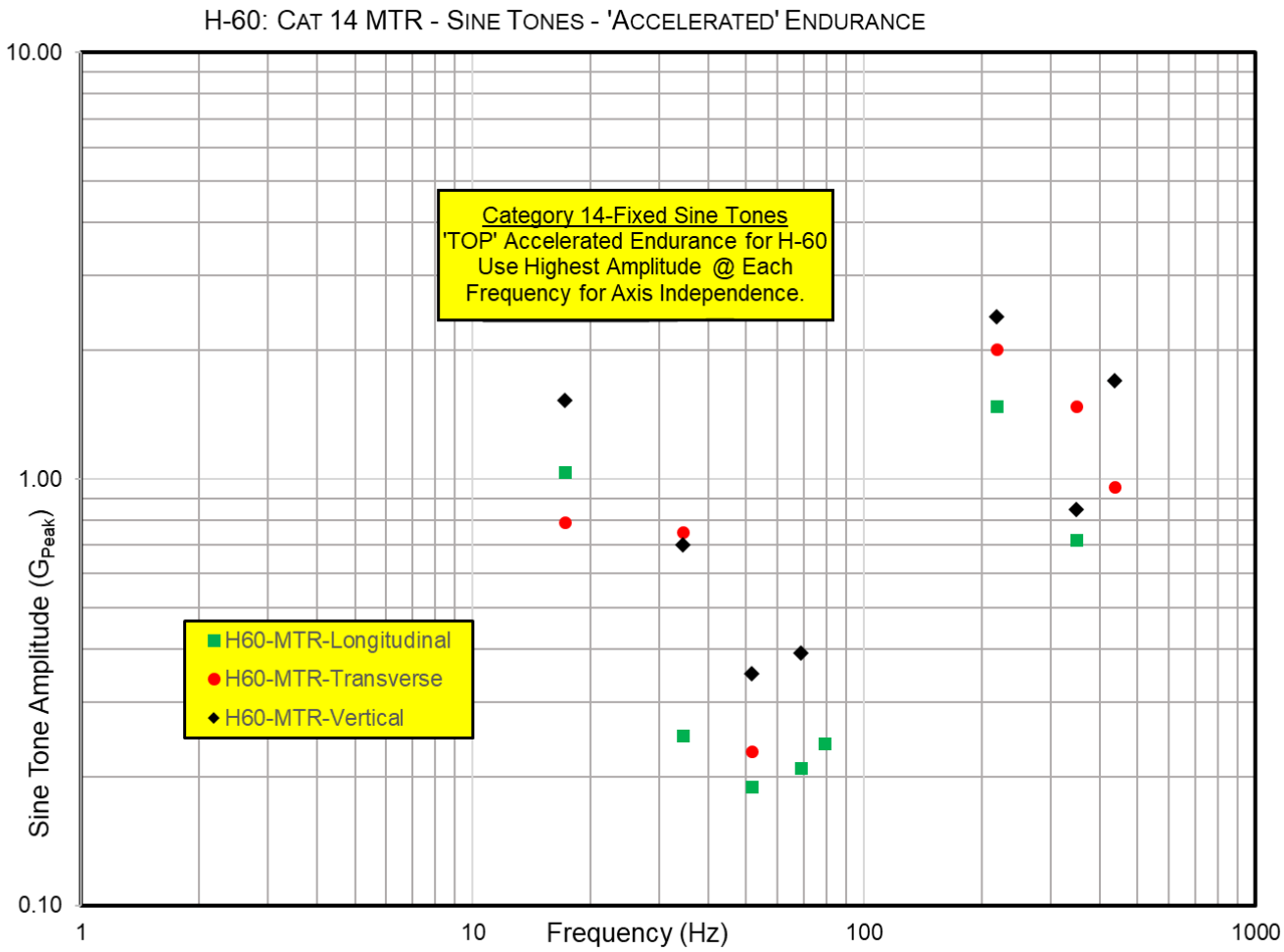
Also plotted (for illustration purposes) is an accelerated endurance profile designed to achieve a 5,000-hour service life demonstration over a 12-hour test period. Using Miner’s rule with a default m-Factor of 8.0, the resulting accelerated test profile has a lower ASD than the one illustrated in step 2 because of the test time increase from 8 hours (when using the TOP accelerated profile) to 12 hours in this example.



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G.3.2.4 Step 4.

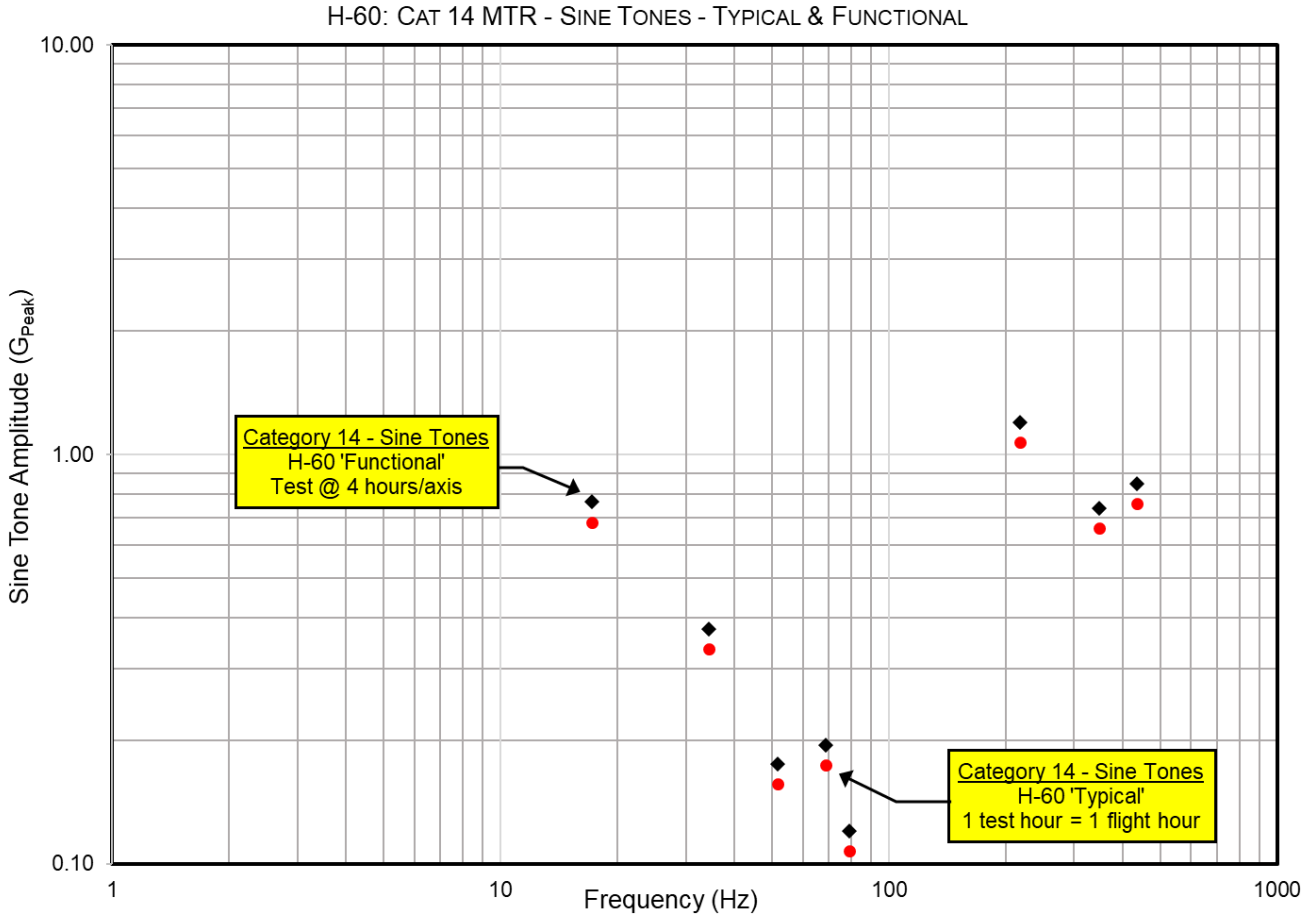
Using an approach similar to the one used to envelope the broadband TOP data, as described in G.3.2.1, the TOP sine tones at the engine/drivetrain location were plotted for the H-60. This process resulted in defining 8 frequencies, each having different amplitudes for the three orthogonal axes that were measured, as shown below. Enveloping the sine tones was therefore a simple process of choosing the highest amplitude at each of the eight frequencies so as to make the sine tone amplitudes axis independent.



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G.3.2.5 Step 5.

The final step in developing the H-60 test profile was to first 'un-accelerate' to 'typical' the maximum amplitude of the sine tones plotted in Step 4 using the scale factor listed in row 10 of the table in G.3.1 (0.447) and to do the same for the functional profile using the scalar in row 13 of that table (0.50). The two resulting sets of sine tones are plotted below. The exact frequencies and amplitudes to be used for testing are specified in TABLE XIII.



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G.3.3 Combining and splitting vibration profiles for test purposes.

(Referenced by: 5.4.3.2.1, K.3, TABLE K-II NOTE 2/)

When multiple vibration sources are imposed on a component (for example Category 14 and Category 22), they can be applied separately to a single serial number component via sequential tests, each for the full specified test time, or the profiles can be combined using the time/amplitude compression equations provided in G.2 'exaggeration' so that the test can be completed in the specified test time, but at a higher amplitude/ASD.

The general process to be used is to:

1) Reduce the test time of the lower profile sinusoidal amplitudes or random ASD values (using the above-mentioned equations) such that the resulting increased amplitudes/ASD values match those of the higher profile values.
2) Add the revised test times of the now-matched amplitude/ASD profiles. Note that test times can only be added when the test profiles are the same with respect to frequency, bandwidth, and amplitude/ASD.
3) Use the same above-mentioned equations to then reduce the total test time to the original specified time/axis by increasing the test profile amplitude/ASD values.

A simplified example demonstrates the basic process. Given the requirement of a 4-hour/axis functional test at 0.030 g^2/Hz over the bandwidth of 10-2000 Hz due to engine vibration (Cat 22), and a 4-hour/axis functional test at 0.025 g^2/Hz over the same bandwidth due to platform vibration (Cat 14), the result of employing the adjustment steps above yields a 4-hour/axis test at 0.033 g^2/Hz. Calculations are illustrated below:

1) Adjusting the test time of the lower profile ASD to match the ASD of the higher profile: $[(0.030\ g^2/Hz) \div (0.025\ g^2/Hz)]^4$ = 2.074 (time compression factor). 4 hours/2.074 = 1.93 hours/axis at 0.030 g^2/Hz which is equivalent to 4 hours/axis at 0.025 g^2/Hz.
2) Adding the test times of the now-matched ASD profiles: 4.0 hours/axis + 1.93 hours/axis = 5.93 hours/axis at 0.030 g^2/Hz.
3) Reducing the total test time of 5.93 hours to the original specified 4 hours/axis: $\sqrt[4]{(5.93/4.0)}$ = 1.103 (ASD adjustment factor). 0.030 g^2/Hz x 1.103 = 0.033 g^2/Hz for 4 hours/axis, or a 10% increase in the ASD to compensate for the platform vibration energy.

For m-Factor = 8.0 (see G.3.1.4), the maximum possible ASD increase for combining two independent vibration profiles having the same specified test time is 19%, which occurs when the ASDs for the two sources are equal, as seen on the following plot, FIGURE G.3.3-1. This plot can be used to quickly identify what frequency ranges need to have their ASD increased when combining vibration profiles with equal specified test times. Inspecting the chart at 0.833 (0.025 g^2/Hz)/(0.030 g^2/Hz) on the X-axis, the Y-axis value is 1.103, which when multiplied by 0.030 g^2/Hz yields 0.033 as the increased ASD for combining the assumed Cat 14 and Cat 22 ASD profiles; the same result as in the above example.

Note that for the sine tones (FIGURE G.3.3-2) the same process is used, however the default compression factor is 7.125 (see G.3.1.4). Consequently, only when the lower sine tone amplitude approaches approximately 89% of the higher value does it have a significant effect on the overall test amplitude. The maximum amplitude increase is 10.5%, which again occurs only when the two sources of vibration are equal, as seen on the plot.

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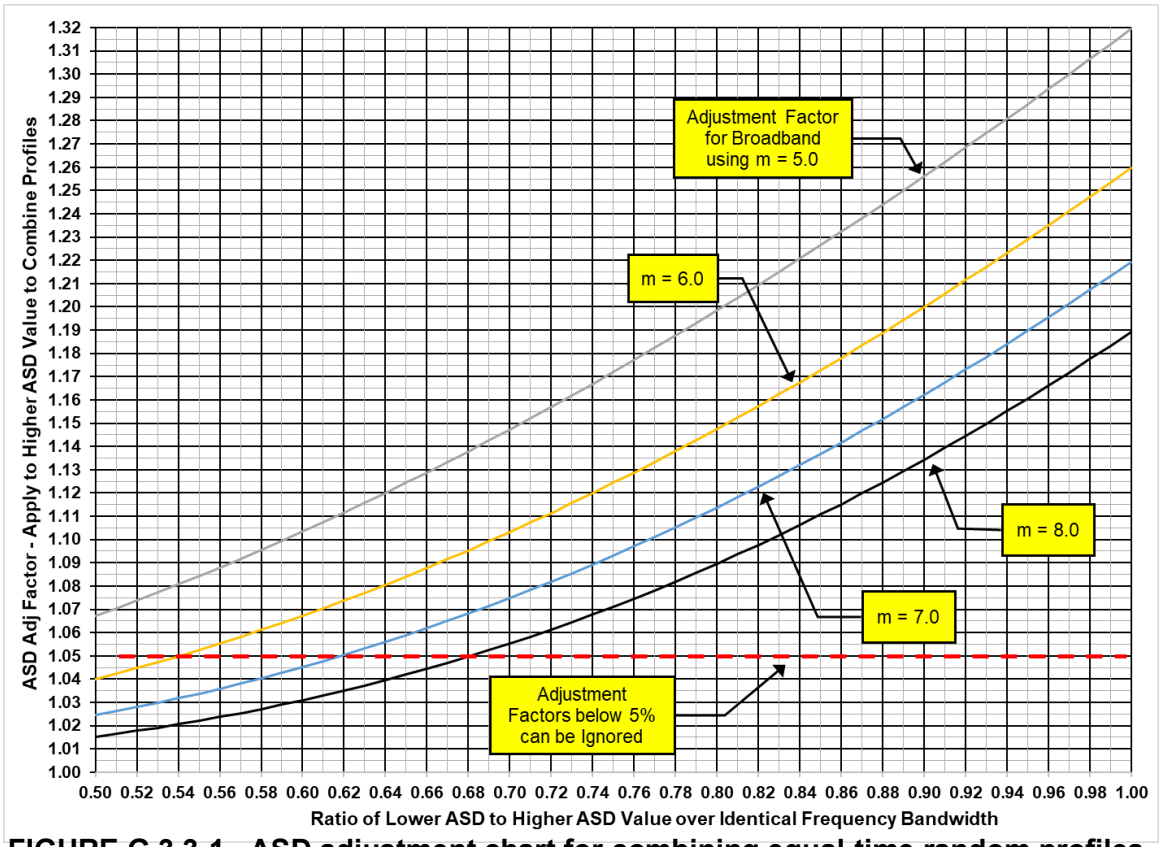


FIGURE G.3.3-1. ASD adjustment chart for combining equal-time random profiles.

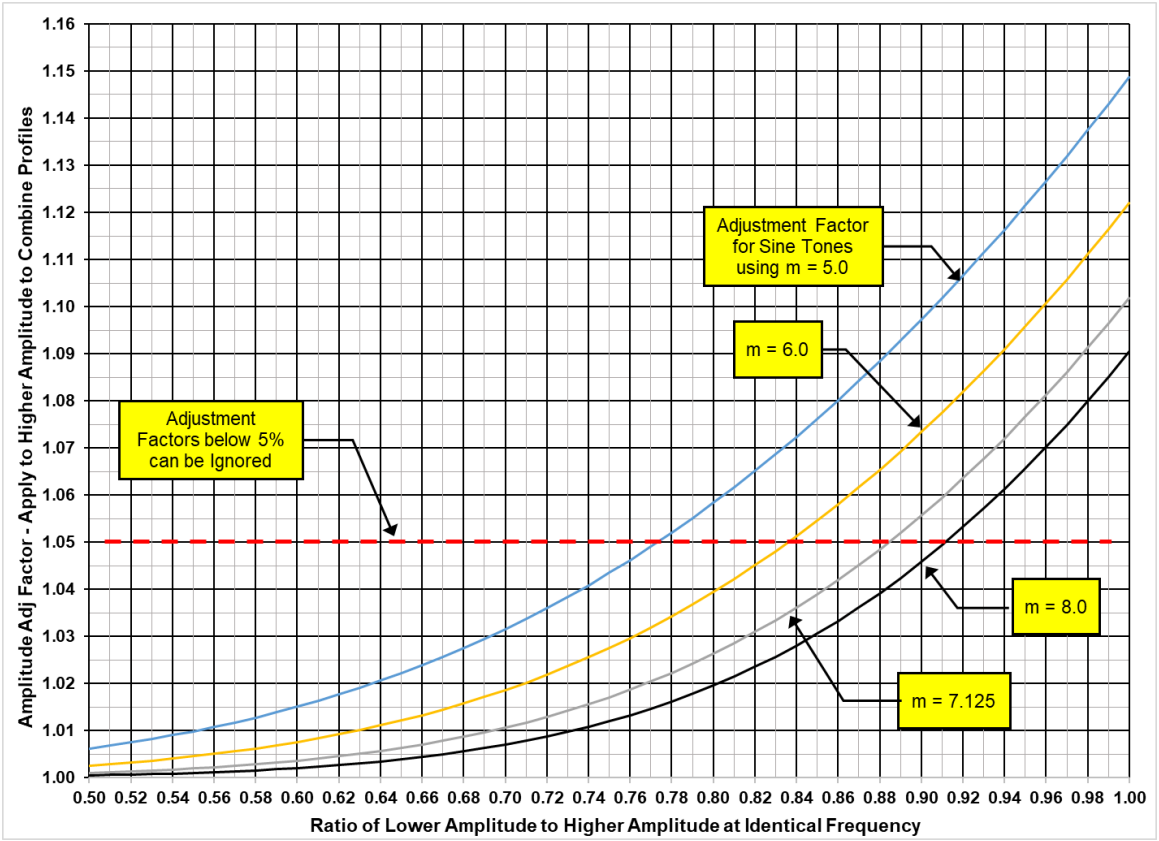


FIGURE G.3.3-2. Amplitude adjustment chart for combining equal-time sine tones.

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MISCELLANEOUS TEST GUIDANCE TOPICS

H.1 SCOPE

This appendix provides guidance and discussion on various testing topics. The information contained herein is intended only to assist the contractor in complying with the requirements defined in the referenced paragraphs; other methods may provide equivalent results.

H.2 GUIDANCE

The topics listed below are discussed with the intent of clarifying testing specified in this standard. Some of this information is the result of answering contractor questions during component development programs, while other information is simply for the benefit of those not familiar with the topic in question.

Nevertheless, regardless of the effort to be clear and concise in specifying the requirements herein, it is recognized that there will always be questions regarding engine control components and accessories qualification, and contractors are therefore encouraged to contact the cognizant Airworthiness Authority to elicit further guidance whenever necessary.

H.2.1 Overheat test guidance

H.2.2 Environmental cycle definition

H.2.3 Component electrical bonding

H.2.4 Explosive atmosphere applicability decision tree

H.2.5 Voltage spike application

H.2.6 Tailoring of CS116

H.2.7 Component categories and classification

H.2.8 Test readiness review sample content

H.2.9 Torsional stability data

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H.2.1 Overheat test guidance.

(Referenced by: 5.6.4.3)

If the ECU is operated in an increasingly hot environment beyond its maximum allowed operating temperature, it is expected that the unit will eventually fail. Because several component parts may fail simultaneously, and simultaneous failures are not usually analyzed, there exists the possibility that the failure mode of the unit may not be 'safe' with respect to the engine or the airframe. The following are some general guidelines on the definition of 'failure', test operating conditions, and test conduct/monitoring.

H.2.1.1 Test failures.

a. For twin-engine helicopters, an engine shutdown is considered a failsafe condition for the purposes of this test. It is recognized that if the helicopter is being flown in a region of the operating envelope where single-engine power is insufficient to maintain continued safe flight, an IFSD would therefore not be considered acceptable (as would be the case for a single-engine machine).

b. If the FADEC system is designed with a 'fail fixed' feature, then by definition any ECU failure which places the control in a 'fixed' power (or fuel flow) condition is also safe, providing the engine can still be shut down by the crew as needed.

c. The engine's power cannot increase beyond any safe operating limit, in particular the maximum allowed gas generator speed and maximum allowed power turbine speed.

d. The engine cannot drive the helicopter's main rotor beyond its safe operating speed.

H.2.1.2 Test operating conditions.

a. The ECU is operated in a closed-loop environment that simulates a worst-case operating condition.

b. Based on the definition of failure in H.2.1.1, the worst-case operating condition would be when the helicopter is being flown single-engine as the possible result of battle damage/fire resulting in an overheat condition on the remaining operational engine.

c. The flight condition should be the lowest single-engine power at which safe flight can be maintained, given a low gross weight helicopter.

H.2.1.3 Test conduct.

a. The ECU should be placed in an oven and brought up to its maximum allowed operating temperature and then allowed to stabilize at that temperature. All the while, the unit should be operated closed-loop at the condition defined in H.2.1.2, and monitored for proper operation.

b. The ambient temperature should then be incremented by 10ºC (or other reasonable increment), allowing the unit to stabilize per some defined criteria.

c. After stabilization has been achieved, the unit should be run for 10 to 20 minutes while verifying continued proper operation, after which the temperature is stepped once more.

d. Steps b and c are repeated until unit failure occurs.

e. Hold at the failure temperature for at least five minutes to verify that no other failure modes occur that are considered critical to continued fail safe operation, including engine re-start, re-acceleration, or termination of fail fixed, if so designed.

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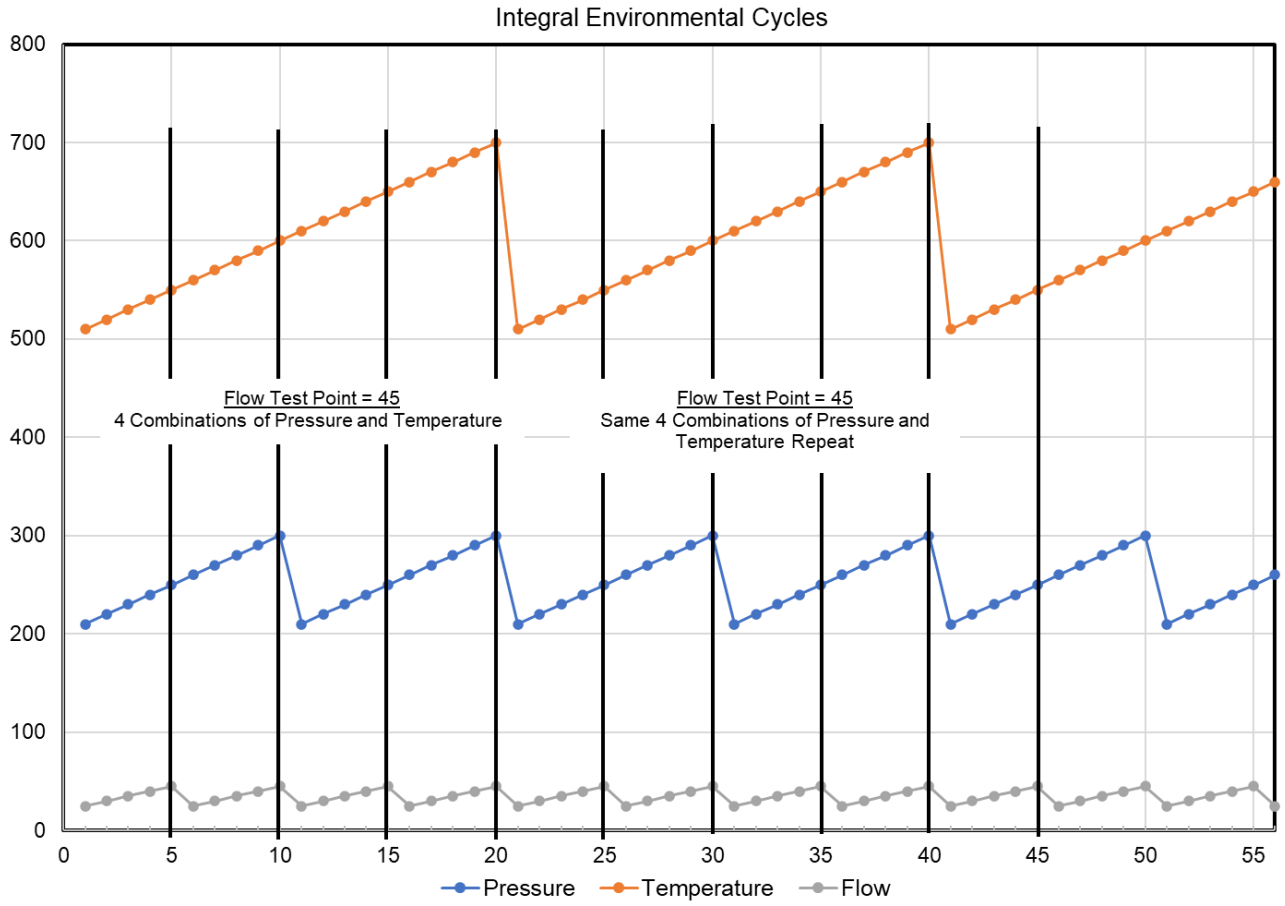
H.2.2 Environmental cycle definition.

(Referenced by: 5.3.1.1d)

Environmental parameters or input variables that are independent of each other, such as pressure altitude, oil temperature, air temperature, etc., should be varied at non-integral multiples of each other so that every input signal or parameter eventually will have functioned at each value of the independent environmental or input variables. The use of integral cycles is not an acceptable approach during testing since it greatly reduces the number of parameter combinations to which the component is exposed. Results of using integral and non-integral test environment parameters are illustrated in the sections which follow.

H.2.2.1 Integral environmental cycles: example.

When the applied environmental cycles are integral multiples of each other, as illustrated in the figure below, it results is a very limited set of unique combinations of pressure and temperature for any value of flow, no matter how long the test runs. In this example, the same combinations repeat every 20 seconds, and thus each flow test point (5 in all) is exposed to only 4 unique combinations of pressure and temperature. Component behavior at other pressures and temperatures, at that flow condition, is unverified by test and therefore unknown other than by inference.

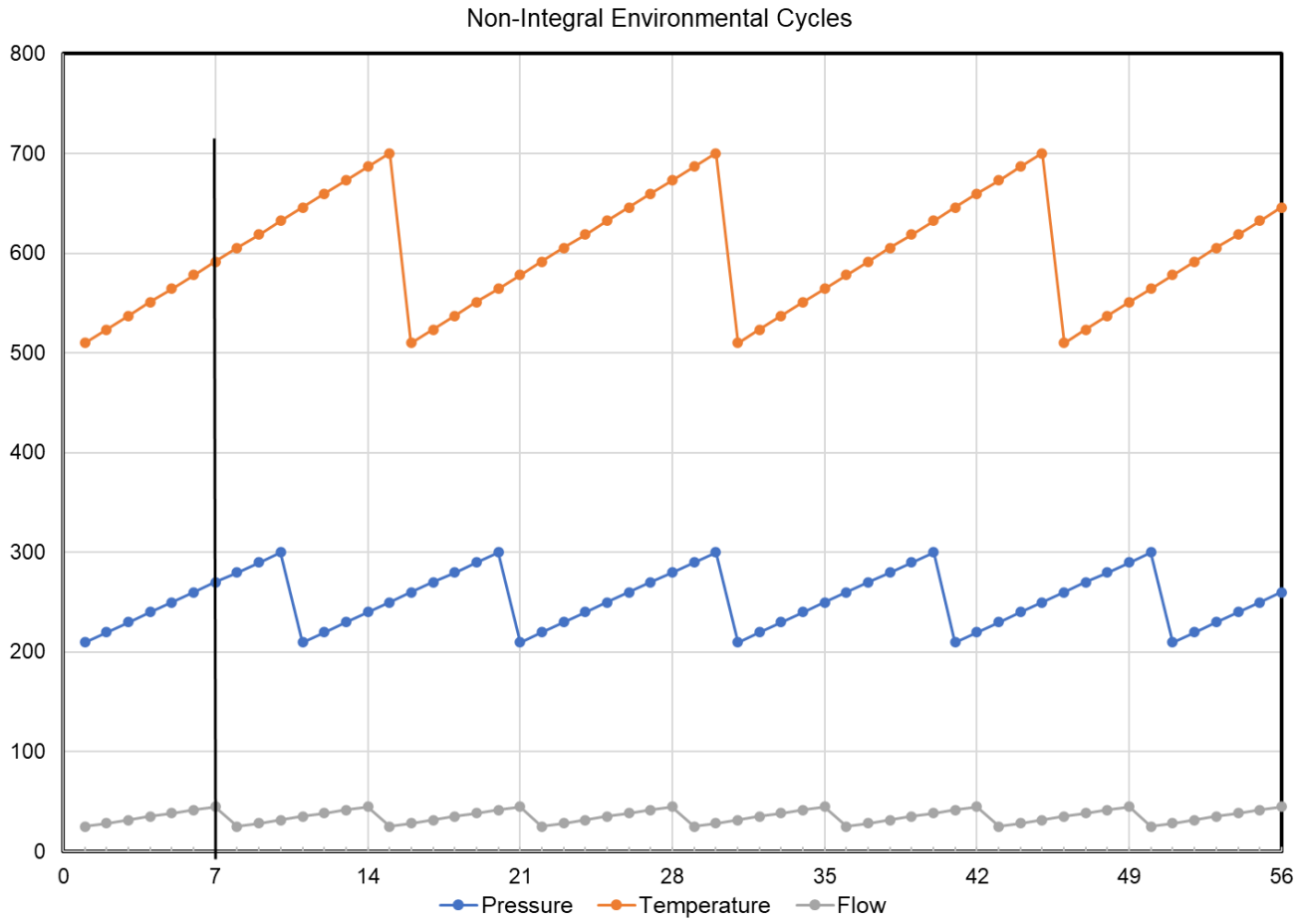


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H.2.2.2 Non-integral environmental cycles: example.

In the plot below, the cycles are not integer multiples of each other and, as a result, there are 10 unique combinations of flow and pressure (repeats every 10 x 7 = 70 seconds) and 15 unique combinations of flow and temperature (repeats every 15 x 7 =105 seconds). This results in a total of 150 unique combinations of flow, pressure, and temperature, and the overall cycle repeats every 1,050 seconds.

It is preferable that the applied environmental or operational parameters are not cyclic at all, so that more unique combinations are encountered during testing, but it is understandable that in some cases not having a repeating cycle may be difficult to implement. Consequently, as a minimum, the cycles need to have a non-integral relationship to each other as shown in the plot below.



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H.2.3 Component electrical bonding.

(Referenced by: 5.7.5)

The engine system needs to implement robust electrical bonding practices to facilitate repeatable field and test performance with respect to:

a. Electromagnetic requirements (including lightning)

b. Current handling for electrical power delivery

c. Shunting electrical power faults

d. Shunting the buildup of static electricity in air, fuel, and oil

e. Preventing unintended sparks in an explosive environment

f. General personnel and aviation safety

g. Reliable performance of the ignition system

Faying surfaces between bonded components should be designed to prevent contamination or corrosion so that declared bonding levels, needed to meet performance requirements, are maintained throughout the expected service life of the component/engine. Interfaces which are incapable of providing the bonding necessary for safe and reliable operation should employ alternate primary means of meeting bonding requirements, such as the use of a ground strap.

Bonding reliability and longevity considers the effects of environmental requirements (such as temperature, humidity, rain, wash, and salt fog) and CBRN requirements (chemical, biological, radiological, and nuclear) imposed on the engine. It also considers the effects of maintenance practices under realistic fielded conditions (for example, how well-bonded joints are cleaned during reassembly).

The interface control document can be used to prescribe a bonding control program of the faying surfaces between the engine and airframe, to be implemented on a periodic basis, such as during ‘phase’ maintenance of the aircraft and/or engine.

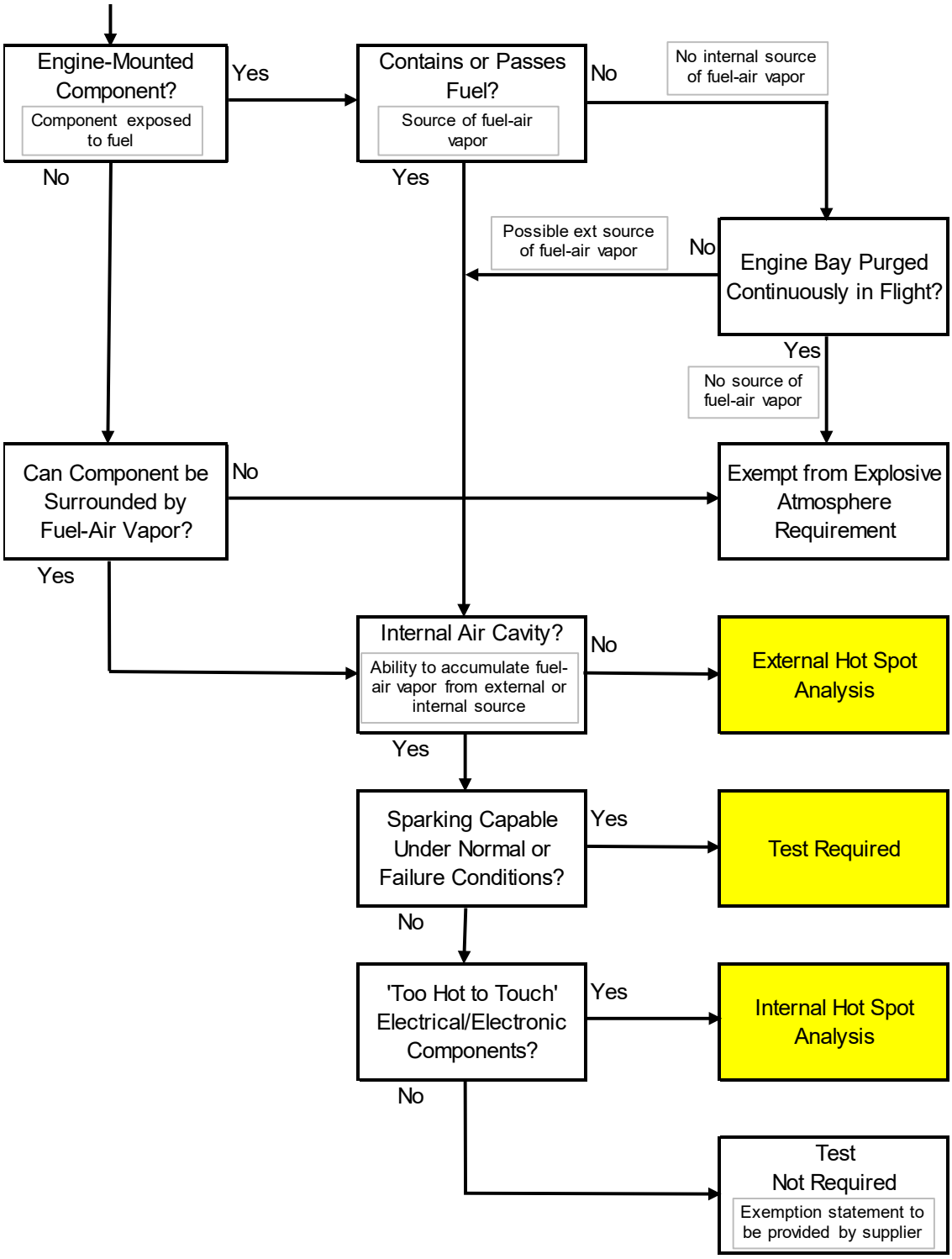
Paragraph 5.11 of MIL-STD-464D describes general bonding requirements. In general, bonds for the management of radio frequency (RF) energy and lightning currents achieve maximum effectiveness at the lowest resistances. It is understood that component materials in an engine environment may not be capable of maintaining bonding requirements of 2.5 milliohms over time. In these cases, the goal is to achieve *attainable and reliable* bonding resistances using MIL-STD-464 for guidance, which will allow the engine to meet all of the performance requirements in 5.7 over the service life of the engine.

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H.2.4 Explosive atmosphere applicability decision tree.

(Referenced by: 5.4.9, TABLE II NOTE 8/)

This tree is meant as an aid in determining the applicability (to both on- and off-engine mounted components) of the MIL-STD-810H(C1) requirement prohibiting a component-initiated explosion when surrounded by a fuel-air vapor mixture. When a component, location, or usage is unique, consultation with the Airworthiness Authority may be necessary to determine test applicability.



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H.2.5 Voltage spike application.

(Referenced by: 5.1.6a)

Voltage spikes are specified in both military and civilian requirements documents. MIL-STD-704A, 6.8.1.1, discusses 600 V spikes lasting up to 50 µs while MIL-STD-704A, Notice 1, Figure 17, provides a graph of pulse width versus amplitude which reaches 600 V at 10 µs. The 50 µs pulse only reaches 200 V amplitude on that graph. The current delivering characteristics of this spike are not discussed, resulting in some component suppliers concluding that this spike needs to be driven by substantial power supplies, which are capable of destroying surge suppressors in the process. That is not the intent of this test.

RTCA/DO-160G, Section 17, Category A, specifies a 10 µs spike with an open circuit voltage of 600 V, delivered by a spike generator with an output impedance of 50 ohms. Therefore, current is limited to 12 Amperes by the 50-ohm output impedance of the generator. However, there is no mention of a 50 µs spike in this section.

The fact that MIL-STD-704A does not specifically state that 600 V is an open circuit requirement is indicative that it is a commonly understood limitation. In addition, the 10 µs requirement fits on the graph provided in MIL-STD-704A.

The spike requirement defined herein (5.1.6a) is a harmonization of the 10 µs spike specified by RTCA/DO-160G and the 50 µs spike discussed in MIL-STD-704A, while setting the amplitudes IAW Figure 17. Both spikes fit on the Figure 17 graph in MIL-STD-704A, Notice 1. While spike generators which can deliver 600 V for 10 µs are common, those which can deliver 200 V for 50 µs are not so ubiquitous. Therefore, application of the 200 V/50 µs spike may be considered for waiver if test equipment limitations present themselves. However, this should not be construed as a blanket authorization to delete this test requirement.

H.2.6 Tailoring of CS116.

(Referenced by: 5.7.2, TABLE XVIII)

CS116 constitutes the current injection component of EMP. In 5.7.2, the test level is specified to be increased by 50% over the test limit specified in CS116-2 in order to mitigate the possible need for a supporting analysis in the case where the predicted platform levels exceed the nominal 100% test level. The 50% increase over nominal is deemed to be sufficient to close the gap between predicted and tested levels in all but the most unusual circumstances.

How to achieve this 50% ‘over test’ concerns both test calibration and imposed current limits. Specifically, this test calibrates an impulse generator with a current injection probe using a fixture terminated by 50 ohms on each end, resulting in a 100-ohm loop impedance.

One end utilizes a 50-ohm termination resistance while the other end comprises an oscilloscope with a 50-ohm input impedance. The current injection probe is clamped around a wire located between the 50-ohm termination resistance and the 50-ohm input impedance to the oscilloscope. After placement, the amplitude of the impulse generator is adjusted until the current requirement (corresponding to the test frequency) is achieved IAW CS116-2. The adjustment for that frequency is then noted for later use.

During test, the impulse generator is adjusted until either the amplitude (noted during calibration) is reached or the current (as measured by a monitoring probe) reaches a limit (whichever is lowest). It is this current limit that is increased by 50% over that specified in CS116.

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H.2.7 Component categories and classification.

(Referenced by: TABLE III, TABLE IV, TABLE VII, TABLE XVI, TABLE XVII, TABLE XX)

The following guidance is provided to assist in categorizing engine control system components to be qualified. In some cases, a component may not be easily slotted into a single category with its associated test requirements. When more than one category definition seems to 'fit' the component, it is appropriate to tailor the requirements by combining and/or deleting the tests from the different categories. The component test matrices are generic in nature and may not cover every conceivable component or performance specification requirement. Ultimately, resolution of any confusion on which test is to be conducted for any specific component is to be concurred by the Airworthiness Authority and detailed in the AQP/AQS.

a. System. 'System', as used herein, is meant to encompass all parts of the engine control or accessory system which interface in one way or another to a main ECU. This includes both on- and off-engine sensors, airframe interfaces such as cockpit displays, data buses, discretes, command inputs, etc., and all of the interconnecting harnesses. For a system test, every pin in every electrical connector has something connected to it. However, not all of the items need to be actual hardware, and can sometimes be simulated when the item itself is not under test.

b. Electronic unit. This category includes analog or digital controllers (such as an engine or propeller ECU), and other electronic units that may use a processor, micro-controller, or programmable logic device (PLD). Other electronic items that do not perform a control function nor contain 'processors' (e.g., an ignition exciter), may also be grouped into this category.

c. Fuel handling. These components include typical pumps (boost or main) and metering devices (FMU or HMU). Other items in this category include fuel filters and related components.

d. Oil handling. Components would include pumps, reservoir tanks, heat exchangers, and oil filters. A fuel/oil cooler would be included in the oil handling group, not the fuel handling group. Sensors associated with the oil system (e.g., debris monitor, oil pressure, oil temperature, etc.) may be tested with this group of components, depending on the complexity. Additional testing may be required under the category of 'complex' sensors.

e. Sensors. Sensors only include those mounted on the engine or in the nacelle; airframe sensors are typically not part of propulsion system qualification. However, any airframe sensors interfacing with the ECU would be part of the 'system' when undergoing system-level tests. This category is further subdivided into 'simple' and 'complex'. A speed monopole is an example of a simple sensor, while a position sensor with moving parts such as a linear variable differential transformer (LVDT) falls into the latter category. Ignitors would also be included in the simple sensor category, as would a 'smart' sensor that might have a self-contained 'chip' to perform some simple function.

f. Actuators. Typical actuators would include an anti-ice/start bleed valve, a separately mounted variable geometry actuator powered by fuel, or an electro-mechanical device such as a solenoid or stepper motor driven shutoff valve.

g. Electrical harness. This category includes all engine electrical harnesses up to the engine compartment firewall. Although a power turbine inlet temperature (PTIT) or T4.5 thermocouple assembly is often referred to as a 'harness', it is also a sensor and therefore tests required for that category should be considered.

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H.2.8 Test readiness review sample content.

(Referenced by: 5.1.1)

Part 1 test plan TRR:

a. Overall test plan

   (1) Test plan review

b. Test discrepancy handling

   (1) Describe the test discrepancy identification, reporting, and remediation process to be followed.

c. Identify part 2 TRRs

   (1) Identify which test procedures should be reviewed in detail, just prior to execution, in order to avoid mistakes that might require costly re-test.

   (2) Establish a general schedule for the part 2 TRRs.

Part 2 test procedure TRRs:

These TRRs are not meant to be formal since the purpose is to walk through the procedure in as much detail as needed so that test mistakes are avoided. The subject of the review is the actual procedure which is to be followed by the test facility personnel, not any of the 'boiler plate'. The test procedure should be reviewed and discussed, section by section, and any areas which may be ambiguous, clarified and agreed with the Airworthiness Authority.

a. The cognizant contractor engineer(s), test facility personnel, and government personnel involved in the test procedure review/approval process should participate in the review.

b. Component maturity level

   (1) Summary of any informal testing performed on the component that is related to the TRR subject testing

   (2) Component conformed to design drawings/specifications, and under configuration control

   (3) Component quantity and availability for test (have units passed ATPs?)

c. Test facility readiness

   (1) Is the test schedule reflective of available resources, and have the assets been reserved to ensure availability during the test period?

   (2) Have the detailed component test procedure(s) been reviewed with the test facility personnel?

   (3) Can calibrated instruments, when specified for use, complete the test schedule within their expiration dates?

   (4) Have test setup 'dry runs' been conducted to assure that complex tests (e.g., contaminated fuel testing, operational vibration at temperature) are being properly executed with respect to automated timing of parameter changes, length of dwells, data acquisition, synchronized event timing, etc.?

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H.2.9 Torsional stability data.

(Referenced by: 5.8.4)

Torsional stability of the helicopter drive system should be completely analyzed, showing both gain margin and phase margin throughout the operational envelope. General guidelines are described in ARP 704A.

Representative open- and closed-loop Bode plots of the power turbine speed governor loop should be included for worst-case gain and phase margin conditions of gas generator speed, gross weight, airspeed, outside air temperature, et.al. Flight test data and simulation results should be submitted to substantiate drive system torsional stability.

The aircraft manufacturer needs to provide sufficient data to permit satisfactory independent assessment of the dynamic characteristics of the helicopter drive system. This data typically includes:

a. A complete drive system simulation diagram similar to FIGURE H.2.9-1.

b. Values of all applicable drive system parameters, including those in TABLE H.2.9-I. This data is either referred to the square of the power turbine speed, or includes the ratio of the speed at which the parameter is evaluated to the speed of the power turbine.

c. Drive system parameters which are not constants need to be defined throughout the operational ranges of gas generator speed, gross weight, airspeed, outside air temperature and other relevant independent variables. Examples of variable drive system parameters are: lag hinge damping and/or elastomeric dampers, and the main and tail rotor aerodynamic damping coefficients.

The engine manufacturer and/or the engine control system designer must provide sufficient data to allow satisfactory independent assessment of frequency response characteristics associated with the engine and engine control system. These include:

a. A complete description of the engine and engine control system transfer functions used for the torsional stability analysis.

b. The polar moments of inertia of the gas generator rotor(s) and the power turbine.

c. Gains and time constants for each control component.

d. Partial derivatives for significant engine parameters as a function of gas generator speed (Ng) with sufficiently fine resolution in Ng to allow accurate determination of locations and magnitudes of minimum gain margin and minimum phase margin.

e. A block diagram of the linearized model used in the stability analysis.

f. A block diagram of the nonlinear model showing the engine systems, engine control system components, and their interface with each other and the drive system with respect to torsional stability. Closed-loop step responses of the nonlinear model power turbine governor loop, corresponding to worst-case conditions should be included. The amplitude of the step input needs to be sufficiently large to exercise the full range of all nonlinear elements in the system model.

g. Assumptions used in the development of the mathematical models of the digital control loops in the system require clear definition. Appropriate values for sample-hold and computational delays associated with the frame rates for each digital control loop, and all digital-to-analog and analog-to-digital conversions require substantiation.

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TABLE H.2.9-I. Helicopter drive system parameters.

Npt1,2	Engine power turbine speed
Dpt1,2	Engine power turbine damping
Jpt1,2	Engine power turbine inertia
Kpt1,2	Engine power turbine shaft spring constant
Khs1,2	High speed shaft spring constant
Dhs,2	High speed shaft damping coefficient
Jgb1,2	Angle gearbox inertia
Kmt1,2	Main transmission input spring constant
Dmt1,2	Main transmission input damping coefficient
Jmt	Main transmission inertia
Ktr	Tail rotor shafting spring constant
Dtr	Tail rotor shafting damping coefficient
Jhtr	Tail rotor hub inertia
Kctr	Tail rotor centrifugal spring constant
Dhtr	Tail rotor lag hinge damping coefficient
Jtr	Tail rotor inertia
Datr	Tail rotor aerodynamic damping coefficient
Ntr	Tail rotor speed
Kmr	Main rotor shaft spring constant
Dmr	Main rotor shaft damping coefficient
Jhmr	Main rotor hub inertia
Kcmr	Main rotor centrifugal spring constant
Dhmr	Main rotor lag hinge damping coefficient
Jmr	Main rotor inertia
Damr	Main rotor aerodynamic damping coefficient
Nmr	Main rotor speed

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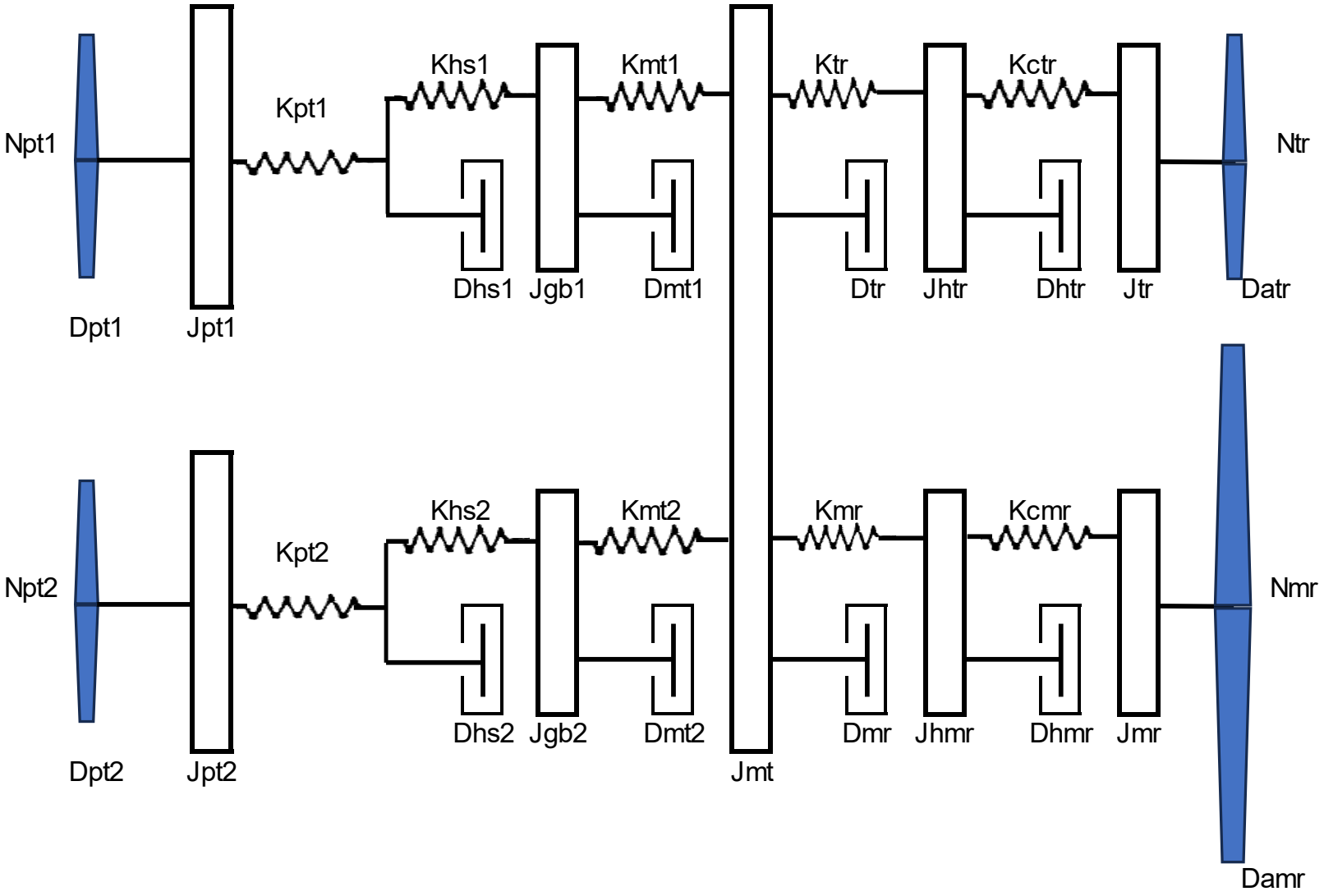


FIGURE H.2.9-1. Helicopter drive system simulation example.

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DOCUMENTATION GUIDELINES

(Referenced by: 5.10)

J.1 SCOPE

This appendix provides guidance on the preparation of test planning and reporting documentation. The information contained herein is intended only to assist the contractor in complying with the requirements defined in the referenced paragraph; other methods may provide equivalent results.

J.2 GUIDANCE

The topics listed below are discussed with the intent of meeting documentation requirements specified in this standard. Information herein is the result of reviewing, rejecting, and ultimately approving component test documentation from countless engine and accessory development programs over the past 30 years.

Contractors need not follow this guidance verbatim since they may have a 'better idea', but this provides a reasonably efficient structure for getting plans and reports to the finish line with the least effort.

J.2.1 Acceptance test procedures
J.2.2 Pass/fail criteria
J.2.3 Test plans/procedures
J.2.4 Reports
- **J.2.4.1** Test
- **J.2.4.2** Similarity
- **J.2.4.3** Analysis

J.2.1 Acceptance test procedures.

At the Airworthiness Authority's option, approval of any individual ATP may be waived until completion of the qualification program. In that case, ATPs should be maintained under configuration control so that all pre- and post-test component performance checks are conducted using the same ATP revision. Use of different ATP versions, pre- and post-test, may complicate the pass/fail decision process, depending on the differences between the two ATPs.

For the purpose of requirements traceability, each ATP pass/fail limit or criteria should identify the specific paragraph of the customer or internal design document where it can be found, or if not specifically identifiable, then how it was derived.

J.2.2 Pass / fail criteria definition and traceability.

In some instances, the pass/fail criteria may be 'relative' rather than ATP absolute. An example might be an ATP requirement for a solenoid coil to measure between 50 ohms and 100 ohms. A pre-test value of 50 ohms and a post-test measured value of 100 ohms would 'pass' the ATP on an absolute basis, but this large change in pre- and post-test values clearly indicates a failure of some type. Pass/fail criteria should consider these types of additional limits beyond simply passing the ATP.

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These criteria should be presented in a table indicating the signal, parameter or characteristic, the pass/fail limit (absolute or relative to the actual value), the source of the pass/fail limit (ATP or other customer document), the paragraph number or other identifying document reference, and the method of calculating or obtaining the pass/fail limit from the performance requirement. The use of ATP 'service' limits versus 'new part' limits as pass/fail criteria will be allowed on a case-by-case basis after review by the Airworthiness Authority. Unless approval is obtained to use service limits, new part limits should apply.

J.2.3 Test plans and procedures.

For each component to be qualified via test, there should be:

a. one plan prepared for the environmental group of tests (5.4). The E^3 plan is to be prepared IAW DI-EMCS-80201 and the software test plans IAW AMACC 15.0.

b. one plan prepared for each group of tests specified in 5.3 and 5.5.2.

c. dedicated plans for each of the tests specified in 5.5 and 5.6.

The test plan organization specified above does not apply to PFR (5.2), for the reason that the number of tests required are very limited in comparison to qualification. Consequently, the contractor is not asked to adhere to these requirements and can propose any test plan/procedure organization and submittal process that is both streamlined and cost effective.

J.2.4 Reports.

J.2.4.1 Test reports.

Test reports should be submitted in accordance with test plan organization; i.e. there should be one test report for every test plan. In some cases, the test report for a given test plan might be split into more than one volume, but there should be no cases where a test report traces to more than one test plan.

J.2.4.2 Similarity argument reports.

Differences between the 'similar' and actual component with regard to the test environment, failure criteria, or test method need to be clearly described and summarized for each test for each component. If test failures occurred during qualification, these are to be documented, including corrective action and successful retest. The test reports for the qualified/certified component should also be submitted so that the similarity report conclusions/recommendations can be independently assessed.

If the component is in service use, the differences between the planned and actual service environments should be clearly described, indicating how the outcome of the tests specified in the AQP might be affected by these service environment differences; that is, can the specified test discriminate between the different service environments? Additionally, the number of in-service unit hours, and the number and type of failures should be provided, along with the predicted and actual mean time between failure (MTBF) for the component, if possible. This information can bolster the claim that component field service militates against the need for additional qualification testing.

Similarity arguments should account for all differences between the two components. Any aspect of the two components that is not identical needs to be discussed and a rationale provided as to how the non-identical aspects might affect qualification test outcome. Any differences in manufacturing processes, cleaning methods, protective coating methods, materials, etc., are considered to be non-identical aspects of the component.

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Documentation includes drawings which compare the two components, part-to-part, clearly depicting the similarities and differences between the parts. For reference, FIGURE J-1 illustrates a 'flowchart' based on the content requirement of the referencing DID in 5.10.4.2.

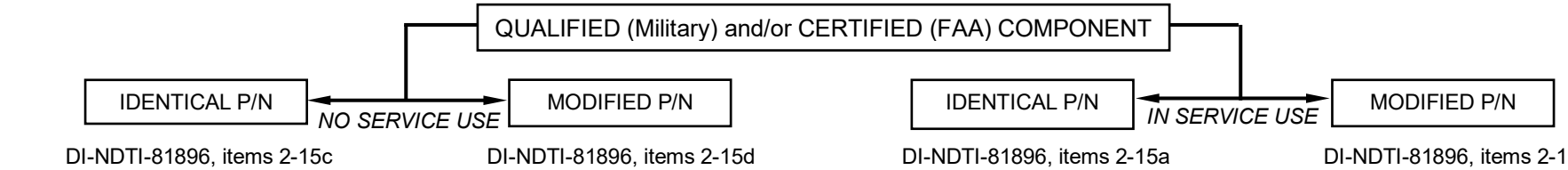
Similarity discussions/descriptions may be supplemented with supporting analyses in accordance with 5.10.4.3 for each non-identical aspect of a similarity claim. When test reports are included, easy to follow references to specific portions of those test reports are needed in the main body of the analysis to assist the reviewer in determining the outcome of the original testing.

J.2.4.3 Analysis reports.

If an analysis is the method chosen to request a test waiver, the analysis should stand alone. That is, the submitted analysis cannot refer to previously accepted analyses or include those as part of the submitted report. The analysis pertains only to the component and the specific test(s) under consideration.

Each report should contain items 2 through 9 of DI-NDTI-81896 (see FIGURE J-1), followed by the analysis for the body of the report. An analysis summary for each component, for each test, should be provided early on, describing the basis or theory of the analysis, how it will be conducted or performed, and what data or information will be provided in the report submitted for Airworthiness Authority approval. Airworthiness Authority concurrence with the summary information will minimize the contractor's risk of non-approval when the analysis report is submitted.

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DI-NDTI-81896, items 2-15c

Physical/Functional Description for Existing Component:
- ❑ Photos/Drawings
- ❑ Schematics
- ❑ Cross sections

Description of Tests Required for Qualification

Summary of Testing Actually Performed:
- ❑ Test name
- ❑ Procedure (military/FAA/in house)
- ❑ Results for each test

+

DI-NDTI-81896, item 16c

Detailed Description of Test Performed:
- ❑ Date of test
- ❑ Test environment/procedure
- ❑ Pass/fail criteria
- ❑ Significant findings/ATP results
- ❑ Comparison of actual test vs. required test

Actual Test Reports as Appendices
- ❑ Annotate as required

DI-NDTI-81896, items 2-15d

Physical/Functional Description for Existing and Modified Component:
- ❑ Photos/Drawings
- ❑ Schematics
- ❑ Cross sections

Description of Tests Required for Qualification

Summary of Testing Actually Performed:
- ❑ Test name
- ❑ Procedure (military/FAA/in house)
- ❑ Results for each test

+

DI-NDTI-81896, item 16c

Detailed Description of Test Performed:
- ❑ Date of test
- ❑ Test environment/procedure
- ❑ Pass/fail criteria
- ❑ Significant findings/ATP results
- ❑ Comparison of actual test vs. required test

Actual Test Reports as Appendices
- ❑ Annotate as required

+

DI-NDTI-81896, item 16d

Discuss Modifications:
- ❑ Rationale for no testing required
- ❑ Side-by-side comparisons
- ❑ Supported by analysis as per 5.10.4.3

DI-NDTI-81896, items 2-15a

Physical/Functional Description for Existing Component:
- ❑ Photos/Drawings
- ❑ Schematics
- ❑ Cross sections

Description of Tests Required for Qualification

Discuss Service Differences:
- ❑ Compare environments
- ❑ Ability of test to discriminate between environments

Service History:
- ❑ Cumulative unit hours
- ❑ Number and type of failures
- ❑ MTBF – predicted vs. actual

DI-NDTI-81896, items 2-15b

Physical/Functional Description for Existing and Modified Component:
- ❑ Photos/Drawings
- ❑ Schematics
- ❑ Cross sections

Description of Tests Required for Qualification

Discuss Service Differences:
- ❑ Compare environments
- ❑ Ability of test to discriminate between environments

Service History:
- ❑ Cumulative unit hours
- ❑ Number and type of failures
- ❑ MTBF – predicted vs. actual

+

DI-NDTI-81896, item 16b

Discuss Modifications:
- ❑ Rationale for no testing required
- ❑ Side-by-side comparisons
- ❑ Supported by analysis as per 5.10.4.3

FIGURE J-1. Qualification by similarity & analysis – process flowchart.
(Referenced by: 5.10.4.2, 5.10.4.3, J.2.4.2, J.2.4.3)

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GUNFIRE SHOCK

(Referenced by: 5.4.5)

K.1 SCOPE

This appendix provides an example of the development of gunfire shock (5.4.5) test profiles for the H-60 platform, using Procedure III as outlined in MIL-STD-810H(C1), Method 519.8, Annex C. Note that gunfire 'shock' is represented for test purposes by a sine-on-random vibration profile, and not shock pulses as defined in 5.4.4.

The information contained herein is intended to assist the contractor in complying with the requirements defined in the referencing paragraph. The aim is to provide sufficient information to allow the contractor to tailor the vibration test profiles for specific gun/platform combinations, as necessary.

K.2 DEFINITIONS

General definitions of the terminology and nomenclature found within this appendix are provided below. Specific definition may vary with contextual usage.

accelerated endurance	A test that has been 'accelerated' to reduce the time needed to prove the endurance of equipment to 'typical' vibration over a service life (i.e., one hour of test time equates to many hours of in-service use at typical vibration levels).
broadband	Random vibration spectrum that is applicable over the complete frequency range of interest for a component.
endurance	The average vibration level that a component is expected to 'endure' over its specified service lifetime. Also referred to as a 'typical', 'mission-weighted', or 'reliability' vibration level. One test-time hour equates to one in-service hour.
exaggeration	The multiple that test levels are above typical or endurance levels is referred to as the 'exaggeration' factor. The exaggeration factor is computed using the time compression algorithms (Miner's rule) specified in MIL-STD-810H(C1), section 2.2, page 514.8A-4 (Eq. 1) and page 514.8A-5 (Eq. 4). Over exaggerating may lead to unrepresentative failures, consequently the test levels (under normal circumstances) should be limited to not greater than: (1) Sinusoids: 2.0 times the functional level. The test time per axis can be calculated as follows: $\text{Test Time}_{\text{per axis}} = \text{Service Life}_{\text{hours}} \times (g_{\text{PEAK Typical Level}} / g_{\text{PEAK Test Level}})^{m}$ (2) Random profiles: 4.0 times the functional level. The test time per axis can be calculated as follows: $\text{Test Time}_{\text{per axis}} = \text{Service Life}_{\text{hours}} \times (g^2/\text{Hz}_{\text{Typical Level}} / g^2/\text{Hz}_{\text{Test Level}})^{m/2}$

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functional	Also known as 'worst-case' levels. These are the highest levels that a component is exposed to in operational service (max allowable engine vibration, exceedance of which requires engine replacement). The component needs to function properly when exposed to these vibration levels. These worst-case levels factor into the constraints used to upper bound the exaggeration factors.
m-Factor	Material exponent (m) used in Miner's Rule equations. Typical values are provided in Table 514.8F-III of MIL-STD-810H(C1). This standard uses a default of 7.125 for sine tones and 8.0 for random vibration. These default values represent a component with a 50/50 mix of steel and aluminum alloy piece parts (see m-Factor cross plot in G.3.1.4). The contractor may propose values based on the material composition of the component under test, with supporting rationale.
random	Vibration profile that is non-deterministic, comprising all sine tones of varying frequencies and amplitudes over a specified bandwidth; expressed in g^2/Hz.
service life	The expected operational usage of the equipment, expressed in engine hours, before the equipment becomes uneconomical to repair due to a failure rate inflection point.
sinusoid	A pure sine tone vibration profile, expressed in peak *g*, used to represent the dominant frequencies resulting from major rotating airframe or engine components. The sine tones may 'wander' about a small band to represent engine or airframe rotor/propeller speed ranges.
typical	The 'typical' level of vibration expected to be encountered over the service life of the equipment. Also known as 'mission-weighted', 'service' or 'endurance' level. This level is the baseline from which exaggerated levels are computed to allow for an accelerated endurance test.
worst-case	The vibration level which the equipment is able to withstand over a short period of time, while still meeting all operational performance requirements. This is not an 'exaggerated' level; one test-time hour per axis at worst-case equates to one in-service unit-hour at worst-case.

K.3 GUNFIRE ESTIMATION PROCESS

(Referenced by: 5.4.5)

When the actual gunfire environment is unknown, it may be estimated using the procedure and formulas detailed in MIL-STD-810H(C1), Method 519.8, Annex C, Part 1, 2.2 Predicting Gunfire Vibration Spectra. To illustrate the results obtained when using this methodology, profiles of gunfire-induced vibration at the H-60 platform engine/drivetrain location, using three different gun types, have been created.

The parameters and values that were used in calculating the above profiles are listed in TABLE K-I for reference. This allows the user to compute different profiles by altering the parameter values, as needed. Users are cautioned that the computed vibration spectra tabulated in TABLE K-II and plotted in companion FIGURE K-2 are not default values.

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Users should derive their own estimate of gunfire-induced vibration, at the location of the equipment being qualified, for the specific intended platform. Contractors should opt for conservatism when making assumptions regarding types of guns to be used, along with the location of said guns, relative to the location of the component in the aircraft or on the engine.

For simplification purposes, test duration for guns having both a low and high firing rate was determined by using the average of the two rates to calculate ammunition load-out depletion time. However, the profile sine tone frequencies and amplitudes use the actual low and high firing rates, and their harmonics, not the averages.

Since gunfire vibration is intermittent over the platform's life, the test times (and amplitudes) have been adjusted to be consistent with the vibration endurance test of 5.9.2. For example, in TABLE K-I the M2 machine gun has an estimated test time of 11.4 hours/axis based on the number of sorties, gun capacity, and average firing rate. However, that is meant to be spread over the platform lifetime. In these examples, a platform lifetime of 5,000 hours has been assumed, which has been compressed to a test time of 8 hours/axis for illustration purposes.

Because the assumed 5,000-hour service life test has been compressed to 8 hours, the gunfire test time of 11.4 hours was also reduced to 8 hours so that it could conveniently run concurrently with the 8-hour/axis endurance test. This reduction in test time requires only a small increase in gunfire vibration amplitude to achieve equivalent damage. This small adjustment (increase or decrease) is the reason that the amplitudes in TABLE K-II do not exactly match the values computed when using the factors in TABLE K-I, as all of those listed durations have been normalized to 8 hours.

Use of multiple guns simultaneously may be assumed in order to provide a maximal vibration environment, which necessitates adding together the broadbands for each, IAW the procedure described in G.3.3 in cases where the lower energy guns represent a significant amplitude percentage with respect to the highest energy gun.

For the sine tones, all unique frequencies are represented in the test profile, since any gun type could use any of its specified firing rates (single or dual). For the H-60, where three dual-rate gun types are assumed, this leads to 24 different frequencies: 3 guns x 2 rates x 4 frequencies (fundamental+3 harmonics). Consequently, only 4 hours per axis, instead of 8 hours, would be required for each of the 24 frequencies because the three gun types are each assumed to fire at each of their two rates for half the total test time.

To simplify testing, some of the 24 sine tones can be eliminated when the tones from one gun coincide with another gun within a reasonable frequency tolerance band, and the lower amplitude tone is less than the higher amplitude tone according to the rule of thumb of FIGURE G.3.3-2. In some cases where a given gun has an identical low rate and high rate harmonic of the same amplitude, that sine tone would have to be tested for the full 8 hours since each rate (even if the same frequency) must be tested for the 4-hour period.

For qualification, the ability of the equipment to withstand gunfire, in addition to normal engine and airframe-induced vibration, must be assessed. For this purpose, gunfire vibration should be run concurrently with the real-time, 4-hour functional vibration test of 5.4.3.2.1 and 5.4.3.2.2 (all sources simultaneously or sequentially applied to the same test unit).

Since gunfire represents only a small fraction of usage time over the life of the equipment (e.g., about 8-10 hours in 5,000), 4 hours of real-time functional testing translates to under 1 minute per axis of gunfire, if assumed to be evenly distributed over the lifetime. Since even usage is not realistic, default test time has been set at 10 (ten) minutes per gun type per axis, or 5 (five) minutes at each firing frequency if the gun is dual rate.

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TABLE K-I. Notional H-60 gunfire estimation parameters @ engine/drivetrain location.

(Referenced by: K.3, TABLE K-II NOTE 3/)

Parameter (Table 519.8C-I)	Units	H-60 Platform					
Gun	Type	M134 7.62mm 'Minigun' Dual Rate (2000-4000 rpm)		M2 50 Cal MG Dual Rate (450-600 rpm)		M240 7.62mm MG Dual Rate (700-900 rpm)	
N	# of guns	1		1		1	
F_1 (Table 519.8C-II)	shots/sec (Hz)	33.33	66.67	7.5	10.0	11.67	15.0
E (Table 519.8C-III)	joules/shot	6,700		26,000		6,700	
H (Fig 519.8C-4)	dB	0		0		0	
M	dB	0		0		0	
W (Fig 519.8C-5)	dB	0		0		0	
J (Fig 519.8C-2/6)	dB	-1		-1		-1	
B_J (Fig 519.8C-7)	dB	-25/-15.5/-11.5		-25/-15.5/-11.5		-25/-15.5/-11.5	
D	cm	120.3		120.3		120.3	
X	ft	0		0		0	
Y	ft	0.6		0.6		0.6	
Z	ft	3.9		3.9		3.9	
h/c	cm/cm	<10		<10		<10	
R_S	cm	15		15		15	
F_1 F_2 F_3 F_4 (Fig 519.8C-1)	Hz	33.33 66.67 100.0 133.33	66.67 133.33 200.0 266.67	7.5 15.0 22.5 30.0	10.0 20.0 30.0 40.0	11.67 23.33 35.0 46.66	15.0 30.0 45.0 60.0
K_1 K_2 K_3 K_4 (Fig 519.8C-8)	dB	-22.40 -21.53 -20.55 -19.30	-21.53 -19.30 -14.80 -12.10	-22.40 -22.40 -22.40 -22.40	-22.40 -22.40 -22.40 -22.40	-22.40 -22.40 -22.40 -22.18	-22.40 -22.40 -22.24 -21.75
Sorties	number	300		300		300	
Capacity	shots	3000		1200		1500	
Avg. Firing Rate	shots/sec	50		8.75		13.33	
Test Time	hours/axis	5.0		11.4		9.4	
Equations Used to Calculate Vibration Profiles							
Broadband ASD T_1, T_2, T_3 (Fig 519.8C-1)	g^2/Hz	$10 \log_{10} (T_i)$ dB = $10 \log_{10}$ (N x F1 x E) +H +M +W +J +B_j -53 dB (referred to 1 g^2/Hz) [p. 519.8C-3, Equation C-1]					
Broadband Breakpoints (Fig 519.8C-1)	Hz	30 (T_1), 300 (T_2), 600 (T_3), 700 (T_3), 2000 (T_1)					
Sine Tone ASD P_1, P_2, P_3, P_4 (Fig 519.8C-1)	g^2/Hz	$10 \log_{10} (P_i)$ dB = $10 \log_{10} (T_3)$ +K_i +17 dB [p. 519.8C-3, Equation C-2]					
Sine Tone Amp. (Table 514.8D-II)	g_{PEAK}	Si = 1.414 x $\sqrt{}$ (BWF x F_i x P_i)					
BWF	octave	$1/6^{th}$ octave = 0.115					

NOTES: 1/ All table and figure references above are to MIL-STD-810H(C1).

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TABLE K-II. H-60 gunfire vibration – engine/drivetrain location.

(Referenced by: K.3)

		H-60 Gunfire Vibration at Engine/Drivetrain Location - Notional						
		Broadband ASD (g^2/Hz)				Sine Amplitude (g_{PEAK})		
		M134	M240	M2		M134	M240	M2
		Test Duration: 8 hours per axis				Test Duration: 4 hours per axis @ each frequency		
	Freq (Hz)				Freq (Hz)			
1	30	0.00375	0.00117	0.00313	7.5			0.188
2	300	0.03342	0.01043	0.02790	10.0			0.217
3	600	0.08394	0.02620	0.07008	11.7		0.143	
4	700	0.08394	0.02620	0.07008	15.0		0.162	
5	2000	0.00375	0.00117	0.00313	15.0			0.265
6					20.0			0.306
7					22.5			0.325
8					23.3		0.202	
9					30.0			0.375
10					30.0			0.375
11					30.0		0.229	
12					33.3	0.428		
13					35.0		0.247	
14					40.0			0.433
15					45.0		0.286	
16					46.7		0.293	
17					60.0		0.349	
18					66.7	0.668		
19					66.7	0.668		
20					100.0	0.916		
21					133.3	1.222		
22					133.3	1.222		
23					200.0	2.513		
24					266.6	3.959		

NOTES: 1/ Test duration is 8 hours per axis broadband, and 4 hours/axis at each sine tone to account for the dual firing rate for each gun.

2/ When combining profiles to account for simultaneous gun usage, in this example the M240 broadband can be ignored as well as several of the sine tones because of low amplitude as compared to other identical or close-proximity frequencies (see G.3.3).

3/ For the 10-minute gunfire 'functional' test (which can be executed simultaneously with the 4-hour vibration functional test of 5.4.3.2.1 or 5.4.3.2.2) the ASD and sine amplitude values would not be those listed above. In this example, they would be the values derived from the calculations using the TABLE K-I estimation parameters, before any adjustment for test time changes. The estimated values are 'functional' values and the 10-minute test must use those values unmodified.

Yellow = Low firing rate
Blue = High firing rate

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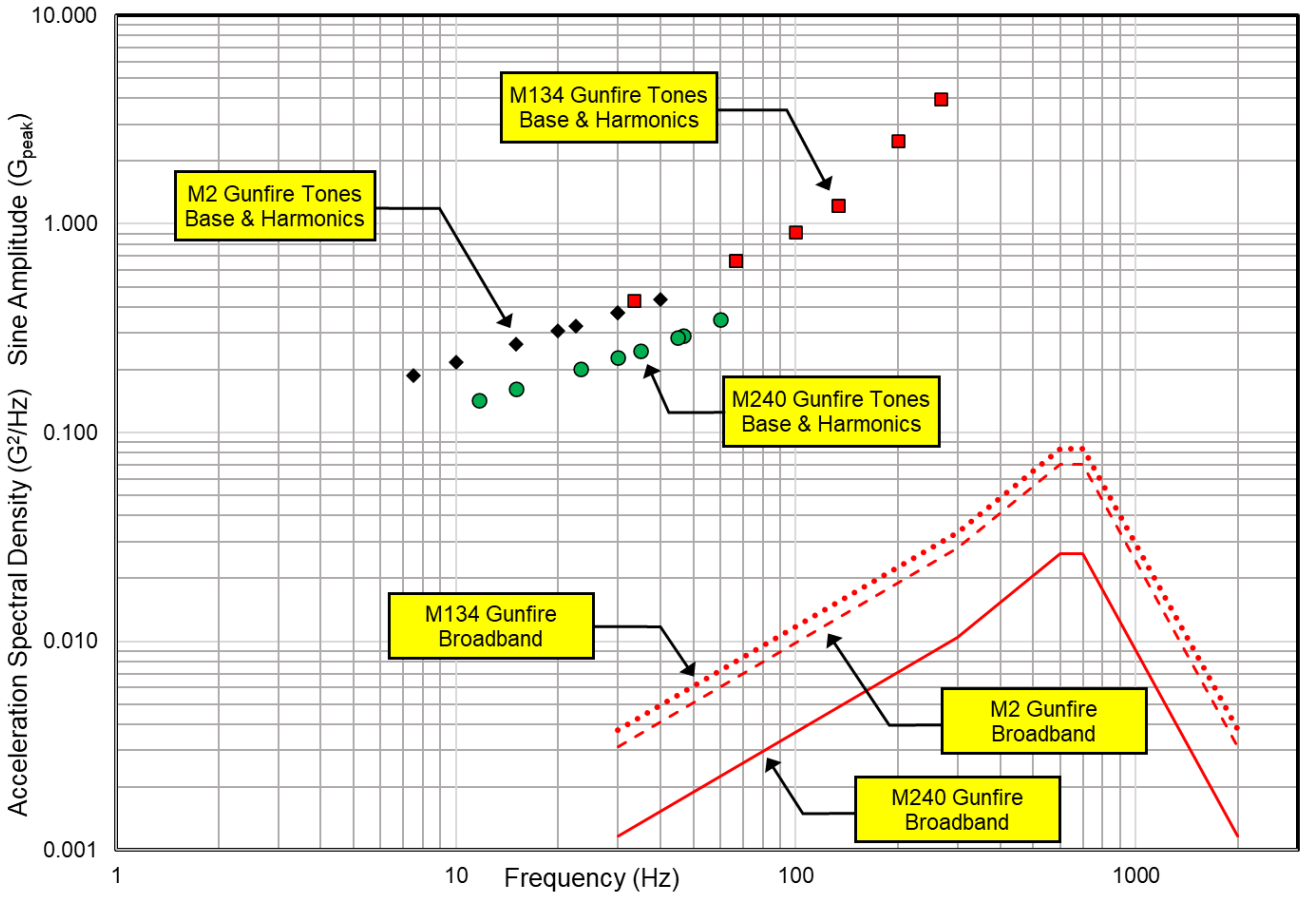


FIGURE K-2. H-60 gunfire vibration – engine/drivetrain location.
(Referenced by: K.3)

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REMARKS

In preparing this standard, every attempt has been made to cover most qualification requirements for components typical of engine control systems and accessories as used on Army Aviation assets. The DEVCOM AvMC has incorporated relevant experience gained over the past 30 years in qualifying control systems and individual accessory components for many engine applications and their derivatives, including the T800, T700, T55, Model 250, T407, YT706, and more recently the T901.

As engine designs and their control systems evolve, tests designed to qualify the components for initial production fielding have sometimes been found to be inadequate because of the component implementation itself (for example, a fuel metering unit with active electronics) or because the test itself has been found to insufficiently stress the design, resulting in the need for component improvement programs to meet satisfactory reliability. This document is the result of this evolution, with testing requirements expanded significantly beyond typical AV-E-8593 or MIL-STD-810H(C1) requirements.

Given the DEVCOM AvMC's history with continually evolving qualification tests based on real-world experience, it should be no surprise if the next engine control system or component to require qualification will have imposed on it modified or additional requirements beyond this standard. This document is meant to be a comprehensive framework upon which a basic qualification plan/specification can be built, allowing preparers/contractors to concentrate on designing or modifying tests that are capable of stressing those aspects of a system or component which may be unique.

Users of this document are cautioned that TABLE III, and the other similar tables that identify tests to be executed by component type, are not to be followed blindly. The DEVCOM AvMC has, to the best of its ability, attempted to define a comprehensive baseline of required testing by component type, but the evolving nature of engine controls and accessories, plus deployment of engines on new platforms utilizing new system architectures, means that some components may not fall neatly into one of the defined types.

Consequently, additional tests may be required beyond those marked by 'X' in the tables. Ultimately, it is the design of the component and the technology utilized that 'selects' the test; that is, what tests, in the suite of those available, stress the failure modes of the component? Is a test incapable of causing unit failure, and therefore not cost-effective to be conducted? Have historical component designs been subjected to certain tests without ever having had a failure provoked? These are the types of questions that the DEVCOM AvMC has asked itself when preparing the tables herein. And they are the same questions that users of this document should ask themselves when preparing an AQP or an AQS.

The authors hope that military, civil, and industry users alike will find this document to be of substantial benefit when developing or responding to qualification requirements for engine controls and accessories. However, we caution everyone to not fall into the trap of blind adherence to the requirements herein. This document is not a substitute for the use of sound engineering judgement or practical considerations in testing.

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CONCLUDING MATERIAL

Custodians:

Army – AV

Preparing Activity:

Army – AV
(Project ENVR-2025-019)

Review Activities:

Army – AM, AVN, TE
Navy – AS
Air Force – 11, 24
FAA – ACO

NOTES: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil/. Special thanks to K A Fry, DEVCOM AvMC, for initiating the creation of this standard.

Code	Government Component
ACO	FAA Aeronautical Center
AS	Naval Air Systems Command
AV	Combat Capabilities Development Command, Aviation and Missile Center
AVN	US Army, PEO Aviation
11	Air Force Life Cycle Management Center – Aircraft Systems
24	Air Force Test Center (AFTC)

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